

# Differentiation 2

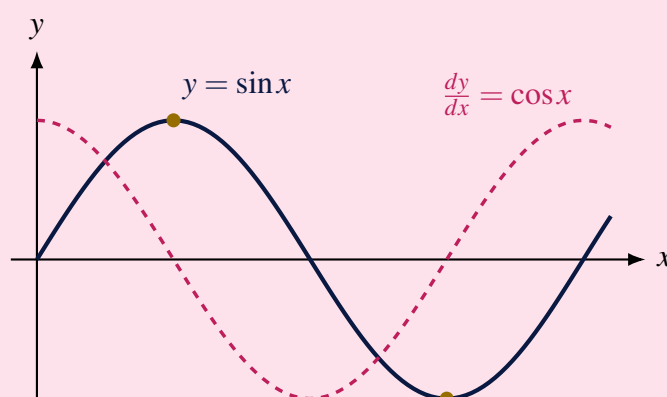
## Cambridge IGCSE Additional Mathematics 0606

### Notes · Questions · Worked Answers

Power rule · Chain, product & quotient rules · Tangents & normals

Small changes · Connected rates of change · Second derivative

Practical max & min ·  $e^x$ ,  $\ln x$ ,  $\sin x$ ,  $\cos x$ ,  $\tan x$  · Past papers 2016–2020



### How to use this booklet

- Read the pink **Key Notes** first. The **Index** lists every question, its source and its page.
- Sections 1–10: **Additional Maths book** (Ex 12.1–12.10, Ch 14) and the past-paper practice set, grouped by topic.
- Section 11: the book's exam exercises (Ex 12.11, 14.5). Sections 12–13: Cambridge 0606 **past papers 2016–2020, most recent first**; the session, paper and question number are shown above each question.
-  No calculator = no calculator (Paper 1, as in the current exam)  Calculator = calculator (Paper 2).
- Each white **Question** page is followed by yellow **Answer** page(s). Past-paper answers are the official mark schemes.

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# Key Notes

## 1. Chain, product and quotient rules

**Chain rule:**  $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$        $\frac{d}{dx}(2x+3)^8 = 8(2x+3)^7 \times 2 = 16(2x+3)^7$   
(power down, reduce the bracket's power by 1, multiply by the derivative of the inside)

**Product rule:**  $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$        $\frac{d}{dx}[x^2(x+1)^3] = 3x^2(x+1)^2 + 2x(x+1)^3 = x(x+1)^2(5x+2)$

**Quotient rule:**  $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$        $\frac{d}{dx}\left(\frac{x+3}{x-1}\right) = \frac{(x-1) - (x+3)}{(x-1)^2} = \frac{-4}{(x-1)^2}$

Factorise answers fully (take out common brackets) — it makes solving  $\frac{dy}{dx} = 0$  easy. Normal gradient  $= -\frac{1}{m}$ .

## 2. Exponentials, logarithms and trig (angles in RADIANS)

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(e^{ax+b}) = ae^{ax+b}$$

$$\frac{d}{dx}(e^{f(x)}) = f'(x)e^{f(x)}$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx} \ln(ax+b) = \frac{a}{ax+b}$$

$$\frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)}$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx} \sin(ax+b) = a \cos(ax+b) \quad \frac{d}{dx} \cos(ax+b) = -a \sin(ax+b) \quad \frac{d}{dx} \tan(ax+b) = a \sec^2(ax+b)$$

• Use log laws **before** differentiating:  $\ln \frac{(2x+1)^3}{x} = 3 \ln(2x+1) - \ln x$ .       $\log_a x = \frac{\ln x}{\ln a}$ .

•  $\sin^3 x = (\sin x)^3$ : chain rule gives  $3 \sin^2 x \cos x$ .       $e^{\ln a} = a$ ,  $\ln e^k = k$ .

## 3. Small increments and approximations

$$\delta y \approx \frac{dy}{dx} \times \delta x$$

$y = x^3 + x^2$ ,  $x$  increases from 2 to 2.05:  $\frac{dy}{dx} = 3x^2 + 2x = 16$ ,  $\delta y \approx 16 \times 0.05 = 0.8$ .

“from  $a$  to  $a+p$ ”  $\Rightarrow$  answer in terms of  $p$ :  $\delta y \approx \left(\frac{dy}{dx}\right)_{x=a} \times p$ . A negative answer means a decrease.

#### 4. Connected rates of change

Link the rates with the chain rule:  $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ ,  $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{dV/dt}{dV/dh}$

1. Write the formula linking the two variables (e.g.  $V = \frac{4}{3}\pi r^3$ ) and differentiate it.
2. Write down the given rate with the correct sign (**decreasing**  $\Rightarrow$  **negative**).
3. Substitute the value at the instant asked (find  $r$  from  $V$  first if needed).

#### 5. Stationary points and the second derivative

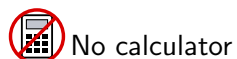
$\frac{dy}{dx} = 0$  at a stationary point.  $\frac{d^2y}{dx^2} < 0$ : **maximum**;  $\frac{d^2y}{dx^2} > 0$ : **minimum**.

- Unknown constants: “stationary point at  $(1, 3)$ ” gives **two** equations:  $\frac{dy}{dx} = 0$  at  $x = 1$  and  $y = 3$  at  $x = 1$ .
- $e^x > 0$  always, so  $e^x(x + 1) = 0 \Rightarrow x = -1$  only. For trig, find all solutions in the given range.
- Gradient is a minimum/maximum where  $\frac{d^2y}{dx^2} = 0$ .

#### 6. Practical maximum and minimum problems

1. Use the given fact (fixed volume, perimeter, area) to write one variable in terms of the other.
2. Substitute to get the quantity as a function of **one** variable — usually a “show that”.
3. Differentiate, set  $= 0$ , solve; then use  $\frac{d^2A}{dx^2}$  to say whether it is a maximum or minimum.
4. Answer the question asked: the stationary **value** (e.g.  $A = 432$ ) and/or the dimensions.

Useful: sector arc  $r\theta$ , area  $\frac{1}{2}r^2\theta$ ; cylinder  $V = \pi r^2h$ ,  $S = 2\pi r^2 + 2\pi rh$ ; cone  $V = \frac{1}{3}\pi r^2h$ ; sphere  $V = \frac{4}{3}\pi r^3$ .

**1**

No calculator

**Add Maths book Ex 12.1 • Q1**Differentiate with respect to  $x$ .

**a**  $x^4$

**b**  $x^9$

**c**  $x^{-3}$

**d**  $x^{-6}$

**e**  $\frac{1}{x}$

**f**  $\frac{1}{x^5}$

**g**  $\sqrt{x}$

**h**  $\sqrt{x^5}$

**i**  $x^{-\frac{1}{5}}$

**j**  $x^{\frac{1}{3}}$

**k**  $\sqrt[3]{x^2}$

**l**  $\frac{1}{\sqrt{x}}$

**m**  $x$

**n**  $x^{\frac{3}{2}}$

**o**  $\sqrt[3]{x^5}$

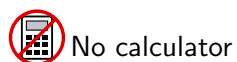
**p**  $x^2 \times x^4$

**q**  $x^2 \times x$

**r**  $\frac{x^4}{x^2}$

**s**  $\frac{x}{\sqrt{x}}$

**t**  $\frac{x\sqrt{x}}{x^3}$

**2**

No calculator

**Add Maths book Ex 12.1 • Q2**Differentiate with respect to  $x$ .

**a**  $2x^3 - 5x + 4$

**b**  $8x^5 - 3x^2 - 2$

**c**  $7 - 2x^3 + 4x$

**d**  $3x^2 + \frac{2}{x} - \frac{1}{x^2}$

**e**  $2x - \frac{1}{x} - \frac{1}{\sqrt{x}}$

**f**  $\frac{x+5}{\sqrt{x}}$

**g**  $\frac{x^2 - 3}{x}$

**h**  $\frac{5x^2 - \sqrt{x}}{x}$

**i**  $\frac{x^2 - x - 1}{\sqrt{x}}$

**j**  $5x^2(x+1)$

**k**  $x^{-2}(2x-5)$

**l**  $\frac{1}{x}(x^2 - 2)$

**m**  $(3x+1)^2$

**n**  $(1-x^3)^2$

**o**  $(2x-1)(3x+4)$

**1** Add Maths book Ex 12.1 · Q1

Write in index form first, e.g.  $\frac{1}{x} = x^{-1}$ ,  $\sqrt{x} = x^{\frac{1}{2}}$ .

**a**  $4x^3$

**f**  $-5x^{-6}$

**k**  $\frac{2}{3}x^{-\frac{1}{3}}$

**p**  $6x^5$

**b**  $9x^8$

**g**  $\frac{1}{2}x^{-\frac{1}{2}}$

**l**  $-\frac{1}{2}x^{-\frac{3}{2}}$

**q**  $3x^2$

**c**  $-3x^{-4}$

**h**  $\frac{5}{2}x^{\frac{3}{2}}$

**m**  $1$

**r**  $2x$

**d**  $-6x^{-7}$

**i**  $-\frac{1}{5}x^{-\frac{6}{5}}$

**n**  $\frac{3}{2}x^{\frac{1}{2}}$

**s**  $\frac{1}{2}x^{-\frac{1}{2}}$

**e**  $-x^{-2}$

**j**  $\frac{1}{3}x^{-\frac{2}{3}}$

**o**  $\frac{5}{3}x^{\frac{2}{3}}$

**t**  $-\frac{3}{2}x^{-\frac{5}{2}}$

**2** Add Maths book Ex 12.1 · Q2

**a**  $6x^2 - 5$

**f**  $\frac{1}{2}x^{-\frac{1}{2}} - \frac{5}{2}x^{-\frac{3}{2}}$

**k**  $-2x^{-2} + 10x^{-3}$

**b**  $40x^4 - 6x$

**g**  $1 + 3x^{-2}$

**l**  $1 + 2x^{-2}$

**c**  $-6x^2 + 4$

**h**  $5 + \frac{1}{2}x^{-\frac{3}{2}}$

**m**  $18x + 6$

**d**  $6x - 2x^{-2} + 2x^{-3}$

**i**  $\frac{3}{2}x^{\frac{1}{2}} - \frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}x^{-\frac{3}{2}}$

**n**  $-6x^2 + 6x^5$

**e**  $2 + x^{-2} + \frac{1}{2}x^{-\frac{3}{2}}$

**j**  $15x^2 + 10x$

**o**  $12x + 5$

e.g. (f)  $\frac{x+5}{\sqrt{x}} = x^{\frac{1}{2}} + 5x^{-\frac{1}{2}}$ ; (m)  $(3x+1)^2 = 9x^2 + 6x + 1$ .



**3**

No calculator

**Add Maths book Ex 12.1 · Q3**

Find the value of  $\frac{dy}{dx}$  at the given point on the curve.

**a**  $y = 3x^2 - 4$  at the point  $(1, -1)$

**b**  $y = 4 - 2x^2$  at the point  $(-1, 2)$

**c**  $y = 2 + \frac{8}{x}$  at the point  $(-2, -2)$

**d**  $y = 5x^3 - 2x^2 - 3$  at the point  $(0, -3)$

**e**  $y = \frac{x+5}{x}$  at the point  $(5, 2)$

**f**  $y = \frac{x-3}{\sqrt{x}}$  at the point  $(9, 2)$

**4**

No calculator

**Add Maths book Ex 12.1 · Q4**

Find the coordinates of the point on the curve  $y = 2x^2 - x - 1$  at which the gradient is 7.

**5**

No calculator

**Add Maths book Ex 12.1 · Q5**

Find the gradient of the curve  $y = \frac{x-4}{x}$  at the point where the curve crosses the  $x$ -axis.

**3** Add Maths book Ex 12.1 · Q3

**a**  $6x \rightarrow \boxed{6}$

**c**  $-8x^{-2} \rightarrow \boxed{-2}$

**e**  $-5x^{-2} \rightarrow \boxed{-0.2}$

**b**  $-4x \rightarrow \boxed{4}$

**d**  $15x^2 - 4x \rightarrow \boxed{0}$

**f**  $\frac{1}{2}x^{-\frac{1}{2}} + \frac{3}{2}x^{-\frac{3}{2}} \rightarrow \boxed{\frac{2}{9}}$

**4** Add Maths book Ex 12.1 · Q4

$4x - 1 = 7 \Rightarrow x = 2, y = 8 - 2 - 1 = 5: \boxed{(2, 5)}$

**5** Add Maths book Ex 12.1 · Q5

$y = 1 - 4x^{-1}$ ; crosses  $x$ -axis at  $x = 4$ .  $\frac{dy}{dx} = 4x^{-2} = \frac{4}{16} = \boxed{0.25}$

6



No calculator

Add Maths book Ex 12.1 · Q6

Find the gradient of the curve  $y = x^3 - 2x^2 + 5x - 3$  at the point where the curve crosses the  $y$ -axis.

7



No calculator

Add Maths book Ex 12.1 · Q7

The curve  $y = 2x^2 + 7x - 4$  and the line  $y = 5$  meet at the points  $P$  and  $Q$ . Find the gradient of the curve at the point  $P$  and at the point  $Q$ .

8



No calculator

Add Maths book Ex 12.1 · Q8

The curve  $y = ax^2 + bx$  has gradient 8 when  $x = 2$  and has gradient  $-10$  when  $x = -1$ .

Find the value of  $a$  and the value of  $b$ .

9



No calculator

Add Maths book Ex 12.1 · Q9

The gradient of the curve  $y = ax + \frac{b}{x}$  at the point  $(-1, -3)$  is  $-7$ .

Find the value of  $a$  and the value of  $b$ .

**6** Add Maths book Ex 12.1 · Q6

$$\frac{dy}{dx} = 3x^2 - 4x + 5 = \boxed{5} \text{ at } x = 0$$

**7** Add Maths book Ex 12.1 · Q7

$$2x^2 + 7x - 9 = 0 \Rightarrow (2x + 9)(x - 1) = 0, x = 1 \text{ or } -4.5. \quad \frac{dy}{dx} = 4x + 7:$$
$$\boxed{11} \text{ and } \boxed{-11}$$

**8** Add Maths book Ex 12.1 · Q8

$$\frac{dy}{dx} = 2ax + b: 4a + b = 8, -2a + b = -10 \Rightarrow 6a = 18: \boxed{a = 3, b = -4}$$

**9** Add Maths book Ex 12.1 · Q9

$$\text{Point: } -a - b = -3 \Rightarrow a + b = 3. \quad \frac{dy}{dx} = a - bx^{-2}: a - b = -7.$$
$$\boxed{a = -2, b = 5}$$

10



No calculator

Add Maths book Ex 12.1 · Q10

Find the coordinates of the points on this curve  $y = \frac{x^3}{3} - \frac{5x^2}{2} + 6x - 1$  where the gradient is 2.

11



No calculator

Add Maths book Ex 12.1 · Q11

The curve  $y = \frac{1}{3}x^3 - 2x^2 - 8x + 5$  and the line  $y = x + 5$  meet at the points  $A$ ,  $B$  and  $C$ .

- a** Find the coordinates of the points  $A$ ,  $B$  and  $C$ .
- b** Find the gradient of the curve at the points  $A$ ,  $B$  and  $C$ .

12



No calculator

Add Maths book Ex 12.1 · Q12

$$y = 4x^3 + 3x^2 - 6x - 1$$

**a** Find  $\frac{dy}{dx}$ .

**b** Find the range of values of  $x$  for which  $\frac{dy}{dx} \geq 0$ .

**10** Add Maths book Ex 12.1 · Q10

$$x^2 - 5x + 6 = 2 \Rightarrow (x - 1)(x - 4) = 0: \boxed{\left(1, 2\frac{5}{6}\right), \left(4, 4\frac{1}{3}\right)}$$

**11** Add Maths book Ex 12.1 · Q11

$$(a) \frac{1}{3}x^3 - 2x^2 - 9x = 0 \Rightarrow x(x - 9)(x + 3) = 0: \boxed{(-3, 2), (0, 5), (9, 14)}$$

$$(b) \frac{dy}{dx} = x^2 - 4x - 8: \boxed{13, -8, 37}$$

**12** Add Maths book Ex 12.1 · Q12

$$(a) \boxed{12x^2 + 6x - 6} \quad (b) 6(2x - 1)(x + 1) \geq 0: \boxed{x \leq -1 \text{ or } x \geq \frac{1}{2}}$$

13



No calculator

Add Maths book Ex 12.1 · Q13

$$y = x^3 + x^2 - 16x - 16$$

**a** Find  $\frac{dy}{dx}$ .

**b** Find the range of values of  $x$  for which  $\frac{dy}{dx} \leq 0$ .

14



No calculator

Add Maths book Ex 12.1 · Q14

A curve has equation  $y = x^5 - 5x^3 + 25x^2 + 145x + 10$ . Show that the gradient of the curve is never negative.

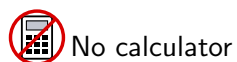
**13** *Add Maths book Ex 12.1 · Q13*

(a)  $3x^2 + 2x - 16$       (b)  $(3x + 8)(x - 2) \leq 0$ :  $-\frac{8}{3} \leq x \leq 2$

**14** *Add Maths book Ex 12.1 · Q14*

$$\begin{aligned}\frac{dy}{dx} &= 5x^4 - 15x^2 + 50x + 145 = 5(x^4 - 3x^2 + 10x + 29) \\ &= 5[(x^2 - 2)^2 + (x + 5)^2]. \quad \text{Squares are } \geq 0 \text{ and cannot both be 0, so} \\ \frac{dy}{dx} &> 0.\end{aligned}$$



**15**

No calculator

**Add Maths book Ex 12.2 · Q1**Differentiate with respect to  $x$ .

**a**  $(x + 2)^9$

**b**  $(3x - 1)^7$

**c**  $(1 - 5x)^6$

**d**  $\left(\frac{1}{2}x - 7\right)^4$

**e**  $\frac{(2x + 1)^6}{3}$

**f**  $2(x - 4)^6$

**g**  $6(5 - x)^5$

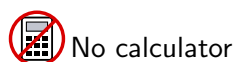
**h**  $\frac{1}{2}(2x + 5)^8$

**i**  $(x^2 + 2)^4$

**j**  $(1 - 2x^2)^7$

**k**  $(x^2 - 3x)^5$

**l**  $\left(x^2 + \frac{2}{x}\right)^4$

**16**

No calculator

**Add Maths book Ex 12.2 · Q2**Differentiate with respect to  $x$ .

**a**  $\frac{1}{(x + 4)}$

**b**  $\frac{3}{(2x - 1)}$

**c**  $\frac{5}{(2 - 3x)}$

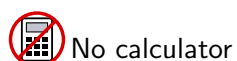
**d**  $\frac{16}{(2x^2 - 5)}$

**e**  $\frac{4}{(x^2 - 2x)}$

**f**  $\frac{1}{(x - 1)^5}$

**g**  $\frac{2}{(5x + 1)^3}$

**h**  $\frac{1}{2(3x - 2)^4}$

**17**

No calculator

**Add Maths book Ex 12.2 · Q3**Differentiate with respect to  $x$ .

**a**  $\sqrt{x + 2}$

**b**  $\sqrt{5x - 1}$

**c**  $\sqrt{2x^2 - 3}$

**d**  $\sqrt{x^3 + 2x}$

**e**  $\sqrt[3]{3 - 2x}$

**f**  $4\sqrt{2x - 1}$

**g**  $\frac{1}{\sqrt{3x - 1}}$

**h**  $\frac{3}{\sqrt[3]{2 - 5x}}$

**15** Add Maths book Ex 12.2 · Q1

$$\frac{d}{dx}(ax + b)^n = an(ax + b)^{n-1}$$

**a**  $9(x + 2)^8$

**e**  $4(2x + 1)^5$

**i**  $8x(x^2 + 2)^3$

**b**  $21(3x - 1)^6$

**f**  $12(x - 4)^5$

**j**  $-28x(1 - 2x^2)^6$

**c**  $-30(1 - 5x)^5$

**g**  $-30(5 - x)^4$

**k**  $5(2x - 3)(x^2 - 3x)^4$

**d**  $2(\frac{1}{2}x - 7)^3$

**h**  $8(2x + 5)^7$

**l**  $8(x - \frac{1}{x^2})(x^2 + \frac{2}{x})^3$

**16** Add Maths book Ex 12.2 · Q2

**a**  $\frac{-1}{(x + 4)^2}$

**c**  $\frac{15}{(2 - 3x)^2}$

**e**  $\frac{-8(x - 1)}{(x^2 - 2x)^2}$

**g**  $\frac{-30}{(5x + 1)^4}$

**b**  $\frac{-6}{(2x - 1)^2}$

**d**  $\frac{-64x}{(2x^2 - 5)^2}$

**f**  $\frac{-5}{(x - 1)^6}$

**h**  $\frac{-6}{(3x - 2)^5}$

**17** Add Maths book Ex 12.2 · Q3

**a**  $\frac{1}{2\sqrt{x+2}}$

**d**  $\frac{3x^2 + 2}{2\sqrt{x^3 + 2x}}$

**g**  $\frac{-3}{2(3x - 1)^{\frac{3}{2}}}$

**b**  $\frac{5}{2\sqrt{5x-1}}$

**e**  $\frac{-2}{3(3 - 2x)^{\frac{2}{3}}}$

**h**  $\frac{5}{(2 - 5x)^{\frac{4}{3}}}$

**c**  $\frac{2x}{\sqrt{2x^2 - 3}}$

**f**  $\frac{4}{\sqrt{2x-1}}$

18



No calculator

Add Maths book Ex 12.2 · Q4

Find the gradient of the curve  $y = (2x - 5)^4$  at the point  $(3, 1)$ .

19



No calculator

Add Maths book Ex 12.2 · Q5

Find the gradient of the curve  $y = \frac{8}{(x - 2)^2}$  at the point where the curve crosses the  $y$ -axis.

20



No calculator

Add Maths book Ex 12.2 · Q6

Find the gradient of the curve  $y = x + \frac{4}{x - 5}$  at the points where the curve crosses the  $x$ -axis.

21



No calculator

Add Maths book Ex 12.2 · Q7

Find the coordinates of the point on the curve  $y = \sqrt{x^2 - 6x + 13}$  where the gradient is 0.

**18** Add Maths book Ex 12.2 · Q4

$$\frac{dy}{dx} = 8(2x - 5)^3 = \boxed{8} \text{ at } x = 3$$

**19** Add Maths book Ex 12.2 · Q5

$$y = 8(x - 2)^{-2}, \quad \frac{dy}{dx} = -16(x - 2)^{-3}. \quad \text{At } x = 0: \frac{-16}{-8} = \boxed{2}$$

**20** Add Maths book Ex 12.2 · Q6

$$x + \frac{4}{x - 5} = 0 \Rightarrow x^2 - 5x + 4 = 0, \quad x = 1, 4. \quad \frac{dy}{dx} = 1 - \frac{4}{(x - 5)^2}: \quad \boxed{0.75} \text{ and } \boxed{-3}$$

**21** Add Maths book Ex 12.2 · Q7

$$\frac{dy}{dx} = \frac{2x - 6}{2\sqrt{x^2 - 6x + 13}} = 0 \Rightarrow x = 3, \quad y = \sqrt{4} = 2: \quad \boxed{(3, 2)}$$

**22**

No calculator

**Add Maths book Ex 12.2 · Q8**

The curve  $y = \frac{a}{\sqrt{bx+1}}$  passes through the point (1, 4) and has gradient  $-\frac{3}{2}$  at this point.

Find the value of  $a$  and the value of  $b$ .

**23**

No calculator

**Add Maths book Ex 12.3 · Q1**

Use the product rule to differentiate each of the following with respect to  $x$ :

**a**  $x(x+4)$

**b**  $2x(3x+5)$

**c**  $x(x+2)^3$

**d**  $x^2(x-1)^3$

**e**  $x\sqrt{x-5}$

**f**  $(x+2)\sqrt{x}$

**g**  $x^2\sqrt{x+3}$

**h**  $\sqrt{x}(3-x^2)^3$

**i**  $(2x+1)(x^2+5)$

**j**  $(x+4)(x-3)^3$

**k**  $(x-1)^2(x+2)^2$

**l**  $(2x+1)^3(x-3)^4$

**24**

No calculator

**Add Maths book Ex 12.3 · Q2**

Find the gradient of the curve  $y = x^2\sqrt{x+2}$  at the point (2, 8).

**25**

No calculator

**Add Maths book Ex 12.3 · Q3**

Find the gradient of the curve  $y = (x-1)^3(x+3)^2$  at the point where  $x = 2$ .

**22** Add Maths book Ex 12.2 · Q8

$$\frac{a}{\sqrt{b+1}} = 4. \quad \frac{dy}{dx} = -\frac{1}{2}ab(bx+1)^{-\frac{3}{2}} = -\frac{3}{2} \text{ at } x = 1.$$

$$\text{Sub } a = 4\sqrt{b+1}: \frac{-2b}{b+1} = -\frac{3}{2} \Rightarrow b = 3, a = 8. \quad \boxed{a = 8, b = 3}$$

**23** Add Maths book Ex 12.3 · Q1

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

$$\mathbf{a} \quad 2x + 4$$

$$\mathbf{b} \quad 12x + 10$$

$$\mathbf{c} \quad (x+2)^2(4x+2)$$

$$\mathbf{d} \quad x(x-1)^2(5x-2)$$

$$\mathbf{e} \quad \frac{3x-10}{2\sqrt{x-5}}$$

$$\mathbf{f} \quad \frac{3x+2}{2\sqrt{x}}$$

$$\mathbf{g} \quad \frac{5x^2 + 12x}{2\sqrt{x+3}}$$

$$\mathbf{h} \quad \frac{(3-x^2)^2(3-13x^2)}{2\sqrt{x}}$$

$$\mathbf{i} \quad 2(3x^2 + x + 5)$$

$$\mathbf{j} \quad (4x+9)(x-3)^2$$

$$\mathbf{k} \quad 2(x-1)(x+2)(2x+1)$$

$$\mathbf{l} \quad 14(x-1)(x-3)^3(2x+1)^2$$

**24** Add Maths book Ex 12.3 · Q2

$$\frac{dy}{dx} = 2x\sqrt{x+2} + \frac{x^2}{2\sqrt{x+2}}. \quad \text{At } x = 2: 8 + 1 = \boxed{9}$$

**25** Add Maths book Ex 12.3 · Q3

$$\frac{dy}{dx} = 3(x-1)^2(x+3)^2 + 2(x-1)^3(x+3). \quad \text{At } x = 2: 75 + 10 = \boxed{85}$$

**26**

No calculator

**Add Maths book Ex 12.3 · Q4**

Find the gradient of the curve  $y = (x + 2)(x - 5)^2$  at the points where the curve meets the  $x$ -axis.

**27**

No calculator

**Add Maths book Ex 12.3 · Q5**

Find the  $x$ -coordinate of the points on the curve  $y = (2x - 3)^3(x + 2)^4$  where the gradient is zero.

**28**

No calculator

**Add Maths book Ex 12.3 · Q6**

Find the  $x$ -coordinate of the point on the curve  $y = (x + 3)\sqrt{4 - x}$  where the gradient is zero.

**29**

No calculator

**Add Maths book Ex 12.4 · Q1**

Use the quotient rule to differentiate each of the following with respect to  $x$ :

**a**  $\frac{1 + 2x}{5 - x}$

**b**  $\frac{3x + 2}{x + 4}$

**c**  $\frac{x - 1}{3x + 4}$

**d**  $\frac{5x - 2}{3 - 8x}$

**e**  $\frac{x^2}{5x - 2}$

**f**  $\frac{x}{x^2 - 1}$

**g**  $\frac{5}{3x - 1}$

**h**  $\frac{x + 4}{x^2 - 2}$

**26** Add Maths book Ex 12.3 · Q4

$$\frac{dy}{dx} = (x-5)^2 + 2(x+2)(x-5) = (x-5)(3x-1). \quad \text{Meets } x\text{-axis at}$$

$$x = -2, 5: \boxed{49} \text{ and } \boxed{0}$$

**27** Add Maths book Ex 12.3 · Q5

$$\frac{dy}{dx} = 6(2x-3)^2(x+2)^4 + 4(2x-3)^3(x+2)^3 = 2(2x-3)^2(x+2)^3(7x) =$$

$$0$$

$$\boxed{x = -2, 0, 1.5}$$

**28** Add Maths book Ex 12.3 · Q6

$$\frac{dy}{dx} = \sqrt{4-x} - \frac{x+3}{2\sqrt{4-x}} = \frac{5-3x}{2\sqrt{4-x}} = 0 \Rightarrow \boxed{x = 1\frac{2}{3}}$$

**29** Add Maths book Ex 12.4 · Q1

$$\frac{d}{dx} \left( \frac{u}{v} \right) = \frac{vu' - uv'}{v^2}$$

$$\mathbf{a} \quad \frac{11}{(5-x)^2}$$

$$\mathbf{d} \quad \frac{-1}{(3-8x)^2}$$

$$\mathbf{g} \quad \frac{-15}{(3x-1)^2}$$

$$\mathbf{b} \quad \frac{10}{(x+4)^2}$$

$$\mathbf{e} \quad \frac{x(5x-4)}{(5x-2)^2}$$

$$\mathbf{h} \quad -\frac{x^2+8x+2}{(x^2-2)^2}$$

$$\mathbf{c} \quad \frac{7}{(3x+4)^2}$$

$$\mathbf{f} \quad -\frac{x^2+1}{(x^2-1)^2}$$



30



No calculator

Add Maths book Ex 12.4 · Q2

Find the gradient of the curve  $y = \frac{x+3}{x-1}$  at the point (2, 5).

31



No calculator

Add Maths book Ex 12.4 · Q3

Find the coordinates of the points on the curve  $y = \frac{x^2}{2x-1}$  where  $\frac{dy}{dx} = 0$ .

32



No calculator

Add Maths book Ex 12.4 · Q4

Find the gradient of the curve  $y = \frac{7x-2}{2x+3}$  at the point where the curve crosses the y-axis.

33



No calculator

Add Maths book Ex 12.4 · Q5

Differentiate with respect to  $x$ :

**a**  $\frac{\sqrt{x}}{2x+1}$

**b**  $\frac{x}{\sqrt{1-2x}}$

**c**  $\frac{x^2}{\sqrt{x^2+2}}$

**d**  $\frac{5\sqrt{x}}{3+x}$

**30** Add Maths book Ex 12.4 · Q2

$$\frac{dy}{dx} = \frac{(x-1) - (x+3)}{(x-1)^2} = \frac{-4}{(x-1)^2} = \boxed{-4} \text{ at } x = 2$$

**31** Add Maths book Ex 12.4 · Q3

$$\frac{dy}{dx} = \frac{2x(2x-1) - 2x^2}{(2x-1)^2} = \frac{2x(x-1)}{(2x-1)^2} = 0: \boxed{(0,0), (1,1)}$$

**32** Add Maths book Ex 12.4 · Q4

$$\frac{dy}{dx} = \frac{7(2x+3) - 2(7x-2)}{(2x+3)^2} = \frac{25}{(2x+3)^2} = \boxed{\frac{25}{9}} \text{ at } x = 0$$

**33** Add Maths book Ex 12.4 · Q5

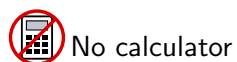
**a**  $\frac{1-2x}{2\sqrt{x}(2x+1)^2}$

**c**  $\frac{x(x^2+4)}{(x^2+2)^{\frac{3}{2}}}$

**b**  $\frac{1-x}{(1-2x)^{\frac{3}{2}}}$

**d**  $\frac{-5(x-3)}{2\sqrt{x}(x+3)^2}$

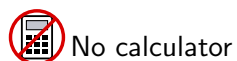
34



Add Maths book Ex 12.4 · Q6

Find the gradient of the curve  $y = \frac{x-2}{\sqrt{x+5}}$  at the point  $(-4, -6)$ .

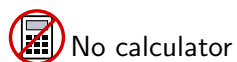
35



Add Maths book Ex 12.4 · Q7

Find the coordinates of the point on the curve  $y = \frac{2(x-5)}{\sqrt{x+1}}$  where the gradient is  $\frac{5}{4}$ .

36



Add Maths book Ex 12.4 · Q8

The line  $5x - 5y = 2$  intersects the curve  $x^2y - 5x + y + 2 = 0$  at three points.

- a Find the coordinates of the points of intersection.
- b Find the gradient of the curve at each of the points of intersection.

**34** Add Maths book Ex 12.4 · Q6

$$\frac{dy}{dx} = \frac{2(x+5) - (x-2)}{2(x+5)^{\frac{3}{2}}} = \frac{x+12}{2(x+5)^{\frac{3}{2}}} = \frac{8}{2} = \boxed{4} \text{ at } x = -4$$

**35** Add Maths book Ex 12.4 · Q7

$$\frac{dy}{dx} = \frac{x+7}{(x+1)^{\frac{3}{2}}} = \frac{5}{4}. \text{ Try } x = 3: \frac{10}{8} = \frac{5}{4} \checkmark y = \frac{2(-2)}{2} = -2: \boxed{(3, -2)}$$

**36** Add Maths book Ex 12.4 · Q8

Line:  $y = \frac{5x-2}{5}$ . Curve:  $y = \frac{5x-2}{x^2+1}$ . Equal  $\Rightarrow 5x-2 = 0$  or  $x^2+1 = 5$ .

(a)  $\boxed{(-2, -2.4), (0.4, 0), (2, 1.6)}$

(b)  $\frac{dy}{dx} = \frac{-5x^2+4x+5}{(x^2+1)^2}: \boxed{-\frac{23}{25}, \frac{125}{29}, -\frac{7}{25}}$

**37**

No calculator

**Add Maths book Ex 12.5 · Q1**Find the equation of the tangent to the curve at the given value of  $x$ .

**a**  $y = x^4 - 3$  at  $x = 1$

**b**  $y = x^2 + 3x + 2$  at  $x = -2$

**c**  $y = 2x^3 + 5x^2 - 1$  at  $x = 1$

**d**  $y = 5 + \frac{2}{x}$  at  $x = -2$

**e**  $y = (x - 3)(2x - 1)^2$  at  $x = 2$

**f**  $y = \frac{x}{x+1}$  at  $x = -3$

**38**

No calculator

**Add Maths book Ex 12.5 · Q2**Find the equation of the normal to the curve at the given value of  $x$ .

**a**  $y = x^2 + 5x$  at  $x = -1$

**b**  $y = 3x^2 - 4x + 1$  at  $x = 2$

**c**  $y = 5x^4 - 7x^2 + 2x$  at  $x = -1$

**d**  $y = 4 - \frac{2}{x^2}$  at  $x = -2$

**e**  $y = 2x(x - 3)^3$  at  $x = 2$

**f**  $y = \frac{x+2}{x-2}$  at  $x = 6$

**39**

No calculator

**Add Maths book Ex 12.5 · Q3**Find the equation of the tangent and the normal to the curve  $y = 5x - \frac{3}{x}$  at the point where  $x = 1$ .

**37** Add Maths book Ex 12.5 · Q1

Find  $y$  and  $m = \frac{dy}{dx}$  at the point, then  $y - y_1 = m(x - x_1)$ .

**a**  $(1, -2), m = 4$ :  $y = 4x - 6$

**d**  $(-2, 4), m = -\frac{1}{2}$ :  $y = -\frac{1}{2}x + 3$

**b**  $(-2, 0), m = -1$ :  $y = -x - 2$

**e**  $(2, -9), m = -3$ :  $y = -3x - 3$

**c**  $(1, 6), m = 16$ :  $y = 16x - 10$

**f**  $(-3, \frac{3}{2}), m = \frac{1}{4}$ :  $y = \frac{1}{4}x + 2\frac{1}{4}$

**38** Add Maths book Ex 12.5 · Q2

Normal gradient  $= -\frac{1}{m}$ .

**a**  $m = 3$ :  $y = -\frac{1}{3}x - 4\frac{1}{3}$

**d**  $m = -\frac{1}{2}$ :  $y = 2x + 7.5$

**b**  $m = 8$ :  $y = -\frac{1}{8}x + 5\frac{1}{4}$

**e**  $m = 10$ :  $y = -0.1x - 3.8$

**c**  $m = -4$ :  $y = \frac{1}{4}x - 3\frac{3}{4}$

**f**  $m = -\frac{1}{4}$ :  $y = 4x - 22$

**39** Add Maths book Ex 12.5 · Q3

$x = 1: y = 2, \frac{dy}{dx} = 5 + \frac{3}{x^2} = 8.$

Tangent  $y = 8x - 6$ ; normal  $y = -\frac{1}{8}x + \frac{17}{8}$

40



No calculator

Add Maths book Ex 12.5 · Q4

The normal to the curve  $y = x^3 - 2x + 1$  at the point  $(2, 5)$  intersects the  $y$ -axis at the point  $P$ .

Find the coordinates of  $P$ .

41



No calculator

Add Maths book Ex 12.5 · Q5

Find the equation of the tangent and the normal to the curve  $y = \frac{x-1}{\sqrt{x+4}}$  at the point where the curve intersects the  $y$ -axis.

42



No calculator

Add Maths book Ex 12.5 · Q6

The tangents to the curve  $y = x^2 - 5x + 4$ , at the points  $(1, 0)$  and  $(3, -2)$ , meet at the point  $Q$ .

Find the coordinates of  $Q$ .

43



No calculator

Add Maths book Ex 12.5 · Q7

The tangent to the curve  $y = 3x^2 - 10x - 8$  at the point  $P$  is parallel to the line  $y = 2x - 5$ .

Find the equation of the tangent at  $P$ .

**40** Add Maths book Ex 12.5 · Q4

$$\frac{dy}{dx} = 3x^2 - 2 = 10, \text{ normal gradient } -\frac{1}{10}: y - 5 = -\frac{1}{10}(x - 2). \quad x = 0:$$

$$\boxed{P(0, 5.2)}$$

**41** Add Maths book Ex 12.5 · Q5

$$x = 0: y = -\frac{1}{2}. \quad \frac{dy}{dx} = \frac{x+9}{2(x+4)^{\frac{3}{2}}} = \frac{9}{16}.$$

$$\text{Tangent } \boxed{y = \frac{9}{16}x - \frac{1}{2}}; \text{ normal } \boxed{y = -\frac{16}{9}x - \frac{1}{2}}$$

**42** Add Maths book Ex 12.5 · Q6

$$\frac{dy}{dx} = 2x - 5: \text{ tangents } y = -3x + 3 \text{ and } y = x - 5. \quad -3x + 3 = x - 5 \Rightarrow$$

$$\boxed{Q(2, -3)}$$

**43** Add Maths book Ex 12.5 · Q7

$$6x - 10 = 2 \Rightarrow x = 2, y = -16. \quad y + 16 = 2(x - 2) \Rightarrow \boxed{y = 2x - 20}$$



44



No calculator

Add Maths book Ex 12.5 · Q8

A curve has equation  $y = x^3 - x + 6$ .

**a** Find the equation of the tangent to this curve at the point  $P(-1, 6)$ .

The tangent at the point  $Q$  is parallel to the tangent at  $P$ .

**b** Find the coordinates of  $Q$ .

**c** Find the equation of the normal at  $Q$ .

45



No calculator

Add Maths book Ex 12.5 · Q9

A curve has equation  $y = 4 + (x - 1)^4$ .

The normal at the point  $P(1, 4)$  and the normal at the point  $Q(2, 5)$  intersect at the point  $R$ .

Find the coordinates of  $R$ .

46



No calculator

Add Maths book Ex 12.5 · Q10

A curve has equation  $y = (2 - \sqrt{x})^4$ .

The normal at the point  $P(1, 1)$  and the normal at the point  $Q(9, 1)$  intersect at the point  $R$ .

**a** Find the coordinates of  $R$ .

**b** Find the area of triangle  $PQR$ .

**44** Add Maths book Ex 12.5 · Q8

$$(a) 3x^2 - 1 = 2: \boxed{y = 2x + 8} \quad (b) 3x^2 - 1 = 2 \Rightarrow x = 1: \boxed{Q(1, 6)}$$

$$(c) \text{Normal gradient } -\frac{1}{2}: \boxed{y = -\frac{1}{2}x + 6\frac{1}{2}}$$

**45** Add Maths book Ex 12.5 · Q9

$$\frac{dy}{dx} = 4(x - 1)^3. \text{ At } P: 0, \text{ so normal is } x = 1. \text{ At } Q: 4, \text{ normal } y - 5 = -\frac{1}{4}(x - 2).$$

$$x = 1: y = 5.25: \boxed{R(1, 5.25)}$$

**46** Add Maths book Ex 12.5 · Q10

$$\frac{dy}{dx} = -\frac{2(2 - \sqrt{x})^3}{\sqrt{x}}: \text{ at } P: -2 \text{ (normal } \frac{1}{2}), \text{ at } Q: \frac{2}{3} \text{ (normal } -\frac{3}{2}).$$

$$\frac{1}{2}(x - 1) = -\frac{3}{2}(x - 9) \Rightarrow (a) \boxed{R(7, 4)} \quad (b) \frac{1}{2} \times 8 \times 3 = \boxed{12 \text{ units}^2}$$

47



No calculator

Add Maths book Ex 12.5 · Q11

A curve has equation  $y = \sqrt{x}(x - 2)^3$ .

The tangent at the point  $P(3, \sqrt{3})$  and the normal at the point  $Q(9, 1)$  intersect at the point  $R$ .

**a** Show that the equation of the tangent at the point  $P(3, \sqrt{3})$  is

$$y = \frac{19\sqrt{3}}{6}x - \frac{17\sqrt{3}}{2}.$$

**b** Find the equation of the normal at the point  $Q(9, 1)$ .

48



No calculator

Add Maths book Ex 12.5 · Q12

The equation of a curve is  $y = \frac{x^2}{x + 2}$ .

The tangent to the curve at the point where  $x = -3$  meets the  $y$ -axis at  $M$ .

The normal to the curve at the point where  $x = -3$  meets the  $x$ -axis at  $N$ .

Find the area of the triangle  $MNO$ , where  $O$  is the origin.

49



No calculator

Add Maths book Ex 12.5 · Q13

The equation of a curve is  $y = \frac{x - 3}{x + 2}$ .

The curve intersects the  $x$ -axis at the point  $P$ .

The normal to the curve at  $P$  meets the  $y$ -axis at the point  $Q$ .

Find the area of the triangle  $POQ$ , where  $O$  is the origin.

**47** Add Maths book Ex 12.5 · Q11

$$\frac{dy}{dx} = \frac{(x-2)^3}{2\sqrt{x}} + 3\sqrt{x}(x-2)^2$$

$$(a) x = 3: \frac{\sqrt{3}}{6} + 3\sqrt{3} = \frac{19\sqrt{3}}{6}; y - \sqrt{3} = \frac{19\sqrt{3}}{6}(x-3) \Rightarrow y = \frac{19\sqrt{3}}{6}x - \frac{17\sqrt{3}}{2} \checkmark$$

$$(b) x = 1: \text{gradient } -\frac{1}{2} + 3 = 2.5, \text{normal } -0.4: \boxed{y = -0.4x - 0.6}$$

(The point is  $Q(1, -1)$ , as in part b.)**48** Add Maths book Ex 12.5 · Q12

$$x = -3: y = -9, \frac{dy}{dx} = \frac{x^2 + 4x}{(x+2)^2} = -3.$$

Tangent  $y = -3x - 18$ :  $M(0, -18)$ . Normal  $y + 9 = \frac{1}{3}(x + 3)$ :  $N(24, 0)$ .

$$\text{Area} = \frac{1}{2} \times 18 \times 24 = \boxed{216 \text{ units}^2}$$

**49** *Add Maths book Ex 12.5 · Q13*

$P(3, 0)$ .  $\frac{dy}{dx} = \frac{5}{(x+2)^2} = \frac{1}{5}$ , normal gradient  $-5$ :  $y = -5x + 15$ ,  
 $Q(0, 15)$ .

$$\text{Area} = \frac{1}{2} \times 3 \times 15 = \boxed{22.5 \text{ units}^2}$$

50



Calculator

**Add Maths book Ex 12.6 · Q1**

Variables  $x$  and  $y$  are connected by the equation  $y = 2x^3 - 3x$ .  
Find the approximate change in  $y$  as  $x$  increases from 2 to 2.01.

51



Calculator

**Add Maths book Ex 12.6 · Q2**

Variables  $x$  and  $y$  are connected by the equation  $y = 5x^2 - \frac{8}{x^3}$ .  
Find the approximate change in  $y$  as  $x$  increases from 1 to 1.02.

52



No calculator

**Add Maths book Ex 12.6 · Q3**

Variables  $x$  and  $y$  are connected by the equation  $x^2y = 400$ .  
Find, in terms of  $p$ , the approximate change in  $y$  as  $x$  increases from 10 to  $10 + p$ , where  $p$  is small.

53



No calculator

**Add Maths book Ex 12.6 · Q4**

Variables  $x$  and  $y$  are connected by the equation  $y = \left(\frac{1}{3}x - 2\right)^6$ .  
Find, in terms of  $p$ , the approximate change in  $y$  as  $x$  increases from 9 to  $9 + p$ , where  $p$  is small.

**50** Add Maths book Ex 12.6 · Q1

$$\delta y \approx \frac{dy}{dx} \times \delta x. \quad \frac{dy}{dx} = 6x^2 - 3 = 21 \text{ at } x = 2; \quad \delta y \approx 21 \times 0.01 = \boxed{0.21}$$

**51** Add Maths book Ex 12.6 · Q2

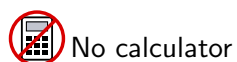
$$\delta y \approx \frac{dy}{dx} \times \delta x. \quad \frac{dy}{dx} = 10x + 24x^{-4} = 34 \text{ at } x = 1; \quad \delta y \approx 34 \times 0.02 = \boxed{0.68}$$

**52** Add Maths book Ex 12.6 · Q3

$$y = 400x^{-2}, \quad \frac{dy}{dx} = -800x^{-3} = -0.8 \text{ at } x = 10; \quad \delta y \approx \boxed{-0.8p}$$

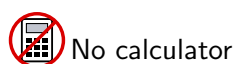
**53** Add Maths book Ex 12.6 · Q4

$$\frac{dy}{dx} = 6\left(\frac{1}{3}x - 2\right)^5 \cdot \frac{1}{3} = 2\left(\frac{1}{3}x - 2\right)^5 = 2 \text{ at } x = 9; \quad \delta y \approx \boxed{2p}$$

**54****Add Maths book Ex 12.6 · Q5**

A curve has equation  $y = (x + 1)(2x - 3)^4$ .

Find, in terms of  $p$ , the approximate change in  $y$  as  $x$  increases from 2 to  $2 + p$ , where  $p$  is small.

**55****Add Maths book Ex 12.6 · Q6**

A curve has equation  $y = (x - 2)\sqrt{2x + 1}$ .

Find, in terms of  $p$ , the approximate change in  $y$  as  $x$  increases from 4 to  $4 + p$ , where  $p$  is small.

**56****Add Maths book Ex 12.6 · Q7**

The periodic time,  $T$  seconds, for a pendulum of length  $L$  cm is

$$T = 2\pi\sqrt{\frac{L}{10}}.$$

Find the approximate increase in  $T$  as  $L$  increases from 40 to 41.



**54** *Add Maths book Ex 12.6 · Q5*

$$\frac{dy}{dx} = (2x - 3)^4 + 8(x + 1)(2x - 3)^3 = 1 + 24 = 25 \text{ at } x = 2; \delta y \approx \boxed{25p}$$

**55** *Add Maths book Ex 12.6 · Q6*

$$\frac{dy}{dx} = \sqrt{2x + 1} + \frac{x - 2}{\sqrt{2x + 1}} = 3 + \frac{2}{3} = \frac{11}{3} \text{ at } x = 4; \delta y \approx \boxed{\frac{11}{3}p}$$

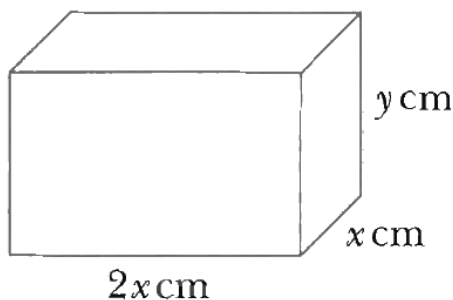
**56** *Add Maths book Ex 12.6 · Q7*

$$T = \frac{2\pi}{\sqrt{10}}L^{\frac{1}{2}}, \quad \frac{dT}{dL} = \frac{\pi}{\sqrt{10L}} = \frac{\pi}{20} \text{ at } L = 40; \delta T \approx \boxed{\frac{\pi}{20}} \approx 0.157 \text{ s}$$

57

 No calculator

Add Maths book Ex 12.6 · Q8



The volume of the solid cuboid is  $360 \text{ cm}^3$  and the surface area is  $A \text{ cm}^2$ .

- a Express  $y$  in terms of  $x$ .
- b Show that  $A = 4x^2 + \frac{1080}{x}$ .
- c Find, in terms of  $p$ , the approximate change in  $A$  as  $x$  increases from 2 to  $2 + p$ , where  $p$  is small. State whether the change is an increase or decrease.

58

 No calculator

Add Maths book Ex 14.4 · Q6

Variables  $x$  and  $y$  are connected by the equation  $y = \sin 2x$ .

Find the approximate increase in  $y$  as  $x$  increases from  $\frac{\pi}{8}$  to  $\frac{\pi}{8} + p$ , where  $p$  is small.

**57** Add Maths book Ex 12.6 · Q8

$$\begin{aligned}
 \text{(a) } 2x^2y &= 360 \Rightarrow \boxed{y = \frac{180}{x^2}} & \text{(b) } A &= 2(2x^2 + 2xy + xy) = 4x^2 + 6x \cdot \\
 \frac{180}{x^2} &= 4x^2 + \frac{1080}{x} \quad \checkmark \\
 \text{(c) } \frac{dA}{dx} &= 8x - \frac{1080}{x^2} = 16 - 270 = -254; \quad \delta A \approx \boxed{-254p}, \text{ a decrease}
 \end{aligned}$$

**58** Add Maths book Ex 14.4 · Q6

$$\frac{dy}{dx} = 2 \cos 2x = 2 \cos \frac{\pi}{4} = \sqrt{2}; \quad \delta y \approx \boxed{\sqrt{2}p}$$

59



No calculator

Add Maths book Ex 14.4 · Q7

Variables  $x$  and  $y$  are connected by the equation  $y = 3 + \ln(2x - 5)$

Find the approximate change in  $y$  as  $x$  increases from 4 to  $4 + p$ , where  $p$  is small.

60



No calculator

Add Maths book Ex 14.4 · Q8

Variables  $x$  and  $y$  are connected by the equation  $y = \frac{\ln x}{x^2 + 3}$ .

Find the approximate change in  $y$  as  $x$  increases from 1 to  $1 + p$ , where  $p$  is small.

61



No calculator

Add Maths book Ex 14.4 · Q9

Variables  $x$  and  $y$  are connected by the equation  $y = 3 + 2x - 5e^{-x}$ .

Find the approximate change in  $y$  as  $x$  increases from  $\ln 2$  to  $\ln 2 + p$ , where  $p$  is small.

**59** *Add Maths book Ex 14.4 · Q7*

$$\frac{dy}{dx} = \frac{2}{2x-5} = \frac{2}{3} \text{ at } x = 4: \delta y \approx \boxed{\frac{2}{3}p}$$

**60** *Add Maths book Ex 14.4 · Q8*

$$\frac{dy}{dx} = \frac{\frac{1}{x}(x^2+3) - 2x \ln x}{(x^2+3)^2} = \frac{4}{16} \text{ at } x = 1: \delta y \approx \boxed{\frac{1}{4}p}$$

**61** *Add Maths book Ex 14.4 · Q9*

$$\frac{dy}{dx} = 2 + 5e^{-x} = 2 + \frac{5}{2} \text{ at } x = \ln 2: \delta y \approx \boxed{\frac{9}{2}p}$$

62



No calculator

Add Maths book Ex 14.4 · Q10

A curve has equation  $y = \frac{\ln(x^2 - 2)}{x^2 - 2}$ .

Find the approximate change in  $y$  as  $x$  increases from  $\sqrt{3}$  to  $\sqrt{3} + p$ , where  $p$  is small.

**62** *Add Maths book Ex 14.4 · Q10*

$$\frac{dy}{dx} = \frac{2x - 2x \ln(x^2 - 2)}{(x^2 - 2)^2}; \text{ at } x = \sqrt{3}: \ln 1 = 0, \frac{2\sqrt{3}}{1}: \delta y \approx \boxed{2\sqrt{3}p}$$

63



No calculator

Add Maths book Ex 12.7 · Q1

Variables  $x$  and  $y$  are connected by the equation  $y = x^2 - 5x$ .

Given that  $x$  increases at a rate of 0.05 units per second, find the rate of change of  $y$  when  $x = 4$ .

64



No calculator

Add Maths book Ex 12.7 · Q2

Variables  $x$  and  $y$  are connected by the equation  $y = x + \sqrt{x - 5}$ .

Given that  $x$  increases at a rate of 0.1 units per second, find the rate of change of  $y$  when  $x = 9$ .

65



No calculator

Add Maths book Ex 12.7 · Q3

Variables  $x$  and  $y$  are connected by the equation  $y = (x - 3)(x + 5)^3$ .

Given that  $x$  increases at a rate of 0.2 units per second, find the rate of change of  $y$  when  $x = -4$ .



**63** Add Maths book Ex 12.7 · Q1

Chain rule:  $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ .

$$\frac{dy}{dx} = 2x - 5 = 3; \quad 3 \times 0.05 = \boxed{0.15} \text{ units/s}$$

**64** Add Maths book Ex 12.7 · Q2

Chain rule:  $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ .

$$\frac{dy}{dx} = 1 + \frac{1}{2\sqrt{x-5}} = 1.25; \quad 1.25 \times 0.1 = \boxed{0.125} \text{ units/s}$$

**65** Add Maths book Ex 12.7 · Q3

Chain rule:  $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ .

$$\frac{dy}{dx} = (x+5)^3 + 3(x-3)(x+5)^2 = 1 - 21 = -20; \quad -20 \times 0.2 = \boxed{-4}$$

units/s

66



No calculator

Add Maths book Ex 12.7 · Q4

Variables  $x$  and  $y$  are connected by the equation  $y = \frac{5}{2x - 1}$ .

Given that  $y$  increases at a rate of 0.1 units per second, find the rate of change of  $x$  when  $x = -2$ .

67



No calculator

Add Maths book Ex 12.7 · Q5

Variables  $x$  and  $y$  are connected by the equation  $y = \frac{2x}{x^2 + 3}$ .

Given that  $x$  increases at a rate of 2 units per second, find the rate of increase of  $y$  when  $x = 1$ .

68



No calculator

Add Maths book Ex 12.7 · Q6

Variables  $x$  and  $y$  are connected by the equation  $y = \frac{2x - 5}{x - 1}$ .

Given that  $x$  increases at a rate of 0.02 units per second, find the rate of change of  $y$  when  $y = 1$ .

**66** Add Maths book Ex 12.7 · Q4

$$\frac{dy}{dx} = \frac{-10}{(2x-1)^2} = -0.4 \text{ at } x = -2. \quad \frac{dx}{dt} = \frac{dy/dt}{dy/dx} = \frac{0.1}{-0.4} = \boxed{-0.25}$$

units/s

**67** Add Maths book Ex 12.7 · Q5

Chain rule:  $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$ .

$$\frac{dy}{dx} = \frac{2(x^2+3) - 4x^2}{(x^2+3)^2} = \frac{4}{16} = 0.25; \quad 0.25 \times 2 = \boxed{0.5} \text{ units/s}$$

**68** Add Maths book Ex 12.7 · Q6

$$y = 1 \Rightarrow 2x - 5 = x - 1 \Rightarrow x = 4. \quad \frac{dy}{dx} = \frac{3}{(x-1)^2} = \frac{1}{3}; \quad \frac{1}{3} \times 0.02 = \boxed{\frac{1}{150}}$$

units/s

69



No calculator

Add Maths book Ex 12.7 · Q7

Variables  $x$  and  $y$  are connected by the equation  $\frac{1}{y} = \frac{1}{8} - \frac{2}{x}$ .

Given that  $x$  increases at a rate of 0.01 units per second, find the rate of change of  $y$  when  $x = 8$ .

70



No calculator

Add Maths book Ex 12.7 · Q8

A square has sides of length  $x$  cm and area  $A$  cm<sup>2</sup>.

The area is increasing at a constant rate of 0.2 cm<sup>2</sup> s<sup>-1</sup>.

Find the rate of increase of  $x$  when  $A = 16$ .

71



No calculator

Add Maths book Ex 12.7 · Q9

A cube has sides of length  $x$  cm and volume  $V$  cm<sup>3</sup>.

The volume is increasing at a rate of 2 cm<sup>3</sup> s<sup>-1</sup>.

Find the rate of increase of  $x$  when  $V = 512$ .

**69** Add Maths book Ex 12.7 · Q7

$$x = 8: \frac{1}{y} = \frac{1}{8} - \frac{1}{4} = -\frac{1}{8}, y = -8. \text{ Differentiate: } -\frac{1}{y^2} \frac{dy}{dx} = \frac{2}{x^2} \Rightarrow \frac{dy}{dx} = -\frac{2y^2}{x^2} = -2; -2 \times 0.01 = \boxed{-0.08} \text{ units/s}$$

**70** Add Maths book Ex 12.7 · Q8

$$A = x^2, x = 4, \frac{dA}{dx} = 8; \frac{dx}{dt} = \frac{0.2}{8} = \boxed{0.025 \text{ cm s}^{-1}}$$

**71** Add Maths book Ex 12.7 · Q9

$$V = x^3, x = 8, \frac{dV}{dx} = 192; \frac{dx}{dt} = \frac{2}{192} = \boxed{\frac{1}{96} \text{ cm s}^{-1}}$$

72



No calculator

Add Maths book Ex 12.7 · Q10

A sphere has radius  $r$  cm and volume  $V$  cm<sup>3</sup>.

The radius is increasing at a rate of  $\frac{1}{\pi}$  cm s<sup>-1</sup>.

Find the rate of increase of the volume when  $V = 972\pi$ .

73



No calculator

Add Maths book Ex 12.7 · Q11

A solid metal cuboid has dimensions  $x$  cm by  $x$  cm by  $5x$  cm.

The cuboid is heated and the volume increases at a rate of  $0.5$  cm<sup>3</sup> s<sup>-1</sup>.

Find the rate of increase of  $x$  when  $x = 4$ .

74



No calculator

Add Maths book Ex 12.7 · Q12

A cone has base radius  $r$  cm and a fixed height 18 cm.

The radius of the base is increasing at a rate of  $0.1$  cm s<sup>-1</sup>.

Find the rate of change of the volume when  $r = 10$ .

**72** Add Maths book Ex 12.7 · Q10

$$\frac{4}{3}\pi r^3 = 972\pi \Rightarrow r = 9; \quad \frac{dV}{dr} = 4\pi r^2 = 324\pi; \quad \frac{dV}{dt} = 324\pi \times \frac{1}{\pi} = \boxed{324 \text{ cm}^3 \text{ s}^{-1}}$$

**73** Add Maths book Ex 12.7 · Q11

$$V = 5x^3, \quad \frac{dV}{dx} = 15x^2 = 240; \quad \frac{dx}{dt} = \frac{0.5}{240} = \boxed{\frac{1}{480} \text{ cm s}^{-1}}$$

**74** Add Maths book Ex 12.7 · Q12

$$V = \frac{1}{3}\pi r^2(18) = 6\pi r^2, \quad \frac{dV}{dr} = 12\pi r = 120\pi; \quad 120\pi \times 0.1 = \boxed{12\pi \text{ cm}^3 \text{ s}^{-1}}$$

**75**

No calculator

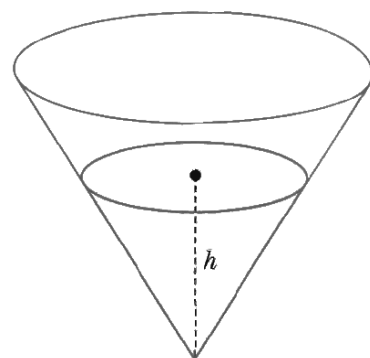
**Add Maths book Ex 12.7 · Q13**

Water is poured into the conical container at a rate of  $5 \text{ cm}^3 \text{ s}^{-1}$ .

After  $t$  seconds, the volume of water in the container,  $V \text{ cm}^3$ , is given by

$V = \frac{1}{12} \pi h^3$ , where  $h \text{ cm}$  is the height of the water in the container.

- a** Find the rate of change of  $h$  when  $h = 5$ .
- b** Find the rate of change of  $h$  when  $h = 10$ .

**76**

No calculator

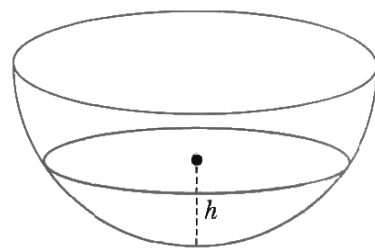
**Add Maths book Ex 12.7 · Q14**

Water is poured into the hemispherical bowl at a rate of  $4\pi \text{ cm}^3 \text{ s}^{-1}$ .

After  $t$  seconds, the volume of water in the bowl,  $V \text{ cm}^3$ , is given by

$V = 8\pi h^2 - \frac{1}{3} \pi h^3$ , where  $h \text{ cm}$  is the height of the water in the bowl.

- a** Find the rate of change of  $h$  when  $h = 2$ .
- b** Find the rate of change of  $h$  when  $h = 4$ .

**77**

No calculator

**Past-paper practice set · Q1**

Sand is poured onto a surface at a rate of  $3\pi \text{ cm}^3 \text{ s}^{-1}$  and forms a right circular cone. The height of the cone is always 3 times the radius. Find the rate of change of the radius 9 seconds after the pouring started.



**75** *Add Maths book Ex 12.7 · Q13*

$$\frac{dV}{dh} = \frac{1}{4}\pi h^2, \quad \frac{dh}{dt} = \frac{5}{\frac{1}{4}\pi h^2} = \frac{20}{\pi h^2}: \quad \text{(a) } h = 5: \boxed{\frac{4}{5\pi} \text{ cm s}^{-1}} \quad \text{(b) } h = 10: \boxed{\frac{1}{5\pi} \text{ cm s}^{-1}}$$

**76** *Add Maths book Ex 12.7 · Q14*

$$\frac{dV}{dh} = 16\pi h - \pi h^2, \quad \frac{dh}{dt} = \frac{4\pi}{\pi(16h - h^2)}: \quad \text{(a) } h = 2: \frac{4}{28} = \boxed{\frac{1}{7} \text{ cm s}^{-1}}$$

$$\text{(b) } h = 4: \frac{4}{48} = \boxed{\frac{1}{12} \text{ cm s}^{-1}}$$

**77** *Past-paper practice set · Q1*

$$h = 3r \Rightarrow V = \pi r^3. \text{ After 9 s, } V = 27\pi \Rightarrow r = 3. \quad \frac{dV}{dr} = 3\pi r^2 = 27\pi$$

$$\frac{dr}{dt} = \frac{3\pi}{27\pi} = \boxed{\frac{1}{9} \text{ cm s}^{-1}}$$

78



No calculator

Add Maths book Ex 12.8 · Q1

Find  $\frac{d^2y}{dx^2}$  for each of the following functions.

**a**  $y = 5x^2 - 7x + 3$     **b**  $y = 2x^3 + 3x^2 - 1$     **c**  $y = 4 - \frac{3}{x^2}$   
**d**  $y = (4x + 1)^5$     **e**  $y = \sqrt{2x + 1}$     **f**  $y = \frac{4}{\sqrt{x + 3}}$

79



No calculator

Add Maths book Ex 12.8 · Q2

Find  $\frac{d^2y}{dx^2}$  for each of the following functions.

**a**  $y = x(x - 4)^3$     **b**  $y = \frac{4x - 1}{x^2}$     **c**  $y = \frac{x + 1}{x - 3}$   
**d**  $y = \frac{x + 2}{x^2 - 1}$     **e**  $y = \frac{x^2}{x - 5}$     **f**  $y = \frac{2x + 5}{3x - 1}$

80



No calculator

Add Maths book Ex 12.8 · Q3

Given that  $f(x) = x^3 - 7x^2 + 2x + 1$ , find

**a**  $f(1)$     **b**  $f'(1)$     **c**  $f''(1)$ .

**78** Add Maths book Ex 12.8 · Q1

**a** 10

**c**  $-\frac{18}{x^4}$

**e**  $-\frac{1}{(2x+1)^{3/2}}$

**b**  $12x+6$

**d**  $320(4x+1)^3$

**f**  $\frac{3}{(x+3)^{5/2}}$

**79** Add Maths book Ex 12.8 · Q2

**a**  $12(x-4)(x-2)$

**d**  $\frac{2(x^3+6x^2+3x+2)}{(x^2-1)^3}$

**b**  $\frac{8x-6}{x^4}$

**e**  $\frac{50}{(x-5)^3}$

**c**  $\frac{8}{(x-3)^3}$

**f**  $\frac{102}{(3x-1)^3}$

**80** Add Maths book Ex 12.8 · Q3

$f'(x) = 3x^2 - 14x + 2$ ,  $f''(x) = 6x - 14$ . (a)  $\boxed{-3}$  (b)  $\boxed{-9}$  (c)  $\boxed{-8}$

81



No calculator

Add Maths book Ex 12.8 · Q4

A curve has equation  $y = 4x^3 + 3x^2 - 6x - 1$ .

**a** Show that  $\frac{dy}{dx} = 0$  when  $x = -1$  and when  $x = 0.5$ .

**b** Find the value of  $\frac{d^2y}{dx^2}$  when  $x = -1$  and when  $x = 0.5$ .

82



No calculator

Add Maths book Ex 12.8 · Q5

A curve has equation  $y = 2x^3 - 15x^2 + 24x + 6$ .

Copy and complete the table to show whether  $\frac{dy}{dx}$  and  $\frac{d^2y}{dx^2}$  are positive (+), negative (−) or zero (0) for the given values of  $x$ .

| $x$                 | 0 | 1 | 2 | 3 | 4 | 5 |
|---------------------|---|---|---|---|---|---|
| $\frac{dy}{dx}$     |   |   |   |   |   |   |
| $\frac{d^2y}{dx^2}$ |   |   |   |   |   |   |

**81** Add Maths book Ex 12.8 · Q4

- (a)  $\frac{dy}{dx} = 12x^2 + 6x - 6 = 6(2x - 1)(x + 1) = 0$  at  $x = -1, 0.5$  ✓
- (b)  $\frac{d^2y}{dx^2} = 24x + 6$ :  $\boxed{-18}$  at  $x = -1$ ,  $\boxed{18}$  at  $x = 0.5$

**82** Add Maths book Ex 12.8 · Q5

$$\frac{dy}{dx} = 6x^2 - 30x + 24 = 6(x - 1)(x - 4), \quad \frac{d^2y}{dx^2} = 12x - 30$$

|                     |   |   |   |   |   |   |
|---------------------|---|---|---|---|---|---|
| $x$                 | 0 | 1 | 2 | 3 | 4 | 5 |
| $\frac{dy}{dx}$     | + | 0 | − | − | 0 | + |
| $\frac{d^2y}{dx^2}$ | − | − | − | + | + | + |

83



No calculator

Add Maths book Ex 12.8 · Q6

A curve has equation  $y = 2x^3 + 3x^2 - 36x + 5$ . Find the range of values of  $x$  for which both  $\frac{dy}{dx}$  and  $\frac{d^2y}{dx^2}$  are both positive.

84



No calculator

Add Maths book Ex 12.8 · Q7

Given that  $y = x^2 - 2x + 5$ , show that  $4 \frac{d^2y}{dx^2} + (x - 1) \frac{dy}{dx} = 2y$ .

85



No calculator

Add Maths book Ex 12.8 · Q8

Given that  $y = 8\sqrt{x}$ , show that  $4x^2 \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} = y$ .

86



No calculator

Add Maths book Ex 12.9 · Q1

Find the coordinates of the stationary points on each of the following curves and determine the nature of each of the stationary points.

**a**  $y = x^2 - 12x + 8$

**b**  $y = (5 + x)(1 - x)$

**c**  $y = x^3 - 12x + 2$

**d**  $y = x^3 + x^2 - 16x - 16$

**e**  $y = x(3 - 4x - x^2)$

**f**  $y = (x - 1)(x^2 - 6x + 2)$

**83** Add Maths book Ex 12.8 · Q6

$$\frac{dy}{dx} = 6(x+3)(x-2) > 0 \Rightarrow x < -3 \text{ or } x > 2; \frac{d^2y}{dx^2} = 12x + 6 > 0 \Rightarrow x > -\frac{1}{2}. \text{ Both: } \boxed{x > 2}$$

**84** Add Maths book Ex 12.8 · Q7

$$y' = 2x - 2, y'' = 2: 4(2) + (x-1)(2x-2) = 8 + 2(x-1)^2 = 2(x^2 - 2x + 5) = 2y \checkmark$$

**85** Add Maths book Ex 12.8 · Q8

$$y = 8x^{\frac{1}{2}}, y' = 4x^{-\frac{1}{2}}, y'' = -2x^{-\frac{3}{2}}: 4x^2(-2x^{-\frac{3}{2}}) + 4x(4x^{-\frac{1}{2}}) = -8x^{\frac{1}{2}} + 16x^{\frac{1}{2}} = 8\sqrt{x} = y \checkmark$$

**86** Add Maths book Ex 12.9 · Q1

**a**  $(6, -28)$  minimum

**b**  $(-2, 9)$  maximum

**c**  $(-2, 18)$  maximum,  $(2, -14)$  minimum

**d**  $(-2\frac{2}{3}, 14\frac{22}{27})$  maximum,  $(2, -36)$  minimum

**e**  $(-3, -18)$  minimum,  $(\frac{1}{3}, \frac{14}{27})$  maximum

**f**  $(\frac{2}{3}, \frac{14}{27})$  maximum,  $(4, -18)$  minimum

Method: solve  $\frac{dy}{dx} = 0$ , then sign of  $\frac{d^2y}{dx^2}$ .

87



No calculator

Add Maths book Ex 12.9 · Q2

Find the coordinates of the stationary points on each of the following curves and determine the nature of each of the stationary points.

**a**  $y = \sqrt{x} + \frac{4}{\sqrt{x}}$

**b**  $y = x^2 - \frac{2}{x}$

**c**  $y = \frac{4}{x} + \sqrt{x}$

**d**  $y = \frac{2x}{x^2 + 9}$

**e**  $y = \frac{x^2}{x + 1}$

**f**  $y = \frac{x^2 - 5x + 3}{x + 1}$

88



No calculator

Add Maths book Ex 12.9 · Q3

The equation of a curve is  $y = \frac{2x + 5}{x + 1}$ .

Find  $\frac{dy}{dx}$  and hence explain why the curve has no turning points.

89



No calculator

Add Maths book Ex 12.9 · Q4

The curve  $y = 2x^3 + ax^2 - 12x + 7$  has a maximum point at  $x = -2$ .

Find the value of  $a$ .



**87** Add Maths book Ex 12.9 · Q2

- a**  $(4, 4)$  minimum
- b**  $(-1, 3)$  minimum
- c**  $(4, 3)$  minimum
- d**  $(-3, -\frac{1}{3})$  minimum,  $(3, \frac{1}{3})$  maximum
- e**  $(-2, -4)$  maximum,  $(0, 0)$  minimum
- f**  $(-4, -13)$  maximum,  $(2, -1)$  minimum

**88** Add Maths book Ex 12.9 · Q3

$$\frac{dy}{dx} = \frac{2(x+1) - (2x+5)}{(x+1)^2} = \frac{-3}{(x+1)^2}. \quad \text{The numerator } -3 \text{ is never zero,}$$

so  $\frac{dy}{dx} \neq 0$ : no turning points.

**89** Add Maths book Ex 12.9 · Q4

$$\frac{dy}{dx} = 6x^2 + 2ax - 12 = 0 \text{ at } x = -2: 24 - 4a - 12 = 0 \Rightarrow \boxed{a = 3}$$

90



No calculator

Add Maths book Ex 12.9 · Q5

The curve  $y = x^3 + ax + b$  has a stationary point at  $(1, 3)$ .

- a** Find the value of  $a$  and the value of  $b$ .
- b** Determine the nature of the stationary point  $(1, 3)$ .
- c** Find the coordinates of the other stationary point on the curve and determine the nature of this stationary point.

91



No calculator

Add Maths book Ex 12.9 · Q6

The curve  $y = x^2 + \frac{a}{x} + b$  has a stationary point at  $(1, -1)$ .

- a** Find the value of  $a$  and the value of  $b$ .
- b** Determine the nature of the stationary point  $(1, -1)$ .

92



No calculator

Add Maths book Ex 12.9 · Q7

The curve  $y = ax + \frac{b}{x^2}$  has a stationary point at  $(-1, -12)$ .

- a** Find the value of  $a$  and the value of  $b$ .
- b** Determine the nature of the stationary point  $(-1, -12)$ .

**90** Add Maths book Ex 12.9 · Q5

(a)  $3 + a = 0$ ,  $1 + a + b = 3$ :  $a = -3$ ,  $b = 5$  (b)  $y'' = 6x = 6 > 0$ :

**minimum**

(c)  $3x^2 - 3 = 0 \Rightarrow x = -1$ :  $(-1, 7)$ ,  $y'' = -6$ : **maximum**

**91** Add Maths book Ex 12.9 · Q6

(a)  $y' = 2x - \frac{a}{x^2}$ :  $2 - a = 0$ ;  $1 + a + b = -1$ :  $a = 2$ ,  $b = -4$  (b)

$y'' = 2 + \frac{2a}{x^3} = 6 > 0$ : **minimum**

**92** Add Maths book Ex 12.9 · Q7

(a)  $y' = a - \frac{2b}{x^3}$ :  $a + 2b = 0$ ;  $-a + b = -12$ :  $a = 8$ ,  $b = -4$  (b)

$y'' = \frac{6b}{x^4} = -24 < 0$ : **maximum**

93



No calculator

Add Maths book Ex 12.9 · Q8

The curve  $y = 2x^3 - 3x^2 + ax + b$  has a stationary point at the point  $(3, -77)$ .

- a Find the value of  $a$  and the value of  $b$ .
- b Find the coordinates of the second stationary point on the curve.
- c Determine the nature of the two stationary points.
- d Find the coordinates of the point on the curve where the gradient is minimum and state the value of the minimum gradient.

94



No calculator

Past-paper practice set · Q4

Sketch the curve  $y = x^3 + 3x^2 - 9x + 3$ , clearly show the  $y$  intercept and all the turning points.

95



No calculator

Past-paper practice set · Q9

Given that  $2x + y = 12$ , find the stationary value of  $x^2 + y^2 + 5xy$  and determine the nature of this stationary point.

**93** *Add Maths book Ex 12.9 · Q8*

(a)  $y' = 6x^2 - 6x + a = 0$  at  $x = 3$ :  $a = -36$ ;  $54 - 27 - 108 + b = -77 \Rightarrow$

$$a = -36, b = 4$$

(b)  $6x^2 - 6x - 36 = 0 \Rightarrow x = -2$ :  $(-2, 48)$  (c)  $y'' = 12x - 6$ :  $(-2, 48)$

**max**,  $(3, -77)$  **min**

(d) Gradient minimum where  $y'' = 0$ :  $x = 0.5$ :  $(0.5, -14.5)$ , minimum gradient  $-37.5$

**94** *Past-paper practice set · Q4*

$\frac{dy}{dx} = 3x^2 + 6x - 9 = 3(x+3)(x-1) = 0$ :  $(-3, 30)$  maximum,  $(1, -2)$  minimum; y-intercept  $(0, 3)$ .

Sketch: a cubic rising through the maximum  $(-3, 30)$ , down through  $(0, 3)$  to the minimum  $(1, -2)$ , then up.

**95** *Past-paper practice set · Q9*

$y = 12 - 2x$ :  $z = x^2 + (12 - 2x)^2 + 5x(12 - 2x) = -5x^2 + 12x + 144$

$\frac{dz}{dx} = -10x + 12 = 0 \Rightarrow x = 1.2$ :  $z = 151.2$ ;  $\frac{d^2z}{dx^2} = -10 < 0$ : **maxi-**

**mum**

96



No calculator

Add Maths book Ex 12.10 · Q1

The sum of two numbers  $x$  and  $y$  is 8.

- a Express  $y$  in terms of  $x$ .
- b i Given that  $P = xy$ , write down an expression for  $P$  in terms of  $x$ .  
ii Find the maximum value of  $P$ .
- c i Given that  $S = x^2 + y^2$ , write down an expression for  $S$ , in terms of  $x$ .  
ii Find the minimum value of  $S$ .

97



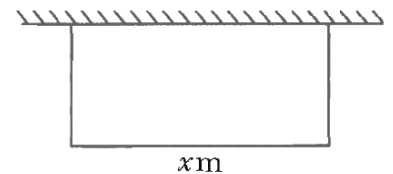
No calculator

Add Maths book Ex 12.10 · Q2

The diagram shows a rectangular garden with a fence on three of its sides and a wall on its fourth side. The total length of the fence is 100 m and the area enclosed is  $A \text{ m}^2$ .

a Show that  $A = \frac{1}{2}x(100 - x)$ .

- b Find the maximum area of the garden enclosed and the value of  $x$  for which this occurs.



**96** Add Maths book Ex 12.10 · Q1

(a)  $y = 8 - x$       (b)(i)  $P = 8x - x^2$       (ii)  $P' = 8 - 2x = 0, x = 4$ :

$P_{\max} = 16$

(c)(i)  $S = x^2 + (8 - x)^2$       (ii)  $S' = 4x - 16 = 0, x = 4$ :  $S_{\min} = 32$

**97** Add Maths book Ex 12.10 · Q2

(a)  $2y + x = 100, A = xy = \frac{1}{2}x(100 - x)$  ✓      (b)  $A' = 50 - x = 0$ :

$x = 50, A = 1250 \text{ m}^2$  ( $A'' = -1 < 0$ , max)

98

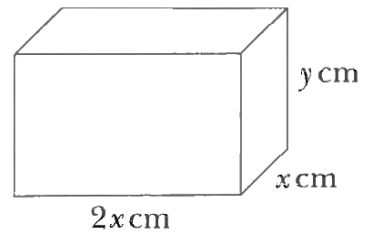


No calculator

Add Maths book Ex 12.10 · Q3

The volume of the solid cuboid is  $576\text{ cm}^3$  and the surface area is  $A\text{ cm}^2$ .

- a Express  $y$  in terms of  $x$ .
- b Show that  $A = 4x^2 + \frac{1728}{x}$ .
- c Find the maximum value of  $A$  and state the dimensions of the cuboid for which this occurs.



99



No calculator

Add Maths book Ex 12.10 · Q4

A cuboid has a total surface area of  $400\text{ cm}^2$  and a volume of  $V\text{ cm}^3$ .  
The dimensions of the cuboid are  $4x\text{ cm}$  by  $x\text{ cm}$  by  $h\text{ cm}$ .

- a Express  $h$  in terms of  $V$  and  $x$ .
- b Show that  $V = 160x - \frac{16}{5}x^3$ .
- c Find the value of  $x$  when  $V$  is a maximum.



**98** Add Maths book Ex 12.10 · Q3

$$(a) 2x^2y = 576 \Rightarrow y = \frac{288}{x^2} \quad (b) A = 4x^2 + 6xy = 4x^2 + \frac{1728}{x} \checkmark$$

$$(c) A' = 8x - \frac{1728}{x^2} = 0 \Rightarrow x = 6: \boxed{A = 432 \text{ cm}^2}, \text{ cuboid } 12 \times 6 \times 8 \text{ cm.}$$

( $A'' > 0$ : this is the **minimum** value of  $A$ .)

**99** Add Maths book Ex 12.10 · Q4

$$(a) \boxed{h = \frac{V}{4x^2}} \quad (b) 2(4x^2 + 5xh) = 400 \Rightarrow h = \frac{200 - 4x^2}{5x}, \quad V = 4x^2h = 160x - \frac{16}{5}x^3 \checkmark$$

$$(c) V' = 160 - \frac{48}{5}x^2 = 0 \Rightarrow x^2 = \frac{50}{3}: \boxed{x = \frac{5\sqrt{6}}{3}}$$

100

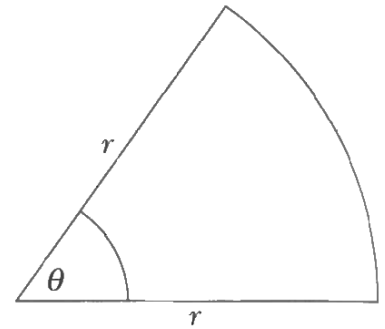


No calculator

Add Maths book Ex 12.10 · Q5

A piece of wire, of length 60 cm, is bent to form a sector of a circle with radius  $r$  cm and sector angle  $\theta$  radians. The total area enclosed by the shape is  $A$  cm<sup>2</sup>.

- Express  $\theta$  in terms of  $r$ .
- Show that  $A = 30r - r^2$ .
- Find  $\frac{dA}{dr}$  and  $\frac{d^2A}{dr^2}$ .
- Find the value for  $r$  for which there is a stationary value of  $A$ .
- Determine the magnitude and nature of this stationary value.



101

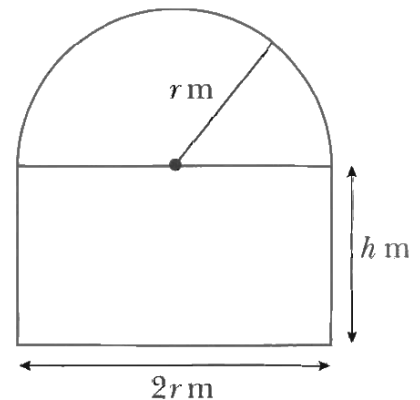


No calculator

Add Maths book Ex 12.10 · Q6

The diagram shows a window made from a rectangle with base  $2r$  m and height  $h$  m and a semicircle of radius  $r$  m. The perimeter of the window is 6 m and the surface area is  $A$  m<sup>2</sup>.

- Express  $h$  in terms of  $r$ .
- Show that  $A = 6r - 2r^2 - \frac{1}{2}\pi r^2$ .
- Find  $\frac{dA}{dr}$  and  $\frac{d^2A}{dr^2}$ .
- Find the value for  $r$  for which there is a stationary value of  $A$ .
- Determine the magnitude and nature of this stationary value.



**100** Add Maths book Ex 12.10 · Q5

(a)  $2r + r\theta = 60 \Rightarrow \theta = \frac{60}{r} - 2$  (b)  $A = \frac{1}{2}r^2\theta = 30r - r^2 \checkmark$   
 (c)  $\frac{dA}{dr} = 30 - 2r, \frac{d^2A}{dr^2} = -2$  (d)  $r = 15$  (e)  $A = 225$ , **maximum**

**101** Add Maths book Ex 12.10 · Q6

(a)  $2r + 2h + \pi r = 6 \Rightarrow h = 3 - \frac{1}{2}r(\pi + 2)$  (b)  $A = 2rh + \frac{1}{2}\pi r^2 = 6r - 2r^2 - \frac{1}{2}\pi r^2 \checkmark$   
 (c)  $6 - 4r - \pi r, -4 - \pi$  (d)  $r = \frac{6}{4+\pi}$  (e)  $A = \frac{18}{4+\pi}$ , **maximum**

102



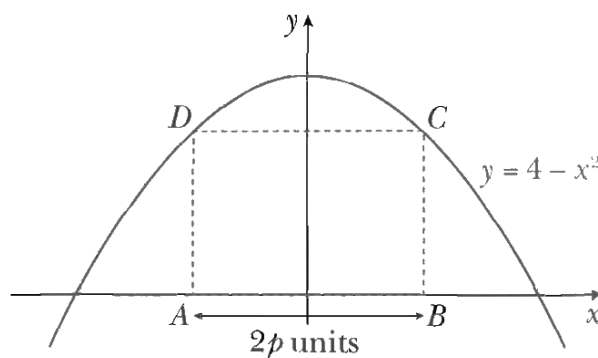
No calculator

Add Maths book Ex 12.10 · Q7

$ABCD$  is a rectangle with base length  $2p$  units, and area  $A$  units<sup>2</sup>.

The points  $A$  and  $B$  lie on the  $x$ -axis and the points  $C$  and  $D$  lie on the curve  $y = 4 - x^2$ .

- Express  $BC$  in terms of  $p$ .
- Show that  $A = 2p(4 - p^2)$ .
- Find the value of  $p$  for which  $A$  has a stationary value.
- Find this stationary value and determine its nature.



103



No calculator

Add Maths book Ex 12.10 · Q8

A solid cylinder has radius  $r$  cm and height  $h$  cm.

The volume of this cylinder is  $250\pi$  cm<sup>3</sup> and the surface area is  $A$  cm<sup>2</sup>.

- Express  $h$  in terms of  $r$ .
- Show that  $A = 2\pi r^2 + \frac{500\pi}{r}$ .
- Find  $\frac{dA}{dr}$  and  $\frac{d^2A}{dr^2}$ .
- Find the value for  $r$  for which there is a stationary value of  $A$ .
- Determine the magnitude and nature of this stationary value.

**102** Add Maths book Ex 12.10 · Q7

(a)  $BC = 4 - p^2$  (b)  $A = 2p(4 - p^2) \checkmark$  (c)  $8 - 6p^2 = 0$ :  $p = \frac{2\sqrt{3}}{3}$   
 (d)  $A = \frac{32\sqrt{3}}{9}$ ,  $A'' = -12p < 0$ : **maximum**

**103** Add Maths book Ex 12.10 · Q8

(a)  $h = \frac{250}{r^2}$  (b)  $A = 2\pi r^2 + 2\pi rh = 2\pi r^2 + \frac{500\pi}{r} \checkmark$   
 (c)  $4\pi r - \frac{500\pi}{r^2}$ ,  $4\pi + \frac{1000\pi}{r^3}$  (d)  $r = 5$  (e)  $A = 150\pi$ , **minimum**

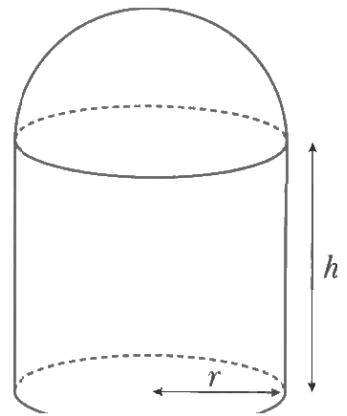
**104**

No calculator

**Add Maths book Ex 12.10 · Q9**

The diagram shows a solid formed by joining a hemisphere of radius  $r$  cm to a cylinder of radius  $r$  cm and height  $h$  cm. The surface area of the solid is  $288\pi$  cm<sup>2</sup> and the volume is  $V$  cm<sup>3</sup>.

- Express  $h$  in terms of  $r$ .
- Show that  $V = 144\pi r - \frac{5}{6}\pi r^3$ .
- Find the exact value of  $r$  such that  $V$  is a maximum.

**105**

Calculator

**Add Maths book Ex 12.10 · Q10**

A piece of wire, of length 50 cm, is cut into two pieces.

One piece is bent to make a square of side  $x$  cm and the other is bent to make a circle of radius  $r$  cm. The total area enclosed by the two shapes is  $A$  cm<sup>2</sup>.

- Express  $r$  in terms of  $x$ .
- Show that  $A = \frac{(\pi + 4)x^2 - 100x + 625}{\pi}$ .
- Find the stationary value of  $A$  and the value of  $x$  for which this occurs.  
Give your answers correct to 3 sf.

**104** Add Maths book Ex 12.10 · Q9

$$(a) 2\pi r^2 + 2\pi rh + \pi r^2 = 288\pi \Rightarrow h = \frac{144}{r} - \frac{3}{2}r$$

$$(b) V = \pi r^2 h + \frac{2}{3}\pi r^3 = 144\pi r - \frac{3}{2}\pi r^3 + \frac{2}{3}\pi r^3 = 144\pi r - \frac{5}{6}\pi r^3 \checkmark$$

$$(c) 144\pi - \frac{5}{2}\pi r^2 = 0 \Rightarrow r^2 = 57.6: r = \frac{12\sqrt{10}}{5}$$

**105** Add Maths book Ex 12.10 · Q10

$$(a) 4x + 2\pi r = 50 \Rightarrow r = \frac{25 - 2x}{\pi} \quad (b) A = x^2 + \pi r^2 = x^2 + \frac{(25 - 2x)^2}{\pi} = \frac{(\pi + 4)x^2 - 100x + 625}{\pi} \checkmark$$

$$(c) \frac{dA}{dx} = \frac{2(\pi + 4)x - 100}{\pi} = 0 \Rightarrow x = \frac{50}{\pi + 4} = 7.00, \quad A = 87.5 \text{ cm}^2$$

(minimum)

106

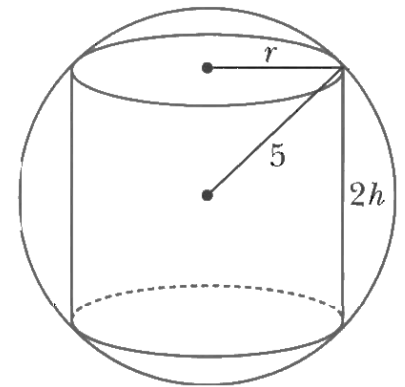


No calculator

Add Maths book Ex 12.10 · Q11

The diagram shows a solid cylinder of radius  $r$  cm and height  $2h$  cm cut from a solid sphere of radius 5 cm. The volume of the cylinder is  $V \text{ cm}^3$ .

- Express  $r$  in terms of  $h$ .
- Show that  $V = 2\pi h(25 - h^2)$ .
- Find the value for  $h$  for which there is a stationary value of  $V$ .
- Determine the nature of this stationary value.



107



No calculator

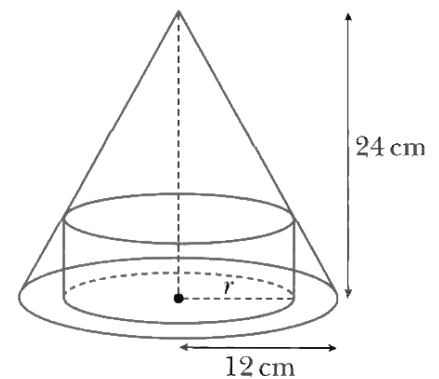
Add Maths book Ex 12.10 · Q12

The diagram shows a hollow cone with base radius 12 cm and height 24 cm.

A solid cylinder stands on the base of the cone and the upper edge touches the inside of the cone.

The cylinder has base radius  $r$  cm, height  $h$  cm and volume  $V \text{ cm}^3$ .

- Express  $h$  in terms of  $r$ .
- Show that  $V = 2\pi r^2(12 - r)$ .
- Find the volume of the largest cylinder which can stand inside the cone.





**106** Add Maths book Ex 12.10 · Q11

$$(a) r^2 + h^2 = 25 \Rightarrow \boxed{r = \sqrt{25 - h^2}} \quad (b) V = \pi r^2(2h) = 2\pi h(25 - h^2)$$

✓

$$(c) 50\pi - 6\pi h^2 = 0: \boxed{h = \frac{5\sqrt{3}}{3}} \quad (d) V'' = -12\pi h < 0: \textbf{maximum}$$

**107** Add Maths book Ex 12.10 · Q12

$$(a) \text{ Similar triangles: } \frac{h}{12-r} = \frac{24}{12} \Rightarrow \boxed{h = 24 - 2r} \quad (b) V = \pi r^2(24 - 2r) = 2\pi r^2(12 - r) \quad \checkmark$$

$$(c) V' = 2\pi(24r - 3r^2) = 0 \Rightarrow r = 8: V_{\max} = 2\pi(64)(4) = \boxed{512\pi \text{ cm}^3}$$

108

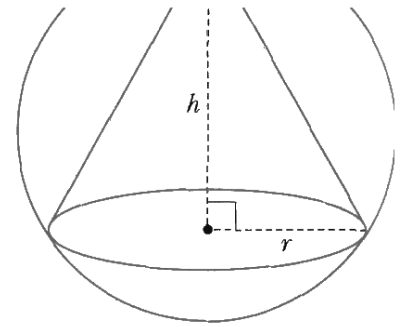


No calculator

Add Maths book Ex 12.10 • Q13

The diagram shows a right circular cone of base radius  $r$  cm and height  $h$  cm cut from a solid sphere of radius 10 cm. The volume of the cone is  $V$  cm<sup>3</sup>.

- a Express  $r$  in terms of  $h$ .
- b Show that  $V = \frac{1}{3}\pi h^2(20 - h)$ .
- c Find the value for  $h$  for which there is a stationary value of  $V$ .
- d Determine the magnitude and nature of this stationary value.



109



Calculator

Past-paper practice set • Q5

A 20 cm piece of wire is cut into 2 pieces. One piece is bent to form a circle while the other piece is bent to form a square. Find the minimum area enclosed by the two pieces.

**108** *Add Maths book Ex 12.10 · Q13*

$$(a) \ r^2 + (h - 10)^2 = 100 \Rightarrow \boxed{r = \sqrt{20h - h^2}} \quad (b) \ V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^2(20 - h) \checkmark$$

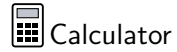
$$(c) \ V' = \frac{1}{3}\pi(40h - 3h^2) = 0: \quad \boxed{h = 13\frac{1}{3}} \quad (d) \ \boxed{V = \frac{32000}{81}\pi}, \quad V'' < 0:$$

**maximum****109** *Past-paper practice set · Q5*

Circle uses  $x$  cm, square  $(20 - x)$  cm:  $A = \left(\frac{20 - x}{4}\right)^2 + \frac{x^2}{4\pi}$

$$\frac{dA}{dx} = -\frac{20 - x}{8} + \frac{x}{2\pi} = 0 \Rightarrow x = 8.80; \quad \boxed{A_{\min} = 14.0 \text{ cm}^2}$$

110



Calculator

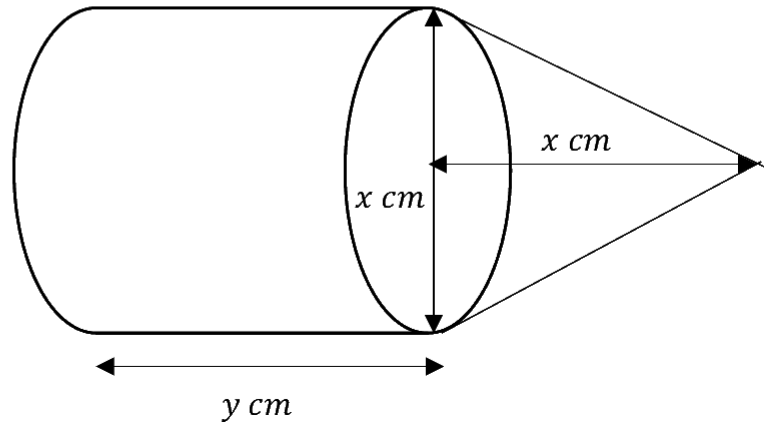
Past-paper practice set · Q6

The diagram shows a solid consisting of a circular cone attached to a cylinder. The diameter of both the cylinder and cone is  $x$  cm, the length of the cylinder is  $y$  cm and the height of the cone is  $x$  cm. Given that the volume of the solid is  $20\pi \text{ cm}^3$ ,

a) Express  $y$  in terms of  $x$

b) Show that the area of the surface is given by  $A = \frac{\pi x^2(3\sqrt{5}-5)}{12} + \frac{160\pi}{x}$ .

c) Find the value of  $x$  for which  $A$  has a stationary value. Determine whether the corresponding value of  $A$  is a maximum or a minimum value.



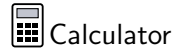
**110** *Past-paper practice set · Q6*

$$(a) 20\pi = \pi\left(\frac{x}{2}\right)^2 y + \frac{1}{3}\pi\left(\frac{x}{2}\right)^2 x \Rightarrow \boxed{y = \frac{80}{x^2} - \frac{x}{3}}$$

$$(b) \text{ slant height } \sqrt{x^2 + \frac{x^2}{4}} = \frac{\sqrt{5}}{2}x; \quad A = \pi\frac{x^2}{4} + \pi xy + \pi\frac{x}{2} \cdot \frac{\sqrt{5}x}{2} = \frac{\pi x^2(3\sqrt{5} - 5)}{12} + \frac{80\pi}{x} \checkmark$$

$$(c) \frac{dA}{dx} = \frac{\pi x(3\sqrt{5} - 5)}{6} - \frac{80\pi}{x^2} = 0 \Rightarrow \boxed{x = 8.25}; \quad \frac{d^2A}{dx^2} > 0: \text{ minimum}$$

111

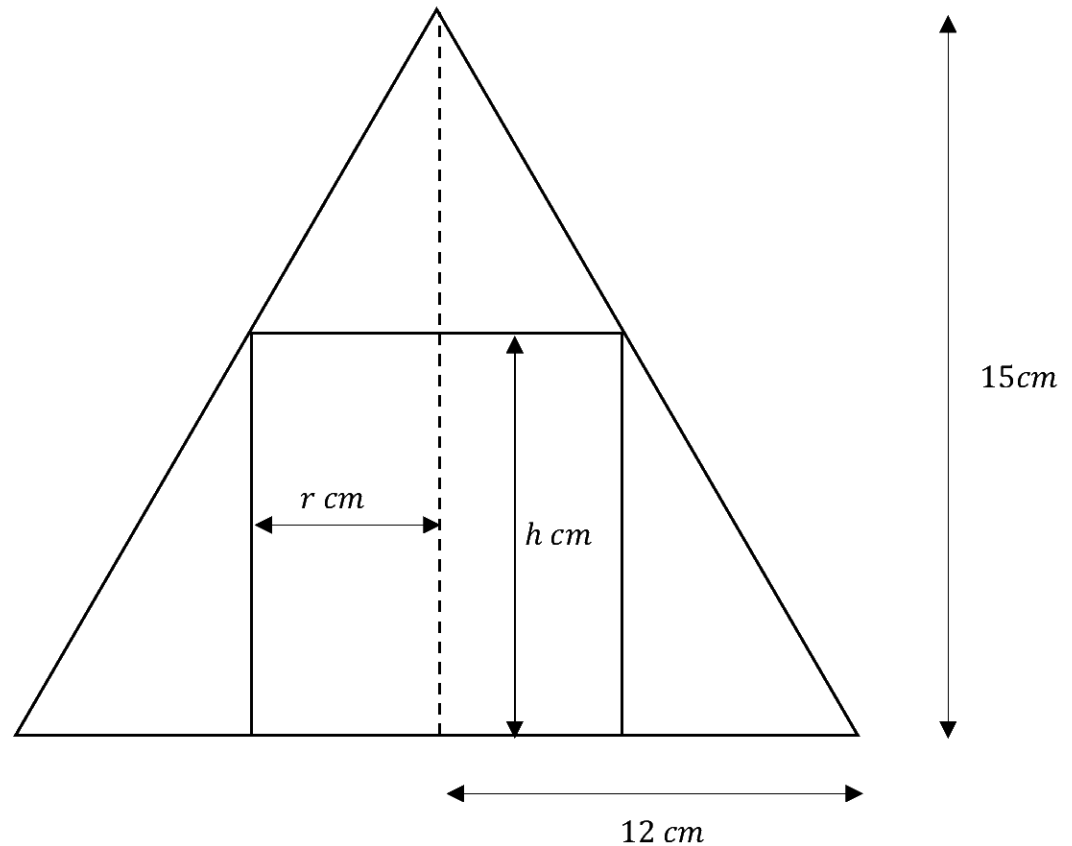


Calculator

Past-paper practice set • Q10

The diagram shows the cross-section of a hollow cone of height 15 cm and radius 12 cm. A solid cylinder of height  $h$  cm and radius  $r$  cm is placed inside the cone such that the upper circular edge of the cylinder is in contact with the inner wall of the cone.

- Show that the volume of the cylinder is given by  $V = 15\pi r^2 - \frac{5}{4}\pi r^3$ .
- Given that  $r$  varies, find the value of  $r$  for which  $V$  has a stationary value.
- Find the stationary value of  $V$  and determine its nature.



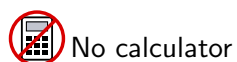
**111** *Past-paper practice set · Q10*

(a) Similar triangles:  $\frac{15-h}{r} = \frac{15}{12} \Rightarrow h = 15 - \frac{5}{4}r$ ;  $V = \pi r^2 h = 15\pi r^2 - \frac{5}{4}\pi r^3$  ✓

(b)  $30\pi r - \frac{15}{4}\pi r^2 = 0 \Rightarrow \boxed{r = 8}$  (c)  $\boxed{V = 320\pi}$ ,  $\frac{d^2V}{dr^2} = -30\pi < 0$ :

**maximum**

112



No calculator

Add Maths book Ex 14.1 · Q1

Differentiate with respect to  $x$ .

**a**  $e^{7x}$

**b**  $e^{3x}$

**c**  $3e^{5x}$

**d**  $2e^{-4x}$

**e**  $6e^{-\frac{x}{2}}$

**f**  $e^{3x+1}$

**g**  $e^{x^2+1}$

**h**  $5x - 3e^{\sqrt{x}}$

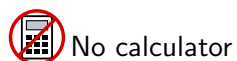
**i**  $2 + \frac{1}{e^{3x}}$

**j**  $2(3 - e^{2x})$

**k**  $\frac{e^x + e^{-x}}{2}$

**l**  $5(x^2 + e^{x^2})$

113



No calculator

Add Maths book Ex 14.1 · Q2

Differentiate with respect to  $x$ .

**a**  $xe^x$

**b**  $x^2e^{2x}$

**c**  $3xe^{-x}$

**d**  $\sqrt{x} e^x$

**e**  $\frac{e^x}{x}$

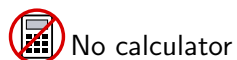
**f**  $\frac{e^{2x}}{\sqrt{x}}$

**g**  $\frac{e^x + 1}{e^x - 1}$

**h**  $xe^{2x} - \frac{e^{2x}}{2}$

**i**  $\frac{x^2e^x - 5}{e^x + 1}$

114



No calculator

Add Maths book Ex 14.1 · Q3

Find the equation of the tangent to

**a**  $y = \frac{5}{e^{2x} + 3}$  at  $x = 0$

**b**  $y = \sqrt{e^{2x} + 1}$  at  $x = \ln 5$

**c**  $y = x^2(1 + e^x)$  at  $x = 1$ .



**112** Add Maths book Ex 14.1 · Q1

$$\frac{d}{dx}e^{f(x)} = f'(x)e^{f(x)}$$

**a**  $7e^{7x}$

**e**  $-3e^{-\frac{x}{2}}$

**i**  $-3e^{-3x}$

**b**  $3e^{3x}$

**f**  $3e^{3x+1}$

**j**  $-4e^{2x}$

**c**  $15e^{5x}$

**g**  $2xe^{x^2+1}$

**k**  $\frac{e^x - e^{-x}}{2}$

**d**  $-8e^{-4x}$

**h**  $5 - \frac{3e^{\sqrt{x}}}{2\sqrt{x}}$

**l**  $10x(1 + e^{x^2})$

**113** Add Maths book Ex 14.1 · Q2

**a**  $xe^x + e^x$

**f**  $\frac{e^{2x}(4x-1)}{2x^{3/2}}$

**b**  $2x^2e^{2x} + 2xe^{2x}$

**g**  $-\frac{2e^x}{(e^x-1)^2}$

**c**  $-3xe^{-x} + 3e^{-x}$

**h**  $2xe^{2x}$

**d**  $\sqrt{x}e^x + \frac{e^x}{2\sqrt{x}}$

**i**  $\frac{e^x(x^2 + 2xe^x + 2x + 5)}{(e^x + 1)^2}$

**e**  $\frac{xe^x - e^x}{x^2}$

**114** Add Maths book Ex 14.1 · Q3

**(a)**  $\boxed{5x + 8y = 10}$

**(b)**  $\boxed{5x - 6y + 15\ln 5 = 0}$

**(c)**  $\boxed{y = 3ex + 2x - 2e - 1}$

e.g. (a)  $y = 1$ ,  $\frac{dy}{dx} = \frac{-10e^{2x}}{(e^{2x}+3)^2} = -\frac{5}{8}$  at  $x = 0$ .

**115**

No calculator

**Add Maths book Ex 14.1 · Q4**

A curve has equation  $y = 5e^{2x} - 4x - 3$ .

The tangent to the curve at the point  $(0, 2)$  meets the  $x$ -axis at the point  $A$ .  
Find the coordinates of  $A$ .

**116**

No calculator

**Add Maths book Ex 14.1 · Q5**

A curve has equation  $y = xe^x$ .

- a** Find, in terms of  $e$ , the coordinates of the stationary point on this curve and determine its nature.
- b** Find, in terms of  $e$ , the equation of the normal to the curve at the point  $P(1, e)$ .
- c** The normal at  $P$  meets the  $x$ -axis at  $A$  and the  $y$ -axis at  $B$ . Find, in terms of  $e$ , the area of triangle  $OAB$ , where  $O$  is the origin.

**117**

No calculator

**Add Maths book Ex 14.2 · Q1**

Differentiate with respect to  $x$ .

**a**  $\ln 5x$

**b**  $\ln 12x$

**c**  $\ln(2x + 3)$

**d**  $2 + \ln(1 - x^2)$

**e**  $\ln(3x + 1)^2$

**f**  $\ln \sqrt{x + 2}$

**g**  $\ln(2 - 5x)^4$

**h**  $2x + \ln\left(\frac{4}{x}\right)$

**i**  $5 - \ln \frac{3}{(2 - 3x)}$

**j**  $\ln(\ln x)$

**k**  $\ln(\sqrt{x} + 1)^2$

**l**  $\ln(x^2 + \ln x)$

**115** Add Maths book Ex 14.1 · Q4

$$\frac{dy}{dx} = 10e^{2x} - 4 = 6 \text{ at } x = 0: \text{ tangent } y = 6x + 2; y = 0: \boxed{A\left(-\frac{1}{3}, 0\right)}$$

**116** Add Maths book Ex 14.1 · Q5

$$(a) \frac{dy}{dx} = e^x(x+1) = 0 \Rightarrow x = -1: \boxed{\left(-1, -\frac{1}{e}\right)}, \text{ minimum}$$

$$(b) \text{ gradient at } P: 2e, \text{ normal: } \boxed{y - e = -\frac{1}{2e}(x - 1)}$$

$$(c) A(2e^2 + 1, 0), B\left(0, e + \frac{1}{2e}\right): \text{ area} = \boxed{\frac{(2e^2 + 1)^2}{4e}}$$

**117** Add Maths book Ex 14.2 · Q1

$$\frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)}$$

$$\mathbf{a} \quad \frac{1}{x}$$

$$\mathbf{e} \quad \frac{6}{3x+1}$$

$$\mathbf{i} \quad \frac{3}{3x-2}$$

$$\mathbf{b} \quad \frac{1}{x}$$

$$\mathbf{f} \quad \frac{1}{2x+4}$$

$$\mathbf{j} \quad \frac{1}{x \ln x}$$

$$\mathbf{c} \quad \frac{2}{2x+3}$$

$$\mathbf{g} \quad \frac{20}{5x-2}$$

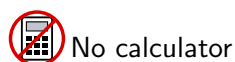
$$\mathbf{k} \quad \frac{1}{(\sqrt{x}+1)\sqrt{x}}$$

$$\mathbf{d} \quad \frac{2x}{x^2-1}$$

$$\mathbf{h} \quad 2 - \frac{1}{x}$$

$$\mathbf{l} \quad \frac{2x^2+1}{x(x^2+\ln x)}$$

118



No calculator

Add Maths book Ex 14.2 · Q2

Differentiate with respect to  $x$ .

**a**  $x \ln x$

**b**  $2x^2 \ln x$

**c**  $(x-1) \ln x$

**d**  $5x \ln x^2$

**e**  $x^2 \ln(\ln x)$

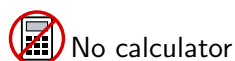
**f**  $\frac{\ln 2x}{x}$

**g**  $\frac{4}{\ln x}$

**h**  $\frac{\ln(2x+1)}{x^2}$

**i**  $\frac{\ln(x^3-1)}{2x+3}$

119

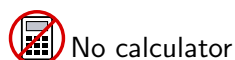


No calculator

Add Maths book Ex 14.2 · Q3

A curve has equation  $y = x^2 \ln 3x$ .Find the value of  $\frac{dy}{dx}$  and  $\frac{d^2y}{dx^2}$  at the point where  $x = 2$ .

120



No calculator

Add Maths book Ex 14.2 · Q4

Use the laws of logarithms to help differentiate these expressions with respect to  $x$ .

**a**  $\ln \sqrt{3x+1}$

**b**  $\ln \frac{1}{(2x-5)}$

**c**  $\ln [x(x-5)^4]$

**d**  $\ln \left( \frac{2x+1}{x-1} \right)$

**e**  $\ln \left( \frac{2-x}{x^2} \right)$

**f**  $\ln \left[ \frac{x(x+1)}{x+2} \right]$

**g**  $\ln \left[ \frac{2x+3}{(x-5)(x+1)} \right]$

**h**  $\ln \left[ \frac{2}{(x+3)^2(x-1)} \right]$

**i**  $\ln \left[ \frac{(x+1)(2x-3)}{x(x-1)} \right]$

**118** Add Maths book Ex 14.2 · Q2

**a**  $1 + \ln x$

**b**  $2x + 4x \ln x$

**c**  $1 - \frac{1}{x} + \ln x$

**d**  $10 + 5 \ln x^2$

**e**  $\frac{x}{\ln x} + 2x \ln(\ln x)$

**f**  $\frac{1 - \ln 2x}{x^2}$

**g**  $-\frac{4}{x(\ln x)^2}$

**h**  $\frac{2}{x^2(2x+1)} - \frac{2 \ln(2x+1)}{x^3}$

**i** 
$$\frac{3x^2(2x+3) - 2(x^3-1)\ln(x^3-1)}{(2x+3)^2(x^3-1)}$$

**119** Add Maths book Ex 14.2 · Q3

$$\frac{dy}{dx} = 2x \ln 3x + x = \boxed{4 \ln 6 + 2}; \quad \frac{d^2y}{dx^2} = 2 \ln 3x + 3 = \boxed{2 \ln 6 + 3}$$

**120** Add Maths book Ex 14.2 · Q4

**a**  $\frac{3}{6x+2}$

**b**  $\frac{2}{5-2x}$

**c**  $\frac{5(x-1)}{x(x-5)}$

**d**  $\frac{-3}{(2x+1)(x-1)}$

**e**  $\frac{x-4}{x(2-x)}$

**f**  $\frac{x^2+4x+2}{x(x+1)(x+2)}$

**g**

$$\frac{2-6x-2x^2}{(x-5)(x+1)(2x+3)}$$

**h**  $\frac{-3x-1}{(x+3)(x-1)}$

**i** 
$$\frac{-x^2+6x-3}{x(x-1)(x+1)(2x-3)}$$

121



No calculator

Add Maths book Ex 14.2 · Q5

Find  $\frac{dy}{dx}$  for each of the following.

**a**  $y = \log_3 x$

**b**  $y = \log_2 x^2$

**c**  $y = \log_4 (5x - 1)$

122



No calculator

Add Maths book Ex 14.2 · Q6

Find  $\frac{dy}{dx}$  for each of the following.

**a**  $e^y = 4x^2 - 1$

**b**  $e^y = 5x^3 - 2x$

**c**  $e^y = (x + 3)(x - 4)$

123



No calculator

Add Maths book Ex 14.2 · Q7

A curve has equation  $x = \frac{1}{2} [e^{y(3x+7)} + 1]$ .

Find the value of  $\frac{dy}{dx}$  when  $x = 1$ .

**121** *Add Maths book Ex 14.2 · Q5*Change base:  $\log_a x = \frac{\ln x}{\ln a}$ .

**a**  $\frac{1}{x \ln 3}$

**b**  $\frac{2}{x \ln 2}$

**c**  $\frac{5}{(5x-1) \ln 4}$

**122** *Add Maths book Ex 14.2 · Q6*Take  $\ln$ :  $y = \ln(\dots)$ .

**a**  $\frac{8x}{4x^2 - 1}$

**b**  $\frac{15x^2 - 2}{5x^3 - 2x}$

**c**  $\frac{2x-1}{(x+3)(x-4)}$

**123** *Add Maths book Ex 14.2 · Q7* $\ln(2x-1) = y(3x+7)$ .  $x=1$ :  $y=0$ . Differentiate:  $\frac{2}{2x-1} = 3y +$ 

$(3x+7)\frac{dy}{dx}$ :  $2 = 10\frac{dy}{dx} \Rightarrow \boxed{\frac{dy}{dx} = 0.2}$

124



No calculator

Past-paper practice set • Q2

Differentiate  $y = \sqrt[3]{\frac{(3x^2+3)^2}{(2x^2+2x+1)}}$  with respect to  $x$ .

125



No calculator

Past-paper practice set • Q7

Differentiate the following with respect to  $x$ .

a)  $y = \ln(2x^2)^{\sin x}$

b)  $y = \frac{\ln(\cos x)}{e^x}$

c)  $y = \frac{(e^{4x^2+5})(e^{3x-2x^2})}{e^{2x^2+4}}$



**124** *Past-paper practice set · Q2*

$$\ln y = \frac{2}{3} \ln(3x^2 + 3) - \frac{1}{3} \ln(2x^2 + 2x + 1)$$

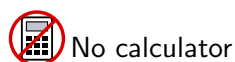
$$\frac{dy}{dx} = \frac{2(2x^3 + 3x^2 - 1)}{(3x^2 + 3)^{\frac{1}{3}}(2x^2 + 2x + 1)^{\frac{4}{3}}}$$

**125** *Past-paper practice set · Q7*

$$(a) y = \sin x \ln(2x^2): \quad \cos x \ln(2x^2) + \frac{2 \sin x}{x} \quad (b) \quad \frac{-\tan x - \ln(\cos x)}{e^x}$$

$$(c) y = e^{(4x^2+5)+(3x-2x^2)-(2x^2+4)} = e^{3x+1}: \quad 3e^{3x+1}$$

126



Add Maths book Ex 14.3 · Q1

Differentiate with respect to  $x$ .

**a**  $2 + \sin x$

**b**  $2 \sin x + 3 \cos x$

**c**  $2 \cos x - \tan x$

**d**  $3 \sin 2x$

**e**  $4 \tan 5x$

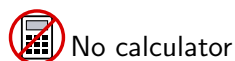
**f**  $2 \cos 3x - \sin 2x$

**g**  $\tan(3x + 2)$

**h**  $\sin\left(2x + \frac{\pi}{3}\right)$

**i**  $2 \cos\left(3x - \frac{\pi}{6}\right)$

127



Add Maths book Ex 14.3 · Q2

Differentiate with respect to  $x$ .

**a**  $\sin^3 x$

**b**  $5 \cos^2(3x)$

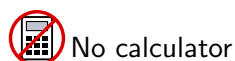
**c**  $\sin^2 x - 2 \cos x$

**d**  $(3 - \cos x)^4$

**e**  $2 \sin^3\left(2x + \frac{\pi}{6}\right)$

**f**  $3 \cos^4 x + 2 \tan^2\left(2x - \frac{\pi}{4}\right)$

128



Add Maths book Ex 14.3 · Q3

Differentiate with respect to  $x$ .

**a**  $x \sin x$

**b**  $2 \sin 2x \cos 3x$

**c**  $x^2 \tan x$

**d**  $x \tan^3\left(\frac{x}{2}\right)$

**e**  $\frac{5}{\cos 3x}$

**f**  $\frac{x}{\cos x}$

**g**  $\frac{\tan x}{x}$

**h**  $\frac{\sin x}{2 + \cos x}$

**i**  $\frac{\sin x}{3x - 1}$

**j**  $\frac{1}{\sin^3 2x}$

**k**  $\frac{3x}{\sin 2x}$

**l**  $\frac{\sin x + \cos x}{\sin x - \cos x}$

**126** Add Maths book Ex 14.3 · Q1

- a**  $\cos x$                       **d**  $6\cos 2x$                       **g**  $3\sec^2(3x+2)$   
**b**  $2\cos x - 3\sin x$               **e**  $20\sec^2 5x$                       **h**  $2\cos\left(2x + \frac{\pi}{3}\right)$   
**c**  $-2\sin x - \sec^2 x$               **f**  $-6\sin 3x - 2\cos 2x$       **i**  $-6\sin\left(3x - \frac{\pi}{6}\right)$

**127** Add Maths book Ex 14.3 · Q2

- a**  $3\sin^2 x \cos x$                       **d**  $4(3 - \cos x)^3 \sin x$   
**b**  $-30\cos 3x \sin 3x$                       **e**  $12\sin^2\left(2x + \frac{\pi}{6}\right) \cos\left(2x + \frac{\pi}{6}\right)$   
**c**  $2\sin x \cos x + 2\sin x$                       **f**  $-12\sin x \cos^3 x + 8\tan\left(2x - \frac{\pi}{4}\right) \sec^2\left(2x - \frac{\pi}{4}\right)$

**128** Add Maths book Ex 14.3 · Q3

- a**  $x\cos x + \sin x$                       **g**  $\frac{x\sec^2 x - \tan x}{x^2}$   
**b**  $-6\sin 2x \sin 3x + 4\cos 2x \cos 3x$                       **h**  $\frac{1 + 2\cos x}{(2 + \cos x)^2}$   
**c**  $x^2 \sec^2 x + 2x \tan x$                       **i**  $\frac{(3x - 1)\cos x - 3\sin x}{(3x - 1)^2}$   
**d**  $\frac{3}{2}x \tan^2 \frac{x}{2} \sec^2 \frac{x}{2} + \tan^3 \frac{x}{2}$                       **j**  $-6\cot 2x \operatorname{cosec}^3 2x$   
**e**  $15\tan 3x \sec 3x$                       **k**  $3\operatorname{cosec} 2x - 6x \cot 2x \operatorname{cosec} 2x$   
**f**  $\sec x + x \tan x \sec x$                       **l**  $\frac{-2}{(\sin x - \cos x)^2}$

129



No calculator

Add Maths book Ex 14.3 · Q4

Differentiate with respect to  $x$ .

**a**  $e^{\cos x}$

**b**  $e^{\cos 5x}$

**c**  $e^{\tan x}$

**d**  $e^{(\sin x + \cos x)}$

**e**  $e^x \sin x$

**f**  $e^x \cos \frac{1}{2}x$

**g**  $e^x (\cos x + \sin x)$

**h**  $x^2 e^{\cos x}$

**i**  $\ln(\sin x)$

**j**  $x^2 \ln(\cos x)$

**k**  $\frac{\sin 3x}{e^{2x-1}}$

**l**  $\frac{x \sin x}{e^x}$

130



No calculator

Add Maths book Ex 14.3 · Q5

Find the gradient of the tangent to

**a**  $y = 2x \cos 3x$  when  $x = \frac{\pi}{3}$

**b**  $y = \frac{2 - \cos x}{3 \tan x}$  when  $x = \frac{\pi}{4}$ .

131



No calculator

Add Maths book Ex 14.3 · Q6

**a** By writing  $\sec x$  as  $\frac{1}{\cos x}$ , find  $\frac{d}{dx}(\sec x)$ .

**b** By writing  $\operatorname{cosec} x$  as  $\frac{1}{\sin x}$ , find  $\frac{d}{dx}(\operatorname{cosec} x)$ .

**c** By writing  $\cot x$  as  $\frac{\cos x}{\sin x}$ , find  $\frac{d}{dx}(\cot x)$ .

**129** Add Maths book Ex 14.3 · Q4

**a**  $-\sin x e^{\cos x}$

**g**  $2e^x \cos x$

**b**  $-5 \sin 5x e^{\cos 5x}$

**h**  $x e^{\cos x} (2 - x \sin x)$

**c**  $\sec^2 x e^{\tan x}$

**i**  $\cot x$

**d**  $(\cos x - \sin x) e^{\sin x + \cos x}$

**j**  $x [2 \ln(\cos x) - x \tan x]$

**e**  $e^x (\sin x + \cos x)$

**k**  $\frac{3 \cos 3x - 2 \sin 3x}{e^{2x-1}}$

**f**  $\frac{1}{2} e^x (2 \cos \frac{1}{2} x - \sin \frac{1}{2} x)$

**l**  $\frac{x \cos x + \sin x - x \sin x}{e^x}$

**130** Add Maths book Ex 14.3 · Q5

(a)  $\frac{dy}{dx} = 2 \cos 3x - 6x \sin 3x$ ; at  $x = \frac{\pi}{3}$ :  $\boxed{-2}$  (b)  $\boxed{\frac{3\sqrt{2}-8}{6}}$

**131** Add Maths book Ex 14.3 · Q6

(a)  $\frac{d}{dx}(\cos x)^{-1} = \frac{\sin x}{\cos^2 x} = \boxed{\tan x \sec x}$  (b)  $\boxed{-\cot x \operatorname{cosec} x}$  (c)  $\boxed{-\operatorname{cosec}^2 x}$

132



No calculator

Add Maths book Ex 14.3 · Q7

Find  $\frac{dy}{dx}$  for each of the following.

**a**  $e^y = \sin 3x$

**b**  $e^y = 3 \cos 2x$

133



No calculator

Add Maths book Ex 14.3 · Q8

A curve has equation  $y = A \sin x + B \sin 2x$ .

The curve passes through the point  $P\left(\frac{\pi}{2}, 3\right)$  and has a gradient of  $\frac{3\sqrt{2}}{2}$  when  $x = \frac{\pi}{4}$ .

Find the value of  $A$  and the value of  $B$ .

134



No calculator

Add Maths book Ex 14.3 · Q9

A curve has equation  $y = A \sin x + B \cos 2x$ .

The curve has a gradient of  $5\sqrt{3}$  when  $x = \frac{\pi}{6}$  and has a gradient of  $6 + 2\sqrt{2}$  when  $x = \frac{\pi}{4}$ .

Find the value of  $A$  and the value of  $B$ .

**132** *Add Maths book Ex 14.3 · Q7*

$$(a) y = \ln \sin 3x: \boxed{3 \cot 3x} \quad (b) y = \ln(3 \cos 2x): \boxed{-2 \tan 2x}$$

**133** *Add Maths book Ex 14.3 · Q8*

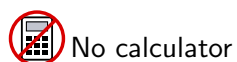
$$P: A \sin \frac{\pi}{2} + B \sin \pi = 3 \Rightarrow \boxed{A = 3}. \quad \frac{dy}{dx} = A \cos x + 2B \cos 2x; \text{ at } x = \frac{\pi}{4}: \\ \cos \frac{\pi}{2} = 0, \text{ so this only gives } \frac{\sqrt{2}}{2}A = \frac{3\sqrt{2}}{2} \text{ (again } A = 3).$$

The answer book gives  $B = -5$ , but with the data as printed  $B$  cannot be found — check the question with your teacher.

**134** *Add Maths book Ex 14.3 · Q9*

$$\frac{dy}{dx} = A \cos x - 2B \sin 2x: \\ x = \frac{\pi}{6}: \frac{\sqrt{3}}{2}A - \sqrt{3}B = 5\sqrt{3} \Rightarrow A - 2B = 10; \quad x = \frac{\pi}{4}: \frac{\sqrt{2}}{2}A - 2B = 6 + 2\sqrt{2} \\ \Rightarrow \boxed{A = 4, B = -3}$$

135

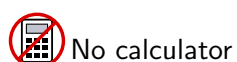


Add Maths book Ex 14.4 · Q1

A curve has equation  $y = 3 \sin\left(2x + \frac{\pi}{2}\right)$ .

Find the equation of the normal to the curve at the point on the curve where  $x = \frac{\pi}{4}$ .

136



Add Maths book Ex 14.4 · Q2

A curve has equation  $y = x \sin 2x$  for  $0 \leq x \leq \pi$  radians.

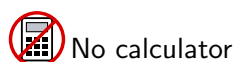
**a** Find the equation of the normal to the curve at the point  $P\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ .

**b** The normal at  $P$  intersects the  $x$ -axis at  $Q$  and the  $y$ -axis at  $R$ .

Find the coordinates of  $Q$  and  $R$ .

**c** Find the area of triangle  $OQR$  where  $O$  is the origin.

137



Add Maths book Ex 14.4 · Q3

A curve has equation  $y = e^{\frac{1}{2}x} + 1$ .

The curve crosses the  $y$ -axis at  $P$ .

The normal to the curve at  $P$  meets the  $x$ -axis at  $Q$ .

Find the coordinates of  $Q$ .



**135** Add Maths book Ex 14.4 · Q1

$$x = \frac{\pi}{4}: y = 3 \sin \pi = 0; \quad \frac{dy}{dx} = 6 \cos(2x + \frac{\pi}{2}) = -6, \text{ normal gradient } \frac{1}{6}:$$

$$\boxed{4x - 24y = \pi}$$

**136** Add Maths book Ex 14.4 · Q2

$$(a) \frac{dy}{dx} = \sin 2x + 2x \cos 2x = 1 \text{ at } P, \text{ normal gradient } -1: \boxed{2x + 2y = \pi}$$

$$(b) \boxed{Q(\frac{\pi}{2}, 0), R(0, \frac{\pi}{2})} \quad (c) \frac{1}{2} \cdot \frac{\pi}{2} \cdot \frac{\pi}{2} = \boxed{\frac{\pi^2}{8} \text{ units}^2}$$

**137** Add Maths book Ex 14.4 · Q3

$$P(0, 2), \frac{dy}{dx} = \frac{1}{2}e^{\frac{1}{2}x} = \frac{1}{2}, \text{ normal gradient } -2: y = -2x + 2: \boxed{Q(1, 0)}$$

138



No calculator

Add Maths book Ex 14.4 · Q4

A curve has equation  $y = 5 - e^{2x}$ .

The curve crosses the  $x$ -axis at  $A$  and the  $y$ -axis at  $B$ .

- a Find the coordinates of  $A$  and  $B$ .
- b The normal to the curve at  $B$  meets the  $x$ -axis at the point  $C$ .  
Find the coordinates of  $C$ .

139



No calculator

Add Maths book Ex 14.4 · Q5

A curve has equation  $y = xe^x$ .

The tangent to the curve at the point  $P(1, e)$  meets the  $y$ -axis at the point  $A$ .

The normal to the curve at  $P$  meets the  $x$ -axis at the point  $B$ .

Find the area of triangle  $OAB$ , where  $O$  is the origin.

140



No calculator

Add Maths book Ex 14.4 · Q11

- 1 Find the coordinates of the stationary points on these curves and determine their nature

a  $y = xe^{\frac{x}{2}}$

b  $y = x^2e^{2x}$

c  $y = e^x - 7x + 2$

d  $y = 5e^{2x} - 10x - 1$

e  $y = (x^2 - 8)e^{-x}$

f  $y = x^2 \ln x$

g  $y = \frac{\ln x}{x^2}$

h  $y = \frac{\ln(x^2 + 1)}{x^2 + 1}$

**138** Add Maths book Ex 14.4 · Q4

(a)  $A\left(\frac{1}{2}\ln 5, 0\right), B(0, 4)$  (b)  $\frac{dy}{dx} = -2e^{2x} = -2$  at  $B$ ; normal  $y = \frac{1}{2}x + 4$ :  
 $C(-8, 0)$

**139** Add Maths book Ex 14.4 · Q5

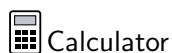
$\frac{dy}{dx} = e^x(1+x) = 2e$  at  $P$ . Tangent:  $y = 2ex - e, A(0, -e)$ . Normal:  
 $y - e = -\frac{1}{2e}(x - 1), B(1 + 2e^2, 0)$ .

Area  $OAB = \frac{1}{2} \times e \times (2e^2 + 1) = \frac{1}{2}e(2e^2 + 1)$

**140** Add Maths book Ex 14.4 · Q11

- a**  $(-2, -2e^{-1})$  minimum
- b**  $(-1, e^{-2})$  maximum,  $(0, 0)$  minimum
- c**  $(\ln 7, 9 - 7\ln 7)$  minimum
- d**  $(0, 4)$  minimum
- e**  $(-2, -4e^2)$  minimum,  $(4, 8e^{-4})$  maximum
- f**  $(e^{-\frac{1}{2}}, -\frac{1}{2}e^{-1})$  minimum
- g**  $(\sqrt{e}, \frac{1}{2}e^{-1})$  maximum
- h**  $(\pm\sqrt{e-1}, e^{-1})$  maxima,  $(0, 0)$  minimum

141



Calculator

Add Maths book Ex 14.4 · Q12

Find the coordinates of the stationary points on these curves and determine their nature.

**a**  $y = 4 \sin x + 3 \cos x$  for  $0 \leq x \leq \frac{\pi}{2}$

**b**  $y = 6 \cos \frac{x}{2} + 8 \sin \frac{x}{2}$  for  $0 \leq x \leq 2\pi$

**c**  $y = 5 \sin \left( 2x + \frac{\pi}{2} \right)$  for  $-\frac{\pi}{6} \leq x \leq \frac{5\pi}{6}$

**d**  $y = \frac{e^x}{\sin x}$  for  $0 < x < \pi$

**e**  $y = 2 \sin x \cos x + 2 \cos x$  for  $0 \leq x \leq \pi$

142



No calculator

Add Maths book Ex 14.4 · Q13

A curve has equation  $y = Ae^{2x} + Be^{-2x}$ .

The gradient of the tangent at the point  $(0, 10)$  is  $-12$ .

**a** Find the value of  $A$  and the value of  $B$ .

**b** Find the coordinates of the turning point on the curve and determine its nature.

**141** Add Maths book Ex 14.4 · Q12

- a**  $(0.927, 5)$  maximum
- b**  $(1.85, 10)$  maximum
- c**  $(0, 5)$  maximum,  $(\frac{\pi}{2}, -5)$  minimum
- d**  $(\frac{\pi}{4}, 3.10)$  minimum
- e**  $(\frac{5\pi}{6}, -\frac{3\sqrt{3}}{2})$  minimum

**142** Add Maths book Ex 14.4 · Q13

- (a)  $A + B = 10, 2A - 2B = -12$ :  $A = 2, B = 8$
- (b)  $4e^{2x} - 16e^{-2x} = 0 \Rightarrow e^{4x} = 4, x = \frac{1}{4}\ln 4$ :  $(\frac{1}{4}\ln 4, 8)$ , **minimum**

143



No calculator

Add Maths book Ex 14.4 · Q14

A curve has equation  $y = x \ln x$ .

The curve crosses the  $x$ -axis at the point  $A$  and has a minimum point at  $B$ .

Find the coordinates of  $A$  and the coordinates of  $B$ .

144



No calculator

Add Maths book Ex 14.4 · Q15

A curve has equation  $y = x^2 e^x$ .

The curve has a minimum point at  $P$  and a maximum point at  $Q$ .

**a** Find the coordinates of  $P$  and the coordinates of  $Q$ .

**b** The tangent to the curve at the point  $A(1, e)$  meets the  $y$ -axis at the point  $B$ .

The normal to the curve at the point  $A(1, e)$  meets the  $y$ -axis at the point  $C$ .

Find the coordinates of  $B$  and the coordinates of  $C$ .

**c** Find the area of triangle  $ABC$ .

**143** Add Maths book Ex 14.4 · Q14

$$A: x \ln x = 0 \Rightarrow \boxed{A(1, 0)}. \quad \frac{dy}{dx} = \ln x + 1 = 0 \Rightarrow x = e^{-1}: \boxed{B(e^{-1}, -e^{-1})}$$

**144** Add Maths book Ex 14.4 · Q15

$$(a) \frac{dy}{dx} = e^x(x^2 + 2x) = 0: \boxed{P(0, 0), Q(-2, 4e^{-2})}$$

$$(b) \text{gradient at } A: 3e. \text{ Tangent } y = 3ex - 2e: \boxed{B(0, -2e)}; \text{ normal } y - e = -\frac{1}{3e}(x - 1): \boxed{C(0, e + \frac{1}{3e})}$$

$$(c) \text{Area} = \frac{1}{2} \times BC \times 1 = \boxed{\frac{1}{2}(3e + \frac{1}{3e})}$$

145

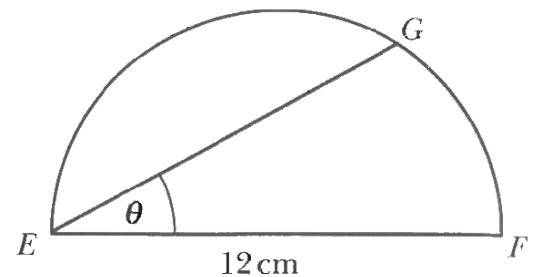
 Calculator

Add Maths book Ex 14.4 · Q16

The diagram shows a semi-circle with diameter  $EF$  of length 12 cm.

Angle  $GEF = \theta$  radians and the shaded region has an area of  $A \text{ cm}^2$ .

- a** Show that  $A = 36\theta + 18 \sin 2\theta$ .
- b** Given that  $\theta$  is increasing at a rate of 0.05 radians per second, find the rate of change of  $A$  when  $\theta = \frac{\pi}{6}$  radians.



146

 No calculator

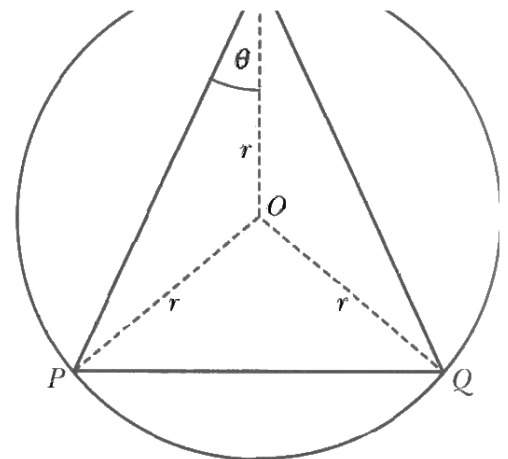
Add Maths book Ex 14.4 · Q17

The diagram shows an isosceles triangle  $PQR$  inscribed in a circle, centre  $O$ , radius  $r \text{ cm}$ .

$PR = QR$  and angle  $ORP = \theta$  radians.

Triangle  $PQR$  has an area of  $A \text{ cm}^2$ .

- a** Show that  $A = r^2 \sin 2\theta + r^2 \sin 2\theta \cos 2\theta$ .
- b** Find the value of  $\theta$  for which  $A$  has a stationary value and determine the nature of this stationary value.





**145** Add Maths book Ex 14.4 · Q16(a) With centre  $O$ : triangle  $EOG = \frac{1}{2}(6)(6)\sin(\pi - 2\theta) = 18\sin 2\theta$ ,sector  $GOF = \frac{1}{2}(36)(2\theta) = 36\theta$  ✓(b)  $\frac{dA}{d\theta} = 36 + 36\cos 2\theta = 54$  at  $\theta = \frac{\pi}{6}$ ;  $\frac{dA}{dt} = 54 \times 0.05 = \boxed{2.7 \text{ cm}^2 \text{ per second}}$ **146** Add Maths book Ex 14.4 · Q17(b)  $\frac{dA}{d\theta} = 2r^2\cos 2\theta + 2r^2\cos 4\theta = 0 \Rightarrow \cos 4\theta = -\cos 2\theta \Rightarrow \boxed{\theta = \frac{\pi}{6}},$ **maximum**

147



Calculator

**Past-paper practice set · Q3**

The normal to the curve  $y = 3x^2 + kx + 2$  at the point  $(-2, 4)$  is parallel to the line  $7y + x = 14$ . Find the value of  $k$  and calculate the coordinates of the point where this normal meets the curve again, giving your answers corrected to 3 s.f.

148



No calculator

**Past-paper practice set · Q8**

The tangent to the curve  $y = \frac{\ln x^2}{x^2}$  at the point where the curve crosses the positive  $x$ -axis and passes through the point  $(2, k)$ . Find the value of  $k$ .

**147** *Past-paper practice set · Q3*

Point  $(-2, 4)$  on curve:  $12 - 2k + 2 = 4 \Rightarrow \boxed{k = 5}$ . Gradient of tangent  $6x + 5 = -7$ , so normal gradient  $\frac{1}{7}$ .

Normal:  $y = \frac{1}{7}x + \frac{30}{7}$ . Meets curve:  $21x^2 + 34x - 16 = 0 \Rightarrow x = -2$  or  $x = \frac{8}{21}$ :  $\boxed{(0.381, 4.34)}$

Note: with  $k = 5$  the normal has gradient  $+\frac{1}{7}$ , so the line in the question should be  $7y - x = 14$ ; the source's printed answer  $(0.286, 3.67)$  used the wrong sign.

**148** *Past-paper practice set · Q8*

Crosses  $x$ -axis:  $\ln x^2 = 0 \Rightarrow x = 1$ .  $\frac{dy}{dx} = \frac{2 - 2\ln x^2}{x^3} = 2$ ; tangent  $y = 2x - 2$ ; through  $(2, k)$ :  $\boxed{k = 2}$

**149**

No calculator

**Oct/Nov 2014 • Paper 11 • (Add Maths book Ex 12.11 Q1)**

Find the coordinates of the stationary point on the curve  $y = x^2 + \frac{16}{x}$ . [4]

**150**

No calculator

**Oct/Nov 2014 • Paper 11 • (Add Maths book Ex 14.5 Q1)**

- a** Find the equation of the tangent to the curve  $y = x^3 - \ln x$  at the point on the curve where  $x = 1$ . [4]
- b** Show that the tangent bisects the line joining the points  $(-2, 16)$  and  $(12, 2)$ . [2]

**151**

Calculator

**Oct/Nov 2014 • Paper 21 • (Add Maths book Ex 12.11 Q2)**

Given that  $y = \frac{x^2}{2 + x^2}$ , show that  $\frac{dy}{dx} = \frac{kx}{(2 + x)^2}$ , where  $k$  is a constant to be found. [3]

**149** Oct/Nov 2014 · Paper 11 · (Add Maths book Ex 12.11 Q1)

$$\frac{dy}{dx} = 2x - \frac{16}{x^2} = 0 \Rightarrow x^3 = 8, x = 2: \boxed{(2, 12)}$$

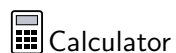
**150** Oct/Nov 2014 · Paper 11 · (Add Maths book Ex 14.5 Q1)

(a)  $\frac{dy}{dx} = 3x^2 - \frac{1}{x} = 2$ , point  $(1, 1)$ :  $\boxed{y = 2x - 1}$

(b) Midpoint of  $(-2, 16), (12, 2)$  is  $(5, 9)$ ;  $2(5) - 1 = 9 \checkmark$

**151** Oct/Nov 2014 · Paper 21 · (Add Maths book Ex 12.11 Q2)

$$\frac{dy}{dx} = \frac{2x(2+x^2) - x^2(2x)}{(2+x^2)^2} = \frac{4x}{(2+x^2)^2}: \boxed{k = 4}$$

**152**

Calculator

**May/June 2014 • Paper 21 • (Add Maths book Ex 12.11 Q3)**

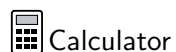
Given that a curve has equation  $y = \frac{1}{x} + 2\sqrt{x}$ , where  $x > 0$ , find

**a**  $\frac{dy}{dx}$ , [2]

**b**  $\frac{d^2y}{dx^2}$ . [2]

Hence or otherwise, find

**c** the coordinates and nature of the stationary point on the curve. [4]

**153**

Calculator

**May/June 2014 • Paper 21 • (Add Maths book Ex 12.11 Q4)**

A sector of a circle of radius  $r$  cm has an angle  $\theta$  radians, where  $\theta < \pi$ . The perimeter of the sector is 30 cm.

**a** Show that the area,  $A$  cm<sup>2</sup>, of the sector is given by  $A = 15r - r^2$ . [3]

**b** Given that  $r$  can vary, find the maximum area of the sector. [3]

**154**

Calculator

**May/June 2014 • Paper 21 • (Add Maths book Ex 14.5 Q2)**

Find  $\frac{dy}{dx}$  when

**a**  $y = \cos 2x \sin\left(\frac{x}{3}\right)$ , [4]

**b**  $y = \frac{\tan x}{1 + \ln x}$ . [4]

**152** May/June 2014 · Paper 21 · (Add Maths book Ex 12.11 Q3)

(a)  $\boxed{-\frac{1}{x^2} + \frac{1}{\sqrt{x}}}$  (b)  $\boxed{\frac{2}{x^3} - \frac{1}{2x^{3/2}}}$  (c)  $x^{\frac{3}{2}} = 1 \Rightarrow x = 1$ :  $\boxed{(1, 3)}$ ,  $y'' = 1.5 > 0$ : **minimum**

**153** May/June 2014 · Paper 21 · (Add Maths book Ex 12.11 Q4)

(a)  $2r + r\theta = 30 \Rightarrow \theta = \frac{30 - 2r}{r}$ ;  $A = \frac{1}{2}r^2\theta = \frac{1}{2}r(30 - 2r) = 15r - r^2 \checkmark$   
 (b)  $15 - 2r = 0$ ,  $r = 7.5$ :  $\boxed{A_{\max} = 56.25 \text{ cm}^2}$

**154** May/June 2014 · Paper 21 · (Add Maths book Ex 14.5 Q2)

(a)  $\boxed{\frac{1}{3} \cos 2x \cos \frac{x}{3} - 2 \sin 2x \sin \frac{x}{3}}$  (b)  $\boxed{\frac{\sec^2 x (1 + \ln x) - \frac{1}{x} \tan x}{(1 + \ln x)^2}}$

**155**

Calculator

**Oct/Nov 2013 • Paper 21 • (Add Maths book Ex 12.11 Q5)**

- a** Given that  $y = \left(\frac{1}{4}x - 5\right)^8$ , find  $\frac{dy}{dx}$ . [2]
- b** Hence find the approximate change in  $y$  as  $x$  increases from 12 to  $12 + p$ , where  $p$  is small. [2]

**156**

Calculator

**May/June 2013 • Paper 21 • (Add Maths book Ex 12.11 Q6)**

Find the equation of the normal to the curve  $y = \frac{x^2 + 8}{x - 2}$  at the point on the curve where  $x = 4$ . [6]

**157**

Calculator

**May/June 2013 • Paper 21 • (Add Maths book Ex 14.5 Q3)**

Variables  $x$  and  $y$  are related by the equation  $y = 10 - 4 \sin^2 x$ , where  $0 \leq x \leq \frac{\pi}{2}$ .  
Given that  $x$  is increasing at a rate of 0.2 radians per second, find the corresponding rate of change of  $y$  when  $y = 8$ . [6]

**158**

No calculator

**Oct/Nov 2012 • Paper 11 • (Add Maths book Ex 14.5 Q4)**

Given that  $y = \frac{x^2}{\cos 4x}$ , find

- a**  $\frac{dy}{dx}$ , [3]
- b** the approximate change in  $y$  when  $x$  increases from  $\frac{\pi}{4}$  to  $\frac{\pi}{4} + p$ , where  $p$  is small. [2]



**155** Oct/Nov 2013 · Paper 21 · (Add Maths book Ex 12.11 Q5)

(a)  $2\left(\frac{1}{4}x - 5\right)^7$  (b) At  $x = 12$ :  $2(-2)^7 = -256$ :  $\delta y \approx -256p$

**156** May/June 2013 · Paper 21 · (Add Maths book Ex 12.11 Q6)

$x = 4$ :  $y = 12$ .  $\frac{dy}{dx} = \frac{2x(x-2) - (x^2+8)}{(x-2)^2} = \frac{-8}{4} = -2$ , normal gradient  $\frac{1}{2}$ :

$y - 12 = \frac{1}{2}(x - 4) \Rightarrow y = \frac{1}{2}x + 10$

**157** May/June 2013 · Paper 21 · (Add Maths book Ex 14.5 Q3)

$y = 8 \Rightarrow \sin^2 x = \frac{1}{2}$ ,  $x = \frac{\pi}{4}$ .  $\frac{dy}{dx} = -8 \sin x \cos x = -4$ ;  $\frac{dy}{dt} = -4 \times 0.2 = -0.8$

**158** Oct/Nov 2012 · Paper 11 · (Add Maths book Ex 14.5 Q4)

(a)  $\frac{2x \cos 4x + 4x^2 \sin 4x}{\cos^2 4x}$  (b) at  $x = \frac{\pi}{4}$ :  $\frac{dy}{dx} = -\frac{\pi}{2}$ :  $\delta y \approx -\frac{\pi}{2}p$

**159**

Calculator

**Oct/Nov 2012 • Paper 21 • (Add Maths book Ex 12.11 Q8)**

The normal to the curve  $y = x^3 + 6x^2 - 34x + 44$  at the point  $P(2, 8)$  cuts the  $x$ -axis at  $A$  and the  $y$ -axis at  $B$ . Show that the mid-point of the line  $AB$  lies on the line  $4y = x + 9$ . [8]

**160**

No calculator

**May/June 2012 • Paper 11 • (Add Maths book Ex 12.11 Q7)**

- a** Find the equation of the tangent to the curve  $y = x^3 + 2x^2 - 3x + 4$  at the point where the curve crosses the  $y$ -axis. [4]
- b** Find the coordinates of the point where the tangent meets the curve again. [3]

**161**

Calculator

**May/June 2012 • Paper 11 • (Add Maths book Ex 14.5 Q5)**

Variables  $x$  and  $y$  are such that  $y = e^{2x} + e^{-2x}$ .

- a** Find  $\frac{dy}{dx}$ . [2]
- b** By using the substitution  $u = e^{2x}$ , find the value of  $y$  when  $\frac{dy}{dx} = 3$ . [4]
- c** Given that  $x$  is decreasing at the rate of  $0.5 \text{ units s}^{-1}$ , find the corresponding rate of change of  $y$  when  $x = 1$ . [3]

**159** Oct/Nov 2012 · Paper 21 · (Add Maths book Ex 12.11 Q8)

$$\frac{dy}{dx} = 3x^2 + 12x - 34 = 2 \text{ at } P; \text{ normal: } y - 8 = -\frac{1}{2}(x - 2) \Rightarrow y = -\frac{1}{2}x + 9.$$

$$A(18, 0), B(0, 9), \text{ midpoint } (9, 4.5): 4(4.5) = 18 = 9 + 9 \checkmark$$

**160** May/June 2012 · Paper 11 · (Add Maths book Ex 12.11 Q7)

$$(a) (0, 4), \frac{dy}{dx} = 3x^2 + 4x - 3 = -3: \boxed{y = -3x + 4} \quad (b) x^3 + 2x^2 = 0:$$

$$\boxed{(-2, 10)}$$

**161** May/June 2012 · Paper 11 · (Add Maths book Ex 14.5 Q5)

$$(a) \boxed{2e^{2x} - 2e^{-2x}} \quad (b) 2u - \frac{2}{u} = 3 \Rightarrow 2u^2 - 3u - 2 = 0, u = 2: y =$$

$$u + \frac{1}{u} = \boxed{2.5}$$

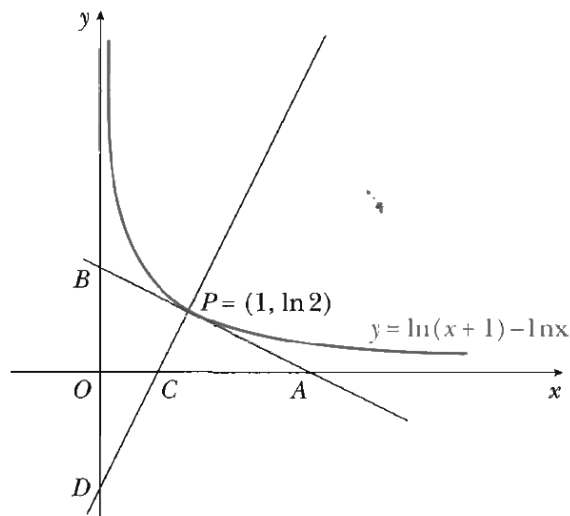
$$(c) \frac{dy}{dt} = (2e^2 - 2e^{-2})(-0.5) = \boxed{-7.25}$$

162



Calculator

Oct/Nov 2011 • Paper 21 • (Add Maths book Ex 14.5 Q6)



The diagram shows part of the curve  $y = \ln(x+1) - \ln x$ . The tangent to the curve at the point  $P(1, \ln 2)$  meets the  $x$ -axis at  $A$  and the  $y$ -axis at  $B$ . The normal to the curve at  $P$  meets the  $x$ -axis at  $C$  and the  $y$ -axis at  $D$ .

- a** Find, in terms of  $\ln 2$ , the coordinates of  $A$ ,  $B$ ,  $C$  and  $D$ . [8]
- b** Given that  $\frac{\text{Area of triangle } BPD}{\text{Area of triangle } APC} = \frac{1}{k}$ , express  $k$  in terms of  $\ln 2$ . [3]

163



No calculator

Oct/Nov 2010 • Paper 11 • (Add Maths book Ex 14.5 Q7)

A curve has equation  $y = 2x \sin x + \frac{\pi}{3}$ . The curve passes through the point  $P\left(\frac{\pi}{2}, a\right)$ .

- a** Find, in terms of  $\pi$ , the value of  $a$ . [1]
- b** Using your value of  $a$ , find the equation of the normal to the curve at  $P$ . [5]

**162** Oct/Nov 2011 · Paper 21 · (Add Maths book Ex 14.5 Q6)

$\frac{dy}{dx} = \frac{1}{x+1} - \frac{1}{x} = -\frac{1}{2}$  at  $P$ . Tangent  $y - \ln 2 = -\frac{1}{2}(x - 1)$ ; normal  $y - \ln 2 = 2(x - 1)$ .

(a)  $A(1 + 2\ln 2, 0), B(0, \frac{1}{2} + \ln 2), C(1 - \frac{1}{2}\ln 2, 0), D(0, -2 + \ln 2)$

(b)  $k = (\ln 2)^2$

**163** Oct/Nov 2010 · Paper 11 · (Add Maths book Ex 14.5 Q7)

(a)  $a = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$  (b)  $\frac{dy}{dx} = 2\sin x + 2x\cos x = 2$ , normal gradient  $-\frac{1}{2}$ :  $2y = \frac{19\pi}{6} - x$

164



No calculator

Oct/Nov 2020 · Paper 11 · Q4

It is given that  $y = \frac{\tan 3x}{\sin x}$ .

(a) Find the exact value of  $\frac{dy}{dx}$  when  $x = \frac{\pi}{3}$ . [4]

(b) Hence find the approximate change in  $y$  as  $x$  increases from  $\frac{\pi}{3}$  to  $\frac{\pi}{3} + h$ , where  $h$  is small. [1]

(c) Given that  $x$  is increasing at the rate of 3 units per second, find the corresponding rate of change in  $y$  when  $x = \frac{\pi}{3}$ , giving your answer in its simplest surd form. [2]

**164** Oct/Nov 2020 · Paper 11 · Q4

|        |                                                                                              |           |                                                                                                                                                                  |
|--------|----------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24 (a) | $\frac{dy}{dx} = \frac{\sin x \times 3\sec^2 3x - \tan 3x \cos x}{\sin^2 x}$                 | <b>3</b>  | <b>B1</b> for $3\sec^2 3x$<br><b>M1</b> for differentiation of a quotient or equivalent product<br><b>A1</b> for all other terms apart from $3\sec^2 3x$ correct |
|        | When $x = \frac{\pi}{3}$ $\frac{dy}{dx} = 2\sqrt{3}$                                         | <b>A1</b> |                                                                                                                                                                  |
| (b)    | $2\sqrt{3}h$                                                                                 | <b>B1</b> | <b>FT</b> on <i>their</i> answer to (a)                                                                                                                          |
| (c)    | $\frac{dy}{dx} \times \frac{dx}{dt} = \frac{dy}{dt}$<br>$2\sqrt{3} \times 3 = \frac{dy}{dt}$ | <b>M1</b> | For correct use of rates of change using <i>their</i> answer to (a)                                                                                              |
|        | $\frac{dy}{dt} = 6\sqrt{3}$                                                                  | <b>A1</b> |                                                                                                                                                                  |

**165**

No calculator

**Oct/Nov 2020 • Paper 11 • Q9**

- (a) Given that  $\int_1^a \left( \frac{1}{x} - \frac{1}{2x+3} \right) dx = \ln 3$ , where  $a > 0$ , find the exact value of  $a$ , giving your answer in simplest surd form. [6]

- (b) Find the exact value of  $\int_0^{\frac{\pi}{3}} \left( \sin \left( 2x + \frac{\pi}{3} \right) - 1 + \cos 2x \right) dx$ . [5]

**166**

No calculator

**Oct/Nov 2020 • Paper 12 • Q7**

A curve has equation  $y = \frac{\ln(3x^2 - 5)}{2x + 1}$  for  $3x^2 > 5$ .

- (a) Find the equation of the normal to the curve at the point where  $x = \sqrt{2}$ . [6]

- (b) Find the approximate change in  $y$  as  $x$  increases from  $\sqrt{2}$  to  $\sqrt{2} + h$ , where  $h$  is small. [1]



## 165 Oct/Nov 2020 · Paper 11 · Q9

|        |                                                     |           |                                                                                                    |
|--------|-----------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------|
| 25 (a) | $\left[ \ln x - \frac{1}{2} \ln(2x+3) \right]_1^a$  | <b>2</b>  | <b>B1</b> for $\ln x$<br><b>B1</b> for $\frac{1}{2} \ln(2x+3)$                                     |
|        | $\ln a - \frac{1}{2} \ln(2a+3) + \frac{1}{2} \ln 5$ | <b>M1</b> | For correct application of limits, must have at least one <b>B1</b>                                |
|        | $\ln a \sqrt{\frac{5}{2a+3}}$                       | <b>M1</b> | <b>Dep</b> on previous M mark, for application of log laws                                         |
|        | $5a^2 - 18a - 27 = 0$                               | <b>M1</b> | <b>Dep</b> on previous M mark for equating to $\ln 3$ and simplification to a 3 term quadratic = 0 |
|        | $a = \frac{9+6\sqrt{6}}{5}$                         | <b>A1</b> | Must have one solution only                                                                        |

|     |                                                                                                                                           |           |                                                                                                                                |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------|
| (b) | $-\frac{1}{2} \cos\left(2x + \frac{\pi}{3}\right) + \frac{1}{2} \sin 2x - x$                                                              | <b>3</b>  | <b>B1</b> for $-\frac{1}{2} \cos\left(2x + \frac{\pi}{3}\right)$<br><b>B1</b> for $+\frac{1}{2} \sin 2x$<br><b>B1</b> for $-x$ |
|     | $\left( -\frac{1}{2} \cos \pi + \frac{1}{2} \sin \frac{2\pi}{3} - \frac{\pi}{3} \right) - \left( -\frac{1}{2} \cos \frac{\pi}{3} \right)$ | <b>M1</b> | For correct use of limits in <i>their</i> integral, must have at least one <b>B1</b> term                                      |
|     | $\frac{3}{4} + \frac{\sqrt{3}}{4} - \frac{\pi}{3}$                                                                                        | <b>A1</b> |                                                                                                                                |

## 166 Oct/Nov 2020 · Paper 12 · Q7

| Question | Answer                                                                                                                                                    | Marks     | Guidance                                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 55 (a)   | $\frac{dy}{dx} = \frac{(2x+1) \frac{6x}{3x^2-5} - 2 \ln(3x^2-5)}{(2x+1)^2}$ or $\frac{dy}{dx} = (2x+1)^{-1} \frac{6x}{3x^2-5} - 2(2x+1)^{-2} \ln(3x^2-5)$ | <b>3</b>  | <b>B1</b> for $\frac{6x}{3x^2-5}$<br><b>M1</b> for attempt at a quotient or equivalent product<br><b>A1</b> for all terms other than $\frac{6x}{3x^2-5}$ correct |
|          | When $x = \sqrt{2}$ , $y = 0$                                                                                                                             | <b>B1</b> | May be implied                                                                                                                                                   |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                   |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p>When <math>x = \sqrt{2}</math>, <math>\frac{dy}{dx} = \frac{6\sqrt{2}}{2\sqrt{2}+1}</math> or <math>\frac{24-6\sqrt{2}}{7}</math><br/>or 2.22 oe</p> <p>Normal: <math>y = -\frac{(2\sqrt{2}+1)}{6\sqrt{2}}(x-\sqrt{2})</math> oe</p> <p>or <math>y = -\frac{7}{24-6\sqrt{2}}(x-\sqrt{2})</math> oe</p> <p>or <math>y = -\frac{1}{2.22}(x-\sqrt{2})</math> oe</p> <p>or <math>y = -\frac{4+\sqrt{2}}{12}(x-\sqrt{2})</math> oe</p> <p>or <math>y = -\frac{9+4\sqrt{2}}{24+6\sqrt{2}}(x-\sqrt{2})</math> oe</p> <p><math>y = -0.451x + 0.638</math></p> | <p><b>2</b></p> <p><b>M1</b> for attempt at normal using <i>their</i> <math>y</math> and <i>their</i> perp gradient<br/><b>A1</b> Allow equivalent surd forms</p> |
| (b) | <p><math>\left(\frac{6\sqrt{2}}{2\sqrt{2}+1}\right)h</math> or <math>\frac{24-6\sqrt{2}}{7}h</math> or other equivalent<br/>surd forms, or <math>2.22h</math></p>                                                                                                                                                                                                                                                                                                                                                                                        | <p><b>B1</b></p> <p><b>FT</b> on <i>their</i> <math>\frac{dy}{dx}</math> from (a)</p>                                                                             |

**167**

No calculator

**Oct/Nov 2020 • Paper 12 • Q9**

A curve has equation  $y = (2x - 1)\sqrt{4x + 3}$ .

(a) Show that  $\frac{dy}{dx} = \frac{4(Ax + B)}{\sqrt{4x + 3}}$ , where  $A$  and  $B$  are constants. [5]

(b) Hence write down the  $x$ -coordinate of the stationary point of the curve. [1]

(c) Determine the nature of this stationary point. [2]

## 167 Oct/Nov 2020 · Paper 12 · Q9

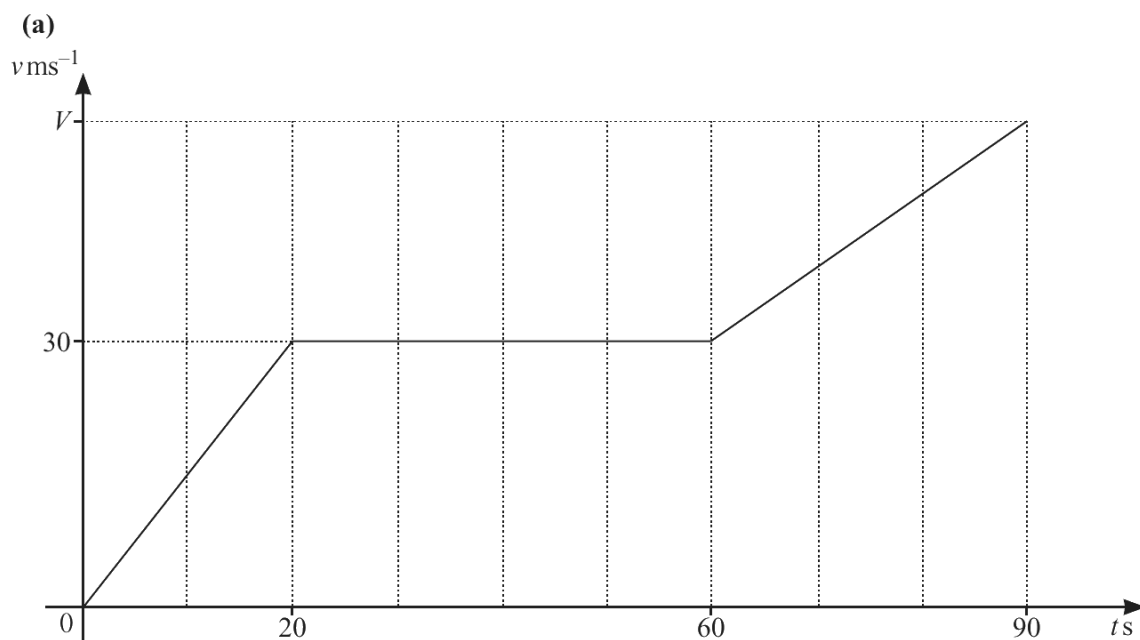
|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |                                                                                                                                                       |                |                 |                 |   |   |   |     |                 |                |                 |     |          |         |          |           |                                            |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------|-----------------|---|---|---|-----|-----------------|----------------|-----------------|-----|----------|---------|----------|-----------|--------------------------------------------|
| 56 (a)          | $\frac{dy}{dx} = (2x-1) \times \frac{1}{2} \times 4(4x+3)^{-\frac{1}{2}} + 2(4x+3)^{\frac{1}{2}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>3</b>       | <b>B1</b> for $\frac{1}{2} \times 4(4x+3)^{-\frac{1}{2}}$ oe<br><b>M1</b> for a correct attempt at a product<br><b>A1</b> for all other terms correct |                |                 |                 |   |   |   |     |                 |                |                 |     |          |         |          |           |                                            |
|                 | $\frac{dy}{dx} = 2(4x+3)^{-\frac{1}{2}}(2x-1+4x+3)$ or equivalent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>M1</b>      | For attempt to simplify to the given form                                                                                                             |                |                 |                 |   |   |   |     |                 |                |                 |     |          |         |          |           |                                            |
|                 | $\frac{dy}{dx} = \frac{4(3x+1)}{(4x+3)^{\frac{1}{2}}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>A1</b>      |                                                                                                                                                       |                |                 |                 |   |   |   |     |                 |                |                 |     |          |         |          |           |                                            |
| (b)             | $-\frac{1}{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>B1</b>      | <b>FT</b> on <i>their</i> $3x+1=0$                                                                                                                    |                |                 |                 |   |   |   |     |                 |                |                 |     |          |         |          |           |                                            |
| (c)             | For a complete method using 2 <sup>nd</sup> derivative, or gradient or $y$ values either side or one side of <i>their</i> stationary point e.g.<br><table border="1" data-bbox="304 994 750 1209"> <tr> <td><math>x</math></td><td><math>&lt;-\frac{1}{3}</math></td><td><math>-\frac{1}{3}</math></td><td><math>&gt;-\frac{1}{3}</math></td></tr> <tr> <td><math>\frac{dy}{dx}</math></td><td>–</td><td>0</td><td>+</td></tr> </table><br><table border="1" data-bbox="304 1243 809 1415"> <tr> <td><math>x</math></td><td><math>&lt;-\frac{1}{3}</math></td><td><math>-\frac{1}{3}</math></td><td><math>&gt;-\frac{1}{3}</math></td></tr> <tr> <td><math>y</math></td><td><math>&lt;-2.15</math></td><td><math>-2.15</math></td><td><math>&gt;-2.15</math></td></tr> </table> | $x$            | $<-\frac{1}{3}$                                                                                                                                       | $-\frac{1}{3}$ | $>-\frac{1}{3}$ | $\frac{dy}{dx}$ | – | 0 | + | $x$ | $<-\frac{1}{3}$ | $-\frac{1}{3}$ | $>-\frac{1}{3}$ | $y$ | $<-2.15$ | $-2.15$ | $>-2.15$ | <b>M1</b> | Must be using values of $x > -\frac{3}{4}$ |
| $x$             | $<-\frac{1}{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $-\frac{1}{3}$ | $>-\frac{1}{3}$                                                                                                                                       |                |                 |                 |   |   |   |     |                 |                |                 |     |          |         |          |           |                                            |
| $\frac{dy}{dx}$ | –                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0              | +                                                                                                                                                     |                |                 |                 |   |   |   |     |                 |                |                 |     |          |         |          |           |                                            |
| $x$             | $<-\frac{1}{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $-\frac{1}{3}$ | $>-\frac{1}{3}$                                                                                                                                       |                |                 |                 |   |   |   |     |                 |                |                 |     |          |         |          |           |                                            |
| $y$             | $<-2.15$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $-2.15$        | $>-2.15$                                                                                                                                              |                |                 |                 |   |   |   |     |                 |                |                 |     |          |         |          |           |                                            |
|                 | Minimum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>A1</b>      | Must be from correct work                                                                                                                             |                |                 |                 |   |   |   |     |                 |                |                 |     |          |         |          |           |                                            |

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No calculator

Oct/Nov 2020 • Paper 12 • Q12



The diagram shows the velocity–time graph of a particle  $P$  that travels 2775 m in 90 s, reaching a final velocity of  $V \text{ ms}^{-1}$ .

(i) Find the value of  $V$ . [3]

(ii) Write down the acceleration of  $P$  when  $t = 40$ . [1]

(b) The acceleration,  $a \text{ ms}^{-2}$ , of a particle  $Q$  travelling in a straight line, is given by  $a = 6 \cos 2t$  at time  $t$  s. When  $t = 0$  the particle is at point  $O$  and is travelling with a velocity of  $10 \text{ ms}^{-1}$ .

(i) Find the velocity of  $Q$  at time  $t$ . [3]

(ii) Find the displacement of  $Q$  from  $O$  at time  $t$ . [3]

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| Question      | Answer                                                                                                                                                                 | Marks     | Guidance                                                                                                                                                               |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 57<br>(a) (i) | Area under graph:<br>$\frac{1}{2}(60+40)\times 30 + \frac{1}{2}(30+V)\times 30 \quad (= 2775)$<br>or $\frac{1}{2}(20\times 30) + (40+30) + \frac{1}{2}(30+V)\times 30$ | <b>2</b>  | <b>M1</b> for attempt to find the area under the graph<br><b>Dep M1</b> on previous M mark for attempt to equate to 2775 and simplify in order to find $V$ or $V - 30$ |
|               | 55                                                                                                                                                                     | <b>A1</b> |                                                                                                                                                                        |
| (a)(ii)       | 0                                                                                                                                                                      | <b>B1</b> |                                                                                                                                                                        |
| (b)(i)        | $v = 3\sin 2t \quad (+c)$                                                                                                                                              | <b>M1</b> | Must have $\pm 3\sin 2t$                                                                                                                                               |
|               | $10 = c$                                                                                                                                                               | <b>M1</b> | <b>Dep</b> for attempt to find $+c$ ,                                                                                                                                  |
|               | $v = 3\sin 2t + 10$                                                                                                                                                    | <b>A1</b> |                                                                                                                                                                        |
| (b)(ii)       | $s = -\frac{3}{2}\cos 2t + 10t + d$                                                                                                                                    | <b>M1</b> | For attempt to integrate <i>their</i> $v$ , must have $\pm \frac{3}{2}\cos 2t$                                                                                         |
|               | $d = \frac{3}{2}$                                                                                                                                                      | <b>M1</b> | <b>Dep</b> on previous M mark for attempt to find $d$ .                                                                                                                |
|               | $s = -\frac{3}{2}\cos 2t + 10t + \frac{3}{2}$                                                                                                                          | <b>A1</b> |                                                                                                                                                                        |

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No calculator

Oct/Nov 2020 • Paper 13 • Q2

- (a) Given that  $y = \frac{e^{2x-3}}{x^2+1}$ , find  $\frac{dy}{dx}$ . [3]

- (b) Hence, given that  $y$  is increasing at the rate of 2 units per second, find the exact rate of change of  $x$  when  $x = 2$ . [3]

**169** Oct/Nov 2020 · Paper 13 · Q2

|        |                                                                                                                                                                                  |           |                                                                                                                                                                                                                                                                               |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 84 (a) | $\frac{dy}{dx} = \frac{(x^2 + 1)2e^{2x-3} - 2xe^{2x-3}}{(x^2 + 1)^2} \text{ oe}$<br>or $\frac{dy}{dx} = \frac{2e^{2x-3}}{(x^2 + 1)} - \frac{2xe^{2x-3}}{(x^2 + 1)^2} \text{ oe}$ | <b>3</b>  | <p><b>B1</b> for <math>\frac{d}{dx}(e^{2x-3}) = 2e^{2x-3}</math> seen in a quotient rule or product rule expression</p> <p><b>M1</b> for correct method for differentiating a quotient or equivalent product</p> <p><b>A1 FT</b> from <i>their</i> <math>2e^{2x-3}</math></p> |
| (b)    | When $x = 2$ , $\frac{dy}{dx} = \frac{6e}{25}$                                                                                                                                   | <b>M1</b> | For evaluation of <i>their</i> $\frac{dy}{dx}$ when $x = 2$                                                                                                                                                                                                                   |
|        | $\frac{6e}{25} \times \frac{dx}{dt} = 2 \text{ oe}$                                                                                                                              | <b>M1</b> | For correct substitution of <i>their</i> evaluated $\frac{dy}{dx}$ and $\frac{dy}{dt} = 2$ in a correct rates of change equation                                                                                                                                              |
|        | $\frac{dx}{dt} = \frac{25}{3e}, \frac{50}{6e}$                                                                                                                                   | <b>A1</b> |                                                                                                                                                                                                                                                                               |



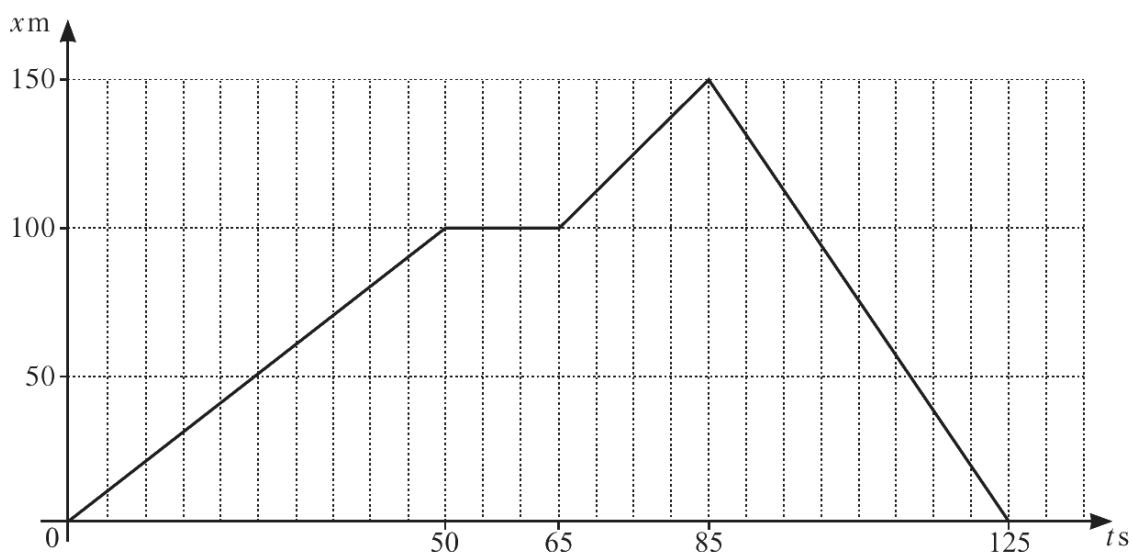
170



No calculator

Oct/Nov 2020 • Paper 13 • Q4

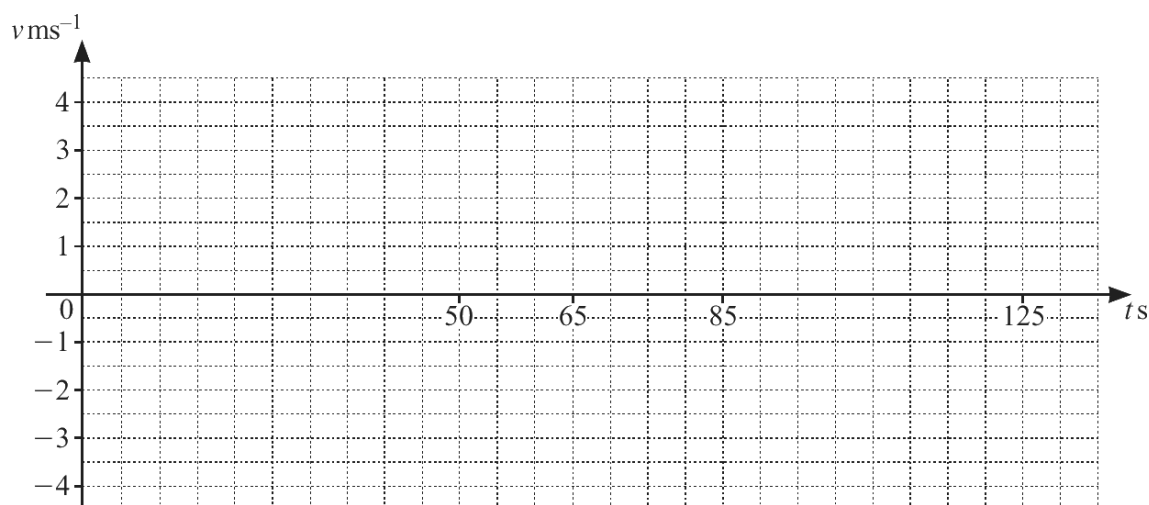
(a)



The diagram shows the  $x$ - $t$  graph for a runner, where displacement,  $x$ , is measured in metres and time,  $t$ , is measured in seconds.

(i) On the axes below, draw the  $v$ - $t$  graph for the runner.

[3]

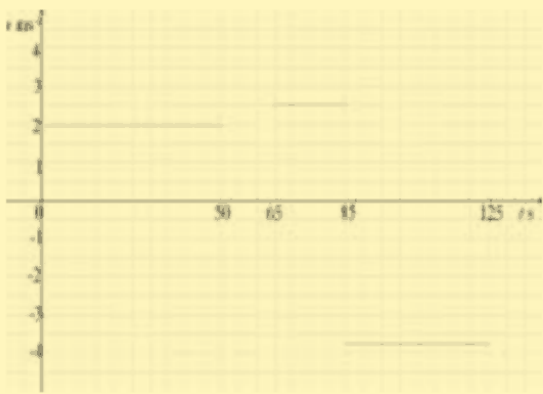


(ii) Find the total distance covered by the runner in 125 s.

[1]

(b) The displacement,  $x$  m, of a particle from a fixed point at time  $t$  s is given by  $x = 6 \cos\left(3t + \frac{\pi}{3}\right)$ . Find the acceleration of the particle when  $t = \frac{2\pi}{3}$ . [3]

**170** Oct/Nov 2020 · Paper 13 · Q4

|              |                                                                                   |           |                                                                                                                                                                                       |
|--------------|-----------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 85<br>(a)(i) |  | <b>3</b>  | <b>B1</b> For $v = 2$ for $0 \leq t \leq 50$<br><b>B1</b> For $v = 2.5$ for $50 \leq t \leq 85$<br><b>B1</b> For $v = 0$ for $85 \leq t \leq 125$ and $v = 0$ for $50 \leq t \leq 85$ |
| (a)(ii)      | 300                                                                               | <b>B1</b> |                                                                                                                                                                                       |
| (b)          | $\frac{dx}{dt} = -18 \sin\left(3t + \frac{\pi}{3}\right)$                         | <b>M1</b> | $\pm 18 \sin\left(3t + \frac{\pi}{3}\right)$                                                                                                                                          |
|              | $\frac{d^2x}{dt^2} = -54 \cos\left(3t + \frac{\pi}{3}\right)$                     | <b>M1</b> | $\pm 54 \cos\left(3t + \frac{\pi}{3}\right)$                                                                                                                                          |
|              | -27 nfw                                                                           | <b>A1</b> |                                                                                                                                                                                       |

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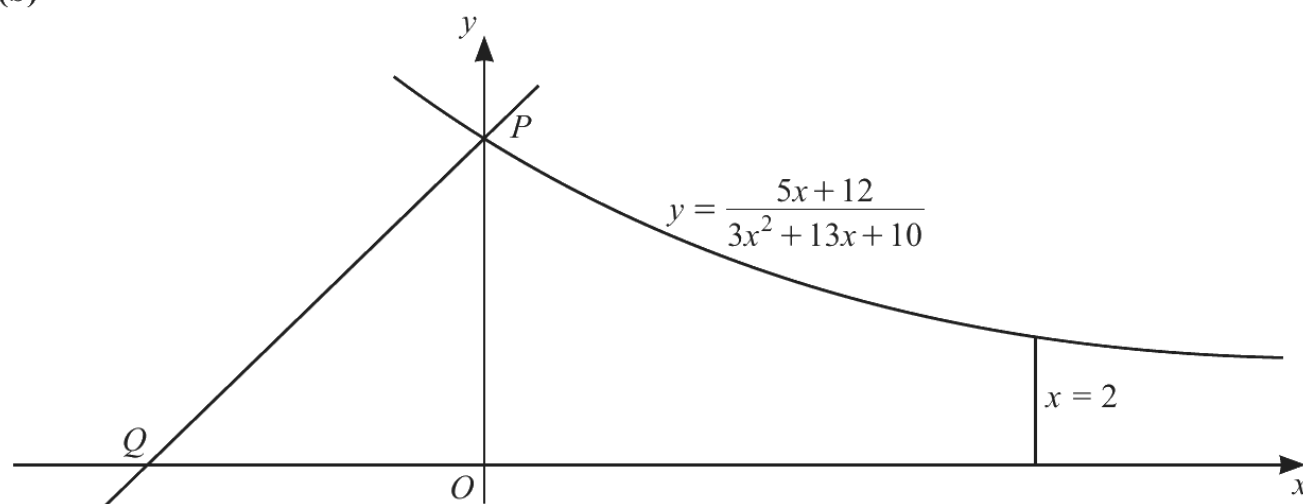


No calculator

Oct/Nov 2020 • Paper 13 • Q10

- (a) Show that  $\frac{1}{x+1} + \frac{2}{3x+10}$  can be written as  $\frac{5x+12}{3x^2+13x+10}$ . [1]

(b)



The diagram shows part of the curve  $y = \frac{5x+12}{3x^2+13x+10}$ , the line  $x = 2$  and a straight line of gradient 1. The curve intersects the  $y$ -axis at the point  $P$ . The line of gradient 1 passes through  $P$  and intersects the  $x$ -axis at the point  $Q$ . Find the area of the shaded region, giving your answer in the form  $a + \frac{2}{3} \ln(b\sqrt{3})$ , where  $a$  and  $b$  are constants. [9]

Additional working space for question 86

**171** Oct/Nov 2020 · Paper 13 · Q10

|        |                                                                                                                                 |           |                                                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------|
| 86 (a) | $\frac{3x+10+2(x+1)}{(x+1)(3x+10)} = \frac{3x+10+2x+2}{(x+1)(3x+10)}$ $= \frac{5x+12}{3x^2+13x+10}$                             | <b>B1</b> | For expansion and simplification to obtain given answer                                                               |
| (b)    | $P\left(0, \frac{6}{5}\right)$ and $Q\left(-\frac{6}{5}, 0\right)$ oe                                                           | <b>B1</b> |                                                                                                                       |
|        | Area of triangle = $\frac{18}{25}$ or 0.72                                                                                      | <b>B1</b> |                                                                                                                       |
|        | Area under curve = $\int_0^2 \frac{1}{x+1} + \frac{2}{3x+10} \, dx$                                                             | <b>M1</b> | For use of part (a) and attempt to integrate to obtain at least one ln term.                                          |
|        | $= \left[ \ln(x+1) + \frac{2}{3} \ln(3x+10) \right]_0^2$                                                                        | <b>2</b>  | <b>B1</b> For $\ln(x+1)$<br><b>B1</b> For $\frac{2}{3} \ln(3x+10)$                                                    |
|        | $= \ln 3 + \frac{2}{3} \ln 16 - \frac{2}{3} \ln 10$                                                                             | <b>M1</b> | For correct use of limits. Dep on previous M mark.                                                                    |
|        | $= \frac{2}{3} \ln 3\sqrt{3} + \frac{2}{3} \ln 16 - \frac{2}{3} \ln 10$                                                         | <b>M1</b> | For use of $\ln 3 = \frac{2}{3} \ln 3\sqrt{3}$                                                                        |
|        | $= \frac{2}{3} \ln 3\sqrt{3} + \frac{2}{3} \ln \left(\frac{16}{10}\right) = \frac{2}{3} \ln \left(\frac{48\sqrt{3}}{10}\right)$ | <b>M1</b> | For use of multiplication and division rule                                                                           |
|        | Total area = $\frac{18}{25} + \frac{2}{3} \ln \left(\frac{24\sqrt{3}}{5}\right)$ oe                                             | <b>A1</b> | For correct answer in the required form dep on the three preceding M marks<br>Must not be obtained using a calculator |

**172**

No calculator

**May/June 2020 • Paper 11 • Q3**

The radius,  $r$  cm, of a circle is increasing at the rate of  $5 \text{ cm s}^{-1}$ . Find, in terms of  $\pi$ , the rate at which the area of the circle is increasing when  $r = 3$ . [4]

**173**

Calculator

**May/June 2020 • Paper 11 • Q5**

Find the equation of the tangent to the curve  $y = \frac{\ln(3x^2 - 1)}{x + 2}$  at the point where  $x = 1$ . Give your answer in the form  $y = mx + c$ , where  $m$  and  $c$  are constants correct to 3 decimal places. [6]

**174**

No calculator

**May/June 2020 • Paper 11 • Q8**

(a) Show that  $\frac{3}{2x-3} + \frac{3}{2x+3}$  can be written as  $\frac{12x}{4x^2 - 9}$ . [2]

(b) Hence find  $\int \frac{12x}{4x^2 - 9} dx$ , giving your answer as a single logarithm and an arbitrary constant. [3]

(c) Given that  $\int_2^a \frac{12x}{4x^2 - 9} dx = \ln 5\sqrt{5}$ , where  $a > 2$ , find the exact value of  $a$ . [4]

**172** May/June 2020 · Paper 11 · Q3

|    |                                                                                              |           |                                                          |
|----|----------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------|
| 10 | $\frac{dr}{dt} = 5$                                                                          | <b>B1</b> |                                                          |
|    | $\frac{dA}{dr} = 2\pi r$                                                                     | <b>B1</b> |                                                          |
|    | $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$ leading to<br>$\frac{dA}{dt} = 10\pi r$ | <b>M1</b> | Use of the chain rule, may be implied by $5 \times 6\pi$ |
|    | $\frac{dA}{dt} = 30\pi$                                                                      | <b>A1</b> |                                                          |

**173** May/June 2020 · Paper 11 · Q5

|    |                                                                        |           |                                                                                                                     |
|----|------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------|
| 11 | $\frac{dy}{dx} = \frac{(x+2)\frac{6x}{3x^2-1} - \ln(3x^2-1)}{(x+2)^2}$ | <b>B1</b> | B1 for $\frac{6x}{3x^2-1}$                                                                                          |
|    |                                                                        | <b>M1</b> | For attempt to differentiate a quotient or an equivalent product, must have correct order of terms and correct sign |
|    |                                                                        | <b>A1</b> |                                                                                                                     |
|    | When $x=1$ , $y = \frac{\ln 2}{3}$ or 0.231(0)                         | <b>B1</b> |                                                                                                                     |
|    | When $x=1$ , $\frac{dy}{dx} = 0.92298$ ,<br>allow 0.923                | <b>B1</b> |                                                                                                                     |
|    | $y = 0.923x - 0.692$                                                   | <b>B1</b> |                                                                                                                     |

**174** May/June 2020 · Paper 11 · Q8

|        |                                  |           |                 |
|--------|----------------------------------|-----------|-----------------|
| 12 (a) | $\frac{3(2x+3)+3(2x-3)}{4x^2-9}$ | <b>M1</b> | Must see for M1 |
|        | $\frac{12x}{4x^2-9}$             | <b>A1</b> |                 |

|     |                                                                                                                                   |           |                                                                      |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------|
| (b) | $\int \frac{3}{2x-3} + \frac{3}{2x+3} \, dx$ $= \frac{3}{2} \ln(2x-3) + \frac{3}{2} \ln(2x+3)$                                    | <b>B2</b> | B1 for each correct term, having made use of (a)                     |
|     | $\frac{3}{2} \ln(4x^2-9) + c \quad \text{or}$ $\frac{3}{2} \ln((2x-3)(2x+3)) + c \quad \text{or}$ $\ln(4x^2-9)^{\frac{3}{2}} + c$ | <b>B1</b> |                                                                      |
| (c) | $\ln(4a^2-9)^{\frac{3}{2}} - \ln 7^{\frac{3}{2}} = \ln 5^{\frac{3}{2}}$                                                           | <b>M1</b> | For correct application of limits, allow equivalent forms            |
|     | $4a^2 - 9 = 35$                                                                                                                   | <b>A1</b> | For a correct method of dealing with logarithms and eliminating them |
|     | $a = \sqrt{11}$                                                                                                                   | <b>M1</b> | For solving a quadratic equation, dep on first M mark                |
|     |                                                                                                                                   | <b>A1</b> |                                                                      |

175



No calculator

May/June 2020 • Paper 11 • Q11

A curve is such that  $\frac{d^2y}{dx^2} = 5 \cos 2x$ . This curve has a gradient of  $\frac{3}{4}$  at the point  $\left(-\frac{\pi}{12}, \frac{5\pi}{4}\right)$ . Find the equation of this curve. [8]

176



No calculator

May/June 2020 • Paper 12 • Q2

The volume,  $V$ , of a sphere of radius  $r$  is given by  $V = \frac{4}{3}\pi r^3$ .

The radius,  $r$  cm, of a sphere is increasing at the rate of  $0.5 \text{ cm s}^{-1}$ . Find, in terms of  $\pi$ , the rate of change of the volume of the sphere when  $r = 0.25$ . [4]



**175** May/June 2020 · Paper 11 · Q11

| Question | Answer                                                                                                                                                                | Marks     | Partial Marks                                            |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------|
| 13       | $\frac{dy}{dx} = \frac{5}{2} \sin 2x (+c)$                                                                                                                            | <b>M1</b> | M1 for $k \sin 2x$                                       |
|          |                                                                                                                                                                       | <b>A1</b> | Condone omission of $c$                                  |
|          | $\frac{3}{4} = \frac{5}{2} \sin\left(-\frac{\pi}{6}\right) + c$                                                                                                       | <b>M1</b> | Dep on first M1 for attempt to find $c$                  |
|          | $c = 2$                                                                                                                                                               | <b>A1</b> |                                                          |
|          | $y = -\frac{5}{4} \cos 2x + 2x (+d)$                                                                                                                                  | <b>M1</b> | M1 for attempt to integrate <i>their</i> $\frac{dy}{dx}$ |
|          |                                                                                                                                                                       | <b>A1</b> | Condone omission of $d$                                  |
|          | $\frac{5\pi}{4} = -\frac{5}{4} \cos\left(-\frac{\pi}{6}\right) - \frac{\pi}{6} + d$                                                                                   | <b>M1</b> | Dep on previous M1 for attempt to find $d$               |
|          | $d = \frac{17\pi}{12} + \frac{5\sqrt{3}}{8}$<br>$y = -\frac{5}{4} \cos 2x + 2x + \frac{17\pi}{12} + \frac{5\sqrt{3}}{8}$ or<br>$y = -\frac{5}{4} \cos 2x + 2x + 5.53$ | <b>A1</b> | Must have the equation for A1                            |

**176** May/June 2020 · Paper 12 · Q2

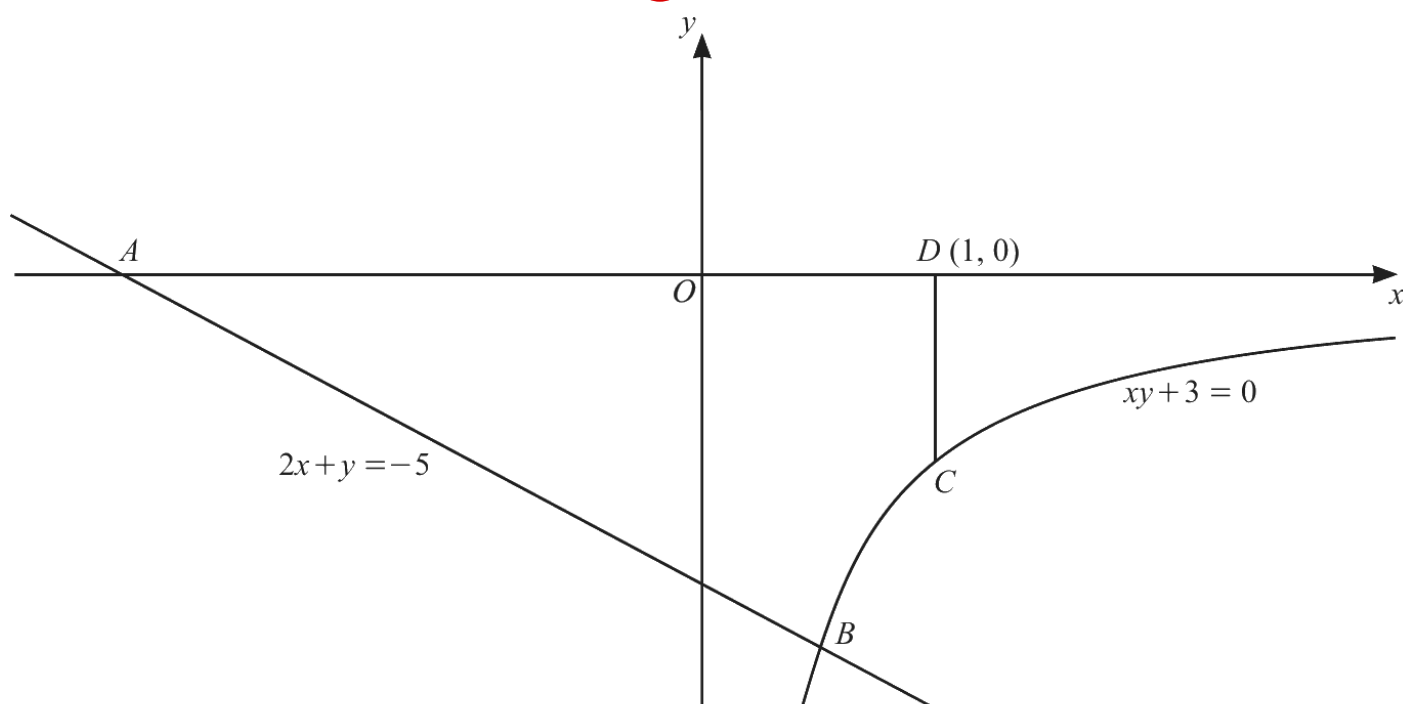
|    |                                                                                   |           |                                                     |
|----|-----------------------------------------------------------------------------------|-----------|-----------------------------------------------------|
| 40 | $\frac{dr}{dt} = 0.5$                                                             | <b>B1</b> |                                                     |
|    | $\frac{dV}{dr} = 4\pi r^2$                                                        | <b>B1</b> |                                                     |
|    | $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$<br>$\frac{dV}{dt} = \pi r^2$ | <b>M1</b> | For attempt to use a correct form of the chain rule |
|    | When $r = \frac{1}{4}$ , $\frac{dV}{dt} = 0.125\pi$                               | <b>A1</b> |                                                     |

177



No calculator

May/June 2020 · Paper 12 · Q6



The diagram shows the straight line  $2x + y = -5$  and part of the curve  $xy + 3 = 0$ . The straight line intersects the  $x$ -axis at the point  $A$  and intersects the curve at the point  $B$ . The point  $C$  lies on the curve. The point  $D$  has coordinates  $(1, 0)$ . The line  $CD$  is parallel to the  $y$ -axis.

(a) Find the coordinates of each of the points  $A$  and  $B$ . [3]

(b) Find the area of the shaded region, giving your answer in the form  $p + \ln q$ , where  $p$  and  $q$  are positive integers. [6]

**177** May/June 2020 · Paper 12 · Q6

|        |                                                                                         |           |                                                                                                    |
|--------|-----------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------|
| 41 (a) | $A \left( -\frac{5}{2}, 0 \right)$                                                      | <b>B1</b> |                                                                                                    |
|        | $x(-5-2x)+3=0$<br>$2x^2+5x-3=0$<br>$(2x-1)(x+3)=0$                                      | <b>M1</b> | For attempt to eliminate one variable, obtain a 3-term quadratic equation = 0 and attempt to solve |
|        | $B \left( \frac{1}{2}, -6 \right)$                                                      | <b>A1</b> | Allow A1 if just the x-coordinates or just the y-coordinates are given                             |
| (b)    | Area of triangle = $\frac{1}{2} \left( \frac{5}{2} + \frac{1}{2} \right) \times 6, = 9$ | <b>M1</b> | For attempt at triangle using <i>their</i> values                                                  |
|        | $\int_{\frac{1}{2}}^1 -\frac{3}{x} dx = [-3 \ln x]_{\frac{1}{2}}^1$                     | <b>M1</b> | For attempt to integrate, must have ln                                                             |
|        | $= 3 \ln \frac{1}{2}$                                                                   | <b>M1</b> | correct application of limits, dep on previous M mark                                              |
|        | $= -3 \ln 2$                                                                            | <b>M1</b> | realisation that value of integral is negative and making the adjustment                           |
|        |                                                                                         | <b>M1</b> | application of log law, dep on previous M mark                                                     |
|        | Area = $9 + \ln 8$                                                                      | <b>A1</b> |                                                                                                    |

178



Calculator

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- (a) Given that  $y = (x^2 - 1)\sqrt{5x + 2}$ , show that  $\frac{dy}{dx} = \frac{Ax^2 + Bx + C}{2\sqrt{5x + 2}}$ , where  $A$ ,  $B$  and  $C$  are integers. [5]

- (b) Find the coordinates of the stationary point of the curve  $y = (x^2 - 1)\sqrt{5x + 2}$ , for  $x > 0$ . Give each coordinate correct to 2 significant figures. [3]

- (c) Determine the nature of this stationary point. [2]

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|        |                                                                                                     |           |                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 42 (a) | $\frac{dy}{dx} = (x^2 - 1) \frac{5}{2} (5x + 2)^{-\frac{1}{2}} + 2x(5x + 2)^{\frac{1}{2}}$          | <b>B1</b> | For $\frac{5}{2}(5x + 2)^{-\frac{1}{2}}$                                                                                                              |
|        |                                                                                                     | <b>M1</b> | For differentiation of a product                                                                                                                      |
|        |                                                                                                     | <b>A1</b> |                                                                                                                                                       |
|        | $\frac{dy}{dx} = \frac{(5x + 2)^{-\frac{1}{2}}}{2} (5(x^2 - 1) + 4x(5x + 2))$<br>or equivalent      | <b>M1</b> | Dep on previous M mark for attempt to simplify                                                                                                        |
|        | $\frac{dy}{dx} = \frac{25x^2 + 8x - 5}{2\sqrt{5x + 2}}$                                             | <b>A1</b> |                                                                                                                                                       |
| (b)    | $25x^2 + 8x - 5 = 0$                                                                                | <b>M1</b> | Equating their numerator in (a) to zero and attempt to solve                                                                                          |
|        | $x = 0.315$                                                                                         | <b>A1</b> |                                                                                                                                                       |
|        | $y = -1.70$                                                                                         | <b>A1</b> |                                                                                                                                                       |
| (c)    | Consideration of gradient or $y$ values either side of stationary point, remembering that $x > 0$ . | <b>M1</b> | Must be a complete method making use of <i>their</i> (a). Allow consideration of $25x^2 + 8x - 5$ as a 'minimum curve'. Accept 2nd derivative method. |
|        | Minimum                                                                                             | <b>A1</b> |                                                                                                                                                       |

179



No calculator

May/June 2020 · Paper 12 · Q9

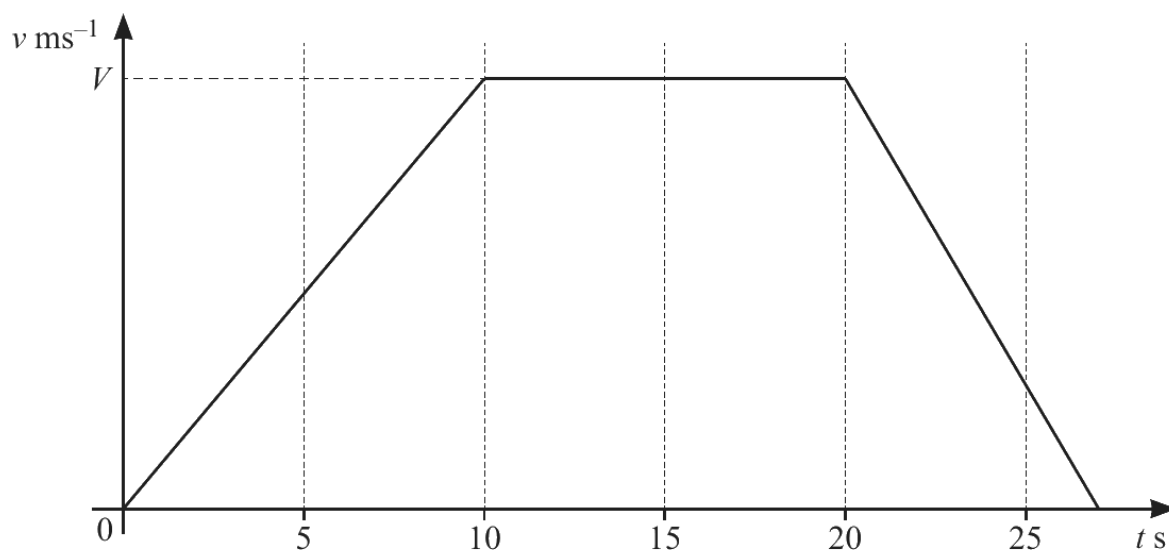
(a) A particle  $P$  moves in a straight line such that its displacement,  $x$  m, from a fixed point  $O$  at time  $t$  s is given by  $x = 10 \sin 2t - 5$ .

(i) Find the speed of  $P$  when  $t = \pi$ . [1]

(ii) Find the value of  $t$  for which  $P$  is first at rest. [2]

(iii) Find the acceleration of  $P$  when it is first at rest. [2]

(b)

*Question continues on the next page →*

The diagram shows the velocity–time graph for a particle  $Q$  travelling in a straight line with velocity  $v\text{ ms}^{-1}$  at time  $t\text{ s}$ . The particle accelerates at  $3.5\text{ ms}^{-2}$  for the first 10 s of its motion and then travels at constant velocity,  $V\text{ ms}^{-1}$ , for 10 s. The particle then decelerates at a constant rate and comes to rest. The distance travelled during the interval  $20 \leq t \leq 25$  is 112.5 m.

(i) Find the value of  $V$ . [1]

(ii) Find the velocity of  $Q$  when  $t = 25$ . [3]

(iii) Find the value of  $t$  when  $Q$  comes to rest. [3]

**179** May/June 2020 · Paper 12 · Q9

|           |                                            |           |                                                                                                         |
|-----------|--------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------|
| 43 (a)(i) | $v = 20 \cos 2t$ when $t = \pi$ , $v = 20$ | <b>B1</b> |                                                                                                         |
| (a)(ii)   | $20 \cos 2t = 0$                           | <b>M1</b> | Equating <i>their</i> (i) to zero, must be a cosine and attempt to solve                                |
|           | $t = \frac{\pi}{4}$                        | <b>A1</b> |                                                                                                         |
| (a)(iii)  | $a = -40 \sin 2t$                          | <b>M1</b> | Attempt to differentiate <i>their</i> $v$ , dep on previous M mark, and use <i>their</i> value for (ii) |
|           | $-40$                                      | <b>A1</b> |                                                                                                         |
| (b)(i)    | 35                                         | <b>B1</b> |                                                                                                         |
| (b)(ii)   | $112.5 = \frac{1}{2}(35 + x) \times 5$     | <b>M1</b> | Use of area under appropriate part of the graph                                                         |
|           |                                            | <b>A1</b> |                                                                                                         |
|           | $x = 10$                                   | <b>A1</b> |                                                                                                         |
| (b)(iii)  | $\frac{25}{5} = \frac{10}{t'}$             | <b>M1</b> | Using a ratio method or otherwise, find extra time to stop = 2s or equivalent                           |
|           | $t' = 2$                                   | <b>A1</b> |                                                                                                         |
|           | 27                                         | <b>A1</b> |                                                                                                         |

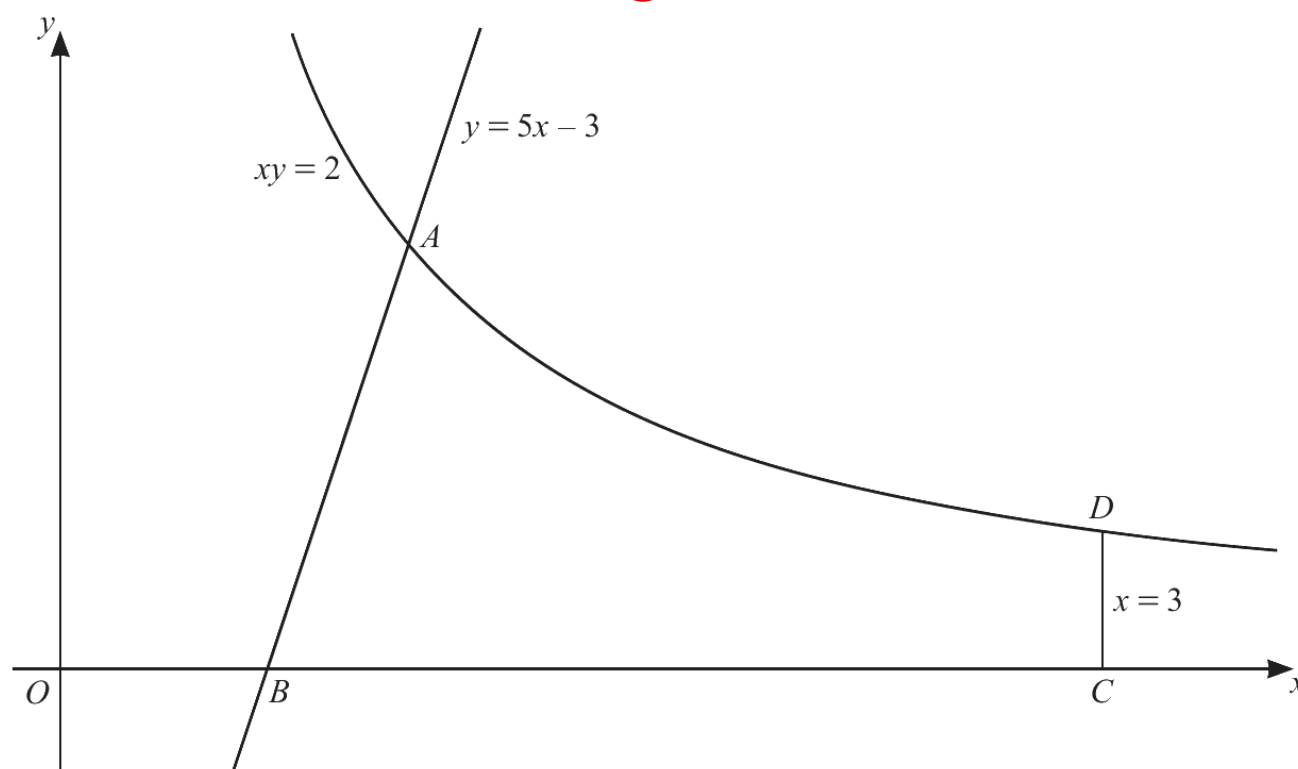


180



No calculator

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The diagram shows part of the curve  $xy = 2$  intersecting the straight line  $y = 5x - 3$  at the point  $A$ . The straight line meets the  $x$ -axis at the point  $B$ . The point  $C$  lies on the  $x$ -axis and the point  $D$  lies on the curve such that the line  $CD$  has equation  $x = 3$ . Find the exact area of the shaded region, giving your answer in the form  $p + \ln q$ , where  $p$  and  $q$  are constants. [8]

Additional working space for question 69.

**180** May/June 2020 · Paper 13 · Q9

| Question | Answer                                                      | Marks     | Partial Marks                                                        |
|----------|-------------------------------------------------------------|-----------|----------------------------------------------------------------------|
| 69       | $x(5x-3)=2$<br>$5x^2-3x-2=0$                                | <b>M1</b> | attempt at a 3-term quadratic equation in one variable with solution |
|          | $x=1, x=-\frac{2}{5}$                                       | <b>A1</b> | Allow if $x=-\frac{2}{5}$ not seen                                   |
|          | $A (1, 2)$                                                  | <b>A1</b> |                                                                      |
|          | $B \left(\frac{3}{5}, 0\right)$                             | <b>B1</b> |                                                                      |
|          | Area of triangle $=\frac{2}{5}$                             | <b>M1</b> | Using <i>their</i> $A$ and $B$                                       |
|          | Area under curve: $\int_1^3 \frac{2}{x} dx = [2 \ln x]_1^3$ | <b>B1</b> | For $[2 \ln x]_1^3$                                                  |
|          | $= 2 \ln 3$                                                 | <b>M1</b> | For use of limits                                                    |
|          | Total area $=\frac{2}{5} + \ln 9$                           | <b>A1</b> |                                                                      |

**181**

No calculator

**May/June 2020 • Paper 13 • Q10**

- (a) Given that  $y = x\sqrt{x+2}$ , show that  $\frac{dy}{dx} = \frac{Ax+B}{2\sqrt{x+2}}$ , where  $A$  and  $B$  are constants. [5]

- (b) Find the exact coordinates of the stationary point of the curve  $y = x\sqrt{x+2}$ . [3]

- (c) Determine the nature of this stationary point. [2]

**182**

No calculator

**Oct/Nov 2019 • Paper 11 • Q4**

It is given that  $y = \frac{\ln(4x^2 + 1)}{2x - 3}$ .

- (i) Find  $\frac{dy}{dx}$ . [3]

- (ii) Find the approximate change in  $y$  as  $x$  increases from 2 to  $2+h$ , where  $h$  is small. [2]

**181** May/June 2020 · Paper 13 · Q10

|        |                                                                                                                             |           |                                                                        |
|--------|-----------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------|
| 70 (a) | $\frac{dy}{dx} = \frac{1}{2}x(x+2)^{-\frac{1}{2}} + (x+2)^{\frac{1}{2}}$                                                    | <b>B1</b> | For $\frac{1}{2}(x+2)^{-\frac{1}{2}}$                                  |
|        |                                                                                                                             | <b>M1</b> | For differentiation of a product                                       |
|        |                                                                                                                             | <b>A1</b> |                                                                        |
|        | $\frac{dy}{dx} = \frac{1}{2}(x+2)^{-\frac{1}{2}}[x+2(x+2)]$                                                                 | <b>M1</b> | For attempt to simplify                                                |
|        | $\frac{dy}{dx} = \frac{3x+4}{2\sqrt{x+2}}$                                                                                  | <b>A1</b> |                                                                        |
| (b)    | $3x+4=0$                                                                                                                    | <b>M1</b> | For setting <i>their</i> numerator in (a) to zero and attempt to solve |
|        | $x = -\frac{4}{3}$                                                                                                          | <b>A1</b> |                                                                        |
|        | $y = -\frac{4\sqrt{6}}{9}$ oe                                                                                               | <b>A1</b> |                                                                        |
| (c)    | Using the gradient method or inspection of y-coordinates either side of stationary point.<br>Allow use of second derivative | <b>M1</b> | complete method                                                        |
|        | Minimum                                                                                                                     | <b>A1</b> | Must be from correct work                                              |

**182** Oct/Nov 2019 · Paper 11 · Q4

|        |                                                                             |           |                                                                             |
|--------|-----------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------|
| 20 (i) | $\frac{d}{dx}(\ln(4x^2+1)) = \frac{8x}{4x^2+1}$                             | <b>B1</b> |                                                                             |
|        | $\frac{dy}{dx} = \frac{(2x-3)\frac{8x}{(4x^2+1)} - 2\ln(4x^2+1)}{(2x-3)^2}$ | <b>M1</b> | For attempt to differentiate a quotient                                     |
|        |                                                                             | <b>A1</b> | For all other terms, not including $\frac{8x}{4x^2+1}$ , correct            |
| (ii)   | When $x=2$ , $\frac{dy}{dx} = \frac{16}{17} - 2\ln 17$<br>$= -4.73$         | <b>M1</b> | For attempt to find value of $\frac{dy}{dx}$ when $x=2$ and multiply by $h$ |
|        | Change in $y = -4.73h$                                                      | <b>A1</b> |                                                                             |

183



No calculator

Oct/Nov 2019 · Paper 11 · Q6

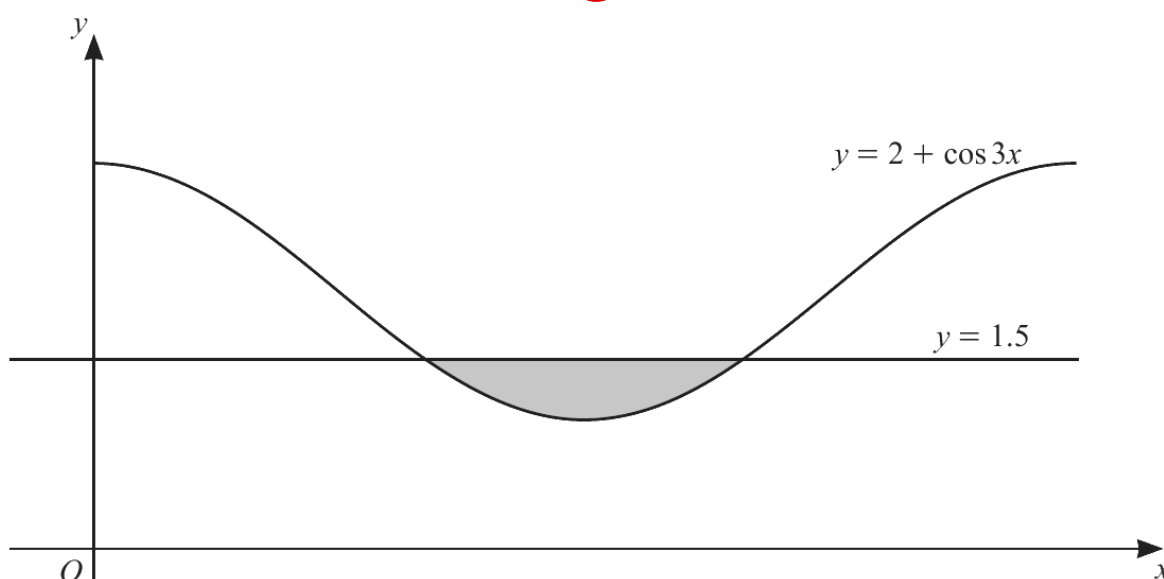
Find the equation of the normal to the curve  $y = \sqrt{8x+5}$  at the point where  $x = \frac{1}{2}$ , giving your answer in the form  $ax + by + c = 0$ , where  $a$ ,  $b$  and  $c$  are integers. [5]

184



No calculator

Oct/Nov 2019 · Paper 11 · Q10



The diagram shows part of the graph of  $y = 2 + \cos 3x$  and the straight line  $y = 1.5$ . Find the exact area of the shaded region bounded by the curve and the straight line. You must show all your working. [9]

Continuation of working space for Question 22

**183** Oct/Nov 2019 · Paper 11 · Q6

|    |                                                            |           |                                                                                                                                  |
|----|------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| 21 | $\frac{dy}{dx} = k(8x + 5)^{-\frac{1}{2}}$                 | <b>M1</b> | For attempt to differentiate, must be in the form $k(8x + 5)^{-\frac{1}{2}}$                                                     |
|    | $\frac{dy}{dx} = 4(8x + 5)^{-\frac{1}{2}}$                 | <b>A1</b> |                                                                                                                                  |
|    | When $x = \frac{1}{2}, y = 3$                              | <b>B1</b> |                                                                                                                                  |
|    | Normal: $y - 3 = -\frac{3}{4}\left(x - \frac{1}{2}\right)$ | <b>M1</b> | For attempt at the normal when $x = \frac{1}{2}$ , using correct process for <i>their</i> $\frac{dy}{dx}$ and <i>their</i> $y$ . |
|    | $6x + 8y - 27 = 0$                                         | <b>A1</b> |                                                                                                                                  |

**184** Oct/Nov 2019 · Paper 11 · Q10

| Question | Answer                                                                                                 | Marks     | Guidance                                                           |
|----------|--------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------|
| 22       | $1.5 = 2 + \cos 3x$<br>$\cos 3x = -0.5$                                                                | <b>M1</b> | For correct attempt to find points of intersection                 |
|          | $3x = \frac{2\pi}{3}, \frac{4\pi}{3}$                                                                  | <b>M1</b> | For dealing with $3x$ correctly                                    |
|          | $x = \frac{2\pi}{9}$ or $40^\circ$                                                                     | <b>A1</b> |                                                                    |
|          | $x = \frac{4\pi}{9}$ or $80^\circ$                                                                     | <b>A1</b> |                                                                    |
|          | <b>Either</b><br>$\int_{\frac{2\pi}{9}}^{\frac{4\pi}{9}} 1.5 - (2 + \cos 3x) \, dx$                    | <b>M1</b> | For subtraction method – condone omission of or incorrect limits   |
|          | $\left[-0.5x - \frac{1}{3}\sin 3x\right]_{\frac{2\pi}{9}}^{\frac{4\pi}{9}}$                            | <b>M1</b> | For attempt to integrate – condone omission of or incorrect limits |
|          | $\left[-0.5x - \frac{1}{3}\sin 3x\right]_{\frac{2\pi}{9}}^{\frac{4\pi}{9}}$                            | <b>A1</b> | All correct – condone omission of or incorrect limits              |
|          | $\left(-\frac{2\pi}{9} + \frac{\sqrt{3}}{6}\right) - \left(-\frac{\pi}{9} - \frac{\sqrt{3}}{6}\right)$ | <b>M1</b> | Dep for application of limits, must be in radians                  |

|  |                                                                                                                    |           |                                                                    |
|--|--------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------|
|  | $\text{Area} = \frac{\sqrt{3}}{3} - \frac{\pi}{9}$                                                                 | <b>A1</b> |                                                                    |
|  | $\left(1.5 \times \frac{2\pi}{9}\right)$                                                                           |           | subtraction subsequently)                                          |
|  | $\left[2x + k \sin 3x\right]_{\frac{2\pi}{9}}^{\frac{4\pi}{9}}$                                                    | <b>M1</b> | For attempt to integrate – condone omission of or incorrect limits |
|  | $\left[2x + \frac{1}{3} \sin 3x\right]_{\frac{2\pi}{9}}^{\frac{4\pi}{9}}$                                          | <b>A1</b> | All correct – condone omission of or incorrect limits              |
|  | $\left(\left(\frac{8\pi}{9} - \frac{\sqrt{3}}{6}\right) - \left(\frac{4\pi}{9} + \frac{\sqrt{3}}{6}\right)\right)$ | <b>M1</b> | Dep for application of limits, must be in radians                  |
|  | $\text{Area} = \frac{\sqrt{3}}{3} - \frac{\pi}{9}$                                                                 | <b>A1</b> |                                                                    |

**185**

No calculator

**Oct/Nov 2019 • Paper 11 • Q12**

A curve is such that  $\frac{d^2y}{dx^2} = 2 \sin\left(x + \frac{\pi}{3}\right)$ . Given that the curve has a gradient of 5 at the point  $\left(\frac{\pi}{3}, \frac{5\pi}{3}\right)$ , find the equation of the curve. [8]

**186**

No calculator

**Oct/Nov 2019 • Paper 12 • Q7**

It is given that  $y = (1 + e^{x^2})(x + 5)$ .

(i) Find  $\frac{dy}{dx}$ . [3]

(ii) Find the approximate change in  $y$  as  $x$  increases from 0.5 to  $0.5 + p$ , where  $p$  is small. [2]

(iii) Given that  $y$  is increasing at a rate of 2 units per second when  $x = 0.5$ , find the corresponding rate of change in  $x$ . [2]



**185** Oct/Nov 2019 · Paper 11 · Q12

|    |                                                                                                                                           |           |                                                                                         |
|----|-------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------|
| 23 | $\frac{dy}{dx} = k \cos\left(x + \frac{\pi}{3}\right) + c$                                                                                | <b>M1</b> | For attempt to integrate                                                                |
|    | $\frac{dy}{dx} = -2 \cos\left(x + \frac{\pi}{3}\right) + c$                                                                               | <b>A1</b> | All correct, condone omission of $+c$                                                   |
|    | $5 = -2 \cos \frac{2\pi}{3} + c$                                                                                                          | <b>M1</b> | Dep for attempt to find $c$                                                             |
|    | $\frac{dy}{dx} = -2 \cos\left(x + \frac{\pi}{3}\right) + 4$                                                                               | <b>A1</b> |                                                                                         |
|    | $y = p \sin\left(x + \frac{\pi}{3}\right) \quad (+qx + d)$                                                                                | <b>M1</b> | attempt to integrate a second time to obtain $y = p \sin\left(x + \frac{\pi}{3}\right)$ |
|    | $y = -2 \sin\left(x + \frac{\pi}{3}\right) + 4x + d$                                                                                      | <b>A1</b> | All correct, condone omission of $+d$                                                   |
|    | $\frac{5\pi}{3} = -2 \sin \frac{2\pi}{3} + \frac{4\pi}{3} + d$                                                                            | <b>M1</b> | Dep for attempt to find a second arbitrary constant                                     |
|    | $y = -2 \sin\left(x + \frac{\pi}{3}\right) + 4x + \frac{\pi}{3} + \sqrt{3}$<br>or $y = -2 \sin\left(x + \frac{\pi}{3}\right) + 4x + 2.78$ | <b>A1</b> |                                                                                         |

**186** Oct/Nov 2019 · Paper 12 · Q7

|        |                                                    |           |                                                                                              |
|--------|----------------------------------------------------|-----------|----------------------------------------------------------------------------------------------|
| 52 (i) | $\frac{dy}{dx} = (e^{x^2} + 1) + 2xe^{x^2}(x + 5)$ | <b>B1</b> | For $2xe^{x^2}$                                                                              |
|        |                                                    | <b>M1</b> | For attempt at differentiating a product or expanding brackets and differentiating a product |
|        |                                                    | <b>A1</b> | For all other terms, apart from $2xe^{x^2}$ , correct                                        |
| (ii)   | When $x = 0.5$ , $\frac{dy}{dx} = 9.35$            | <b>M1</b> | For attempt to find <i>their</i> $\frac{dy}{dx}$ when $x = 0.5$ and multiplication by $p$    |
|        | Approximate change = $9.35p$                       | <b>A1</b> |                                                                                              |

|       |                                                                                       |                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| (iii) | $\frac{dy}{dx} \times \frac{dx}{dt} = \frac{dy}{dt}$ $9.346 \times \frac{dx}{dt} = 2$ | <b>M1</b> For use of correct rates of change equation using <i>their</i> $\frac{dy}{dx}$ when $x = 0.5$ and $\frac{dy}{dt} = 2$ |
|       | $\frac{dx}{dt} = 0.214$                                                               | <b>A1</b> FT on $\frac{2}{\text{their } 9.346}$<br>Must be correct to at least 3 sf                                             |

**187**

No calculator

**Oct/Nov 2019 • Paper 12 • Q9**

A solid circular cylinder has a base radius of  $r$  cm and a height of  $h$  cm. The cylinder has a volume of  $1200\pi \text{ cm}^3$  and a total surface area of  $S \text{ cm}^2$ .

(i) Show that  $S = 2\pi r^2 + \frac{2400\pi}{r}$ . [3]

(ii) Given that  $h$  and  $r$  can vary, find the stationary value of  $S$  and determine its nature. [5]

**188**

No calculator

**Oct/Nov 2019 • Paper 12 • Q11**

A curve is such that  $\frac{d^2y}{dx^2} = 2(3x-1)^{-\frac{2}{3}}$ . Given that the curve has a gradient of 6 at the point (3, 11), find the equation of the curve. [8]

## 187 Oct/Nov 2019 · Paper 12 · Q9

|                 |                                                                                                                                                                                                                                                                                                              |           |                                                                                                              |      |          |                 |   |   |   |           |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------|------|----------|-----------------|---|---|---|-----------|
| 53 (i)          | $\pi r^2 h = 1200\pi$                                                                                                                                                                                                                                                                                        | <b>B1</b> |                                                                                                              |      |          |                 |   |   |   |           |
|                 | $h = \frac{1200}{r^2}$ or $\pi r h = \frac{1200\pi}{r}$ and substitution into <i>their</i> $S$                                                                                                                                                                                                               | <b>B1</b> | Must have attempt to use in an equation for $S$                                                              |      |          |                 |   |   |   |           |
|                 | $S = 2\pi r^2 + \left(2\pi r \times \frac{1200}{r^2}\right)$<br>leading to given answer                                                                                                                                                                                                                      | <b>B1</b> |                                                                                                              |      |          |                 |   |   |   |           |
| (ii)            | $\frac{dS}{dr} = 4\pi r - \frac{2400\pi}{r^2}$                                                                                                                                                                                                                                                               | <b>M1</b> | Must obtain the form $Ar + \frac{B}{r^2}$                                                                    |      |          |                 |   |   |   |           |
|                 | When $\frac{dS}{dr} = 0$ , $r = \sqrt[3]{600}$ , 8.43                                                                                                                                                                                                                                                        | <b>M1</b> | Dep for equating to zero and attempt to solve to obtain $r = \dots$                                          |      |          |                 |   |   |   |           |
|                 |                                                                                                                                                                                                                                                                                                              | <b>A1</b> | For correct $r$                                                                                              |      |          |                 |   |   |   |           |
|                 | $S_{\min} = 1340$ or 1341                                                                                                                                                                                                                                                                                    | <b>A1</b> |                                                                                                              |      |          |                 |   |   |   |           |
|                 | <b>Either</b> $\frac{d^2S}{dr^2} = 4\pi + \frac{4800\pi}{r^3}$<br>$\frac{d^2S}{dr^2} > 0$ so minimum                                                                                                                                                                                                         | <b>B1</b> | For a correct method to reach a correct conclusion<br>If $r$ is not calculated, then must state that $r > 0$ |      |          |                 |   |   |   |           |
|                 | <b>Or</b> Consideration of gradient e.g.<br><table border="1" data-bbox="322 1176 716 1285"> <tr> <td><math>r</math></td><td><math>&lt; 8.43</math></td><td>8.43</td><td><math>&gt; 8.43</math></td></tr> <tr> <td><math>\frac{dS}{dr}</math></td><td>–</td><td>0</td><td>+</td></tr> </table> Minimum point | $r$       | $< 8.43$                                                                                                     | 8.43 | $> 8.43$ | $\frac{dS}{dr}$ | – | 0 | + | <b>B1</b> |
| $r$             | $< 8.43$                                                                                                                                                                                                                                                                                                     | 8.43      | $> 8.43$                                                                                                     |      |          |                 |   |   |   |           |
| $\frac{dS}{dr}$ | –                                                                                                                                                                                                                                                                                                            | 0         | +                                                                                                            |      |          |                 |   |   |   |           |

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|    |                                                           |           |                                                                                      |
|----|-----------------------------------------------------------|-----------|--------------------------------------------------------------------------------------|
| 54 | $\frac{dy}{dx} = 2(3x-1)^{\frac{1}{3}} + c$               | <b>M1</b> | For $\left(\frac{dy}{dx} = \right) a(3x-1)^{\frac{1}{3}}$ , condone omission of $+c$ |
|    |                                                           | <b>A1</b> | All correct, condone omission of $c$                                                 |
|    | $6 = 4 + c$                                               | <b>M1</b> | Dep for attempt to find $c$                                                          |
|    | $\left(\frac{dy}{dx} = \right) 2(3x-1)^{\frac{1}{3}} + 2$ | <b>A1</b> | All correct, may be implied by $c = 2$                                               |

|  |                                                |           |                                                                                                                  |
|--|------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------|
|  | $y = \frac{1}{2}(3x-1)^{\frac{4}{3}} + 2x + d$ | <b>M1</b> | For attempt to integrate <i>their</i> $\frac{dy}{dx}$ to obtain the form $y = b(3x-1)^{\frac{4}{3}} (+mx + d)$   |
|  |                                                | <b>A1</b> | All correct, condone omission of $d$                                                                             |
|  | $11 = 14 + d$                                  | <b>M1</b> | Dep for attempt to find $d$ , a second arbitrary constant, having used an arbitrary constant for $\frac{dy}{dx}$ |
|  | $y = \frac{1}{2}(3x-1)^{\frac{4}{3}} + 2x - 3$ | <b>A1</b> |                                                                                                                  |

189



No calculator

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81 (i) Differentiate  $y = (3x^2 - 1)^{-\frac{1}{3}}$  with respect to  $x$ . [2]

(ii) Find the approximate change in  $y$  as  $x$  increases from  $\sqrt{3}$  to  $\sqrt{3} + p$ , where  $p$  is small. [1]

(iii) Find the equation of the normal to the curve  $y = (3x^2 - 1)^{-\frac{1}{3}}$  at the point where  $x = \sqrt{3}$ . [3]

190



No calculator

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(i) Differentiate  $(x^2 + 3)\ln(x^2 + 3)$  with respect to  $x$ . [3]

(ii) Hence find  $\int x \ln(x^2 + 3) dx$ . [2]

**189** Oct/Nov 2019 · Paper 13 · Q3

|        |                                                                                                    |           |                                                                                                                     |
|--------|----------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------|
| 81 (i) | $\frac{dy}{dx} = kx(3x^2 - 1)^{\frac{4}{3}}$                                                       | <b>M1</b> |                                                                                                                     |
|        | $\frac{dy}{dx} = -\frac{1}{3} \times 6x(3x^2 - 1)^{\frac{4}{3}}$                                   | <b>A1</b> |                                                                                                                     |
| (ii)   | When $x = \sqrt{3}$ , $\frac{dy}{dx} = -\frac{\sqrt{3}}{8}$<br>$-\frac{\sqrt{3}}{8}p$ or $-0.217p$ | <b>B1</b> | <b>FT</b> on <i>their</i> $\frac{dy}{dx}$ of the form $kx(3x^2 - 1)^{\frac{4}{3}}$                                  |
| (iii)  | When $x = \sqrt{3}$ , $y = \frac{1}{2}$                                                            | <b>B1</b> | for $y = \frac{1}{2}$                                                                                               |
|        | Normal: $y - \frac{1}{2} = \frac{8}{\sqrt{3}}(x - \sqrt{3})$                                       | <b>M1</b> | Dep on M1 in part(i). An equation of the normal using <i>their</i> normal gradient, $\sqrt{3}$ and <i>their</i> $y$ |
|        |                                                                                                    | <b>A1</b> | allow unsimplified                                                                                                  |

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|        |                                                                                        |           |                                                                   |
|--------|----------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------|
| 82 (i) | $\frac{d}{dx}(\ln(x^2 + 3)) = \frac{2x}{(x^2 + 3)}$                                    | <b>B1</b> |                                                                   |
|        | $\frac{dy}{dx} = (x^2 + 3) \frac{2x}{(x^2 + 3)} + 2x \ln(x^2 + 3)$                     | <b>M1</b> | for product rule                                                  |
|        |                                                                                        | <b>A1</b> | <b>FT</b> <i>their</i> $\frac{2x}{(x^2 + 3)}$                     |
| (ii)   | $(x^2 + 3) \ln(x^2 + 3) = \int 2x + 2x \ln(x^2 + 3) dx$                                | <b>M1</b> | for using <i>their</i> result from (i) for $2x + kx \ln(x^2 + 3)$ |
|        | $\int x \ln(x^2 + 3) dx$<br>$= \frac{1}{2}(x^2 + 3) \ln(x^2 + 3) - \frac{x^2}{2} (+c)$ | <b>A1</b> |                                                                   |

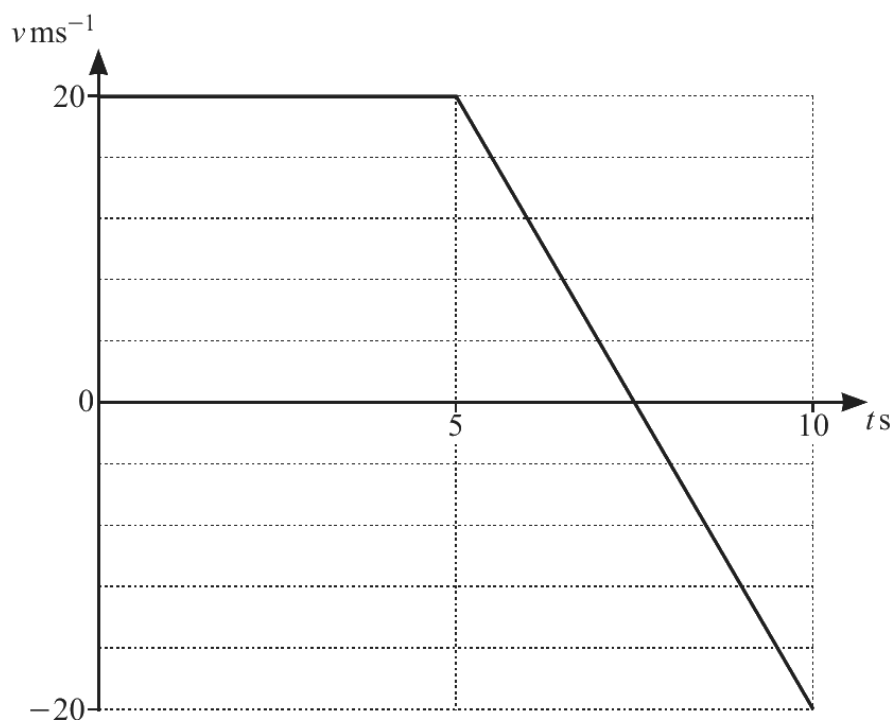
191

(a)



No calculator

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The velocity-time graph for a particle  $P$  is shown by the two straight lines in the diagram.

- (i) Find the deceleration of  $P$  for  $5 \leq t \leq 10$ . [2]
- (ii) Write down the value of  $t$  when the speed of  $P$  is zero. [1]
- (iii) Find the distance  $P$  has travelled for  $0 \leq t \leq 10$ . [2]

Question continues on the next page →



- (b) A particle  $Q$  has a displacement of  $x$  m from a fixed point  $O$ ,  $t$  s after leaving  $O$ . The velocity,  $v \text{ ms}^{-1}$ , of  $Q$  at time  $t$  s is given by  $v = 6e^{2t} + 1$ .

(i) Find an expression for  $x$  in terms of  $t$ . [3]

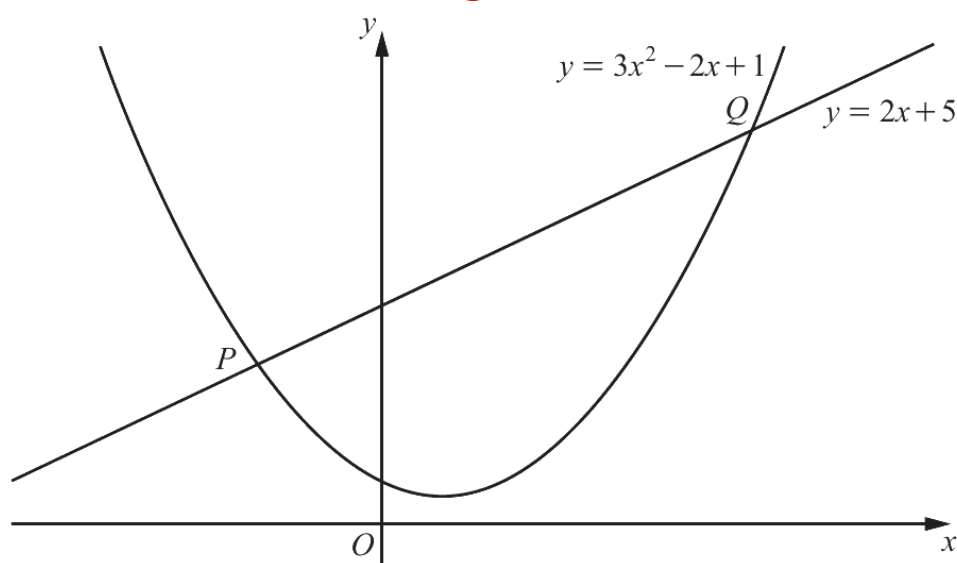
(ii) Find the value of  $t$  when the acceleration of  $Q$  is  $24 \text{ ms}^{-2}$ . [3]

192



No calculator

May/June 2019 · Paper 11 · Q6



The diagram shows the curve  $y = 3x^2 - 2x + 1$  and the straight line  $y = 2x + 5$  intersecting at the points  $P$  and  $Q$ . Showing all your working, find the area of the shaded region. [8]

**191** Oct/Nov 2019 · Paper 13 · Q10

|          |                                                                                                                                                                                                       |           |                                                                |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------|
| (a)(i)   | $\pm \frac{20 - -20}{5}$                                                                                                                                                                              | <b>M1</b> | for finding the gradient of the relevant part                  |
|          | 8                                                                                                                                                                                                     | <b>A1</b> |                                                                |
| (a)(ii)  | 7.5                                                                                                                                                                                                   | <b>B1</b> |                                                                |
| (a)(iii) | $\frac{1}{2}(5 + 7.5)20 + \left(\frac{1}{2} \times 2.5 \times 20\right)$<br>or<br>$20 \times 5 + \left(\frac{1}{2} \times 2.5 \times 20\right) + \left(\frac{1}{2} \times 2.5 \times 20\right)$<br>oe | <b>M1</b> | for a correct expression for total area using <i>their</i> 7.5 |
|          | 150                                                                                                                                                                                                   | <b>A1</b> |                                                                |
| (b)(i)   | $x = 3e^{2t} + t + c$                                                                                                                                                                                 | <b>M1</b> | for $ke^{2t} + t$ Condone omission of $c$                      |
|          | $0 = 3e^0 + 0 + c$<br>When $t = 0$ , $x = 0$ so $c = -3$                                                                                                                                              | <b>M1</b> | Dep for substitution to find $c$                               |
|          | $x = 3e^{2t} + t - 3$                                                                                                                                                                                 | <b>A1</b> |                                                                |
| (b)(ii)  | $\frac{dy}{dt} = 12e^{2t}$ so $12e^{2t} = 24$                                                                                                                                                         | <b>M1</b> | for $ke^{2t}$ equated to 24                                    |
|          | $2t = \ln 2$                                                                                                                                                                                          | <b>M1</b> | Dep for correct order of operations to obtain $2t$             |
|          | $t = \frac{1}{2} \ln 2$ , $\ln \sqrt{2}$ or 0.347                                                                                                                                                     | <b>A1</b> |                                                                |

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|   |                                                           |           |                                                                                                                   |
|---|-----------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------|
| 7 | <u>Method 1</u><br>$3x^2 - 2x + 1 = 2x + 5$<br>leading to | <b>M1</b> | equating the equations of the line and the curve and rearranging to obtain a three term quadratic equated to zero |
|   | $3x^2 - 4x - 4 = 0$                                       | <b>A1</b> |                                                                                                                   |
|   | $x = -\frac{2}{3}$ and $x = 2$                            | <b>A1</b> |                                                                                                                   |
|   | $\int_{-\frac{2}{3}}^2 (2x + 5 - (3x^2 - 2x + 1)) \, dx$  | <b>M1</b> | subtraction (either way round)                                                                                    |

|                                                                                                 |           |                                                     |
|-------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------|
| $\int_{-\frac{2}{3}}^2 (4 + 4x - 3x^2) dx$                                                      | <b>M1</b> | integration to $Ax + Bx^2 + Cx^3$                   |
| $\left[4x + 2x^2 - x^3\right]_{-\frac{2}{3}}^2$                                                 | <b>A1</b> | for $4x + 2x^2 - x^3$ oe                            |
| $(8 + 8 - 8) - \left(-\frac{8}{3} + \frac{8}{9} + \frac{8}{27}\right)$<br>$= 8 - \frac{40}{27}$ | <b>M1</b> | <b>Dep</b> on preceding M1<br>correct use of limits |
| $= \frac{256}{27}$ or 9.48 or $9\frac{13}{27}$                                                  | <b>A1</b> |                                                     |

|                                                                                                                |           |                                                                                                        |
|----------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------|
| $3x^2 - 2x + 1 = 2x + 5$<br>leading to                                                                         | <b>M1</b> | equating the line and the curve and<br>rearranging to obtain a three term quadratic<br>equated to zero |
| $3x^2 - 4x - 4 = 0$                                                                                            | <b>A1</b> |                                                                                                        |
| $x = -\frac{2}{3}$ and $x = 2$                                                                                 | <b>A1</b> |                                                                                                        |
| Area of trapezium $= \frac{1}{2} \left( \frac{11}{3} + 9 \right) \times \frac{8}{3}$                           | <b>B1</b> | area of the trapezium, allow unsimplified                                                              |
| Area under curve $= \int_{-\frac{2}{3}}^2 3x^2 - 2x + 1 dx$                                                    | <b>M1</b> | integration to $Ax + Bx^2 + Cx^3$                                                                      |
| $= \left[ x^3 - x^2 + x \right]_{-\frac{2}{3}}^2$                                                              | <b>A1</b> | for $x^3 - x^2 + x$                                                                                    |
| $= \left( 8 - 4 + 2 \right) - \left( -\frac{8}{27} - \frac{4}{9} - \frac{2}{3} \right)$<br>$6 - \frac{38}{27}$ | <b>M1</b> | <b>DepM1</b> for correct use of limits.                                                                |
| Shaded Area $= \frac{152}{9} - \frac{200}{27}$<br>$= \frac{256}{27}$ or 9.48 or $9\frac{13}{27}$               | <b>A1</b> |                                                                                                        |

**193**

No calculator

**May/June 2019 · Paper 11 · Q9**

In this question all lengths are in centimetres.

A closed cylinder has base radius  $r$ , height  $h$  and volume  $V$ . It is given that the total surface area of the cylinder is  $600\pi$  and that  $V$ ,  $r$  and  $h$  can vary.

(i) Show that  $V = 300\pi r - \pi r^3$ . [3]

(ii) Find the stationary value of  $V$  and determine its nature. [5]

**194**

No calculator

**May/June 2019 · Paper 11 · Q11**

It is given that  $y = (x^2 + 1)(2x - 3)^{\frac{1}{2}}$ .

(i) Show that  $\frac{dy}{dx} = \frac{Px^2 + Qx + 1}{(2x - 3)^{\frac{1}{2}}}$ , where  $P$  and  $Q$  are integers. [5]

(ii) Hence find the equation of the normal to the curve  $y = (x^2 + 1)(2x - 3)^{\frac{1}{2}}$  at the point where  $x = 2$ , giving your answer in the form  $ax + by + c = 0$ , where  $a$ ,  $b$  and  $c$  are integers. [4]

## 193 May/June 2019 · Paper 11 · Q9

|       |                                                                                                                                                                |           |                                                                                                                 |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------|
| 8 (i) | <u>Method 1</u><br>$600\pi = 2\pi r^2 + 2\pi rh$                                                                                                               | <b>B1</b> |                                                                                                                 |
|       | $h = \frac{600\pi - 2\pi r^2}{2\pi r}$                                                                                                                         | <b>M1</b> | making $h$ subject from a two term expression for SA.                                                           |
|       | $V = \pi r^2 h$<br>$V = \pi r^2 \left( \frac{600\pi - 2\pi r^2}{2\pi r} \right)$<br>$V = \pi r^2 \left( \frac{300}{r} - r \right)$<br>$V = 300\pi r - \pi r^3$ | <b>A1</b> | correct substitution and manipulation to obtain given answer                                                    |
| (i)   | <u>Method 2</u><br>$600\pi = 2\pi r^2 + 2\pi rh$                                                                                                               | <b>B1</b> |                                                                                                                 |
|       | $600\pi r = 2\pi r^3 + 2\pi r^2 h$                                                                                                                             | <b>M1</b> | multiplying both sides by $r$                                                                                   |
|       | $\frac{600\pi r - 2\pi r^3}{2} = \pi r^2 h$<br>$V = \pi r^2 h$<br>$V = 300\pi r - \pi r^3$                                                                     | <b>A1</b> | correct manipulation to obtain $\pi r^2 h$                                                                      |
| (ii)  | $\frac{dV}{dr} = 300\pi - 3\pi r^2$                                                                                                                            | <b>M1</b> | differentiation of given formula to $A + Br^2$                                                                  |
|       | When $\frac{dV}{dr} = 300\pi - 3\pi r^2 = 0$                                                                                                                   | <b>M1</b> | equating to zero and attempt to solve                                                                           |
|       | $r = 10$                                                                                                                                                       | <b>A1</b> |                                                                                                                 |
|       | $V = 2000\pi$ or 6280 or 6283                                                                                                                                  | <b>A1</b> |                                                                                                                 |
|       | $\frac{d^2V}{dr^2} = -6\pi r$ , $\frac{d^2V}{dr^2} < 0$<br>so maximum                                                                                          | <b>B1</b> | cao for $\frac{d^2V}{dr^2} = -6\pi r$ , $\frac{d^2V}{dr^2} = -60\pi$ or other correct method leading to maximum |

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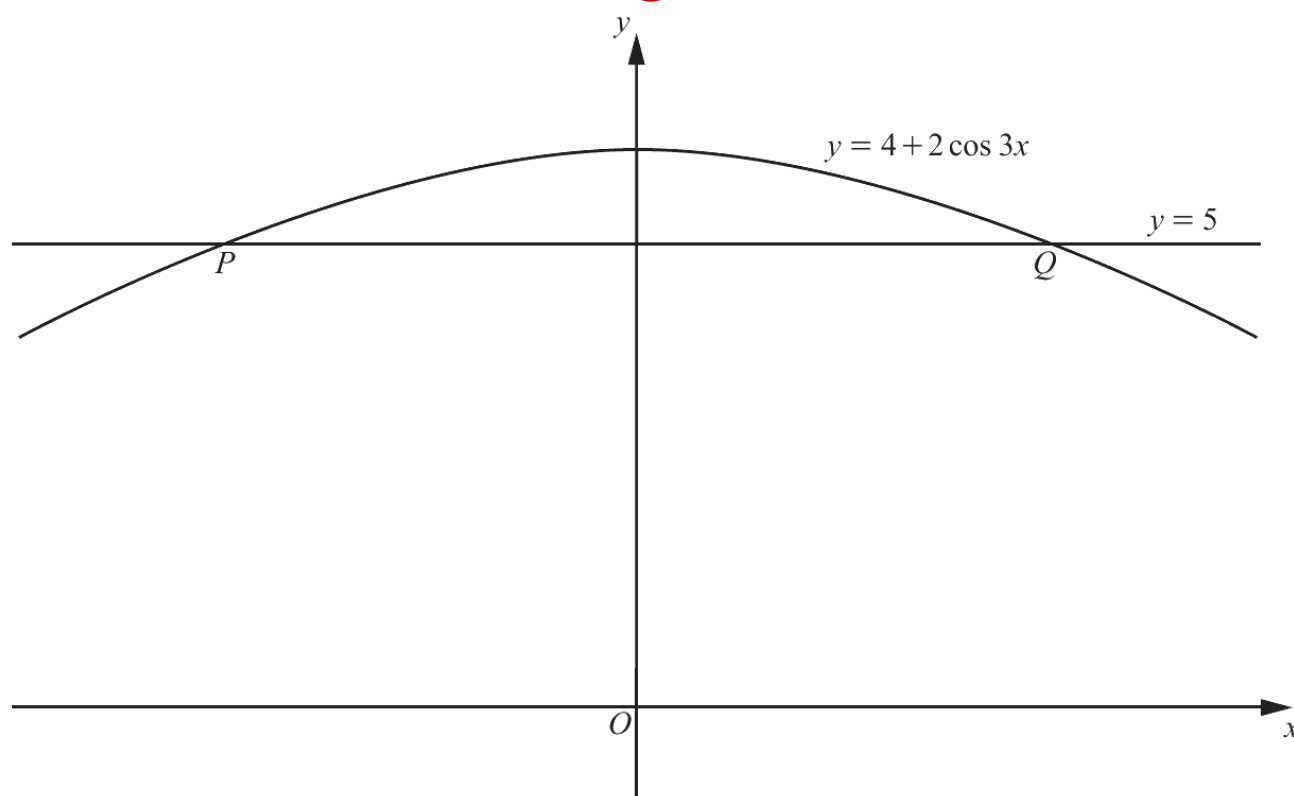
| Question | Answer                                                                         | Marks     | Guidance                                                                                                                 |
|----------|--------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------|
| 9 (i)    | $\frac{dy}{dx} = (x^2 + 1)(2x - 3)^{-\frac{1}{2}} + 2x(2x - 3)^{-\frac{1}{2}}$ | <b>M1</b> | differentiation of a product                                                                                             |
|          |                                                                                | <b>B1</b> | for $\frac{d}{dx}(2x - 3)^{-\frac{1}{2}} = \frac{1}{2} \times 2(2x - 3)^{-\frac{1}{2}}$ oe                               |
|          |                                                                                | <b>A1</b> | all else correct i.e.<br>$\frac{dy}{dx} = (x^2 + 1)f(x) + 2x(2x - 3)^{-\frac{1}{2}}$                                     |
|          | $= (2x - 3)^{-\frac{1}{2}}(x^2 + 1 + 2x(2x - 3))$                              | <b>M1</b> | correctly taking out a factor of $(2x - 3)^{-\frac{1}{2}}$<br>or correctly using $(2x - 3)^{\frac{1}{2}}$ as denominator |
|          | $= \frac{5x^2 - 6x + 1}{(2x - 3)^{\frac{1}{2}}}$                               | <b>A1</b> |                                                                                                                          |
| (ii)     | When $x = 2$ , $y = 5$                                                         | <b>B1</b> |                                                                                                                          |
|          | $\frac{dy}{dx} = 9$ , so gradient of normal $= -\frac{1}{9}$                   | <b>M1</b> | substitution to obtain gradient and correct method for gradient of normal                                                |
|          | Equation of normal $y - 5 = -\frac{1}{9}(x - 2)$                               | <b>M1</b> | <b>DepM1</b> for equation of normal                                                                                      |
|          | $x + 9y - 47 = 0$ or $-x - 9y + 47 = 0$                                        | <b>A1</b> | Must be in this form                                                                                                     |

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No calculator

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The diagram shows the curve  $y = 4 + 2 \cos 3x$  intersecting the line  $y = 5$  at the points  $P$  and  $Q$ .

(i) Find, in terms of  $\pi$ , the  $x$ -coordinate of  $P$  and of  $Q$ .

[3]

(ii) Find the exact area of the shaded region. You must show all your working.

[6]

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| Question | Answer                                                                                                   | Marks     | Guidance                                                                                                             |
|----------|----------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------|
| 37 (i)   | $5 = 4 + 2\cos 3x$                                                                                       | <b>M1</b> | For attempt to solve trig equation to obtain one correct solution                                                    |
|          | $\frac{\pi}{9}$                                                                                          | <b>A1</b> |                                                                                                                      |
|          | $-\frac{\pi}{9}$                                                                                         | <b>A1</b> |                                                                                                                      |
| (ii)     | <b>Either:</b> $\int_{-\frac{\pi}{9}}^{\frac{\pi}{9}} 4 + 2\cos 3x - 5 \, dx$                            | <b>M1</b> | For use of subtraction method                                                                                        |
|          | $\left[ \frac{2}{3}\sin 3x - x \right]_{-\frac{\pi}{9}}^{\frac{\pi}{9}}$                                 | <b>M1</b> | For attempt to integrate to obtain the form $a\sin 3x + bx$                                                          |
|          |                                                                                                          | <b>B1</b> | For $\frac{2}{3}\sin 3x$                                                                                             |
|          |                                                                                                          | <b>B1</b> | For $-x$ , may be implied by $4x - 5x$                                                                               |
|          | $\left( \frac{\sqrt{3}}{3} - \frac{\pi}{9} \right) - \left( -\frac{\sqrt{3}}{3} + \frac{\pi}{9} \right)$ | <b>M1</b> | <b>Dep</b> on previous M mark for correct application of <i>their</i> limits in radians from (i) retaining exactness |
|          | Shaded area = $\frac{2\sqrt{3}}{3} - \frac{2\pi}{9}$ oe isw                                              | <b>A1</b> |                                                                                                                      |

|  |                                                                                                 |           |                                                             |
|--|-------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------|
|  | <b>Or:</b> Area of rectangle = $5 \times \frac{\pi}{9}$                                         |           | exact radians                                               |
|  | Area under curve =<br>$\left[ 4x + \frac{2}{3}\sin 3x \right]_{-\frac{\pi}{9}}^{\frac{\pi}{9}}$ | <b>M1</b> | For attempt to integrate to obtain the form $a\sin 3x + bx$ |
|  |                                                                                                 | <b>B1</b> | For $\frac{2}{3}\sin 3x$                                    |
|  |                                                                                                 | <b>B1</b> | For $4x$                                                    |



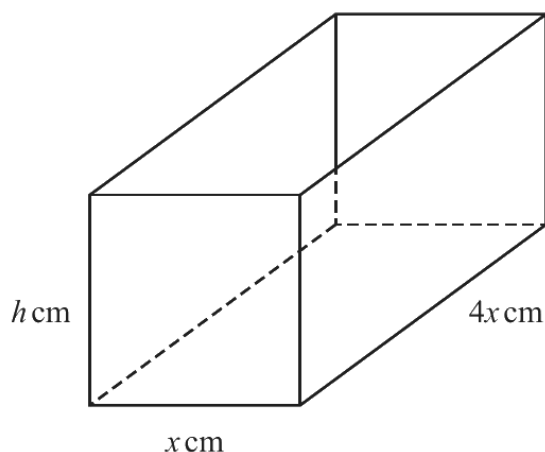
|  |                                                                                                                                                              |           |                                                                                                                                   |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
|  | $\left(\frac{\sqrt{3}}{3} + \frac{4\pi}{9}\right) - \left(-\frac{\sqrt{3}}{3} - \frac{4\pi}{9}\right)$ $\left(= \frac{2\sqrt{3}}{3} + \frac{8\pi}{9}\right)$ | <b>M1</b> | <b>Dep</b> on previous M mark for correct application of <i>their</i> limits in exact radians from <b>(i)</b> retaining exactness |
|  | Shaded area = $\frac{2\sqrt{3}}{3} - \frac{2\pi}{9}$ oe isw                                                                                                  | <b>A1</b> |                                                                                                                                   |

196



No calculator

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The diagram shows an open container in the shape of a cuboid of width  $x$  cm, length  $4x$  cm and height  $h$  cm. The volume of the container is  $800 \text{ cm}^3$ .

- (i) Show that the external surface area,  $S \text{ cm}^2$ , of the open container is such that  $S = 4x^2 + \frac{2000}{x}$ . [4]

- (ii) Given that  $x$  can vary, find the stationary value of  $S$  and determine its nature. [5]

197



No calculator

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The normal to the curve  $y = (x-2)(3x+1)^{\frac{2}{3}}$  at the point where  $x = \frac{7}{3}$ , meets the  $y$ -axis at the point  $P$ . Find the exact coordinates of the point  $P$ . [7]

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|        |                                                                                        |           |                                                                                                                                      |
|--------|----------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------|
| 38 (i) | $800 = 4x^2h$                                                                          | <b>B1</b> |                                                                                                                                      |
|        | $h = \frac{800}{4x^2}$ oe or $xh = \frac{800}{4x}$ oe                                  | <b>B1</b> |                                                                                                                                      |
|        | $(S =) 2hx + 8xh + 4x^2$ oe                                                            | <b>M1</b> | Allow if $h$ is substituted at this point                                                                                            |
|        | $S = 4x^2 + \left(\frac{2000}{x}\right)$                                               | <b>A1</b> | Leading to AG, must have $S =$ or surface area = at some point and no errors                                                         |
| (ii)   | $\left(\frac{dS}{dx} = \right) 8x - \frac{2000}{x^2}$                                  | <b>B1</b> | For correct differentiation                                                                                                          |
|        | When $\frac{dS}{dx} = 0$ , $x = \sqrt[3]{250}$ oe (6.30)                               | <b>M1</b> | For equating to zero and attempt to solve, must get as far as $x = \dots$ , must be using the form $ax + \frac{b}{x^2}$              |
|        |                                                                                        | <b>A1</b> | For correct positive $x$                                                                                                             |
|        | $S = 476$ only                                                                         | <b>A1</b> |                                                                                                                                      |
|        | $\frac{d^2S}{dx^2} = 8 + \frac{4000}{x^3}$<br>$\frac{d^2S}{dx^2} > 0$ or 24 so minimum | <b>B1</b> | For a correct convincing method, with enough detail to reach a correct conclusion of a minimum. Must be using $x = \sqrt[3]{250}$ oe |

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|    |                                                                                                              |           |                                                |
|----|--------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|
| 39 |                                                                                                              | <b>M1</b> | For attempt at differentiating a product       |
|    |                                                                                                              | <b>B1</b> | For $\frac{2}{3} \times 3(3x+1)^{\frac{1}{3}}$ |
|    | $\left(\frac{dy}{dx} = \right) (x-2) \times \frac{2}{3} \times 3(3x+1)^{\frac{1}{3}} + (3x+1)^{\frac{2}{3}}$ | <b>A1</b> | For all other terms correct                    |
|    | $y = \frac{4}{3}$                                                                                            | <b>B1</b> |                                                |

|                                                                                   |           |                                                                                                               |
|-----------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------|
| When $x = \frac{7}{3}$ , $\frac{dy}{dx} = \frac{13}{3}$                           | <b>M1</b> | For attempt at normal equation using $-\frac{1}{\text{their } m}$ and <i>their</i> $y$ when $x = \frac{7}{3}$ |
| Equation of normal: $y - \frac{4}{3} = -\frac{3}{13}\left(x - \frac{7}{3}\right)$ | <b>A1</b> | For correct normal equation, may be implied by a correct final answer                                         |
| At y-axis, $y = \frac{73}{39}$<br>$\left(0, \frac{73}{39}\right)$ isw             | <b>A1</b> |                                                                                                               |

198



No calculator

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It is given that  $y = \frac{\ln(2x^3 + 5)}{x - 1}$  for  $x > 1$ .

- (i) Find the value of  $\frac{dy}{dx}$  when  $x = 2$ . You must show all your working. [4]

- (ii) Find the approximate change in  $y$  as  $x$  increases from 2 to  $2 + p$ , where  $p$  is small. [1]

199



No calculator

May/June 2019 • Paper 13 • Q10

A curve is such that  $\frac{d^2y}{dx^2} = (2x + 3)^{-\frac{1}{2}}$ . The curve has a gradient of 5 at the point where  $x = 3$  and passes through the point  $\left(\frac{1}{2}, -\frac{1}{3}\right)$ .

- (i) Find the equation of the curve. [7]

- (ii) Find the equation of the normal to the curve at the point where  $x = 3$ , giving your answer in the form  $ax + by + c = 0$ , where  $a$ ,  $b$  and  $c$  are integers. [4]

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|        |                                                                                                                |           |                                         |
|--------|----------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------|
| 67 (i) |                                                                                                                | <b>B1</b> | For $\frac{6x^3}{(2x^3+5)}$             |
|        |                                                                                                                | <b>M1</b> | For attempt to differentiate a quotient |
|        | $\frac{dy}{dx} = \frac{(x-1)\frac{6x^2}{(2x^3+5)} - \ln(2x^3+5)}{(x-1)^2}$                                     | <b>A1</b> | For all other terms correct             |
|        | When $x = 2$ ,<br>$\frac{dy}{dx} = \frac{24}{21} - \ln 21 \text{ or } \frac{8}{7} - \ln 21, \text{ or } -1.90$ | <b>A1</b> |                                         |
| (ii)   | $-1.90p$ oe                                                                                                    | <b>B1</b> |                                         |

**199** May/June 2019 · Paper 13 · Q10

|        |                                                   |           |                                                        |
|--------|---------------------------------------------------|-----------|--------------------------------------------------------|
| 68 (i) |                                                   | <b>M1</b> | Attempt to integrate to obtain $k(2x+3)^{\frac{1}{2}}$ |
|        | $\frac{dy}{dx} = (2x+3)^{\frac{1}{2}} \quad (+c)$ | <b>A1</b> | All correct, condone omission of $+c$                  |
|        | $5 = 3 + c$                                       | <b>M1</b> | <b>Dep</b><br>For attempt at $c$                       |
|        | $\frac{dy}{dx} = (2x+3)^{\frac{1}{2}} + 2$        | <b>M1</b> | For a further attempt to integrate                     |
|        | $y = \frac{1}{3}(2x+3)^{\frac{3}{2}} + 2x(+d)$    | <b>A1</b> | All correct, condone omission of $+d$                  |
|        | $-\frac{1}{3} = \frac{8}{3} + 1 + d$              | <b>M1</b> | For attempt at $d$                                     |
|        | $y = \frac{1}{3}(2x+3)^{\frac{3}{2}} + 2x - 4$    | <b>A1</b> | Must have $y =$                                        |
| (ii)   | When $x = 3, y = 11$                              | <b>M1</b> | For attempt to find $y$ using <i>their</i> (i)         |
|        |                                                   | <b>M1</b> | <b>Dep</b><br>For attempt at normal                    |
|        | Normal: $y - 11 = -\frac{1}{5}(x - 3)$            | <b>A1</b> | All correct unsimplified                               |

$$x + 5y - 58 = 0$$

**A1** For correct form

**200**

No calculator

**Oct/Nov 2018 · Paper 11 · Q6**

A curve has equation  $y = \frac{\ln(2x^2 + 3)}{5x + 2}$ .

(i) Show that  $\frac{dy}{dx} = -\frac{5}{4} \ln 3$  when  $x = 0$ . [4]

(ii) Hence find the equation of the tangent to the curve at the point where  $x = 0$ . [2]

**201**

No calculator

**Oct/Nov 2018 · Paper 11 · Q8**

(i) Find  $\frac{d}{dx}(5x^2 + 4)^{\frac{3}{2}}$ . [2]

(ii) Hence find  $\int x(5x^2 + 4)^{\frac{1}{2}} dx$ . [2]

Given that  $\int_0^a x(5x^2 + 4)^{\frac{1}{2}} dx = \frac{19}{15}$ ,

(iii) find the value of the positive constant  $a$ . [4]



**200** Oct/Nov 2018 · Paper 11 · Q6

|        |                                                                                                               |           |                                                                |
|--------|---------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------|
| 17 (i) | For $\frac{4x}{2x^2 + 3}$                                                                                     | <b>B1</b> |                                                                |
|        |                                                                                                               | <b>M1</b> | For attempt to differentiate a quotient or appropriate product |
|        | $\frac{dy}{dx} = \frac{(5x+2)\frac{4x}{2x^2+3} - 5\ln(2x^2+3)}{(5x+2)^2}$                                     | <b>A1</b> | All other terms correct                                        |
|        | When $x=0$ $\frac{dy}{dx} = \frac{-5\ln 3}{4}$                                                                | <b>A1</b> | For <b>given answer</b>                                        |
| (ii)   | $y = \frac{1}{2}\ln 3$ or 0.549                                                                               | <b>B1</b> | May be implied by tangent equation, allow 0.55                 |
|        | Equation of tangent<br>$y = \left(-\frac{5}{4}\ln 3\right)x + \frac{1}{2}\ln 3$<br>or<br>$y = -1.37x + 0.549$ | <b>B1</b> |                                                                |

**201** Oct/Nov 2018 · Paper 11 · Q8

|        |                                                                                                      |           |                                                                                              |
|--------|------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------|
| 18 (i) |                                                                                                      | <b>M1</b> | For attempt to use chain rule to obtain $kx(5x^2 + 4)^{\frac{1}{2}}$ where $k$ is a constant |
|        | $\frac{3}{2}(10x)(5x^2 + 4)^{\frac{1}{2}}$                                                           | <b>A1</b> | Allow unsimplified                                                                           |
| (ii)   |                                                                                                      | <b>M1</b> | For attempt to use part (i) if in correct form of $m(5x^2 + 4)^{\frac{3}{2}}$                |
|        | $\frac{1}{15}(5x^2 + 4)^{\frac{3}{2}} (+c)$                                                          | <b>A1</b> | <b>FT</b> on <i>their</i> $\frac{1}{k}(5x^2 + 4)^{\frac{3}{2}}$                              |
| (iii)  |                                                                                                      | <b>M1</b> | For use of limits if <i>their</i> (ii)<br>Must be in the form $m(5x^2 + 4)^{\frac{3}{2}}$    |
|        | $\frac{1}{15}\left((5a^2 + 4)^{\frac{3}{2}} - 4^{\frac{3}{2}}\right) \left[ = \frac{19}{15} \right]$ | <b>A1</b> |                                                                                              |

|  |                                 |           |                                                                                          |
|--|---------------------------------|-----------|------------------------------------------------------------------------------------------|
|  | $(5a^2 + 4)^{\frac{3}{2}} = 27$ | <b>M1</b> | <b>Dep</b><br>For complete and correct method to deal with the<br>power of $\frac{3}{2}$ |
|  | leading to $a = 1$              | <b>A1</b> |                                                                                          |

**202**

No calculator

**Oct/Nov 2018 · Paper 11 · Q9**

Variables  $s$  and  $t$  are such that  $s = 4t + 3e^{-t}$ .

(i) Find the value of  $s$  when  $t = 0$ . [1]

(ii) Find the exact value of  $t$  when  $\frac{ds}{dt} = 2$ . [4]

(iii) Find the approximate increase in  $s$  when  $t$  increases from  $\ln 5$  to  $\ln 5 + h$ , where  $h$  is small. [3]

**203**

No calculator

**Oct/Nov 2018 · Paper 12 · Q2**

Find the equation of the curve which has a gradient of 4 at the point  $(0, -3)$  and is such that

$$\frac{d^2y}{dx^2} = 5 + e^{2x}. \quad [5]$$

**202** Oct/Nov 2018 · Paper 11 · Q9

|        |                                                        |           |                                                                                           |
|--------|--------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|
| 19 (i) | 3                                                      | <b>B1</b> |                                                                                           |
| (ii)   |                                                        | <b>M1</b> | For attempt to differentiate to obtain $a + be^{-t}$                                      |
|        | $\frac{ds}{dt} = 4 - 3e^{-t}$                          | <b>A1</b> | All correct                                                                               |
|        | $2 = 4 - 3e^{-t}$                                      | <b>M1</b> | <b>Dep</b> for correct attempt to solve equation involving exponential where $e^{-t} > 0$ |
|        | leading to $t = \ln \frac{3}{2}$ or $-\ln \frac{2}{3}$ | <b>A1</b> | Must be an exact form                                                                     |
| (iii)  | When $t = \ln 5$ , $\frac{ds}{dt} = \frac{17}{5}$      | <b>M1</b> | For attempt to find value of $\frac{ds}{dt}$ when $t = \ln 2$                             |
|        |                                                        | <b>M1</b> | <b>Dep</b> for attempt to use method of small changes                                     |
|        | $\partial s = \frac{17h}{5}$                           | <b>A1</b> |                                                                                           |

**203** Oct/Nov 2018 · Paper 12 · Q2

|    |                                                                        |           |                                                                                                                                        |
|----|------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------|
| 49 | $\frac{dy}{dx} = 5x + \frac{1}{2}e^{2x} \quad (+c)$                    | <b>M1</b> | For attempt to integrate to get $\frac{dy}{dx}$ in the form $5x + pe^{2x}$ .<br>Condone omission of $+c$                               |
|    | When $x=0$ , $\frac{dy}{dx} = 4$ so $c = \frac{7}{2}$                  | <b>M1</b> | <b>Dep</b><br>For attempt to get value of $c$                                                                                          |
|    | $y = \frac{5x^2}{2} + \frac{1}{4}e^{2x} + \frac{7}{2}x \quad (+d)$     | <b>M1</b> | <b>Dep</b> on first M1 only<br>For attempt to get $y$ in the form including $\frac{5x^2}{2} + pe^{2x}$ .<br>Condone omission of $+d$ . |
|    | When $x=0$ , $y = -3$ so $d = -\frac{13}{4}$                           | <b>M1</b> | <b>Dep</b> on previous <b>DepM1</b><br>For attempt to obtain $d$ , allow if $c$ not found                                              |
|    | $y = \frac{5x^2}{2} + \frac{1}{4}e^{2x} + \frac{7}{2}x - \frac{13}{4}$ | <b>A1</b> | Must have an equation                                                                                                                  |

**204**

No calculator

**Oct/Nov 2018 • Paper 12 • Q4**

$$y = x^3 \ln(2x + 1)$$

- (i) Find the value of  $\frac{dy}{dx}$  when  $x = 0.3$ . You must show all your working. [4]

- (ii) Hence find the approximate increase in  $y$  when  $x$  increases from 0.3 to  $0.3 + h$ , where  $h$  is small. [1]

**205**

No calculator

**Oct/Nov 2018 • Paper 12 • Q6**

- (i) Find  $\frac{d}{dx}(5x^2 - 125)^{\frac{2}{3}}$ . [2]

- (ii) Using your answer to part (i), find  $\int x(5x^2 - 125)^{-\frac{1}{3}} dx$ . [2]

- (iii) Hence find  $\int_6^{10} x(5x^2 - 125)^{-\frac{1}{3}} dx$ . [2]

**204** Oct/Nov 2018 · Paper 12 · Q4

|        |                                                       |           |                                                        |
|--------|-------------------------------------------------------|-----------|--------------------------------------------------------|
| 50 (i) |                                                       | <b>B1</b> | For $\frac{2}{2x+1}$                                   |
|        |                                                       | <b>M1</b> | For attempt to differentiate a product                 |
|        | $\frac{dy}{dx} = x^3 \frac{2}{2x+1} + 3x^2 \ln(2x+1)$ | <b>A1</b> | For all other terms correct                            |
|        | When $x = 0.3$ , $\frac{dy}{dx} = 0.161$              | <b>A1</b> | For awrt 0.161                                         |
| (ii)   | $0.161h$                                              | <b>B1</b> | <b>FT</b> on <i>their</i> numerical answer to part (i) |

**205** Oct/Nov 2018 · Paper 12 · Q6

|        |                                                                                                          |           |                                                                                                                                                     |
|--------|----------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 51 (i) |                                                                                                          | <b>M1</b> | For $kx(5x-125)^{\frac{1}{3}}$                                                                                                                      |
|        | $\frac{2}{3} \times 10x(5x^2-125)^{\frac{1}{3}}$<br>$\left(\frac{20}{3}x(5x^2-125)^{\frac{1}{3}}\right)$ | <b>A1</b> | Allow unsimplified                                                                                                                                  |
| (ii)   |                                                                                                          | <b>M1</b> | For $m(5x^2-125)^{\frac{2}{3}} (+c)$                                                                                                                |
|        | $\frac{3}{20}(5x^2-125)^{\frac{2}{3}} (+c)$                                                              | <b>A1</b> | <b>FT</b> on <i>their</i> $k$ from part (i)                                                                                                         |
| (iii)  | $\frac{3}{20}\left((375)^{\frac{2}{3}} - (55)^{\frac{2}{3}}\right)$                                      | <b>M1</b> | <b>Dep</b> on previous <b>M1</b><br>For use of limits in <i>their</i> answer to part (ii), must be in the form<br>$m(5x^2-125)^{\frac{2}{3}} (+c),$ |
|        | $= 5.63$                                                                                                 | <b>A1</b> | Allow greater accuracy                                                                                                                              |

**206**

No calculator

**Oct/Nov 2018 · Paper 13 · Q4**

In this question, the units of  $x$  are radians and the units of  $y$  are centimetres.

It is given that  $y = (1 + \cos 3x)^{10}$ .

- (i) Find the value of  $\frac{dy}{dx}$  when  $x = \frac{\pi}{2}$ . [4]

Given also that  $y$  is increasing at a rate of  $6 \text{ cm s}^{-1}$  when  $x = \frac{\pi}{2}$ ,

- (ii) find the corresponding rate of change of  $x$ . [2]

**206** Oct/Nov 2018 · Paper 13 · Q4

|        |                                                                                         |           |                                                       |
|--------|-----------------------------------------------------------------------------------------|-----------|-------------------------------------------------------|
| 77 (i) |                                                                                         | <b>M1</b> | for $10(1 + \cos 3x)^9 f(x)$                          |
|        |                                                                                         | <b>M1</b> | for $k \sin 3x(1 + \cos 3x)^9$                        |
|        | $\frac{dy}{dx} = -30 \sin 3x(1 + \cos 3x)^9$                                            | <b>A1</b> |                                                       |
|        | When $x = \frac{\pi}{2}$ , $\frac{dy}{dx} = 30$                                         | <b>A1</b> |                                                       |
| (ii)   | Use of $\frac{dy}{dx} \times \frac{dx}{dt} = \frac{dy}{dt}$<br>with $\frac{dy}{dt} = 6$ | <b>M1</b> | <i>their</i> $\frac{dy}{dx} \times \frac{dx}{dt} = 6$ |
|        | $\frac{dx}{dt} = \frac{1}{5}$ or 0.2                                                    | <b>A1</b> | <b>FT</b> from <i>their</i> answer from part (i)      |



207



No calculator

Oct/Nov 2018 · Paper 13 · Q6

A particle  $P$  is moving in a straight line such that its displacement,  $s$  m, from a fixed point  $O$  at time  $t$  s, is given by  $s = 12e^{-0.5t} + 4t - 12$ .

(i) Find the value of  $t$  when  $P$  is instantaneously at rest. [3]

(ii) Find an expression for the acceleration of  $P$  at time  $t$  s. [2]

(iii) Find the value of  $s$  when the acceleration of  $P$  is  $0.3 \text{ ms}^{-2}$ . [3]

(iv) Explain why the acceleration of the particle will always be positive. [1]

208



Calculator

Oct/Nov 2018 · Paper 13 · Q9

Find the equation of the normal to the curve  $y = \frac{\ln(3x^2 + 1)}{x^2}$  at the point where  $x = 2$ , giving your answer in the form  $y = mx + c$ , where  $m$  and  $c$  are correct to 2 decimal places. You must show all your working. [8]

**207** Oct/Nov 2018 · Paper 13 · Q6

|        |                                                                                                               |           |                                                                                             |
|--------|---------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------|
| 78 (i) | $\frac{ds}{dt} = -6e^{-0.5t} + 4$                                                                             | <b>M1</b> | for $ke^{-0.5t} + 4$                                                                        |
|        | When $\frac{ds}{dt} = 0$ , $e^{-0.5t} = \frac{2}{3}$<br>$-0.5t = \ln \frac{2}{3}$<br>$t = -2 \ln \frac{2}{3}$ | <b>M1</b> | <b>Dep</b><br>equating to zero and correct order of operations to solve for $t$             |
|        | $t = 0.811$                                                                                                   | <b>A1</b> |                                                                                             |
| (ii)   |                                                                                                               | <b>M1</b> | for $ke^{-0.5t}$                                                                            |
|        | $\frac{d^2s}{dt^2} = 3e^{-0.5t}$                                                                              | <b>A1</b> |                                                                                             |
| (iii)  | $3e^{-0.5t} = 0.3$<br>$e^{-0.5t} = 0.1$<br>$t = \frac{\ln 0.1}{-0.5}$                                         | <b>M1</b> | correct order of operations and correct use of $\ln$ to solve<br>$ke^{-0.5t} = 0.3$ for $t$ |
|        | $s = 12e^{-0.5 \times 4.605} + 4 \times 4.605 - 12$                                                           | <b>M1</b> | <b>Dep</b><br>use of $t$ to obtain $s$                                                      |
|        | $s = 7.62$                                                                                                    | <b>A1</b> |                                                                                             |
| (iv)   | $e^{-0.5t}$ is always positive or $e^{-0.5t}$ can never be zero or negative                                   | <b>B1</b> | correct comment about $e^{-0.5t}$                                                           |

**208** Oct/Nov 2018 · Paper 13 · Q9

| Question | Answer                                                                                                                                                                                           | Marks     | Guidance                                 |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------|
| 79       | $\frac{d}{dx}(\ln(3x^2 + 1)) = \frac{6x}{3x^2 + 1}$                                                                                                                                              | <b>B1</b> | for $\frac{6x}{3x^2 + 1}$                |
|          | $\frac{dy}{dx} = \frac{x^2 \frac{6x}{3x^2 + 1} - 2x \ln(3x^2 + 1)}{x^4}$<br>or<br>$\frac{dy}{dx} = \left(\frac{-2}{x^3}\right) \ln(3x^2 + 1) + \left(\frac{1}{x^2}\right) \frac{6x}{(3x^2 + 1)}$ | <b>M1</b> | differentiation of a quotient or product |

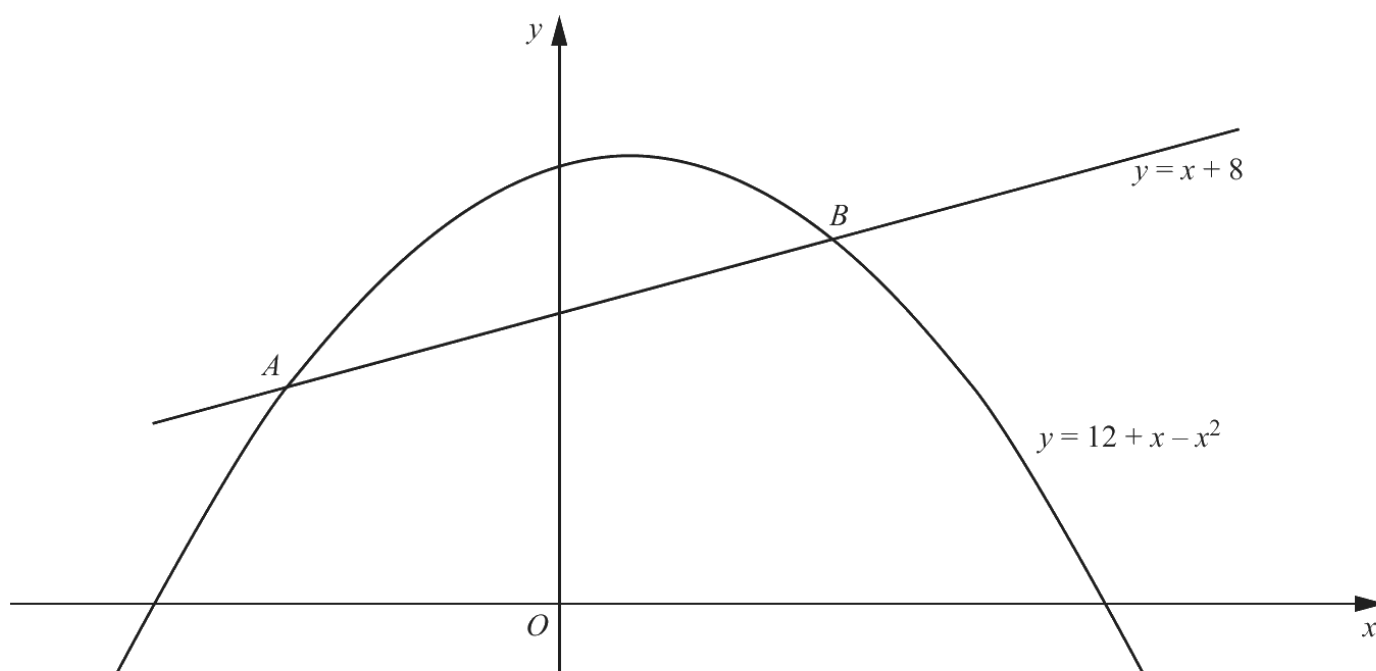
|                                                                                                                                |           |                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------|
| $\frac{x^2 f(x) - 2x \ln(3x^2 + 1)}{x^4}$ or for $\left(-\frac{2}{x^3}\right) \ln(3x^2 + 1) + \left(\frac{1}{x^2}\right) f(x)$ | <b>A1</b> |                                                                   |
| When $x = 2$ , $\frac{dy}{dx} = -0.410$                                                                                        | <b>A1</b> |                                                                   |
| Gradient of perp = 2.436...                                                                                                    | <b>M1</b> | use of $-\frac{1}{m}$ with a gradient obtained by differentiation |
| When $x = 2$ , $y = 0.641$ or $\frac{1}{4} \ln 13$                                                                             | <b>B1</b> |                                                                   |
| Normal: $y - 0.641 = 2.436(x - 2)$                                                                                             | <b>M1</b> | <b>Dep</b>                                                        |
| $y = 2.44x - 4.23$                                                                                                             | <b>A1</b> |                                                                   |

209



No calculator

Oct/Nov 2018 • Paper 13 • Q10



The diagram shows the curve  $y = 12 + x - x^2$  intersecting the line  $y = x + 8$  at the points  $A$  and  $B$ .

(i) Find the coordinates of the points  $A$  and  $B$ . [3]

(ii) Find  $\int (12 + x - x^2) dx$ . [2]

(iii) Showing all your working, find the area of the shaded region. [4]

## 209 Oct/Nov 2018 · Paper 13 · Q10

|        |                                                                                                                              |           |                                                                                                                                                                                                        |
|--------|------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 80 (i) | $x + 8 = 12 + x - x^2$<br>$x^2 = 4$ , $x = \pm 2$<br>or<br>$y^2 - 16y + 60 = 0$<br>$y = 6$ or $y = 10$                       | <b>M1</b> | correct method of solution to obtain $x$ or $y$                                                                                                                                                        |
|        | $x = 2$ , $y = 10$<br>$x = -2$ , $y = 6$                                                                                     | <b>A2</b> | <b>A1</b> for $x = -2$ and $x = 2$ or for $y = 6$ and $y = 10$<br>or for either point from a correctly solved equation.                                                                                |
| (ii)   |                                                                                                                              | <b>M1</b> | for $12x + px^2 + qx^3$ (+c)                                                                                                                                                                           |
|        | $12x + \frac{x^2}{2} - \frac{x^3}{3}$ (+c)                                                                                   | <b>A1</b> |                                                                                                                                                                                                        |
| (iii)  | $\left[12x + \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2}^2 - \left(\frac{1}{2}(6+10) \times 4\right)$                          | <b>B1</b> | <b>FT</b><br>area of the trapezium unsimplified<br>$\left(\frac{1}{2}(6+10) \times 4\right)$ or<br>$\left[\frac{2^2}{2} + 8 \times 2\right] - \left[\frac{(-2)^2}{2} + 8 \times (-2)\right]$<br>(= 32) |
|        | $\left[12 \times 2 + \frac{2^2}{2} - \frac{2^3}{3}\right] - \left[12 \times -2 + \frac{(-2)^2}{2} - \frac{(-2)^3}{3}\right]$ | <b>M1</b> | correct use of correct limits for area under the curve using <i>their</i> integral of the form $12x + px^2 + qx^3$                                                                                     |
|        | $= \frac{128}{3}$ oe                                                                                                         | <b>A1</b> |                                                                                                                                                                                                        |
|        | $= \frac{32}{3}$ oe                                                                                                          | <b>A1</b> |                                                                                                                                                                                                        |
|        | $\int_{-2}^2 12 + x - x^2 - x - 8 \, dx$<br>$= \int_{-2}^2 4 - x^2 \, dx$                                                    | <b>M1</b> | subtraction of the two equations with intent to integrate the result                                                                                                                                   |
|        | $= \left[4x - \frac{x^3}{3}\right]_{-2}^2$                                                                                   | <b>A1</b> |                                                                                                                                                                                                        |

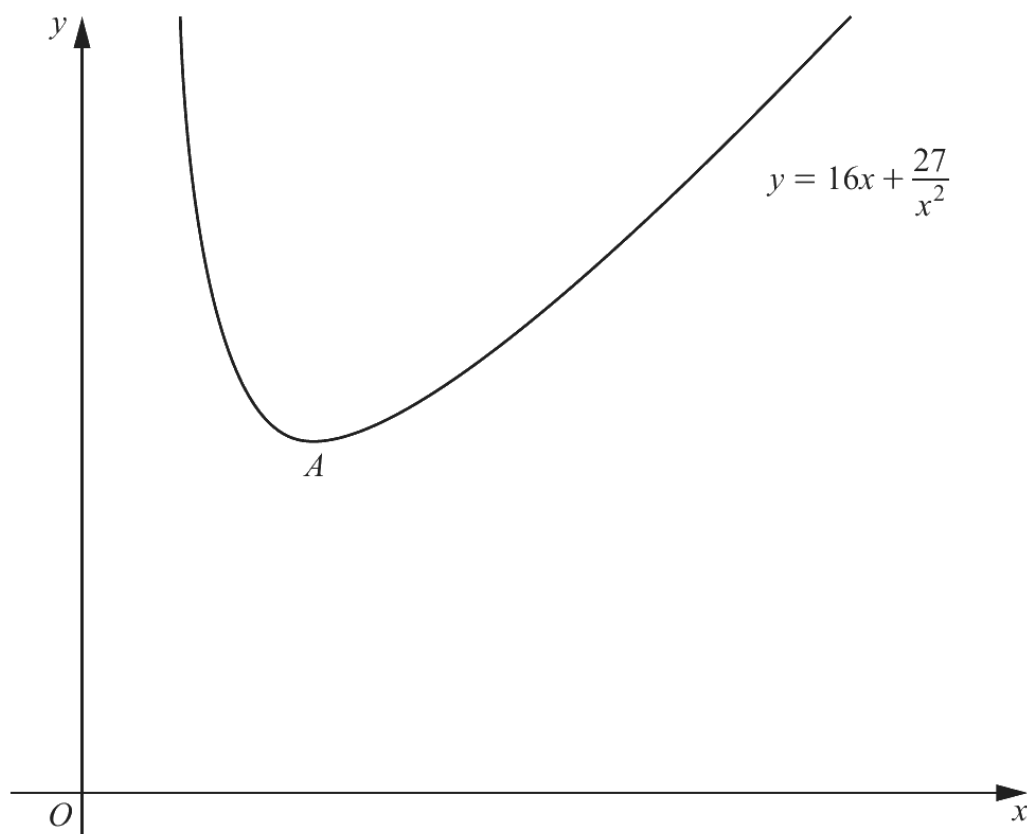
|  |                                                                                  |           |                                                 |
|--|----------------------------------------------------------------------------------|-----------|-------------------------------------------------|
|  | $\left[4 \times 2 - \frac{8}{3}\right] - \left[4 \times -2 + \frac{8}{3}\right]$ | <b>M1</b> | <b>Dep</b><br>for correct application of limits |
|  | $= \frac{32}{3}$ oe                                                              | <b>A1</b> |                                                 |

210



No calculator

May/June 2018 • Paper 11 • Q11



The diagram shows part of the graph of  $y = 16x + \frac{27}{x^2}$ , which has a minimum at  $A$ .

(i) Find the coordinates of  $A$ .

[4]

The points  $P$  and  $Q$  lie on the curve  $y = 16x + \frac{27}{x^2}$  and have  $x$ -coordinates 1 and 3 respectively.

(ii) Find the area enclosed by the curve and the line  $PQ$ . You must show all your working.

[6]

**210** May/June 2018 · Paper 11 · Q11

|       |                                       |           |                                            |
|-------|---------------------------------------|-----------|--------------------------------------------|
| 5 (i) | $\frac{dy}{dx} = 16 - \frac{54}{x^3}$ | <b>M1</b> | for $\frac{dy}{dx} = 16 \pm \frac{p}{x^3}$ |
|       | Equating to zero and obtaining $x^3$  | <b>M1</b> | <b>M1dep</b>                               |
|       | $x = \frac{3}{2}, y = 36$             | <b>A2</b> | <b>A1</b> for each                         |

|      |                                                                                         |           |                                                  |
|------|-----------------------------------------------------------------------------------------|-----------|--------------------------------------------------|
| (ii) | <b>EITHER:</b><br>When $x = 1, y = 43$<br>When $x = 3, y = 51$                          | <b>B1</b> | <b>B1</b> for both                               |
|      | $\left(\frac{1}{2}(43+51) \times 2\right) - \int_1^3 16x + \frac{27}{x^2} dx$           |           |                                                  |
|      | Area of trapezium = $\left(\frac{1}{2}(43+51) \times 2\right)$ oe                       | <b>B1</b> | <b>FT</b> from <i>their P</i> and <i>their Q</i> |
|      | Integration to find area under curve                                                    | <b>M1</b> | for $\left[px^2 + \frac{q}{x}\right]$            |
|      | $= \left[8x^2 - \frac{27}{x}\right]$                                                    | <b>A1</b> | Integration correct                              |
|      | $= \left[8 \times 3^2 - \frac{27}{3}\right] - \left[8 \times 1^2 - \frac{27}{1}\right]$ | <b>M1</b> | <b>M1dep</b> for application of limits           |
|      | Required area = $94 - 82$<br>= 12                                                       | <b>A1</b> |                                                  |

|  |                                                       |           |                                                                   |
|--|-------------------------------------------------------|-----------|-------------------------------------------------------------------|
|  | When $x = 1, y = 43$<br>When $x = 3, y = 51$          |           |                                                                   |
|  | Equation of $PQ$ : $y = 4x + 39$                      | <b>B1</b> | Equation of line <b>FT</b> from <i>their P</i> and <i>their Q</i> |
|  | Integration of their $4x + 39 - 16x - \frac{27}{x^2}$ | <b>M1</b> | for $\left[px + qx^2 + \frac{r}{x}\right]$                        |
|  | $= \left[39x - 6x^2 + \frac{27}{x}\right]$            | <b>A1</b> | All correct                                                       |



|  |                                                                                                                           |           |                                        |
|--|---------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------|
|  | $= \left[ 39 \times 3 - 6 \times 3^2 + \frac{27}{3} \right]$ $- \left[ 39 \times 1 - 6 \times 1^2 + \frac{27}{1} \right]$ | <b>M1</b> | <b>M1dep</b> for application of limits |
|  | Required area = $72 - 60$<br>= $12$                                                                                       | <b>A1</b> |                                        |

**211**

No calculator

**May/June 2018 • Paper 11 • Q12**

A curve is such that  $\frac{d^2y}{dx^2} = (2x - 5)^{-\frac{1}{2}}$ . Given that the curve has a gradient of 6 at the point  $\left(\frac{9}{2}, \frac{2}{3}\right)$ , find the equation of the curve. [8]

**212**

No calculator

**May/June 2018 • Paper 12 • Q4**

A particle  $P$  moves so that its displacement,  $x$  metres from a fixed point  $O$ , at time  $t$  seconds, is given by  $x = \ln(5t + 3)$ .

(i) Find the value of  $t$  when the displacement of  $P$  is 3m. [2]

(ii) Find the velocity of  $P$  when  $t = 0$ . [2]

(iii) Explain why, after passing through  $O$ , the velocity of  $P$  is never negative. [1]

(iv) Find the acceleration of  $P$  when  $t = 0$ . [2]

**211** May/June 2018 · Paper 11 · Q12

| Question | Answer                                                  | Marks     | Partial Marks                                                                                                                |
|----------|---------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------|
| 6        | $\frac{dy}{dx} = (2x-5)^{\frac{1}{2}} \quad (+c)$       | <b>M1</b> | for $k(2x-5)^{\frac{1}{2}}$ ,                                                                                                |
|          | for $(2x-5)^{\frac{1}{2}}$                              | <b>A1</b> |                                                                                                                              |
|          | Substitution to obtain arbitrary constant               | <b>M1</b> | <b>M1 dep</b><br>Using $\frac{dy}{dx} = 6$ when $x = \frac{9}{2}$                                                            |
|          | $\left(\frac{dy}{dx}\right) = (2x-5)^{\frac{1}{2}} + 4$ | <b>A1</b> | for correct $\frac{dy}{dx}$                                                                                                  |
|          | Integration of <i>their</i> $k(2x-5)^{\frac{1}{2}} + c$ | <b>M1</b> | <b>M1 dep</b> on first M1 for integration of $k(2x-5)^{\frac{1}{2}}$ to obtain $m(2x-5)^{\frac{3}{2}}$                       |
|          | $y = \frac{1}{3}(2x-5)^{\frac{3}{2}} + 4x \quad (+d)$   | <b>A1</b> | for $\frac{1}{3}(2x-5)^{\frac{3}{2}} + 4x$<br><b>FT</b> <i>their</i> (non -zero) constant                                    |
|          | Finding constant                                        | <b>M1</b> | <b>M1 dep</b> for obtaining arbitrary constant for $m(2x-5)^{\frac{3}{2}} + nx + d$ using $x = \frac{9}{2}, y = \frac{2}{3}$ |
|          | $y = \frac{1}{3}(2x-5)^{\frac{3}{2}} + 4x - 20$         | <b>A1</b> | for correct equation                                                                                                         |

**212** May/June 2018 · Paper 12 · Q4

|        |                                                               |           |                        |
|--------|---------------------------------------------------------------|-----------|------------------------|
| 34 (i) | $3 = \ln(5t+3)$<br>$e^3 = 5t+3$ or better                     | <b>B1</b> |                        |
|        | $t = 3.42$                                                    | <b>B1</b> |                        |
| (ii)   | $\frac{dx}{dt} = \frac{5}{5t+3}$                              | <b>M1</b> | for $\frac{k_1}{5t+3}$ |
|        | When $t = 0$ , $\frac{dx}{dt} = \frac{5}{3}$ , 1.67 or better | <b>A1</b> | all correct            |

|       |                                                                                                             |           |                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------|
| (iii) | If $t > 0$ each term in $\frac{k_1}{5t+3} > 0$ so never negative oe                                         | <b>B1</b> | dep on <b>M1</b> in (ii)<br><br><b>FT</b> on <i>their</i> $\frac{k_1}{5t+3}$ , provided $k_1 > 0$ |
| (iv)  | $\frac{d^2x}{dt^2} = \frac{k_2}{(5t+3)^2}$                                                                  | <b>M1</b> |                                                                                                   |
|       | $\frac{d^2x}{dt^2} = -\frac{25}{(5t+3)^2}$<br>When $t = 0$ , $\frac{d^2x}{dt^2} = -\frac{25}{9}$ or $-2.78$ | <b>A1</b> | all correct                                                                                       |

**213**

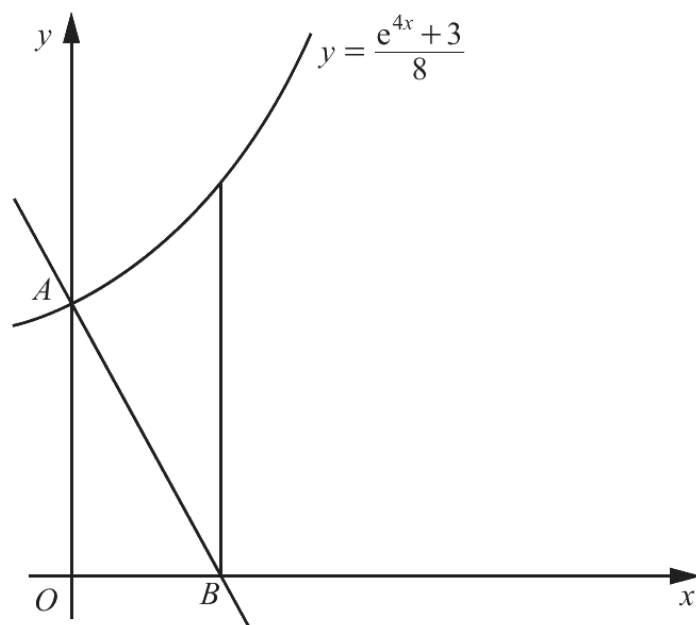
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**May/June 2018 · Paper 12 · Q6**

Find the coordinates of the stationary point of the curve  $y = \frac{x+2}{\sqrt{2x-1}}$ . [6]

**214**

No calculator

**May/June 2018 · Paper 12 · Q11**

The diagram shows the graph of the curve  $y = \frac{e^{4x} + 3}{8}$ . The curve meets the  $y$ -axis at the point  $A$ .

The normal to the curve at  $A$  meets the  $x$ -axis at the point  $B$ . Find the area of the shaded region enclosed by the curve, the line  $AB$  and the line through  $B$  parallel to the  $y$ -axis. Give your answer in the form  $\frac{e}{a}$ , where  $a$  is a constant. You must show all your working.

[10]

**Continuation of working space for Question 36**

**213** May/June 2018 · Paper 12 · Q6

|    |                                                                                                                                                                                                |           |                                     |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------|
| 35 | attempt to differentiate quotient or equivalent product                                                                                                                                        | <b>M1</b> |                                     |
|    | $\frac{d}{dx}(2x-1)^{\frac{1}{2}} = (2x-1)^{-\frac{1}{2}}$ for a quotient<br>$\frac{d}{dx}(2x-1)^{-\frac{1}{2}} = -(2x-1)^{-\frac{3}{2}}$ for a product                                        | <b>B1</b> |                                     |
|    | either $\frac{dy}{dx} = \frac{\sqrt{2x-1} - (x+2)\left[(2x-1)^{-\frac{1}{2}}\right]}{(\sqrt{2x-1})^2}$<br>or $\frac{dy}{dx} = (2x-1)^{-\frac{1}{2}} - (x+2)\left[(2x-1)^{-\frac{3}{2}}\right]$ | <b>A1</b> | All other terms correct             |
|    | When $\frac{dy}{dx} = 0$ , $2x-1 = x+2$                                                                                                                                                        | <b>M1</b> | equate to zero and attempt to solve |
|    | $x = 3$                                                                                                                                                                                        | <b>A1</b> |                                     |
|    | $y = \sqrt{5}, \frac{5}{\sqrt{5}}, 2.24$                                                                                                                                                       | <b>A1</b> |                                     |

## 214 May/June 2018 · Paper 12 · Q11

| Question | Answer                                                                              | Marks     | Partial Marks                                                                                                                |
|----------|-------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------|
| 36       | When $x = 0$ , $y = \frac{1}{2}$                                                    | <b>B1</b> | For $y = \frac{1}{2}$                                                                                                        |
|          | $\frac{dy}{dx} = \frac{1}{2}e^{4x}$                                                 | <b>B1</b> |                                                                                                                              |
|          | $\frac{dy}{dx} = \frac{1}{2}$ , Gradient of normal = $-2$                           | <b>B1</b> | <b>FT</b> on <i>their</i> $\frac{dy}{dx}$ , must be numeric                                                                  |
|          | either: Normal $y - \frac{1}{2} = -2x$<br>or: Gradient of normal = $-\frac{OA}{OB}$ | <b>M1</b> | For an attempt at a normal equation passing through <i>their</i> $\left(0, \frac{1}{2}\right)$ and a substitution of $y = 0$ |
|          | When $y = 0$ , $x = \frac{1}{4}$                                                    | <b>A1</b> |                                                                                                                              |
|          | EITHER:<br>$\int_0^{\frac{1}{4}} \frac{1}{8}e^{4x} + \frac{3}{8} dx$                | <b>M1</b> | For attempt to integrate to obtain $k_1e^{4x} + \frac{3}{8}x$ , $k_1 \neq \frac{1}{8}$ , $k_1 \neq \frac{1}{2}$              |

|  |                                                                    |           |                                             |
|--|--------------------------------------------------------------------|-----------|---------------------------------------------|
|  | $\left[ \frac{1}{32}e^{4x} + \frac{3x}{8} \right]_0^{\frac{1}{4}}$ | <b>A1</b> | For correct integration                     |
|  | Use of limits                                                      | <b>M1</b> | <b>M1dep</b>                                |
|  | For area of triangle = $\frac{1}{16}$                              | <b>B1</b> | <b>FT</b> on <i>their</i> $x = \frac{1}{4}$ |
|  | $= \frac{e}{32}$                                                   | <b>A1</b> | final answer in correct form                |

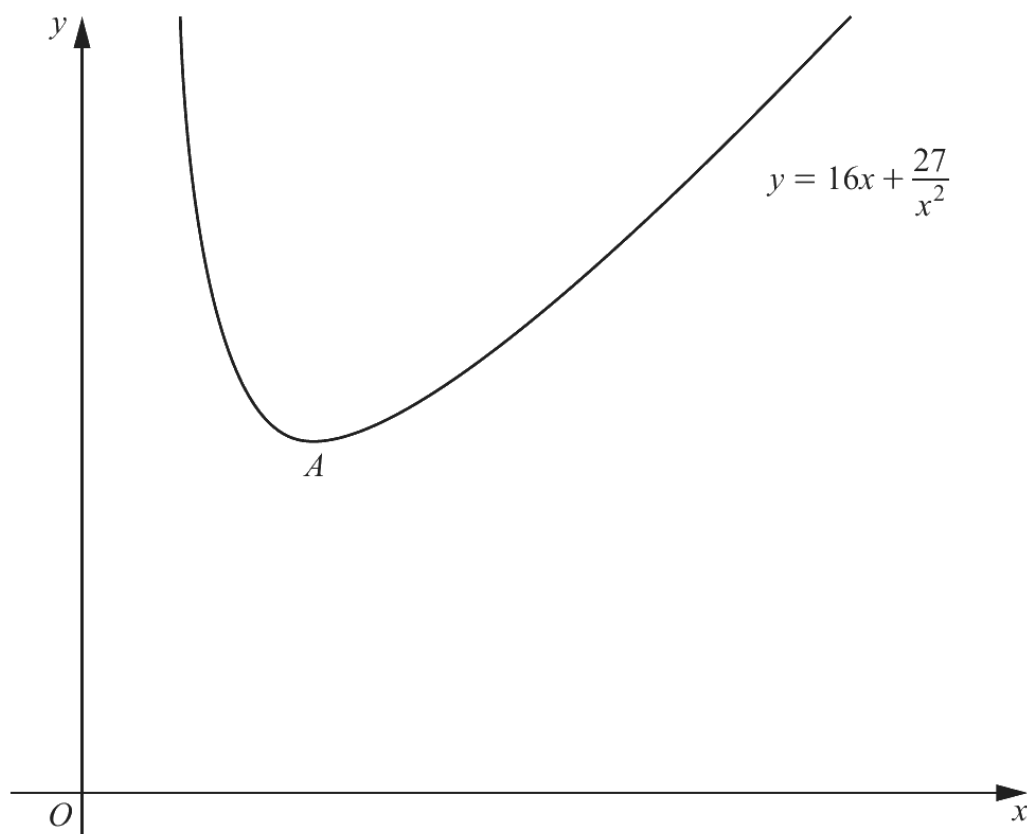
|  |                                                                                     |           |                                                                                                                                      |
|--|-------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------|
|  | OR:<br>$\int_0^{\frac{1}{4}} \frac{1}{8}e^{4x} + \frac{3}{8} - \frac{1}{2} + 2x dx$ | <b>M1</b> | For attempt at subtraction and attempt to integrate<br>to obtain $k_1e^{4x} + \frac{3}{8}x + k_2x + k_3x^2$ , $k_1 \neq \frac{1}{8}$ |
|  | $\left[ \frac{1}{32}e^{4x} - \frac{1}{8}x + x^2 \right]_0^{\frac{1}{4}}$            | <b>A2</b> | -1 for each error for integration                                                                                                    |
|  | for use of limits                                                                   | <b>M1</b> | <b>M1dep</b>                                                                                                                         |
|  | $= \frac{e}{32}$                                                                    | <b>A1</b> | final answer in correct form                                                                                                         |

215



No calculator

May/June 2018 • Paper 13 • Q11



The diagram shows part of the graph of  $y = 16x + \frac{27}{x^2}$ , which has a minimum at  $A$ .

- (i) Find the coordinates of  $A$ .

[4]

The points  $P$  and  $Q$  lie on the curve  $y = 16x + \frac{27}{x^2}$  and have  $x$ -coordinates 1 and 3 respectively.

- (ii) Find the area enclosed by the curve and the line  $PQ$ . You must show all your working.

[6]



**215** May/June 2018 · Paper 13 · Q11

|        |                                       |           |                                            |
|--------|---------------------------------------|-----------|--------------------------------------------|
| 65 (i) | $\frac{dy}{dx} = 16 - \frac{54}{x^3}$ | <b>M1</b> | for $\frac{dy}{dx} = 16 \pm \frac{p}{x^3}$ |
|        | Equating to zero and obtaining $x^3$  | <b>M1</b> | <b>M1dep</b>                               |
|        | $x = \frac{3}{2}, y = 36$             | <b>A2</b> | <b>A1</b> for each                         |

|      |                                                                                         |           |                                                  |
|------|-----------------------------------------------------------------------------------------|-----------|--------------------------------------------------|
| (ii) | <b>EITHER:</b><br>When $x = 1, y = 43$<br>When $x = 3, y = 51$                          | <b>B1</b> | <b>B1</b> for both                               |
|      | $\left(\frac{1}{2}(43+51) \times 2\right) - \int_1^3 16x + \frac{27}{x^2} dx$           |           |                                                  |
|      | Area of trapezium = $\left(\frac{1}{2}(43+51) \times 2\right)$ oe                       | <b>B1</b> | <b>FT</b> from <i>their P</i> and <i>their Q</i> |
|      | Integration to find area under curve                                                    | <b>M1</b> | for $\left[px^2 + \frac{q}{x}\right]$            |
|      | $= \left[8x^2 - \frac{27}{x}\right]$                                                    | <b>A1</b> | Integration correct                              |
|      | $= \left[8 \times 3^2 - \frac{27}{3}\right] - \left[8 \times 1^2 - \frac{27}{1}\right]$ | <b>M1</b> | <b>M1dep</b> for application of limits           |
|      | Required area = $94 - 82$<br>= 12                                                       | <b>A1</b> |                                                  |

|  |                                                       |           |                                                                   |
|--|-------------------------------------------------------|-----------|-------------------------------------------------------------------|
|  | When $x = 1, y = 43$<br>When $x = 3, y = 51$          |           |                                                                   |
|  | Equation of $PQ$ : $y = 4x + 39$                      | <b>B1</b> | Equation of line <b>FT</b> from <i>their P</i> and <i>their Q</i> |
|  | Integration of their $4x + 39 - 16x - \frac{27}{x^2}$ | <b>M1</b> | for $\left[px + qx^2 + \frac{r}{x}\right]$                        |
|  | $= \left[39x - 6x^2 + \frac{27}{x}\right]$            | <b>A1</b> | All correct                                                       |

|  |                                                                                                                           |           |                                        |
|--|---------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------|
|  | $= \left[ 39 \times 3 - 6 \times 3^2 + \frac{27}{3} \right]$ $- \left[ 39 \times 1 - 6 \times 1^2 + \frac{27}{1} \right]$ | <b>M1</b> | <b>M1dep</b> for application of limits |
|  | Required area = $72 - 60$<br>= $12$                                                                                       | <b>A1</b> |                                        |

**216**

No calculator

**May/June 2018 • Paper 13 • Q12**

A curve is such that  $\frac{d^2y}{dx^2} = (2x - 5)^{-\frac{1}{2}}$ . Given that the curve has a gradient of 6 at the point  $\left(\frac{9}{2}, \frac{2}{3}\right)$ , find the equation of the curve. [8]

**216** May/June 2018 · Paper 13 · Q12

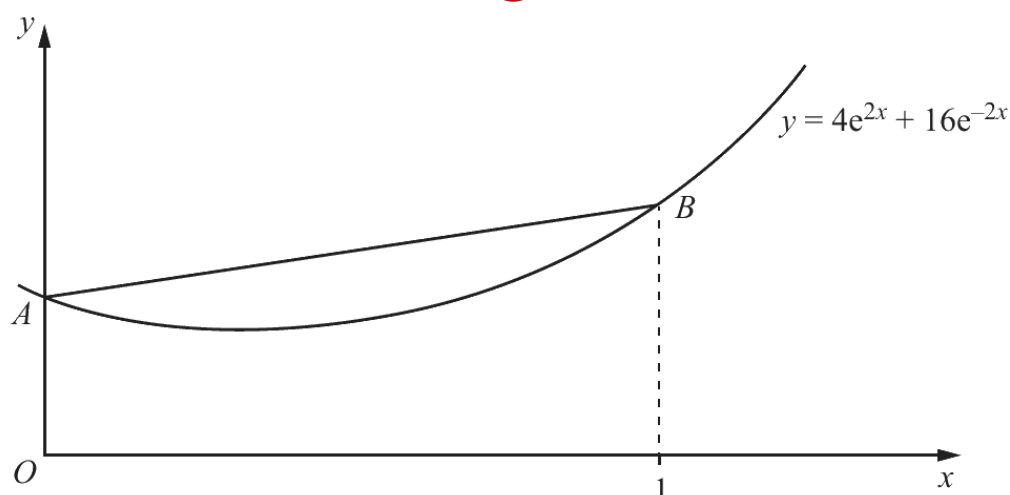
| Question | Answer                                                  | Marks     | Partial Marks                                                                                                                |
|----------|---------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------|
| 66       | $\frac{dy}{dx} = (2x-5)^{\frac{1}{2}} \quad (+c)$       | <b>M1</b> | for $k(2x-5)^{\frac{1}{2}}$ ,                                                                                                |
|          | for $(2x-5)^{\frac{1}{2}}$                              | <b>A1</b> |                                                                                                                              |
|          | Substitution to obtain arbitrary constant               | <b>M1</b> | <b>M1 dep</b><br>Using $\frac{dy}{dx} = 6$ when $x = \frac{9}{2}$                                                            |
|          | $\left(\frac{dy}{dx}\right) = (2x-5)^{\frac{1}{2}} + 4$ | <b>A1</b> | for correct $\frac{dy}{dx}$                                                                                                  |
|          | Integration of <i>their</i> $k(2x-5)^{\frac{1}{2}} + c$ | <b>M1</b> | <b>M1 dep</b> on first M1 for integration of $k(2x-5)^{\frac{1}{2}}$ to obtain $m(2x-5)^{\frac{3}{2}}$                       |
|          | $y = \frac{1}{3}(2x-5)^{\frac{3}{2}} + 4x \quad (+d)$   | <b>A1</b> | for $\frac{1}{3}(2x-5)^{\frac{3}{2}} + 4x$<br><b>FT</b> <i>their</i> (non -zero) constant                                    |
|          | Finding constant                                        | <b>M1</b> | <b>M1 dep</b> for obtaining arbitrary constant for $m(2x-5)^{\frac{3}{2}} + nx + d$ using $x = \frac{9}{2}, y = \frac{2}{3}$ |
|          | $y = \frac{1}{3}(2x-5)^{\frac{3}{2}} + 4x - 20$         | <b>A1</b> | for correct equation                                                                                                         |

217



No calculator

Oct/Nov 2017 · Paper 11 · Q5



The diagram shows part of the graph of  $y = 4e^{2x} + 16e^{-2x}$  meeting the  $y$ -axis at the point  $A$  and the line  $x = 1$  at the point  $B$ .

- (i) Find the coordinates of  $A$ . [1]
- (ii) Find the  $y$ -coordinate of  $B$ . [1]
- (iii) Find  $\int (4e^{2x} + 16e^{-2x}) dx$ . [2]
- (iv) Hence find the area of the shaded region enclosed by the curve and the line  $AB$ . You must show all your working. [4]

**217** Oct/Nov 2017 · Paper 11 · Q5

|        |                                                                   |           |                                                                                                 |
|--------|-------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------|
| 15 (i) | (0, 20)                                                           | <b>B1</b> |                                                                                                 |
| (ii)   | 31.7                                                              | <b>B1</b> |                                                                                                 |
| (iii)  | $2e^{2x} - 8e^{-2x} \quad (+c)$                                   | <b>B2</b> | B1 for each correct term                                                                        |
| (iv)   | Area of trapezium = $\frac{1}{2}(20 + 31.7)$<br>= 25.86 or 25.85  | <b>B1</b> |                                                                                                 |
|        | $\left[ 2e^{2x} - 8e^{-2x} \right]_0^1 = (2e^2 - 8e^{-2}) - (-6)$ | <b>M1</b> | substitution of both limits, must have come from integration of the form $ae^{2x} + be^{-2x}$ . |
|        | 19.7                                                              | <b>A1</b> |                                                                                                 |
|        | Required area = 6.15, 6.16, 6.17                                  | <b>A1</b> |                                                                                                 |

**218**

No calculator

**Oct/Nov 2017 • Paper 11 • Q7**

- (i) Write  $\ln\left(\frac{2x+1}{2x-1}\right)$  as the difference of two logarithms. [1]

A curve has equation  $y = \ln\left(\frac{2x+1}{2x-1}\right) + 4x$  for  $x > \frac{1}{2}$ .

- (ii) Using your answer to part (i) show that  $\frac{dy}{dx} = \frac{ax^2+b}{4x^2-1}$ , where  $a$  and  $b$  are integers. [4]

- (iii) Hence find the  $x$ -coordinate of the stationary point on the curve. [2]

- (iv) Determine the nature of this stationary point. [2]

**219**

No calculator

**Oct/Nov 2017 • Paper 12 • Q4**

Given that  $y = \frac{\ln(3x^2+2)}{x^2+1}$ , find the value of  $\frac{dy}{dx}$  when  $x = 2$ , giving your answer as  $a + b \ln 14$ , where  $a$  and  $b$  are fractions in their simplest form. [6]

**218** Oct/Nov 2017 · Paper 11 · Q7

|        |                                                                         |            |                                                                  |
|--------|-------------------------------------------------------------------------|------------|------------------------------------------------------------------|
| 16 (i) | $\ln(2x+1) - \ln(2x-1)$                                                 | <b>B1</b>  |                                                                  |
| (ii)   | attempt to differentiate                                                | <b>M1</b>  |                                                                  |
|        | $\frac{dy}{dx} = \frac{2}{2x+1} - \frac{2}{2x-1} + 4$                   | <b>A1</b>  | all correct                                                      |
|        | attempt to obtain in required form                                      | <b>DM1</b> |                                                                  |
|        | $= \frac{16x^2 - 8}{4x^2 - 1}$                                          | <b>A1</b>  | A1 all correct                                                   |
| (iii)  | When $\frac{dy}{dx} = 0$ , $16x^2 - 8 = 0$                              | <b>M1</b>  | setting $\frac{dy}{dx} = 0$ and attempt to solve                 |
|        | $x = \frac{1}{\sqrt{2}}$ only                                           | <b>A1</b>  |                                                                  |
| (iv)   | $\frac{d^2y}{dx^2} = \frac{32x(4x^2-1) - 8x(16x^2-8)}{(4x^2-1)^2}$      | <b>M1</b>  | attempt at second derivative and conclusion or equivalent method |
|        | When $x = \frac{1}{\sqrt{2}}$ , $\frac{d^2y}{dx^2}$ is + ve, so minimum | <b>A1</b>  |                                                                  |

**219** Oct/Nov 2017 · Paper 12 · Q4

| Question | Answer                                                                                    | Marks     | Guidance                                                  |
|----------|-------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------|
| 47       | correct differentiation of $\ln(3x^2 + 2)$                                                | <b>B1</b> |                                                           |
|          | attempt to differentiate a quotient or a product                                          | <b>M1</b> |                                                           |
|          | $\frac{dy}{dx} = \frac{(x^2+1)\left(\frac{6x}{3x^2+2}\right) - 2x\ln(3x^2+2)}{(x^2+1)^2}$ | <b>A1</b> | all other terms correct.                                  |
|          | When $x = 2$ , $\frac{dy}{dx} = \frac{5\left(\frac{12}{14}\right) - 4\ln 14}{25}$         | <b>M1</b> | <b>M1dep</b> for substitution and attempt to simplify     |
|          | $= \frac{6}{35} - \frac{4}{25}\ln 14$                                                     | <b>A2</b> | <b>A1</b> for each correct term, must be in simplest form |



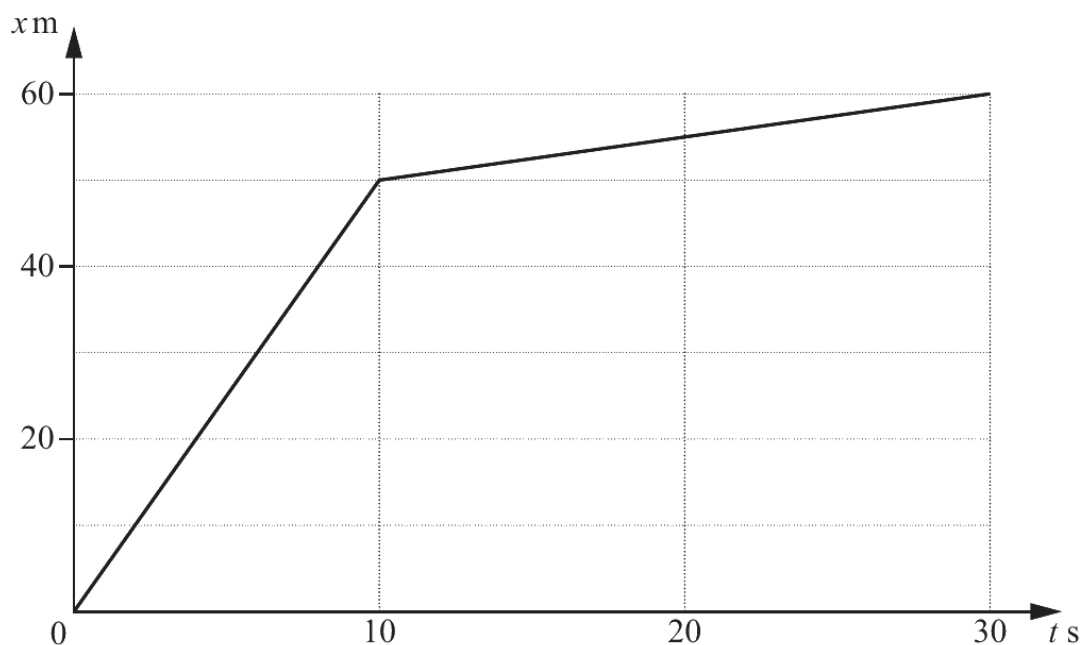
220

(a)

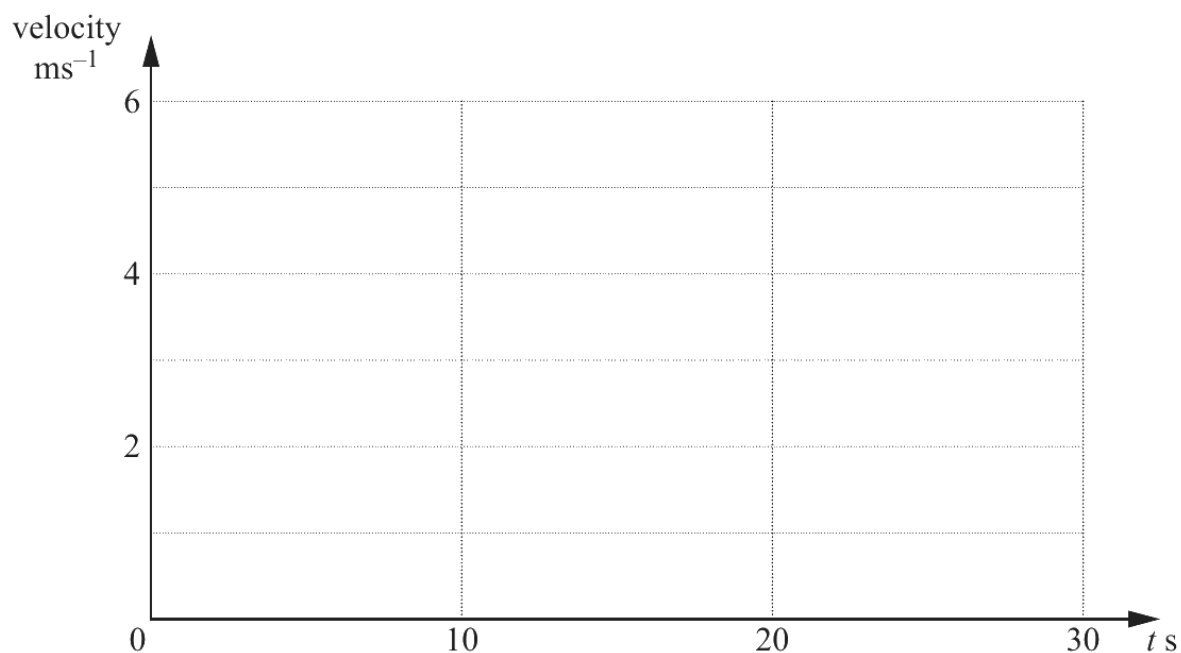


No calculator

Oct/Nov 2017 · Paper 12 · Q9



The diagram shows the displacement-time graph of a particle  $P$  which moves in a straight line such that,  $t \text{ s}$  after leaving a fixed point  $O$ , its displacement from  $O$  is  $x \text{ m}$ . On the axes below, draw the velocity-time graph of  $P$ .



[3]

*Question continues on the next page →*

(b) A particle  $Q$  moves in a straight line such that its velocity,  $v \text{ ms}^{-1}$ ,  $t$  s after passing through a fixed point  $O$ , is given by  $v = 3e^{-5t} + \frac{3t}{2}$ , for  $t \geq 0$ .

(i) Find the velocity of  $Q$  when  $t = 0$ . [1]

(ii) Find the value of  $t$  when the acceleration of  $Q$  is zero. [3]

(iii) Find the distance of  $Q$  from  $O$  when  $t = 0.5$ . [4]

**221**

No calculator

**Oct/Nov 2017 • Paper 13 • Q2**

A curve is such that its gradient at the point  $(x, y)$  is given by  $10e^{5x} + 3$ . Given that the curve passes through the point  $(0, 9)$ , find the equation of the curve. [4]

220 Oct/Nov 2017 · Paper 12 · Q9

|          |                                                               |           |                                                                                                                                                                                    |
|----------|---------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 48 (a)   |                                                               | <b>B3</b> | <b>B1</b> for line joining (0,5) and (10,5)<br><b>B1</b> for a line joining (10,0.5) and (30,0.5)<br><b>B1</b> all correct with <b>no solid</b> line joining (10,5) to (10,0.5)    |
| (b)(i)   | 3                                                             | <b>B1</b> |                                                                                                                                                                                    |
| (b)(ii)  | $\frac{dv}{dt} = -15e^{-5t} + \frac{3}{2}$                    | <b>M1</b> | attempt to differentiate, must be in the form $ae^{-5t} + b$                                                                                                                       |
|          | When $\frac{dv}{dt} = 0$ , $e^{-5t} = 0.1$                    | <b>M1</b> | <b>M1dep</b> for equating to zero and attempt to solve, must be of the form $ae^{-5t} = b$ , $b > 0$ to obtain an equation in the form $-5t = k$ where $k$ is a logarithm or $< 0$ |
|          | $t = 0.461$                                                   | <b>A1</b> |                                                                                                                                                                                    |
| (b)(iii) | <b>Either</b>                                                 |           |                                                                                                                                                                                    |
|          | attempt to integrate, must be in the form $ce^{-5t} + dt^2$   | <b>M1</b> |                                                                                                                                                                                    |
|          | $s = -\frac{3}{5}e^{-5t} + \frac{3}{4}t^2 (+c)$               | <b>A1</b> |                                                                                                                                                                                    |
|          | When $t = 0$ , $s = 0$ so $c = \frac{3}{5}$                   | <b>M1</b> | <b>M1dep</b> for attempt to find $c$ and substitute $t = 0.5$                                                                                                                      |
|          | $s = 0.738$                                                   | <b>A1</b> |                                                                                                                                                                                    |
|          | <b>Or</b>                                                     |           |                                                                                                                                                                                    |
|          | attempt to integrate, must be in the form $ce^{-5t} + dt^2$   | <b>M1</b> |                                                                                                                                                                                    |
|          | $\left[ -\frac{3}{5}e^{-5t} + \frac{3}{4}t^2 \right]_0^{0.5}$ | <b>A1</b> |                                                                                                                                                                                    |
|          | correct use of limits                                         | <b>M1</b> | <b>M1dep</b>                                                                                                                                                                       |
|          | leading to $s = 0.738$                                        | <b>A1</b> |                                                                                                                                                                                    |

**221** Oct/Nov 2017 · Paper 13 · Q2

|    |                                                                                    |           |                         |
|----|------------------------------------------------------------------------------------|-----------|-------------------------|
| 74 | $\frac{dy}{dx} = 10e^{5x} + 3$<br>an attempt at integration in form $ae^{5x} + bx$ | <b>M1</b> |                         |
|    | $y = \frac{10}{5}e^{5x} + 3x (+c)$                                                 | <b>A1</b> | condone omission of $c$ |
|    | attempt to find $c$ using $x = 0, y = 9$                                           | <b>M1</b> | <b>M1dep</b>            |
|    | $y = 2e^{5x} + 3x + 7$                                                             | <b>A1</b> |                         |

**222**

No calculator

**Oct/Nov 2017 • Paper 13 • Q5**

(i) Find  $\int (7x - 10)^{-\frac{3}{5}} dx$ .

[2]

(ii) Given that  $\int_6^a (7x - 10)^{-\frac{3}{5}} dx = \frac{25}{14}$ , find the exact value of  $a$ .

[3]

**223**

No calculator

**Oct/Nov 2017 • Paper 13 • Q8**

76 It is given that  $y = (x - 4)(3x - 1)^{\frac{5}{3}}$ .

(i) Show that  $\frac{dy}{dx} = (3x - 1)^{\frac{2}{3}}(Ax + B)$ , where  $A$  and  $B$  are integers to be found.

[5]

(ii) Hence find, in terms of  $h$ , where  $h$  is small, the approximate change in  $y$  when  $x$  increases from 3 to  $3 + h$ .

[3]

**224**

No calculator

**May/June 2017 • Paper 11 • Q7**

3 Show that the curve  $y = (3x^2 + 8)^{\frac{5}{3}}$  has only one stationary point. Find the coordinates of this stationary point and determine its nature.

[8]

**222** Oct/Nov 2017 · Paper 13 · Q5

|        |                                                                                                                                                                                                           |           |                                                                                                                                              |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 75 (i) | $\frac{5}{14}(7x-10)^{\frac{2}{5}}$                                                                                                                                                                       | <b>B2</b> | <b>B1</b> for $k(7x-10)^{\frac{2}{5}}$                                                                                                       |
| (ii)   | $\frac{5}{14} \left[ (7x-10)^{\frac{2}{5}} \right]_6^a = \frac{25}{14}$ $\frac{5}{14}(7a-10)^{\frac{2}{5}} - \frac{5}{14}(7 \times 6 - 10)^{\frac{2}{5}} = \frac{25}{14}$ $(7a-10)^{\frac{2}{5}} - 4 = 5$ | <b>M1</b> | correct application of limits for $k(7x-10)^{\frac{2}{5}}$                                                                                   |
|        | $a = \frac{9^{\frac{5}{2}} + 10}{7}$                                                                                                                                                                      | <b>M1</b> | <b>M1dep</b> for evaluation of $(7 \times 6 - 10)^{\frac{2}{5}}$ and correct order of operations to find $a$ , including dealing with power. |
|        | $a = \frac{253}{7}$ or $36\frac{1}{7}$                                                                                                                                                                    | <b>A1</b> |                                                                                                                                              |

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|        |                                                                                                               |           |                                                                                                                                     |
|--------|---------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------|
| 76 (i) | attempt to differentiate a product                                                                            | <b>M1</b> |                                                                                                                                     |
|        | $\frac{dy}{dx} = \left( (x-4) \times \frac{5}{3} \times 3(3x-1)^{\frac{2}{3}} \right) + (3x-1)^{\frac{5}{3}}$ | <b>A2</b> | <b>A1</b> for $(+) \left( (x-4) \times \frac{5}{3} \times 3(3x-1)^{\frac{2}{3}} \right)$<br><b>A1</b> for $(+)(3x-1)^{\frac{5}{3}}$ |
|        | $= (3x-1)^{\frac{2}{3}} ((5x-20) + (3x-1))$                                                                   | <b>M1</b> | use of $(3x-1)^{\frac{5}{3}} = (3x-1)^{\frac{2}{3}}(3x-1)$                                                                          |
|        | $= (3x-1)^{\frac{2}{3}}(8x-21)$                                                                               | <b>A1</b> |                                                                                                                                     |
| (ii)   | When $x=3$ , $\frac{dy}{dx} = 8^{\frac{2}{3}} \times 3$                                                       | <b>M1</b> | $(3 \times 3 - 1)^{\frac{2}{3}} \times k$ or $(9-1)^{\frac{2}{3}} \times k$ or $4 \times k$<br>(where $k$ is any number)            |
|        | $\partial y = 8^{\frac{2}{3}} \times 3 \times h$                                                              | <b>M1</b> | <b>M1dep</b> for their $((9-1)^{\frac{2}{3}} \times k) \times h$                                                                    |
|        | $\partial y = 12h$                                                                                            | <b>A1</b> |                                                                                                                                     |

**224** May/June 2017 · Paper 11 · Q7

$\frac{dy}{dx} = \frac{5}{3}(3x^2 + 8)^{\frac{2}{3}}(6x) = 10x(3x^2 + 8)^{\frac{2}{3}}$ . Since  $3x^2 + 8 > 0$ ,  $\frac{dy}{dx} = 0$  only

when  $x = 0$ : only one stationary point.

$y = 8^{\frac{5}{3}} = 32$ :  $\boxed{(0, 32)}$ .  $\frac{d^2y}{dx^2} = 10(3x^2 + 8)^{\frac{2}{3}} + \dots = 40 > 0$  at  $x = 0$ :

**minimum**

**225**

No calculator

**May/June 2017 • Paper 11 • Q10**

- (a) Given that  $y = \frac{e^{3x}}{4x^2 + 1}$ , find  $\frac{dy}{dx}$ . [3]

- (b) Variables  $x$ ,  $y$  and  $t$  are such that  $y = 4 \cos\left(x + \frac{\pi}{3}\right) + 2\sqrt{3} \sin\left(x + \frac{\pi}{3}\right)$  and  $\frac{dy}{dt} = 10$ .

- (i) Find the value of  $\frac{dy}{dx}$  when  $x = \frac{\pi}{2}$ . [3]

- (ii) Find the value of  $\frac{dx}{dt}$  when  $x = \frac{\pi}{2}$ . [2]

**226**

No calculator

**May/June 2017 • Paper 12 • Q2**

- It is given that  $y = \frac{(5x^2 + 4)^{\frac{1}{2}}}{x + 1}$ . Showing all your working, find the exact value of  $\frac{dy}{dx}$  when  $x = 3$ . [5]



**225** May/June 2017 · Paper 11 · Q10

|         |                                                                                                                                           |           |                                                                             |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------|
| 4 (a)   | differentiation of a quotient or equivalent product                                                                                       | <b>M1</b> |                                                                             |
|         | correct differentiation of $e^{3x}$                                                                                                       | <b>B1</b> |                                                                             |
|         | $\frac{dy}{dx} = \frac{3e^{3x}(4x^2+1) - 8xe^{3x}}{(4x^2+1)^2}$ or $\frac{dy}{dx} = \frac{3e^{3x}}{4x^2+1} - \frac{8xe^{3x}}{(4x^2+1)^2}$ | <b>A1</b> | everything else correct including brackets where needed, allow unsimplified |
| (b)(i)  | one term differentiated correctly                                                                                                         | <b>M1</b> |                                                                             |
|         | $\frac{dy}{dx} = -4\sin\left(x + \frac{\pi}{3}\right) + 2\sqrt{3}\cos\left(x + \frac{\pi}{3}\right)$                                      | <b>A1</b> | all correct                                                                 |
|         | When $x = \frac{\pi}{2}$ , $\frac{dy}{dx} = -5$                                                                                           | <b>A1</b> |                                                                             |
| (b)(ii) | $\frac{dy}{dx} \times \frac{dx}{dt} = \frac{dy}{dt}$ $-5 \times \frac{dx}{dt} = 10 \text{ oe}$                                            | <b>M1</b> | correct use of rates of change                                              |
|         | $\frac{dy}{dt} = -2$                                                                                                                      | <b>A1</b> | <b>FT</b> answer to (i)                                                     |

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|    |                                                                                                         |            |                  |
|----|---------------------------------------------------------------------------------------------------------|------------|------------------|
| 31 | attempt at differentiating a quotient, must have minus sign and $(x+1)^2$ in the denominator            | <b>M1</b>  |                  |
|    | for $(5x^2+4)^{-\frac{1}{2}}$                                                                           | <b>B1</b>  |                  |
|    | for $\frac{1}{2}(10x)(5x^2+4)^{-\frac{1}{2}}$                                                           | <b>DB1</b> |                  |
|    | $\frac{dy}{dx} = \frac{(x+1)\frac{1}{2}(10x)(5x^2+4)^{-\frac{1}{2}} - (5x^2+4)^{\frac{1}{2}}}{(x+1)^2}$ | <b>A1</b>  | all else correct |
|    | When $x = 3$ , $\frac{dy}{dx} = \frac{11}{112}$                                                         | <b>A1</b>  | must be exact    |

|                                                                                                             |            |                                    |
|-------------------------------------------------------------------------------------------------------------|------------|------------------------------------|
| <b>Alternative</b>                                                                                          |            |                                    |
| $y = (5x^2 + 4)^{\frac{1}{2}}(x+1)^{-1}$                                                                    | <b>M1</b>  | attempt to differentiate a product |
| for $(5x^2 + 4)^{-\frac{1}{2}}$                                                                             | <b>B1</b>  |                                    |
| for $\frac{1}{2}(10x)(5x^2 + 4)^{-\frac{1}{2}}$                                                             | <b>DB1</b> |                                    |
| $\frac{dy}{dx} = \frac{1}{2}10x(5x^2 + 4)^{-\frac{1}{2}}(x+1)^{-1} + (5x^2 + 4)^{\frac{1}{2}}(-(x+1)^{-2})$ | <b>A1</b>  | all else correct                   |
| When $x = 3$ , $\frac{dy}{dx} = \frac{11}{112}$                                                             | <b>A1</b>  | A1 must be exact                   |

**227**

No calculator

**May/June 2017 · Paper 12 · Q5**

A particle  $P$  moves in a straight line, such that its displacement,  $x$  m, from a fixed point  $O$ ,  $t$  s after passing  $O$ , is given by  $x = 4 \cos(3t) - 4$ .

(i) Find the velocity of  $P$  at time  $t$ . [1]

(ii) Hence write down the maximum speed of  $P$ . [1]

(iii) Find the smallest value of  $t$  for which the acceleration of  $P$  is zero. [3]

(iv) For the value of  $t$  found in part (iii), find the distance of  $P$  from  $O$ . [1]

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|        |                                       |           |                                                                       |
|--------|---------------------------------------|-----------|-----------------------------------------------------------------------|
| 32 (i) | $v = -12 \sin 3t$                     | <b>B1</b> |                                                                       |
| (ii)   | 12                                    | <b>B1</b> | <b>FT</b> on <i>their</i> (i) of the form $k \sin 3t$ , must be $ k $ |
| (iii)  | $a = -36 \cos 3t$                     | <b>B1</b> | allow unsimplified                                                    |
|        | $3t = \frac{\pi}{2}$ , 1.57 or better | <b>B1</b> |                                                                       |
|        | $t = \frac{\pi}{6}$ or 0.524          | <b>B1</b> |                                                                       |
| (iv)   | 4 cao                                 | <b>B1</b> | may be obtained from knowledge of cosine curve                        |

**228**

No calculator

**May/June 2017 • Paper 12 • Q11**

The curve  $y = f(x)$  passes through the point  $\left(\frac{1}{2}, \frac{7}{2}\right)$  and is such that  $f'(x) = e^{2x-1}$ .

(i) Find the equation of the curve.

[4]

(ii) Find the value of  $x$  for which  $f''(x) = 4$ , giving your answer in the form  $a + b \ln \sqrt{2}$ , where  $a$  and  $b$  are constants.

[4]

**229**

No calculator

**May/June 2017 • Paper 13 • Q5**

The normal to the curve  $y = \sqrt{4x+9}$ , at the point where  $x = 4$ , meets the  $x$ - and  $y$ -axes at the points  $A$  and  $B$ . Find the coordinates of the mid-point of the line  $AB$ .

[7]

**230**

No calculator

**May/June 2017 • Paper 13 • Q9**

It is given that  $\int_{-k}^k (15e^{5x} - 5e^{-5x}) dx = 6$ .

(i) Show that  $e^{5k} - e^{-5k} = 3$ .

[5]

(ii) Hence, using the substitution  $y = e^{5k}$ , or otherwise, find the value of  $k$ .

[3]

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|        |                                             |            |                                            |
|--------|---------------------------------------------|------------|--------------------------------------------|
| 33 (i) | $me^{2x-1}$ where $m$ is numeric constant   | <b>M1</b>  |                                            |
|        | $f(x) = \frac{1}{2}e^{2x-1} (+c)$           | <b>A1</b>  | condone omission of $+c$                   |
|        | $\frac{7}{2} = \frac{1}{2} + c$             | <b>DM1</b> | correct attempt to find arbitrary constant |
|        | $f(x) = \frac{1}{2}e^{2x-1} + 3$            | <b>A1</b>  | must be an equation                        |
| (ii)   | $ke^{2x-1}$ where $k$ is a numeric constant | <b>M1</b>  |                                            |
|        | $f''(x) = 2e^{2x-1}$                        | <b>A1</b>  |                                            |
|        | $2x-1 = \ln\left(\frac{4}{k}\right)$        | <b>DM1</b> | attempt to equate to 4 and use logarithms  |
|        | $x = \frac{1}{2} + \ln\sqrt{2}$             | <b>A1</b>  |                                            |

**229** May/June 2017 · Paper 13 · Q5

| Question | Answer                                                                          | Marks     | Partial Marks                                     |
|----------|---------------------------------------------------------------------------------|-----------|---------------------------------------------------|
| 60       | When $x = 4$ , $y = 5$                                                          | <b>B1</b> | for $y$                                           |
|          | $\frac{dy}{dx} = \frac{1}{2} \times 4(4x+9)^{-\frac{1}{2}}$                     | <b>B1</b> | for $2(4x+9)^{-\frac{1}{2}}$ , allow unsimplified |
|          | When $x = 4$ , $\frac{dy}{dx} = \frac{2}{5}$ ,<br>so perp grad = $-\frac{5}{2}$ | <b>M1</b> | obtaining numerical gradient for normal           |
|          | Equation of normal<br>$y - 5 = -\frac{5}{2}(x - 4)$<br>$(2y = 30 - 5x)$         | <b>M1</b> | for equation of normal                            |
|          | $A(6, 0)$ , $B(0, 15)$                                                          | <b>A2</b> | <b>A1</b> for each                                |
|          | Midpoint $\left(3, \frac{15}{2}\right)$                                         | <b>B1</b> | <b>FT</b> on <i>their</i> $x/y$ intercepts        |

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|        |                                                                           |            |                                                                                |
|--------|---------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------|
| 61 (i) | $\left[ 3e^{5x} + e^{-5x} \right]_{-k}^k = 6$                             | <b>B2</b>  | <b>B1</b> for each term integrated correctly                                   |
|        | $\left( 3e^{5k} + e^{-5k} \right) - \left( 3e^{-5k} + e^{5k} \right) = 6$ | <b>M1</b>  | for use of limits with $ae^{5x} + be^{-5x}$                                    |
|        | $2e^{5k} - 2e^{-5k} = 6$                                                  | <b>A1</b>  | correct unsimplified                                                           |
|        | $e^{5k} - e^{-5k} = 3$                                                    | <b>A1</b>  | correct simplification to obtain given answer                                  |
| (ii)   | $y^2 - 3y - 1 = 0$                                                        | <b>M1</b>  | for correct attempt to obtain a quadratic equation in terms of $y$ or $e^{5x}$ |
|        | $y = \frac{3 \pm \sqrt{9+4}}{2}$ , $y = e^{5k} = 3.303$<br>only           | <b>DM1</b> | for attempt to solve quadratic equation and solve for $k$                      |
|        | $k = 0.239$                                                               | <b>A1</b>  | <b>A0</b> if more than one solution is given                                   |

**231**

No calculator

**May/June 2017 • Paper 13 • Q10**

It is given that  $y = (10x + 2)\ln(5x + 1)$ .

- (i) Find  $\frac{dy}{dx}$ . [4]

- (ii) Hence show that  $\int \ln(5x + 1) dx = \frac{(ax + b)}{5} \ln(5x + 1) - x + c$ , where  $a$  and  $b$  are integers and  $c$  is a constant of integration. [3]

- (iii) Hence find  $\int_0^{\frac{1}{5}} \ln(5x + 1) dx$ , giving your answer in the form  $\frac{d + \ln f}{5}$ , where  $d$  and  $f$  are integers. [2]



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|        |                                                                         |           |                                                   |
|--------|-------------------------------------------------------------------------|-----------|---------------------------------------------------|
| 62 (i) | for attempt to differentiate a product                                  | <b>M1</b> |                                                   |
|        | $\frac{5}{5x+1}$                                                        | <b>B2</b> | <b>B1</b> for $\frac{1}{5x+1}$                    |
|        | $\frac{dy}{dx} = (10x+2) \times \frac{5}{5x+1} + 10 \ln(5x+1)$          | <b>A1</b> | all else correct                                  |
| (ii)   | $(10x+2) \times \frac{5}{5x+1} = 10$                                    | <b>B1</b> | simplification to obtain 10, allow if seen in (i) |
|        | $10 \int \ln(5x+1) \, dx$<br>$= (10x+2) \ln(5x+1) - 10x$                | <b>M1</b> | use of result from part (i)                       |
|        | $\int \ln(5x+1) \, dx$<br>$= \frac{(5x+1)}{5} \ln(5x+1) - x$            | <b>A1</b> |                                                   |
| (iii)  | $\left[ (x+0.2) \ln(5x+1) - x \right]_0^{\frac{1}{5}}$                  | <b>M1</b> | use of limits in result from (ii)                 |
|        | $= -\frac{1}{5} + \frac{2}{5} \ln 2 = \frac{-1 + \ln 4}{5} \text{ cao}$ | <b>A1</b> |                                                   |

**232**

No calculator

**May/June 2017 • Paper 13 • Q11**

A curve has equation  $y = 6x - x\sqrt{x}$ .

(i) Find the coordinates of the stationary point of the curve. [4]

(ii) Determine the nature of this stationary point. [2]

(iii) Find the approximate change in  $y$  when  $x$  increases from 4 to  $4 + h$ , where  $h$  is small. [3]

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|        |                                                    |            |                                       |
|--------|----------------------------------------------------|------------|---------------------------------------|
| 63 (i) | attempt to differentiate                           | <b>M1</b>  |                                       |
|        | $\frac{dy}{dx} = 6 - \frac{3}{2}x^{\frac{1}{2}}$   | <b>A1</b>  |                                       |
|        | When $\frac{dy}{dx} = 0$                           | <b>M1</b>  | equating to zero and attempt to solve |
|        | $x = 16, y = 32$                                   | <b>A1</b>  | both correct                          |
| (ii)   | $\frac{d^2y}{dx^2} = -\frac{3}{4}x^{-\frac{1}{2}}$ | <b>B1</b>  | correct differentiation               |
|        | This is negative so a maximum point                | <b>DB1</b> | correct conclusion                    |
| (iii)  | When $x = 4, \frac{dy}{dx} = 3$                    | <b>B1</b>  |                                       |
|        | $\partial y \approx \frac{dy}{dx} \times h$        | <b>M1</b>  | use of small increases                |
|        | $\approx 3h$                                       | <b>A1</b>  | <b>FT</b> <i>their</i> (iii)          |

**233**

No calculator

**May/June 2017 • Paper 13 • Q12**

A particle moves in a straight line, such that its velocity,  $v \text{ ms}^{-1}$ ,  $t$  s after passing a fixed point  $O$ , is given by  $v = 2 + 6t + 3 \sin 2t$ .

(i) Find the acceleration of the particle at time  $t$ . [2]

(ii) Hence find the smallest value of  $t$  for which the acceleration of the particle is zero. [2]

(iii) Find the displacement,  $x$  m from  $O$ , of the particle at time  $t$ . [5]

**234**

No calculator

**Oct/Nov 2016 • Paper 11 • Q5**

(i) Find the equation of the normal to the curve  $y = \frac{1}{2} \ln(3x + 2)$  at the point  $P$  where  $x = -\frac{1}{3}$ . [4]

The normal to the curve at the point  $P$  intersects the  $y$ -axis at the point  $Q$ . The curve  $y = \frac{1}{2} \ln(3x + 2)$  intersects the  $y$ -axis at the point  $R$ .

(ii) Find the area of the triangle  $PQR$ . [3]

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|        |                                                     |           |                                         |
|--------|-----------------------------------------------------|-----------|-----------------------------------------|
| 64 (i) | attempt to differentiate                            | <b>M1</b> |                                         |
|        | $6 \cos 2t + 6$                                     | <b>A1</b> |                                         |
| (ii)   | $\cos 2t = -1$                                      | <b>M1</b> | attempt to equate (i) to zero and solve |
|        | $t = \frac{\pi}{2}$                                 | <b>A1</b> |                                         |
| (iii)  | attempt to integrate                                | <b>M1</b> |                                         |
|        | $x = -\frac{3}{2} \cos 2t + 3t^2 + 2t \quad (+c)$   | <b>A2</b> | -1 for each error                       |
|        | When $t = 0$ , $x = 0$ , so $c = \frac{3}{2}$       | <b>M1</b> | attempt to find $c$                     |
|        | $x = \frac{3}{2} - \frac{3}{2} \cos 2t + 3t^2 + 2t$ | <b>A1</b> |                                         |

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| Question | Answer                                                                                                      | Marks                  | Part Marks                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------|
| 14 (i)   | $\frac{dy}{dx} = \frac{3}{2(3x+2)}$                                                                         | <b>B1</b>              | for correct derivative of log function                                                                     |
|          | When $x = -\frac{1}{3}$ , $y = 0$ , $\frac{dy}{dx} = \frac{3}{2}$                                           | <b>B1</b>              | for $y = 0$                                                                                                |
|          | Equation of normal: $y = -\frac{2}{3}\left(x + \frac{1}{3}\right)$                                          | <b>M1</b><br><b>A1</b> | <b>M1</b> for attempt at a gradient of a perpendicular from differentiation and the equation of the normal |
|          |                                                                                                             |                        |                                                                                                            |
| (ii)     | $Q\left(0, -\frac{2}{9}\right)$ or $(0, 0.22)$ or better                                                    | <b>B1 ft</b>           | Follow through on <i>their c</i> from part (i)                                                             |
|          | $R\left(0, \frac{1}{2} \ln 2\right)$ or $(0, 0.35)$ or better                                               | <b>B1</b>              |                                                                                                            |
|          | Area of $PQR = \frac{1}{2} \left( \frac{1}{2} \ln 2 + \frac{2}{9} \right) \times \frac{1}{3}$<br>$= 0.0948$ | <b>B1</b>              | Allow 0.095                                                                                                |

235



No calculator

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- (i) Find the equation of the normal to the curve  $y = \frac{1}{2} \ln(3x + 2)$  at the point  $P$  where  $x = -\frac{1}{3}$ . [4]

The normal to the curve at the point  $P$  intersects the  $y$ -axis at the point  $Q$ . The curve  $y = \frac{1}{2} \ln(3x + 2)$  intersects the  $y$ -axis at the point  $R$ .

- (ii) Find the area of the triangle  $PQR$ . [3]

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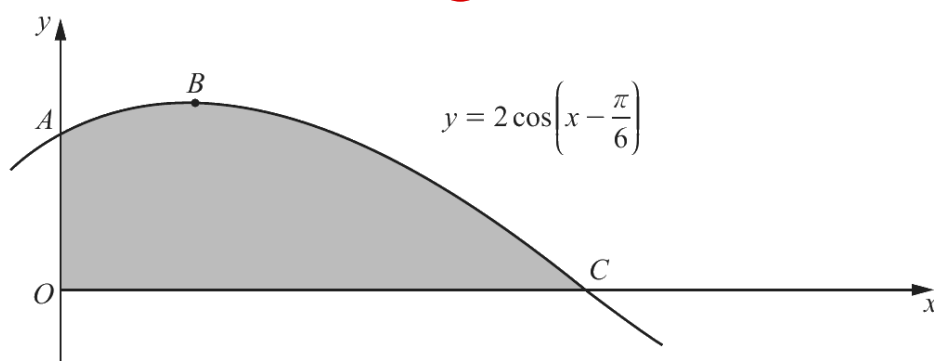
| Question | Answer                                                                                                  | Marks                  | Part Marks                                                                                                 |
|----------|---------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------|
| 44 (i)   | $\frac{dy}{dx} = \frac{3}{2(3x+2)}$                                                                     | <b>B1</b>              | for correct derivative of log function                                                                     |
|          | When $x = -\frac{1}{3}$ , $y = 0$ , $\frac{dy}{dx} = \frac{3}{2}$                                       | <b>B1</b>              | for $y = 0$                                                                                                |
|          | Equation of normal: $y = -\frac{2}{3}\left(x + \frac{1}{3}\right)$                                      | <b>M1</b><br><b>A1</b> | <b>M1</b> for attempt at a gradient of a perpendicular from differentiation and the equation of the normal |
| (ii)     | $Q\left(0, -\frac{2}{9}\right)$ or (0, 0.22) or better                                                  | <b>B1 ft</b>           | Follow through on <i>their c</i> from part (i)                                                             |
|          | $R\left(0, \frac{1}{2}\ln 2\right)$ or (0, 0.35) or better                                              | <b>B1</b>              |                                                                                                            |
|          | Area of $PQR = \frac{1}{2}\left(\frac{1}{2}\ln 2 + \frac{2}{9}\right) \times \frac{1}{3}$<br>$= 0.0948$ | <b>B1</b>              | Allow 0.095                                                                                                |

236



No calculator

Oct/Nov 2016 · Paper 12 · Q7



The diagram shows part of the graph of  $y = 2 \cos\left(x - \frac{\pi}{6}\right)$ . The graph intersects the y-axis at the point  $A$ , has a maximum point at  $B$  and intersects the x-axis at the point  $C$ .

(i) Find the coordinates of  $A$ . [1]

(ii) Find the coordinates of  $B$ . [2]

(iii) Find the coordinates of  $C$ . [2]

(iv) Find  $\int 2 \cos\left(x - \frac{\pi}{6}\right) dx$ . [1]

(v) Hence find the area of the shaded region. [2]



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| Question | Answer                                                                                         | Marks                  | Part Marks                                                                                          |
|----------|------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------|
| 45 (i)   | $(0, \sqrt{3})$ or $(0, 1.73)$ or better                                                       | <b>B1</b>              |                                                                                                     |
| (ii)     | $\left(\frac{\pi}{6}, 2\right)$ or $(0.524, 2)$ or better                                      | <b>B1, B1</b>          | <b>B1</b> for each                                                                                  |
| (iii)    | $\cos\left(x - \frac{\pi}{6}\right) = 0$<br>$x = \frac{2\pi}{3}$ oe or 2.09 or better          | <b>M1</b><br><b>A1</b> | for correct attempt to solve trigonometric equation                                                 |
| (iv)     | $2\sin\left(x - \frac{\pi}{6}\right)$ (+c)                                                     | <b>B1</b>              |                                                                                                     |
| (v)      | Area = $\left[2\sin\left(x - \frac{\pi}{6}\right)\right]_0^{\frac{2\pi}{3}}$<br>= 2 + 1<br>= 3 | <b>M1</b><br><b>A1</b> | for correct use of <b>their</b> limits, in radians,<br>into $k\sin\left(x - \frac{\pi}{6}\right)$ . |

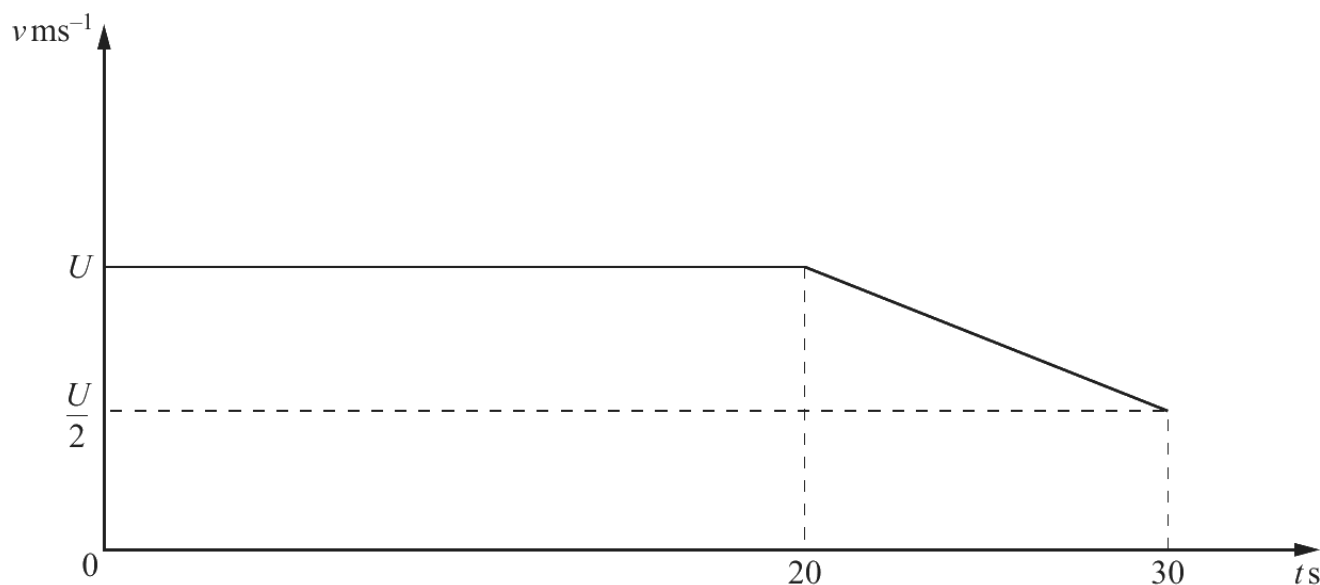
237



No calculator

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(a)



The diagram shows part of the velocity-time graph for a particle, moving at  $v \text{ ms}^{-1}$  in a straight line,  $t \text{ s}$  after passing through a fixed point. The particle travels at  $U \text{ ms}^{-1}$  for 20 s and then decelerates uniformly for 10 s to a velocity of  $\frac{U}{2} \text{ ms}^{-1}$ . In this 30 s interval, the particle travels 165 m.

(i) Find the value of  $U$ . [3]

(ii) Find the acceleration of the particle between  $t = 20$  and  $t = 30$ . [2]

Question continues on the next page →

- (b) A particle  $P$  travels in a straight line such that,  $t$  s after passing through a fixed point  $O$ , its velocity,  $v \text{ ms}^{-1}$ , is given by  $v = \left( e^{\frac{t^2}{8}} - 4 \right)^3$ .

(i) Find the speed of  $P$  at  $O$ . [1]

(ii) Find the value of  $t$  for which  $P$  is instantaneously at rest. [2]

(iii) Find the acceleration of  $P$  when  $t = 1$ . [4]

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|           |                                                                                                                             |                                        |                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 46 (a)(i) | $20U + \frac{1}{2}\left(U + \frac{U}{2}\right)10 = 165$<br><br>leading to $U = 6$                                           | <b>M1</b><br><b>DM1</b><br><b>A1</b>   | for realising that area under the graph is needed and attempt to find an area for equating their area to 165 and attempt to solve   |
| (ii)      | Gradient of line: $-0.3$                                                                                                    | <b>M1, A1</b>                          | <b>M1</b> for use of the gradient, must be negative                                                                                 |
| (b)(i)    | 27                                                                                                                          | <b>B1</b>                              |                                                                                                                                     |
| (ii)      | $t^2 = 8 \ln 4$<br>$t = 3.33$ or better                                                                                     | <b>M1</b><br><b>A1</b>                 | for a correct attempt to solve $e^{\frac{t^2}{8}} = 4$                                                                              |
| (iii)     | $\text{acceleration} = 3\frac{2t}{8}e^{\frac{t^2}{8}}\left(e^{\frac{t^2}{8}} - 4\right)^2$<br><br>When $t = 1$ , $a = 6.98$ | <b>M1, A1</b><br><br><br><b>M1, A1</b> | <b>M1</b> for a correct attempt to differentiate using the chain rule<br><br><br><b>M1</b> for use of $t = 1$ in their acceleration |

238



No calculator

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(i) Find  $\frac{d}{dx}(\ln(3x^2 - 11))$ .

[2]

(ii) Hence show that  $\int \frac{x}{3x^2 - 11} dx = p \ln(3x^2 - 11) + c$ , where  $p$  is a constant to be found, and  $c$  is a constant of integration.

[1]

(iii) Given that  $\int_2^a \frac{x}{3x^2 - 11} dx = \ln 2$ , where  $a > 2$ , find the value of  $a$ .

[4]

**238** Oct/Nov 2016 · Paper 13 · Q6

|        |                                                                                                                            |                                                    |                                                                                                                                                                                                   |
|--------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 71 (i) | $\frac{6x}{3x^2 - 11}$                                                                                                     | <b>M1</b><br><b>A1</b>                             | <b>M1</b> for $\frac{mx}{3x^2 - 11}$                                                                                                                                                              |
| (ii)   | $p = \frac{1}{6}$                                                                                                          | <b>B1</b>                                          | <b>FT</b> for $p = \frac{1}{m}$                                                                                                                                                                   |
| (iii)  | $\frac{1}{6} \ln(3a^2 - 11) - \frac{1}{6} \ln 1 = \ln 2$<br>$\ln(3a^2 - 11) = \ln 2^6$<br>$3a^2 - 11 = 64$<br>$a = 5$ only | <b>M1</b><br><b>DM1</b><br><b>DM1</b><br><b>A1</b> | for correct use of limits in $p \ln(3x^2 - 11)$ May be implied by following equation<br>for dealing with logs correctly<br>for solution of $3a^2 - 11 = k$<br>for 5 obtained from an exact method |

**239**

No calculator

**Oct/Nov 2016 • Paper 13 • Q10**

A curve  $y = f(x)$  is such that  $f'(x) = 6x - 8e^{2x}$ .

- (i) Given that the curve passes through the point  $P(0, -3)$ , find the equation of the curve. [5]

The normal to the curve  $y = f(x)$  at  $P$  meets the line  $y = 2 - 3x$  at the point  $Q$ .

- (ii) Find the area of the triangle  $OPQ$ , where  $O$  is the origin. [5]

**240**

No calculator

**Oct/Nov 2016 • Paper 13 • Q11**

A particle moving in a straight line has a velocity of  $v \text{ ms}^{-1}$  such that,  $t$  s after leaving a fixed point,  $v = 4t^2 - 8t + 3$ .

- (i) Find the acceleration of the particle when  $t = 3$ . [2]

- (ii) Find the values of  $t$  for which the particle is momentarily at rest. [2]

- (iii) Find the total distance the particle has travelled when  $t = 1.5$ . [5]

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| Question | Answer                                                                                                                                                                                  | Marks                                                                      | Part Marks                                                                                                                                                                                                                                                                                                    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 72 (i)   | $f(x) = 3x^2 - 4e^{2x} (+c)$<br><br>passing through $(0, -3)$<br>$-3 = 3 \times 0 - 4e^0 + c$<br>$f(x) = 3x^2 - 4e^{2x} + 1$                                                            | <b>M1</b><br><b>A1</b><br><b>A1</b><br><b>DM1</b><br><br><b>A1</b>         | for one correct term<br>for one correct term $3x^2$ or $-4e^{2x}$<br>for a second correct term with no extras<br>for correct method to find $c$ .<br><br>for correct equation                                                                                                                                 |
| (ii)     | $f'(0) = -8$<br><br>Normal: $y + 3 = \frac{1}{8}x$<br>$8y + 24 = x$<br>$y = 2 - 3x$<br><br>leads to $x = \frac{8}{5}$ oe<br>Area = $= \frac{1}{2} \times 3 \times \frac{8}{5} = 2.4$ oe | <b>B1</b><br><br><b>M1</b><br><b>DM1</b><br><br><b>A1</b><br><br><b>B1</b> | for $m = \frac{1}{8}$<br>for equation of normal using $m = \frac{1}{8}$<br>for solving normal equation simultaneously with $y = 2 - 3x$ to get a value of $x$<br><br>for $x = \frac{8}{5}$ , 1.6 oe<br><b>FT</b> for a numerical answer equal to $\left  \frac{1}{2} \times 3 \times \text{their } x \right $ |



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|        |                                                                                                                                                                                                         |                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 73 (i) | $a = 8t - 8$<br>When $t = 3$ , $a = 16$                                                                                                                                                                 | <b>B1</b><br><b>B1</b>                                                                                                         | for $8t - 8$<br>for 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| (ii)   | 0.5, 1.5                                                                                                                                                                                                | <b>B1,B1</b>                                                                                                                   | <b>B1</b> for each                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| (iii)  | $s = \frac{4}{3}t^3 - 4t^2 + 3t$<br>when $t = \frac{1}{2}$ , $s = \frac{2}{3}$<br><br>when $t = \frac{3}{2}$ , $s = 0$<br><br>total distance travelled = $\frac{4}{3}$<br><br><b>Alternative method</b> | <b>M1</b><br><b>A1</b><br><br><b>DM1</b><br><br><b>DM1</b><br><br><b>A1</b><br><br><b>M1A1</b><br><b>DM1</b><br><br><b>DM1</b> | for at least two terms correct<br>all correct<br><br>for calculating displacement when either $t = \frac{1}{2}$<br>or $t = \frac{3}{2}$<br><br>for calculating displacement at $t = \frac{1}{2}$ and<br>doubling.<br>for $\frac{4}{3}$ oe allow 1.33<br><br>As before<br><b>DM1</b> for calculating displacement when $t = 0.5$ or<br>for calculating distance travelled between $t = 0.5$<br>and $t = 1.5$<br><b>DM1</b> for doubling distance travelled between<br>$t = 0.5$ and $t = 1.5$ or for adding that distance to<br>displacement at $t = 0.5$ |

|  |  |           |                                           |
|--|--|-----------|-------------------------------------------|
|  |  | <b>A1</b> | <b>A1</b> for $\frac{4}{3}$ oe allow 1.33 |
|--|--|-----------|-------------------------------------------|

**241**

No calculator

**May/June 2016 • Paper 11 • Q3**

Find the equation of the normal to the curve  $y = \ln(2x^2 - 7)$  at the point where the curve crosses the positive  $x$ -axis. Give your answer in the form  $ax + by + c = 0$ , where  $a$ ,  $b$  and  $c$  are integers. [5]

**242**

No calculator

**May/June 2016 • Paper 11 • Q5**

**Do not use a calculator in this question.**

(i) Show that  $\frac{d}{dx}\left(\frac{e^{4x}}{4} - xe^{4x}\right) = pxe^{4x}$ , where  $p$  is an integer to be found. [4]

(ii) Hence find the exact value of  $\int_0^{\ln 2} xe^{4x} dx$ , giving your answer in the form  $a\ln 2 + \frac{b}{c}$ , where  $a$ ,  $b$  and  $c$  are integers to be found. [4]

**243**

No calculator

**May/June 2016 • Paper 12 • Q6**

Show that  $\frac{d}{dx}(e^{3x}\sqrt{4x+1})$  can be written in the form  $\frac{e^{3x}(px+q)}{\sqrt{4x+1}}$ , where  $p$  and  $q$  are integers to be found. [5]

**241** May/June 2016 · Paper 11 · Q3

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                        | Marks                                                                       | Guidance                                                                                                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | <p>On <math>x</math>-axis, <math>2x^2 - 7 = 1</math><br/> <math>x = 2</math></p> <p><math>\frac{dy}{dx} = \frac{4x}{2x^2 - 7}</math></p> <p>When <math>x = 2</math>, <math>\frac{dy}{dx} = 8</math></p> <p>Gradient of normal <math>= -\frac{1}{8}</math></p> <p>Equation of normal <math>y = -\frac{1}{8}(x - 2)</math></p> <p>Required form <math>x + 8y - 2 = 0</math></p> | <p><b>M1</b><br/><b>A1</b><br/><b>B1</b></p> <p><b>M1</b><br/><b>A1</b></p> | <p>for equating to 1</p> <p>for attempt at perpendicular through <i>their</i> <math>(2, 0)</math>, must be using <math>y = 0</math><br/> must be equated to zero with integer coefficients</p> |

**242** May/June 2016 · Paper 11 · Q5

$$(i) \frac{d}{dx} \left( \frac{e^{4x}}{4} - xe^{4x} \right) = e^{4x} - (e^{4x} + 4xe^{4x}) = -4xe^{4x}: \quad \boxed{p = -4}$$

$$(ii) \int_0^{\ln 2} xe^{4x} dx = \left[ -\frac{1}{4} \left( \frac{e^{4x}}{4} - xe^{4x} \right) \right]_0^{\ln 2} = (4 \ln 2 - 1) - \left( -\frac{1}{16} \right) = \boxed{4 \ln 2 - \frac{15}{16}}$$

$$a = 4, b = -15, c = 16$$

**243** May/June 2016 · Paper 12 · Q6

$$\frac{d}{dx} (e^{3x} \sqrt{4x+1}) = 3e^{3x} \sqrt{4x+1} + e^{3x} \cdot \frac{2}{\sqrt{4x+1}} = \frac{e^{3x} (3(4x+1) + 2)}{\sqrt{4x+1}} = \frac{e^{3x} (12x+5)}{\sqrt{4x+1}}$$

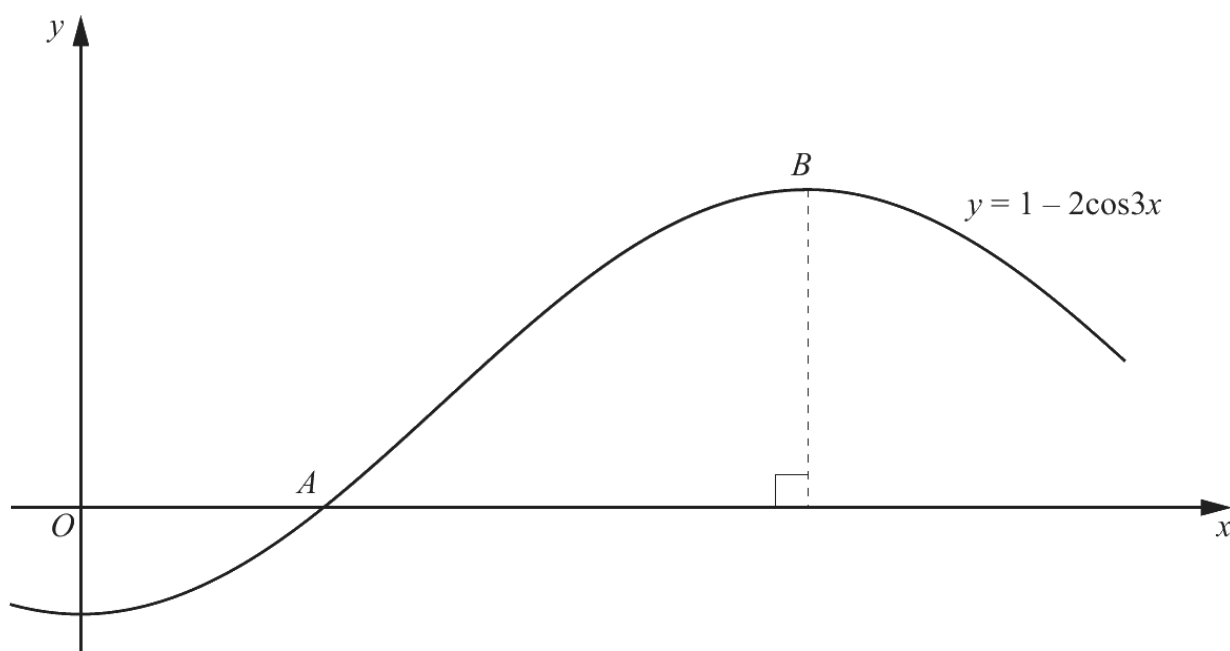
$$\boxed{p = 12, q = 5}$$

244



No calculator

May/June 2016 · Paper 12 · Q7



The diagram shows part of the graph of  $y = 1 - 2\cos 3x$ , which crosses the  $x$ -axis at the point  $A$  and has a maximum at the point  $B$ .

(i) Find the coordinates of  $A$ . [2]

(ii) Find the coordinates of  $B$ . [2]

(iii) Showing all your working, find the area of the shaded region bounded by the curve, the  $x$ -axis and the perpendicular from  $B$  to the  $x$ -axis. [4]

**244** May/June 2016 · Paper 12 · Q7

|        |                                                                                                                                                                                                                                                                                                         |                                                   |                                                                                                                                                            |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 27 (i) | $\cos 3x = \frac{1}{2}, \quad x = \frac{\pi}{9} \text{ or } 0.349, 20^\circ,$<br>allow 0.35                                                                                                                                                                                                             | <b>M1</b><br><b>A1</b>                            | for correct attempt to solve the trigonometric equation                                                                                                    |
| (ii)   | $B\left(\frac{\pi}{3}, 3\right)$ or $(1.05, 3), (60^\circ, 3)$                                                                                                                                                                                                                                          | <b>B1B1</b>                                       | <b>B1</b> for each, must be in correct position or in terms of $x =$ and $y =$                                                                             |
| (iii)  | $\int_{\frac{\pi}{9}}^{\frac{\pi}{3}} 1 - 2 \cos 3x \, dx = \left[ x - \frac{2}{3} \sin 3x \right]_{\frac{\pi}{9}}^{\frac{\pi}{3}}$ $= \frac{\pi}{3} - \left( \frac{\pi}{9} - \left( \frac{2}{3} \times \frac{\sqrt{3}}{2} \right) \right)$ $= \frac{2\pi}{9} + \frac{\sqrt{3}}{3} \text{ oe or } 1.28$ | <b>M1</b><br><b>A1</b><br><b>DM1</b><br><b>A1</b> | for $x \pm a \sin 3x$ attempt to integrate at least one term<br>for correct integration<br>for correct use of limits from (i) and (ii), must be in radians |

**245**

No calculator

**May/June 2016 • Paper 12 • Q9**

A curve passes through the point  $\left(2, -\frac{4}{3}\right)$  and is such that  $\frac{dy}{dx} = (3x + 10)^{-\frac{1}{2}}$ .

- (i) Find the equation of the curve.

[4]

The normal to the curve, at the point where  $x = 5$ , meets the line  $y = -\frac{5}{3}$  at the point  $P$ .

- (ii) Find the  $x$ -coordinate of  $P$ .

[6]

**245** May/June 2016 · Paper 12 · Q9

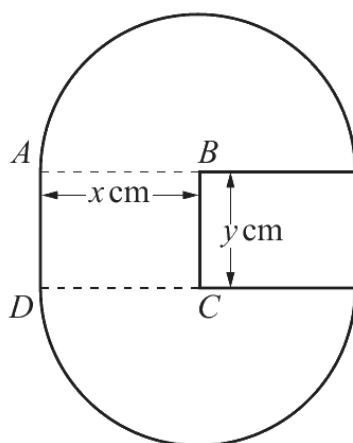
|        |                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                     |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28 (i) | $y = \frac{2}{3}(3x+10)^{\frac{1}{2}} + c$<br>passes through $\left(2, -\frac{4}{3}\right)$ , so $c = -4$<br>$y = \frac{2}{3}(3x+10)^{\frac{1}{2}} - 4$ oe                      | <b>B1</b> for $p(3x+10)^{\frac{1}{2}}$ where $p$ is a constant<br><b>B1</b> for $\frac{2}{3}(3x+10)^{\frac{1}{2}}$ oe unsimplified<br><b>M1</b> for attempt to find $c$ , must have attempt to integrate, must have the first <b>B1</b><br><b>A1</b>                                                                |
| (ii)   | When $x = 5$ ,<br>$y = -\frac{2}{3}$<br>perpendicular gradient $= -5$<br><br>Equation of normal: $y + \frac{2}{3} = -5(x - 5)$<br>When $y = -\frac{5}{3}$ ,<br><br>$x = 5.2$ oe | <b>B1</b><br><b>B1</b><br><b>M1</b> for attempt at the normal using <i>their</i> perpendicular gradient and <i>their</i> $y$ value (but not $y = -\frac{4}{3}$ or $-\frac{5}{3}$ ).<br><b>A1</b><br><br><b>DM1</b> for use of $y = -\frac{5}{3}$ in their normal equation to get as far as $x = \dots$<br><b>A1</b> |

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No calculator

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The diagram shows a badge, made of thin sheet metal, consisting of two semi-circular pieces, centres  $B$  and  $C$ , each of radius  $x$  cm. They are attached to each other by a rectangular piece of thin sheet metal,  $ABCD$ , such that  $AB$  and  $CD$  are the radii of the semi-circular pieces and  $AD = BC = y$  cm.

- (i) Given that the area of the badge is  $20 \text{ cm}^2$ , show that the perimeter,  $P$  cm, of the badge is given

$$\text{by } P = 2x + \frac{40}{x}. \quad [4]$$

- (ii) Given that  $x$  can vary, find the minimum value of  $P$ , justifying that this value is a minimum. [5]



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|        |                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                         |                                                                                                                                                                                                                                                                                        |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29 (i) | <p>Area: <math>20 = \pi x^2 + xy</math></p> $y = \frac{20 - \pi x^2}{x}$ $P = 2\pi x + 2x + 2y$ $= 2\pi x + 2x + 2\left(\frac{20}{x} - \pi x\right)$ $= 2x + \frac{40}{x}$ <p>Alternative method:</p> $20 = \pi x^2 + xy$ $P = 2\pi x + 2y + 2x$ $= \frac{2}{x}(\pi x^2 + xy) + 2x$ $= \frac{2}{x}(20) + 2x$ $= 2x + \frac{40}{x}$ | <p><b>B1</b></p> <p><b>B1</b></p> <p><b>M1</b></p> <p><b>A1</b></p> <p><b>B1</b></p> <p><b>M1</b></p> <p><b>B1</b></p> <p><b>A1</b></p> | <p>for attempt to use perimeter and obtain in terms of <math>x</math> only</p> <p>all steps seen, www AG</p> <p>for attempt to use perimeter and write in <math>\frac{\pi x^2 + xy}{x}</math></p> <p>for replacing <math>\pi x^2 + xy</math> with 20</p> <p>all steps seen, www AG</p> |
| (ii)   | $\frac{dP}{dx} = 2 - \frac{40}{x^2}$ <p>When <math>\frac{dP}{dx} = 0</math>,</p> $x = 2\sqrt{5} \quad \text{allow } 4.47, \sqrt{20}$ <p>leading to <math>P = 8\sqrt{5}</math>, allow 17.9</p> $\frac{d^2P}{dx^2} = \frac{80}{x^3}$ <p>, always positive so a minimum</p>                                                           | <p><b>M1</b></p> <p><b>DM1</b></p> <p><b>A1</b></p> <p><b>A1</b></p> <p><b>A1</b></p>                                                   | <p>for attempt to differentiate</p> <p>for equating to zero and attempt to solve at least as far as <math>x^2 =</math></p> <p>for this statement or use of gradient inspection either side of correct <math>x</math></p>                                                               |

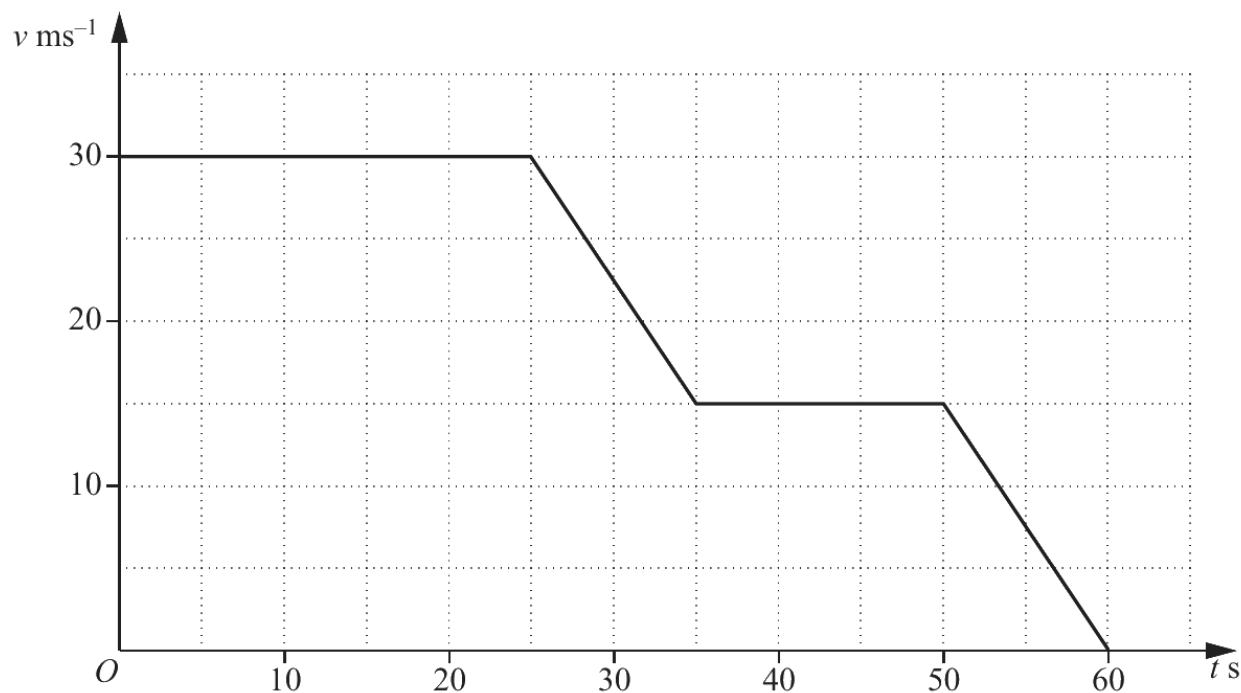
247

(a)



No calculator

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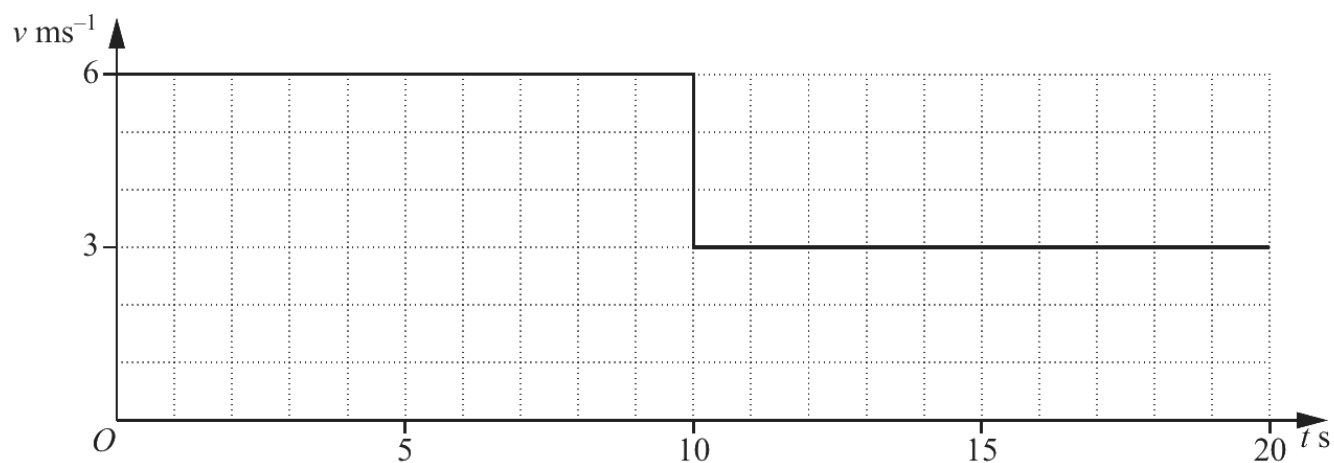
The diagram shows the velocity-time graph of a particle  $P$  moving in a straight line with velocity  $v \text{ ms}^{-1}$  at time  $t \text{ s}$  after leaving a fixed point.

(i) Find the distance travelled by the particle  $P$ . [2]

(ii) Write down the deceleration of the particle when  $t = 30$ . [1]

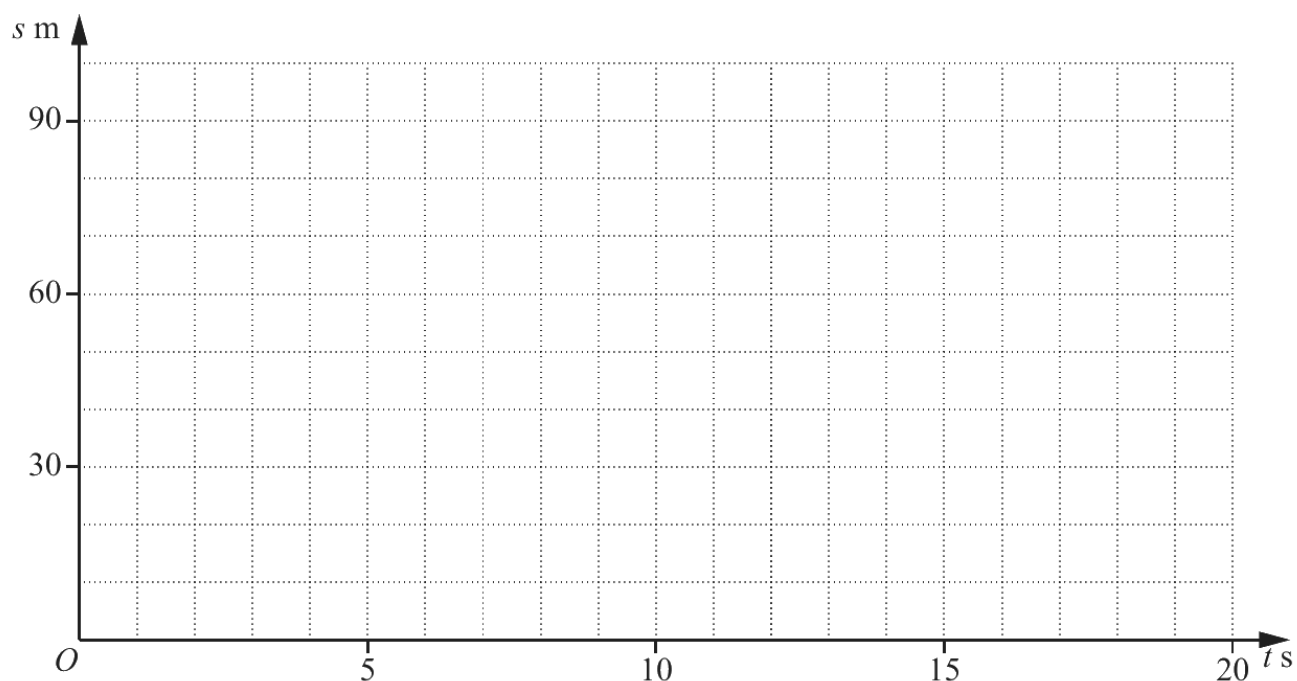
(b) The diagram shows a velocity-time graph of a particle  $Q$  moving in a straight line with velocity  $v \text{ ms}^{-1}$ , at time  $t \text{ s}$  after leaving a fixed point.

Question continues on the next page →



The displacement of  $Q$  at time  $t$  s is  $s$  m. On the axes below, draw the corresponding displacement-time graph for  $Q$ .

[2]



- (c) The velocity,  $v \text{ ms}^{-1}$ , of a particle  $R$  moving in a straight line,  $t$  s after passing through a fixed point  $O$ , is given by  $v = 4e^{2t} + 6$ .

- (i) Explain why the particle is never at rest.

[1]

*Question continues on the next page →*

(ii) Find the exact value of  $t$  for which the acceleration of  $R$  is  $12 \text{ ms}^{-2}$ . [2]

(iii) Showing all your working, find the distance travelled by  $R$  in the interval between  $t = 0.4$  and  $t = 0.5$ . [4]

**248**

No calculator

**May/June 2016 • Paper 13 • Q3**

Find the equation of the normal to the curve  $y = \ln(2x^2 - 7)$  at the point where the curve crosses the positive  $x$ -axis. Give your answer in the form  $ax + by + c = 0$ , where  $a$ ,  $b$  and  $c$  are integers. [5]

**249**

No calculator

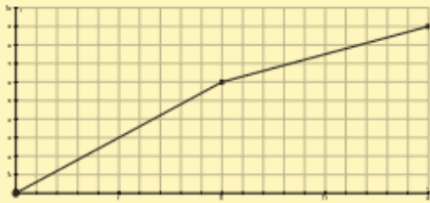
**May/June 2016 • Paper 13 • Q5**

**Do not use a calculator in this question.**

(i) Show that  $\frac{d}{dx} \left( \frac{e^{4x}}{4} - x e^{4x} \right) = p x e^{4x}$ , where  $p$  is an integer to be found. [4]

(ii) Hence find the exact value of  $\int_0^{\ln 2} x e^{4x} dx$ , giving your answer in the form  $a \ln 2 + \frac{b}{c}$ , where  $a$ ,  $b$  and  $c$  are integers to be found. [4]

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|           |                                                                                                                                  |                                                   |                                                                                                                                                                                                                                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30 (a)(i) | Distance = area under graph<br><br>= 1275                                                                                        | <b>M1</b>                                         | for attempt to find the area, one correct area seen ( triangle, rectangle or trapezium) as part of a sum.                                                                                                                                                                                              |
| (ii)      | deceleration is 1.5 oe                                                                                                           | <b>A1</b>                                         |                                                                                                                                                                                                                                                                                                        |
| (b)       |                                                 | <b>B1</b>                                         | for a straight line between (0,0) and (10,60)                                                                                                                                                                                                                                                          |
| (c)(i)    | $e^{2t}$ is always positive or oe                                                                                                | <b>B1FT</b>                                       | <b>FT</b> a straight line between (10, 60) and (20, 90), a displacement vector $\begin{pmatrix} 10 \\ 30 \end{pmatrix}$ from <i>their</i> (10, <i>their</i> 60)                                                                                                                                        |
| (ii)      | $a = 8e^{2t}$<br>$e^{2t} = \frac{3}{2}$<br>$t = \frac{1}{2} \ln \frac{3}{2}$ , $\ln \sqrt{\frac{3}{2}}$ or $\frac{1}{2} \ln 1.5$ | <b>B1</b>                                         |                                                                                                                                                                                                                                                                                                        |
| (iii)     | $s = [2e^{2t} + 6t]_{0.4}^{0.5}$<br><br>$= (2e + 3) - (2e^{0.8} + 2.4)$<br>$(= 8.436 - 6.851)$<br>$= 1.59$ , allow 1.58          | <b>M1</b><br><b>A1</b><br><b>DM1</b><br><b>A1</b> | for attempt to differentiate, must be of the form $pe^{2t}$ , equate to 12 and solve.<br><br>Allow fractions equivalent to $\frac{3}{2}$<br><br>for attempt to integrate to get $qe^{2t} + 6t$<br>all correct<br>for correct use of limits or considering distances separately, ignore attempts at $c$ |

**248** May/June 2016 · Paper 13 · Q3

| Question | Answer                                                                                                                                                                                                                                              | Marks                                                                                     | Guidance                                                                                                                                                                                  |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 58       | On $x$ -axis, $2x^2 - 7 = 1$<br>$x = 2$<br><br>$\frac{dy}{dx} = \frac{4x}{2x^2 - 7}$<br>When $x = 2$ , $\frac{dy}{dx} = 8$<br>Gradient of normal = $-\frac{1}{8}$<br>Equation of normal $y = -\frac{1}{8}(x - 2)$<br>Required form $x + 8y - 2 = 0$ | <b>M1</b><br><b>A1</b><br><br><b>B1</b><br><br><br><br><br><br><b>M1</b><br><br><b>A1</b> | for equating to 1<br><br><br><br><br><br><br><br><br>for attempt at perpendicular through <i>their</i> (2, 0), must be using $y = 0$<br>must be equated to zero with integer coefficients |

**249** May/June 2016 · Paper 13 · Q5

|        |                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                      |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 59 (i) | $\frac{d}{dx}\left(\frac{e^{4x}}{4} - xe^{4x}\right) = e^{4x} - \left((x \times 4e^{4x}) + e^{4x}\right)$ $= -4xe^{4x}$                                                                                            | <b>B1</b> for $\frac{d}{dx}\left(\frac{e^{4x}}{4}\right) = e^{4x}$<br><b>M1</b> for attempt to differentiate a product<br><b>A1</b> for a correct product<br><b>A1</b> for correct final answer                                                                                      |
| (ii)   | $\int_0^{\ln 2} xe^{4x} \, dx = -\frac{1}{4} \left[ \frac{e^{4x}}{4} - xe^{4x} \right]_0^{\ln 2}$ $= -\frac{1}{4} \left( \left( \frac{16}{4} - 16 \ln 2 \right) - \frac{1}{4} \right)$ $= 4 \ln 2 - \frac{15}{16}$ | <b>B1FT</b> <b>FT</b> for use of <i>their</i> $\frac{1}{p} \times \left( \frac{e^{4x}}{4} - xe^{4x} \right)$ , must be numerical $p$ , but $\neq 0$<br><b>B1</b> for $e^{4 \ln 2} = 16$<br><b>M1</b> for correct use of limits, must be an integral of the correct form<br><b>A1</b> |

**250** Calculator**Oct/Nov 2020 • Paper 21 • Q4**

It is given that  $y = \ln(\sin x + 3 \cos x)$  for  $0 < x < \frac{\pi}{2}$ .

(a) Find  $\frac{dy}{dx}$ . [3]

(b) Find the value of  $x$  for which  $\frac{dy}{dx} = -\frac{1}{2}$ . [3]

**251** Calculator**Oct/Nov 2020 • Paper 21 • Q9**

A particle moves in a straight line such that,  $t$  seconds after passing a fixed point  $O$ , its displacement from  $O$  is  $s$  m, where  $s = e^{2t} - 10e^t - 12t + 9$ .

(a) Find expressions for the velocity and acceleration at time  $t$ . [3]

(b) Find the time when the particle is instantaneously at rest. [3]

(c) Find the acceleration at this time. [2]

**250** Oct/Nov 2020 · Paper 21 · Q4

|        |                                                             |           |                                                                                                                                             |
|--------|-------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 28 (a) | $\frac{dy}{dx} = \frac{\cos x - 3\sin x}{\sin x + 3\cos x}$ | <b>3</b>  | <b>M1</b> for attempt at chain rule must have function in numerator and denominator<br><b>A1</b> for denominator<br><b>A1</b> for numerator |
| (b)    | $-2 \cos x - 3 \cos x = \sin x - 6 \sin x$                  | <b>M1</b> | Expand and collect terms in $\sin x$ and $\cos x$                                                                                           |
|        | $1 = \tan x$                                                | <b>M1</b> | Use $\frac{\sin x}{\cos x} = \tan x$                                                                                                        |
|        | $x = \frac{\pi}{4}$                                         | <b>A1</b> | Must be radians                                                                                                                             |

**251** Oct/Nov 2020 · Paper 21 · Q9

|        |                                                                                   |           |                                                                                                              |
|--------|-----------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------|
| 29 (a) | $v = 2e^{2t} - 10e^t - 12$<br>$a = 4e^{2t} - 10e^t$                               | <b>3</b>  | <b>M1</b> for correctly differentiating $e^{2t}$ .<br><b>A1</b> for $v$ correct<br><b>A1</b> for $a$ correct |
| (b)    | $v = 0 \rightarrow e^{2t} - 5e^t - 6 = 0$<br>$\rightarrow (e^t + 1)(e^t - 6) = 0$ | <b>M1</b> | Factorise quadratic<br>Solve and discard $e^t = -1$                                                          |
|        | $e^t = 6$                                                                         | <b>A1</b> |                                                                                                              |
|        | $t = \ln 6 = 1.79$                                                                | <b>A1</b> |                                                                                                              |
| (c)    | $t = \ln 6 \rightarrow a = 4 \times 36 - 10 \times 6 = 84$                        | <b>2</b>  | <b>M1</b> for inserting <i>their</i> value of $t$ into $a$                                                   |



**252**

Calculator

**Oct/Nov 2020 • Paper 21 • Q10**

The gradient of the normal to a curve at the point  $(x, y)$  is given by  $\frac{x}{x+1}$ .

- (a) Given that the curve passes through the point  $(1, 4)$ , show that its equation is  $y = 5 - \ln x - x$ . [5]

- (b) Find, in the form  $y = mx + c$ , the equation of the tangent to the curve at the point where  $x = 3$ . [3]

**253**

Calculator

**Oct/Nov 2020 • Paper 21 • Q11**

The equation of a curve is  $y = x\sqrt{16-x^2}$  for  $0 \leq x \leq 4$ .

- (a) Find the exact coordinates of the stationary point of the curve. [6]

- (b) Find  $\frac{d}{dx}(16-x^2)^{\frac{3}{2}}$  and hence evaluate the area enclosed by the curve  $y = x\sqrt{16-x^2}$  and the lines  $y = 0$ ,  $x = 1$  and  $x = 3$ . [5]

**252** Oct/Nov 2020 · Paper 21 · Q10

|        |                                                                                            |           |                                                                                    |
|--------|--------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------|
| 30 (a) | $\frac{dy}{dx} = -\frac{(1+x)}{x} = -\left(\frac{1}{x} + 1\right)$                         | <b>2</b>  | <b>M1</b> for using $m_1 \times m_2 = -1$                                          |
|        | $y = -\ln x - x + C$                                                                       | <b>2</b>  | <b>M1</b> for integrating $\frac{1}{x}$<br><b>A1</b> for all correct including $C$ |
|        | $4 = -\ln 1 - 1 + C$<br>$C = 5 \rightarrow y = 5 - \ln x - x$                              | <b>A1</b> | Insert (1, 4) and arrive at correct answer. AG                                     |
| (b)    | $x = 3 \rightarrow y = 2 - \ln 3$<br>and $\frac{dy}{dx} = -\frac{1}{3} - 1 = -\frac{4}{3}$ | <b>B1</b> |                                                                                    |
|        | $\frac{y - (2 - \ln 3)}{x - 3} = -\frac{4}{3}$                                             | <b>M1</b> |                                                                                    |
|        | $y = -\frac{4}{3}x + 6 - \ln 3$<br>or $y = -1.33x + 4.90$                                  | <b>A1</b> |                                                                                    |

**253** Oct/Nov 2020 · Paper 21 · Q11

|        |                                                                                                                                  |          |                                                                                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31 (a) | $\frac{dy}{dx} = x \times \frac{1}{2}(16 - x^2)^{-\frac{1}{2}} \times (-2x) + (16 - x^2)^{\frac{1}{2}}$                          | <b>3</b> | <b>B1</b> for $\frac{d}{dx}(16 - x^2)^{\frac{1}{2}}$<br>$= \frac{1}{2}(16 - x^2)^{-\frac{1}{2}} \times (-2x)$<br><b>M1</b> for product rule<br><b>A1</b> for all correct |
|        | $\frac{dy}{dx} = 0 \rightarrow (16 - x^2)^{\frac{1}{2}} = \frac{x^2}{(16 - x^2)^{\frac{1}{2}}}$<br>$x^2 = 8$<br>$(2\sqrt{2}, 8)$ | <b>3</b> | <b>M1</b> for setting $\frac{dy}{dx} = 0$ and attempt to solve<br><b>M1</b> for obtaining $x^2 = k$<br><b>A1</b>                                                         |
| (b)    | $\frac{3}{2}(16 - x^2)^{\frac{1}{2}} \times (-2x)$                                                                               | <b>2</b> | <b>M1</b> for attempt at chain rule<br><b>A1</b> for all correct unsimplified                                                                                            |

|                                                                                                                                                                                       |                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| $\text{Area} = \int_1^3 x(16-x^2)^{\frac{1}{2}} dx = \left[ -\frac{1}{3}(16-x^2)^{\frac{3}{2}} \right]_1^3$ $= -\frac{1}{3} \left[ 7^{\frac{3}{2}} - 15^{\frac{3}{2}} \right] = 13.2$ | <b>3</b><br><b>M1</b> for obtaining $k(16-x^2)^{\frac{3}{2}}$<br><b>A1</b> for obtaining $k = -\frac{1}{3}$<br><b>A1</b> for 13.2 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|

**254**

No calculator

**Oct/Nov 2020 · Paper 22 · Q5****DO NOT USE A CALCULATOR IN THIS QUESTION.**

- (a) Find the equation of the tangent to the curve  $y = x^3 - 6x^2 + 3x + 10$  at the point where  $x = 1$ . [4]

- (b) Find the coordinates of the point where this tangent meets the curve again. [5]

**255**

Calculator

**Oct/Nov 2020 · Paper 22 · Q6**

Find the exact value of  $\int_2^4 \frac{(x+1)^2}{x^2} dx$ .

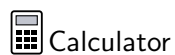
[6]

**254** Oct/Nov 2020 · Paper 22 · Q5

|        |                                                                                        |           |                                                                                                                                                                                                               |
|--------|----------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 57 (a) | $x = 1 \rightarrow y = 8$                                                              | <b>B1</b> |                                                                                                                                                                                                               |
|        | $\frac{dy}{dx} = 3x^2 - 12x + 3$                                                       | <b>M1</b> | Attempt to differentiate. Powers reduced by 1 in all four terms.                                                                                                                                              |
|        | $x = 1 \rightarrow \frac{dy}{dx} = -6$                                                 | <b>A1</b> |                                                                                                                                                                                                               |
|        | $\frac{y-8}{x-1} = -6 \rightarrow y = -6x + 14$                                        | <b>A1</b> | Either form.<br>isw                                                                                                                                                                                           |
| (b)    | $x^3 - 6x^2 + 9x - 4 = 0$<br>$(x-1)(x^2 - 5x + 4) = 0$<br>or $(x-4)(x^2 - 2x + 1) = 0$ | <b>2</b>  | <b>M1</b> for equating <i>their</i> tangent to curve and simplifying to 4 term cubic.<br><b>M1Dep</b> for finding a factor or stating that $(x-1)$ is a factor or makes at least 3 attempts to find a factor. |
|        | $(x-1)(x-1)(x-4) = 0$                                                                  | <b>2</b>  | <b>A1</b> for $(x-1)$ or $x = 1$ can be implied.<br>nfw<br><b>A1</b> for $(x-4)$ or $x = 4$ not repeated.<br>nfw                                                                                              |
|        | $x = 4 \rightarrow y = -10$ only                                                       | <b>A1</b> | nfw                                                                                                                                                                                                           |

**255** Oct/Nov 2020 · Paper 22 · Q6

|    |                                                                                       |           |                                                                                                                                  |
|----|---------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| 58 | $\frac{(x+1)^2}{x^2} = \frac{x^2 + 2x + 1}{x^2} = 1 + \frac{2}{x} + \frac{1}{x^2}$    | <b>2</b>  | <b>B1</b> for expanding numerator seen anywhere.<br><b>M1</b> for attempt to divide <i>their</i> three term numerator by $x^2$ . |
|    | $\int 1 + \frac{2}{x} + \frac{1}{x^2} dx = x + 2 \ln x - \frac{1}{x} + (c)$           | <b>2</b>  | <b>A2/1/0</b> minus 1 each error or omission.                                                                                    |
|    | $\left[ 4 - 2 \ln 4 - \frac{1}{4} \right] - \left[ 2 + 2 \ln 2 - \frac{1}{2} \right]$ | <b>M1</b> | <b>Dep</b> insert 4 and 2 into <i>their</i> three or two term integrand and subtract correctly.                                  |
|    | $= \frac{9}{4} + 2 \ln 2$                                                             | <b>A1</b> | oe must be exact two terms.<br>isw                                                                                               |

**256****Oct/Nov 2020 • Paper 23 • Q4**

It is given that  $y = \ln(1 + \sin x)$  for  $0 < x < \pi$ .

(a) Find  $\frac{dy}{dx}$ . [2]

(b) Find the value of  $\frac{dy}{dx}$  when  $x = \frac{\pi}{6}$ , giving your answer in the form  $\frac{1}{\sqrt{a}}$ , where  $a$  is an integer. [2]

(c) Find the values of  $x$  for which  $\frac{dy}{dx} = \tan x$ . [5]

**256** Oct/Nov 2020 · Paper 23 · Q4

|        |                                                                    |           |                                                                                                                                              |
|--------|--------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 78 (a) | $\frac{dy}{dx} = \frac{1}{1 + \sin x}$                             | <b>M1</b> |                                                                                                                                              |
|        | $\times \cos x = \frac{\cos x}{1 + \sin x}$                        | <b>A1</b> |                                                                                                                                              |
| (b)    | insert $\frac{\pi}{6}$ into <i>their</i> $\frac{dy}{dx}$           | <b>M1</b> |                                                                                                                                              |
|        | $\frac{1}{\sqrt{3}}$                                               | <b>A1</b> | not $\frac{\sqrt{3}}{3}$                                                                                                                     |
| (c)    | $\text{their } \frac{\cos x}{1 + \sin x} = \frac{\sin x}{\cos x}$  | <b>M1</b> | replace $\tan x$ with $\frac{\sin x}{\cos x}$                                                                                                |
|        | use $\cos^2 x = 1 - \sin^2 x$<br>( $2 \sin^2 x + \sin x - 1 = 0$ ) | <b>M1</b> | earned when equation reduced to a quadratic in $\sin x$                                                                                      |
|        | $(2 \sin x - 1)(\sin x + 1) = 0$                                   | <b>M1</b> | solve three term quadratic in $\sin x$                                                                                                       |
|        | $x = \frac{\pi}{6}$                                                | <b>A1</b> | or 0.524 or better radians only<br>if M0 M0 M0 and (a) and (b) correct,<br>allow SC2 for<br>$\tan x = \frac{1}{\sqrt{3}}, x = \frac{\pi}{6}$ |
|        | $x = \frac{5\pi}{6}$                                               | <b>A1</b> | or 2.62 or better radians only<br>A0 if extra solution(s) in range                                                                           |

257

 Calculator

Oct/Nov 2020 • Paper 23 • Q7

A curve has equation  $y = x \cos x$ .

(a) Find  $\frac{dy}{dx}$ . [2]

(b) Find the equation of the normal to the curve at the point where  $x = \pi$ , giving your answer in the form  $y = mx + c$ . [4]

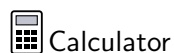
(c) Using your answer to **part (a)**, find the exact value of  $\int_0^{\frac{\pi}{6}} x \sin x \, dx$ . [5]



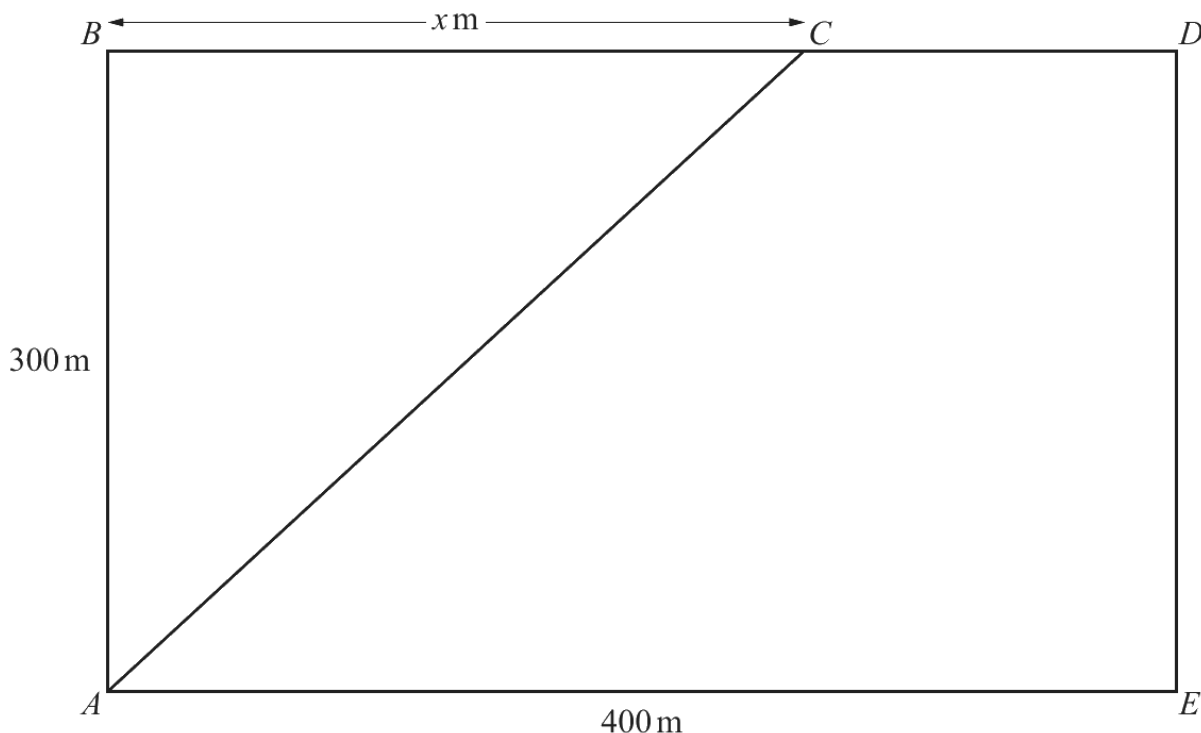
**257** Oct/Nov 2020 · Paper 23 · Q7

|        |                                                                                           |           |                                                                                                   |
|--------|-------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------|
| 79 (a) | $-x \sin x + \cos x$ isw                                                                  | <b>B2</b> | accept unsimplified<br>if incorrect allow B1 for<br>$\frac{d}{dx}(\cos x) = -\sin x$ clearly seen |
| (b)    | $x = \pi, y = -\pi$                                                                       | <b>B1</b> | or $-3.14$ or better                                                                              |
|        | $x = \pi, \frac{dy}{dx} = -1$                                                             | <b>B1</b> | from correct $\frac{dy}{dx}$                                                                      |
|        | gradient of normal = 1                                                                    | <b>M1</b> | use $m_1 m_2 = -1$ with <i>their</i> grad of tangent                                              |
|        | $y = x - 2\pi$ cso                                                                        | <b>A1</b> | or $y = x - 6.28$ or better<br>fully correct solution                                             |
| (c)    | $\int \text{their}(a) = x \cos x$<br>$\left( \int -\sin x + \cos x dx = x \cos x \right)$ | <b>M1</b> | *                                                                                                 |
|        | $\int \cos x dx = \sin x$                                                                 | <b>B1</b> | clearly seen anywhere                                                                             |
|        | $-x \cos x + \sin x$                                                                      | <b>A1</b> | implies previous marks if (a) is correct                                                          |
|        | insert $\frac{\pi}{6}$ into <i>their</i> integral                                         | <b>M1</b> | * dep                                                                                             |
|        | $\frac{1}{2} - \frac{\pi\sqrt{3}}{12}$                                                    | <b>A1</b> | reject decimals                                                                                   |

258



Oct/Nov 2020 • Paper 23 • Q9



The rectangle  $ABCDE$  represents a ploughed field where  $AB = 300$  m and  $AE = 400$  m. Joseph needs to walk from  $A$  to  $D$  in the least possible time. He can walk at  $0.9 \text{ ms}^{-1}$  on the ploughed field and at  $1.5 \text{ ms}^{-1}$  on any part of the path  $BCD$  along the edge of the field. He walks from  $A$  to  $C$  and then from  $C$  to  $D$ . The distance  $BC = x$  m.

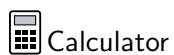
(a) Find, in terms of  $x$ , the total time,  $T$ s, Joseph takes for the journey. [3]

(b) Given that  $x$  can vary, find the value of  $x$  for which  $T$  is a minimum and hence find the minimum value of  $T$ . [6]

**258** Oct/Nov 2020 · Paper 23 · Q9

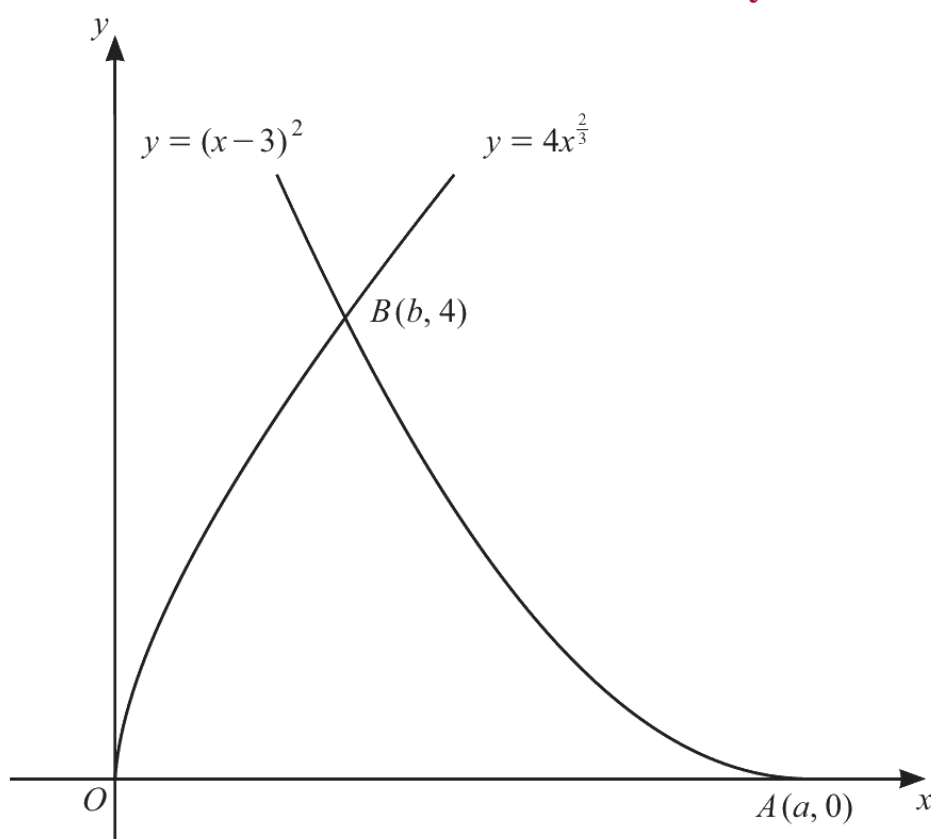
|        |                                                                                                   |           |                                                                                                                  |
|--------|---------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------|
| 80 (a) | $(AC =) \sqrt{300^2 + x^2}$ seen isw                                                              | <b>B1</b> |                                                                                                                  |
|        | time for $AC = \frac{\sqrt{300^2 + x^2}}{0.9}$ oe<br>or time for $CD = \frac{400 - x}{1.5}$ oe    | <b>M1</b> | using clearly indicated value for <i>their</i> AC or <i>their</i> CD                                             |
|        | $T = \frac{\sqrt{300^2 + x^2}}{0.9} + \frac{400 - x}{1.5}$ oe seen<br>isw                         | <b>A1</b> |                                                                                                                  |
| (b)    | $\frac{dT}{dx} = \frac{1}{2} \frac{(300^2 + x^2)^{-\frac{1}{2}}}{0.9} \times 2x - \frac{2}{3}$ oe | <b>B2</b> | accept unsimplified;<br>if incorrect allow B1 for correct<br>differentiation of $(300^2 + x^2)^{\pm\frac{1}{2}}$ |
|        | set <i>their</i> $\frac{dT}{dx} = 0$                                                              | <b>M1</b> | $\frac{dT}{dx}$ must be a function of $x$                                                                        |
|        | $25x^2 = 9(300^2 + x^2)$ oe                                                                       | <b>A1</b> | equation in $x^2$ with square root removed                                                                       |
|        | $x = 225$ (m)                                                                                     | <b>A1</b> |                                                                                                                  |
|        | $T = 533$ (s) or $1600/3$ (exact value)                                                           | <b>A1</b> | or 8 min 53 s                                                                                                    |

259



Calculator

May/June 2020 · Paper 21 · Q10



The diagram shows part of the graphs of  $y = 4x^{\frac{2}{3}}$  and  $y = (x-3)^2$ . The graph of  $y = (x-3)^2$  meets the  $x$ -axis at the point  $A(a, 0)$  and the two graphs intersect at the point  $B(b, 4)$ .

(a) Find the value of  $a$  and of  $b$ .

[2]

(b) Find the area of the shaded region.

[5]

**259** May/June 2020 · Paper 21 · Q10

|        |                                                                                                                                                     |           |                  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------|
| 14 (a) | $a = 3, b = 1$                                                                                                                                      | <b>B2</b> | B1 for each      |
| (b)    | $\int_0^{\text{their } b} 4x^{\frac{2}{3}} dx + \int_{\text{their } b}^{\text{their } a} (x-3)^2 dx$                                                | <b>M1</b> |                  |
|        | $\left[ \frac{3}{5} \times 4x^{\frac{5}{3}} \right]_0^{\text{their } b} + \left[ \frac{(x-3)^3}{3} \right]_{\text{their } b}^{\text{their } a}$ soi | <b>M2</b> | M1 for each, soi |
|        | $\frac{12}{5}(\text{their } b) - \frac{12}{5}(0) +$<br>$\frac{(\text{their } a - 3)^3}{3} - \frac{(\text{their } b - 3)^3}{3}$                      | <b>M1</b> |                  |
|        | $\frac{76}{15}$ or $5\frac{1}{15}$ or<br>5.07 or 5.06 rot to four or more figs;<br>cao                                                              | <b>A1</b> |                  |

260



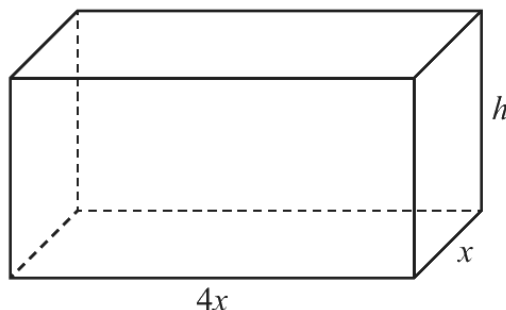
Calculator

May/June 2020 · Paper 21 · Q12

- (a) Find the  $x$ -coordinates of the stationary points of the curve  $y = e^{3x}(2x+3)^6$ . [6]

- (b) A curve has equation  $y = f(x)$  and has exactly two stationary points. Given that  $f''(x) = 4x - 7$ ,  $f'(0.5) = 0$  and  $f'(3) = 0$ , use the second derivative test to determine the nature of each of the stationary points of this curve. [2]

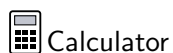
- (c) In this question all lengths are in centimetres.



The diagram shows a solid cuboid with height  $h$  and a rectangular base measuring  $4x$  by  $x$ . The volume of the cuboid is  $40 \text{ cm}^3$ . Given that  $x$  and  $h$  can vary and that the surface area of the cuboid has a minimum value, find this value. [5]

**260** May/June 2020 · Paper 21 · Q12

|        |                                                                                                               |           |                                                        |
|--------|---------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------|
| 15 (a) | $\frac{d(e^{3x})}{dx} = 3e^{3x}$                                                                              | <b>B1</b> |                                                        |
|        | $\frac{d(2x+3)^6}{dx} = k(2x+3)^5$                                                                            | <b>M1</b> |                                                        |
|        | <i>their</i> $(3e^{3x})(2x+3)^6 + (e^{3x})(\text{their } 12(2x+3)^5)$                                         | <b>M1</b> |                                                        |
|        | $(3e^{3x})(2x+3)^6 + (e^{3x})(12(2x+3)^5)$                                                                    | <b>A1</b> |                                                        |
|        | $(3e^{3x})(2x+3)^5(2x+7) = 0$                                                                                 | <b>M1</b> |                                                        |
|        | $x = -1.5, -3.5$                                                                                              | <b>A1</b> |                                                        |
| (b)    | $x = 0.5 \quad f''(0.5)[-5] < 0 \Rightarrow \text{max}$<br>$x = 3 \quad f''(3)[5] > 0 \Rightarrow \text{min}$ | <b>B2</b> | <b>B1</b> for either one correct                       |
| (c)    | $h = \frac{10}{x^2}$                                                                                          | <b>B1</b> |                                                        |
|        | $S = 8x^2 + 10x \left( \text{their } \frac{10}{x^2} \right)$                                                  | <b>B1</b> |                                                        |
|        | $\frac{dS}{dx} = 16x - 100x^{-2}$ oe                                                                          | <b>M1</b> |                                                        |
|        | $16x - 100x^{-2} = 0, \quad x = \sqrt[3]{\frac{25}{4}}$ oe                                                    | <b>A1</b> | <b>FT</b> <i>their</i> $\frac{dS}{dx} = 0$ if possible |
|        | 81.4 or 81.4325... rot to four or more figs                                                                   | <b>A1</b> |                                                        |

**261****May/June 2020 • Paper 22 • Q1**

Variables  $x$  and  $y$  are such that  $y = \sin x + e^{-x}$ . Use differentiation to find the approximate change in  $y$  as  $x$  increases from  $\frac{\pi}{4}$  to  $\frac{\pi}{4} + h$ , where  $h$  is small. [4]

**262****May/June 2020 • Paper 22 • Q6**

(a) Find the equation of the tangent to the curve  $2y = \tan 2x + 7$  at the point where  $x = \frac{\pi}{8}$ .

Give your answer in the form  $ax - y = \frac{\pi}{b} + c$ , where  $a$ ,  $b$  and  $c$  are integers. [5]

(b) This tangent intersects the  $x$ -axis at  $P$  and the  $y$ -axis at  $Q$ . Find the length of  $PQ$ . [2]

**263****May/June 2020 • Paper 22 • Q7**

Giving your answer in its simplest form, find the exact value of

(a)  $\int_0^4 \frac{10}{5x+2} dx$ , [4]

(b)  $\int_0^{\ln 2} (e^{4x+2})^2 dx$ . [5]



**261** May/June 2020 · Paper 22 · Q1

| Question | Answer                                                                    | Marks     | Partial Marks                       |
|----------|---------------------------------------------------------------------------|-----------|-------------------------------------|
| 44       | $\frac{dy}{dx} = \cos x - e^{-x}$                                         | <b>B2</b> | <b>B1</b> for $\cos x$ or $-e^{-x}$ |
|          | $\delta y = \text{their } \frac{dy}{dx} \Big _{x=\frac{\pi}{4}} \times h$ | <b>M1</b> |                                     |
|          | $0.251h$                                                                  | <b>A1</b> |                                     |

**262** May/June 2020 · Paper 22 · Q6

|        |                                                                                 |           |                                         |
|--------|---------------------------------------------------------------------------------|-----------|-----------------------------------------|
| 45 (a) | $\frac{dy}{dx} = \sec^2 2x$                                                     | <b>B1</b> |                                         |
|        | $\text{their } \frac{dy}{dx} \Big _{x=\frac{\pi}{8}} = \text{their } 2$         | <b>B1</b> | <b>FT</b> $\text{their } \frac{dy}{dx}$ |
|        | $x = \frac{\pi}{8}, \quad y = 4$                                                | <b>B1</b> |                                         |
|        | $y - \text{their } 4 = (\text{their } 2) \left( x - \frac{\pi}{8} \right)$ oe   | <b>M1</b> |                                         |
|        | $2x - y = \frac{\pi}{4} - 4$                                                    | <b>A1</b> |                                         |
| (b)    | $\sqrt{\left(\frac{\pi}{8} - 2\right)^2 + \left(4 - \frac{\pi}{4}\right)^2}$ oe | <b>M1</b> |                                         |
|        | 3.59 or 3.59[03...] rot to four or more figs                                    | <b>A1</b> |                                         |

**263** May/June 2020 · Paper 22 · Q7

|        |                                                 |           |                                                  |
|--------|-------------------------------------------------|-----------|--------------------------------------------------|
| 46 (a) | $2\ln(5x + 2)$                                  | <b>B2</b> | <b>B1</b> for $k\ln(5x + 2)$                     |
|        | $2(\ln(22) - \ln(2))$ oe soi                    | <b>M1</b> |                                                  |
|        | $2\ln 11$ or $\ln 121$ or $\ln 11^2$            | <b>A1</b> |                                                  |
| (b)    | $\int e^{8x+4} dx$                              | <b>M1</b> |                                                  |
|        | $\left[ \frac{1}{8} e^{8x+4} \right]_0^{\ln 2}$ | <b>M1</b> |                                                  |
|        | $\frac{1}{8}(e^{\ln 2^8} \times e^4 - e^4)$ oe  | <b>M2</b> | <b>M1</b> for $\frac{1}{8}(e^{\ln 2^8+4} - e^4)$ |
|        | $\frac{255}{8}e^4$ or exact equivalent          | <b>A1</b> |                                                  |

**264** Calculator**May/June 2020 • Paper 23 • Q8**

(a) Differentiate  $y = \tan(x+4) - 3 \sin x$  with respect to  $x$ .

[2]

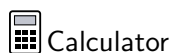
(b) Variables  $x$  and  $y$  are such that  $y = \frac{\ln(2x+5)}{2e^{3x}}$ . Use differentiation to find the approximate change in  $y$  as  $x$  increases from 1 to  $1+h$ , where  $h$  is small.

[6]

**264** May/June 2020 · Paper 23 · Q8

|        |                                                                                                                           |           |                                                                 |
|--------|---------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------|
| 65 (a) | $\sec^2(x+4) - 3\cos x$                                                                                                   | <b>B2</b> | <b>B1</b> for each                                              |
| (b)    | $\frac{d(\ln(2x+5))}{dx} = \frac{2}{2x+5}$                                                                                | <b>B1</b> |                                                                 |
|        | $\frac{d(2e^{3x})}{dx} = 6e^{3x}$                                                                                         | <b>B1</b> |                                                                 |
|        | $\frac{dy}{dx} =$<br>$\frac{2e^{3x}\left(\text{their } \frac{2}{2x+5}\right) - \text{their } 6e^{3x} \ln(2x+5)}{4e^{6x}}$ | <b>M1</b> | <b>FT</b> <i>their</i> derivatives of $\ln(2x+5)$ and $2e^{3x}$ |
|        | $\frac{dy}{dx} =$<br>$\frac{2e^{3x}\left(\frac{2}{2x+5}\right) - 6e^{3x} \ln(2x+5)}{4e^{6x}}$                             | <b>A1</b> |                                                                 |
|        | $\delta y = \text{their } \frac{dy}{dx} \Big _{x=1} \times h$                                                             | <b>M1</b> |                                                                 |
|        | $-0.138h$                                                                                                                 | <b>A1</b> |                                                                 |

265

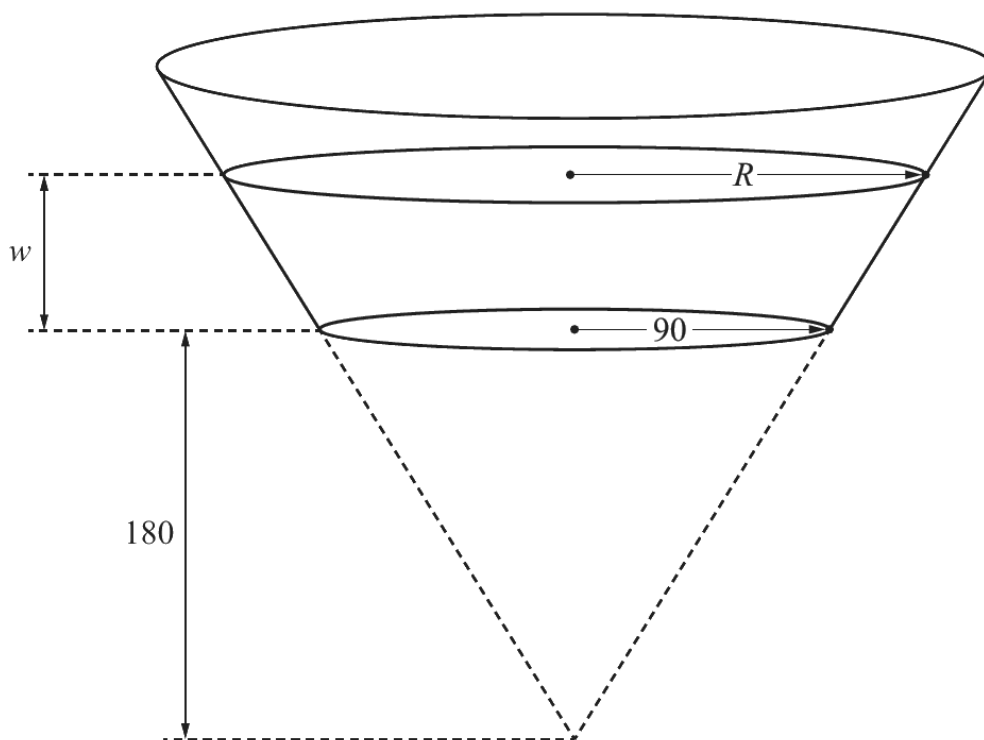


Calculator

May/June 2020 • Paper 23 • Q11

In this question all lengths are in centimetres.

The volume,  $V$ , of a cone of height  $h$  and base radius  $r$  is given by  $V = \frac{1}{3}\pi r^2 h$ .



The diagram shows a large hollow cone from which a smaller cone of height 180 and base radius 90 has been removed. The remainder has been fitted with a circular base of radius 90 to form a container for water. The depth of water in the container is  $w$  and the surface of the water is a circle of radius  $R$ .

- (a) Find an expression for  $R$  in terms of  $w$  and show that the volume  $V$  of the water in the container is given by  $V = \frac{\pi}{12}(w + 180)^3 - 486000\pi$ . [3]

- (b) Water is poured into the container at a rate of  $10\,000\text{ cm}^3\text{s}^{-1}$ . Find the rate at which the depth of the water is increasing when  $w = 10$ . [4]

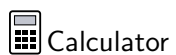
**265** May/June 2020 · Paper 23 · Q11

|        |                                                                                  |           |  |
|--------|----------------------------------------------------------------------------------|-----------|--|
| 66 (a) | $R = \frac{1}{2}(w+180)$                                                         | <b>B1</b> |  |
|        | $V = \frac{1}{3}\pi(\text{their } R)^2(w+180)$<br>$-\frac{1}{3}\pi(90)^2(180)$   | <b>M1</b> |  |
|        | Correct completion to given answer:<br>$V = \frac{\pi}{12}(w+180)^3 - 486000\pi$ | <b>A1</b> |  |

|     |                                                                                                 |           |  |
|-----|-------------------------------------------------------------------------------------------------|-----------|--|
| (b) | $\frac{dV}{dw} = 3\frac{\pi}{12}(w+180)^2$ oe                                                   | <b>B1</b> |  |
|     | $\frac{dw}{dt} = \frac{dw}{dV} \times \frac{dV}{dt}$ soi                                        | <b>M1</b> |  |
|     | $\frac{dw}{dt} = \frac{1}{\text{their } \left(\frac{dV}{dw}\right)\bigg _{w=10}} \times 10000$  | <b>M1</b> |  |
|     | 0.353 [cms <sup>-1</sup> ]<br>or 0.3526[97...] [cms <sup>-1</sup> ] rot to four or more<br>figs | <b>A1</b> |  |

266



May/June 2020 · Paper 23 · Q12

(a) (i) Given that  $f(x) = \frac{1}{\cos x}$ , show that  $f'(x) = \tan x \sec x$ . [3]

(ii) Hence find  $\int (3 \tan x \sec x - \sqrt[4]{e^{3x}}) dx$ . [3]

(b) Given that  $\int_2^5 \frac{p}{px+10} dx = \ln 2$ , find the value of the positive constant  $p$ . [5]

**266** May/June 2020 · Paper 23 · Q12

|           |                                                                                         |           |                                               |
|-----------|-----------------------------------------------------------------------------------------|-----------|-----------------------------------------------|
| 67 (a)(i) | $\frac{-(-\sin x)}{\cos^2 x}$ oe                                                        | <b>B2</b> | <b>B1</b> for $\frac{-\sin x}{\cos^2 x}$ oe   |
|           | Correct completion to given answer:<br>$\tan x \sec x$                                  | <b>B1</b> | dep on all previous marks having been awarded |
| (a)(ii)   | $\sqrt[4]{e^{3x}} = e^{\frac{3x}{4}}$ oe                                                | <b>B1</b> |                                               |
|           | $\frac{3}{\cos x} - \int e^{\frac{3x}{4}} dx = \frac{3}{\cos x} - ke^{\frac{3x}{4}}$ oe | <b>M1</b> |                                               |
|           | $\frac{3}{\cos x} - \frac{4}{3}e^{\frac{3x}{4}} + c$ oe                                 | <b>A1</b> |                                               |
| (b)       | $[\ln(px+10)]_2^5 = \ln 2$                                                              | <b>M1</b> |                                               |
|           | $\ln(5p+10) - \ln(2p+10) = \ln 2$                                                       | <b>M1</b> |                                               |
|           | $\ln\left(\frac{5p+10}{2p+10}\right) = \ln 2$                                           | <b>M1</b> |                                               |
|           | $5p+10 = 2(2p+10)$                                                                      | <b>M1</b> |                                               |
|           | $p = 10$                                                                                | <b>A1</b> |                                               |



**267**

Calculator

**Oct/Nov 2019 • Paper 21 • Q5**

A particle is moving in a straight line such that  $t$  seconds after passing a fixed point  $O$  its displacement,  $s$  m, is given by  $s = 3 \sin 2t + 4 \cos 2t - 4$ .

(i) Find expressions for the velocity and acceleration of the particle at time  $t$ . [3]

(ii) Find the first time when the particle is instantaneously at rest. [3]

(iii) Find the acceleration of the particle at the time found in **part (ii)**. [2]

**267** Oct/Nov 2019 · Paper 21 · Q5

|        |                                             |           |                                       |
|--------|---------------------------------------------|-----------|---------------------------------------|
| 26 (i) | Differentiate                               | <b>M1</b> | Obtain $2\cos 2t$ or $-2\sin 2t$      |
|        | $v = 6\cos 2t - 8\sin 2t$                   | <b>A1</b> |                                       |
|        | $a = -12\sin 2t - 16\cos 2t$                | <b>A1</b> |                                       |
| (ii)   | Equate $v$ to 0 and attempt to solve        | <b>M1</b> |                                       |
|        | $\tan 2t = 0.75$                            | <b>A1</b> | or $\sin 2t = 0.6$ or $\cos 2t = 0.8$ |
|        | $t = 0.32(2)$                               | <b>A1</b> | Must be in radians                    |
| (iii)  | Insert value of $t$ into expression for $a$ | <b>M1</b> | Radians or degrees                    |
|        | $a = -20$                                   | <b>A1</b> | Must have used radians                |

**268** Calculator**Oct/Nov 2019 • Paper 21 • Q8**

The equation of a curve is given by  $y = xe^{-2x}$ .

(i) Find  $\frac{dy}{dx}$ . [3]

(ii) Find the exact coordinates of the stationary point on the curve  $y = xe^{-2x}$ . [2]

(iii) Find, in terms of  $e$ , the equation of the tangent to the curve  $y = xe^{-2x}$  at the point  $\left(1, \frac{1}{e^2}\right)$ . [2]

(iv) Using your answer to **part (i)**, find  $\int xe^{-2x} dx$ . [3]

**269** Calculator**Oct/Nov 2019 • Paper 22 • Q2**

Given that  $y = 2 \sin 3x + \cos 3x$ , show that  $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 3y = k \sin 3x$ , where  $k$  is a constant to be determined. [5]

**268** Oct/Nov 2019 · Paper 21 · Q8

|        |                                                                                    |           |                     |
|--------|------------------------------------------------------------------------------------|-----------|---------------------|
| 27 (i) | $-2e^{-2x}$ seen                                                                   | <b>B1</b> |                     |
|        | Product rule                                                                       | <b>M1</b> | Clear attempt       |
|        | $e^{-2x}(1-2x)$                                                                    | <b>A1</b> |                     |
| (ii)   | Set $\frac{dy}{dx} = 0$ and attempt to solve                                       | <b>M1</b> | Must have two terms |
|        | $\left(\frac{1}{2}, \frac{1}{2e}\right)$                                           | <b>A1</b> |                     |
| (iii)  | Attempt to find $\frac{dy}{dx}$ at $x=1$                                           | <b>M1</b> |                     |
|        | $y - \frac{1}{e^2} = \frac{-1}{e^2}(x-1)$ or $y = -\frac{1}{e^2}x + \frac{2}{e^2}$ | <b>A1</b> |                     |
| (iv)   | Integrate <b>part(i)</b><br>$xe^{-2x} = \int (-2xe^{-2x} + e^{-2x}) dx$            | <b>M1</b> |                     |
|        | Integrate $e^{-2x}$ and make $\int xe^{-2x} dx$ the subject                        | <b>M1</b> |                     |
|        | $\frac{-xe^{-2x}}{2} - \frac{e^{-2x}}{4} + c$                                      | <b>A1</b> |                     |

**269** Oct/Nov 2019 · Paper 22 · Q2

|    |                                               |           |                                                                                                            |
|----|-----------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------|
| 54 | $\frac{dy}{dx} = 6 \cos 3x$                   | <b>B1</b> |                                                                                                            |
|    | $-3 \sin 3x$                                  | <b>B1</b> |                                                                                                            |
|    | $\frac{d^2y}{dx^2} = -18 \sin 3x - 9 \cos 3x$ | <b>B1</b> | <b>FT</b> Correct derivative of <i>their</i> $\frac{dy}{dx}$                                               |
|    | Insert and collect like terms                 | <b>M1</b> | Must insert for $y$ , <i>their</i> $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ correctly resulting in 6 terms. |
|    | $k = -15$                                     | <b>A1</b> | Allow $-15 \sin 3x$ seen nfw                                                                               |

**270**

Calculator

**Oct/Nov 2019 • Paper 22 • Q5**

At the point where  $x = 1$  on the curve  $y = \frac{k}{(x+1)^2}$ , the normal has a gradient of  $\frac{1}{3}$ .

(i) Find the value of the constant  $k$ .

[4]

(ii) Using your value of  $k$ , find the equation of the tangent to the curve at  $x = 2$ .

[3]

**270** Oct/Nov 2019 · Paper 22 · Q5

|        |                                                                                      |           |                                                                                                                 |
|--------|--------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------|
| 55 (i) | $\frac{dy}{dx} = \frac{-2k}{(x+1)^3}$                                                | <b>B1</b> | oe Unsimplified                                                                                                 |
|        | Gradient of normal = $\frac{(x+1)^3}{2k}$<br>or<br>Gradient of tangent = $-3$        | <b>M1</b> | Gradient of normal =<br>$\frac{-1}{\text{gradient of tangent}}$                                                 |
|        | $\frac{8}{2k} = \frac{1}{3}$ or $\frac{2k}{8} = -3$                                  | <b>M1</b> | Equate gradient of normal to $\frac{1}{3}$ at<br>$x = 1$<br>or equate gradient of tangent to $-3$ at<br>$x = 1$ |
|        | $k = 12$                                                                             | <b>A1</b> |                                                                                                                 |
| (ii)   | $x = 2 \rightarrow \frac{dy}{dx} = -\frac{8}{9}$ or <i>their</i> $\frac{-2k}{27}$    | <b>B1</b> | <b>FT</b>                                                                                                       |
|        | $y = \frac{4}{3}$ or <i>their</i> $\frac{k}{9}$                                      | <b>B1</b> | <b>FT</b>                                                                                                       |
|        | $\frac{y - \frac{4}{3}}{x - 2} = -\frac{8}{9}$ or $y = -\frac{8}{9}x + \frac{28}{9}$ | <b>B1</b> | isw                                                                                                             |

271



Calculator

Oct/Nov 2019 · Paper 22 · Q10

(i) Given that  $y = \frac{\ln x}{x^2}$ , find  $\frac{dy}{dx}$ .

[3]

(ii) Find the coordinates of the stationary point on the curve  $y = \frac{\ln x}{x^2}$ .

[3]

(iii) Using your answer to **part (i)**, find  $\int \frac{\ln x}{x^3} dx$ .

[3]

(iv) Hence evaluate  $\int_1^2 \frac{\ln x}{x^3} dx$ .

[2]

271 Oct/Nov 2019 · Paper 22 · Q10

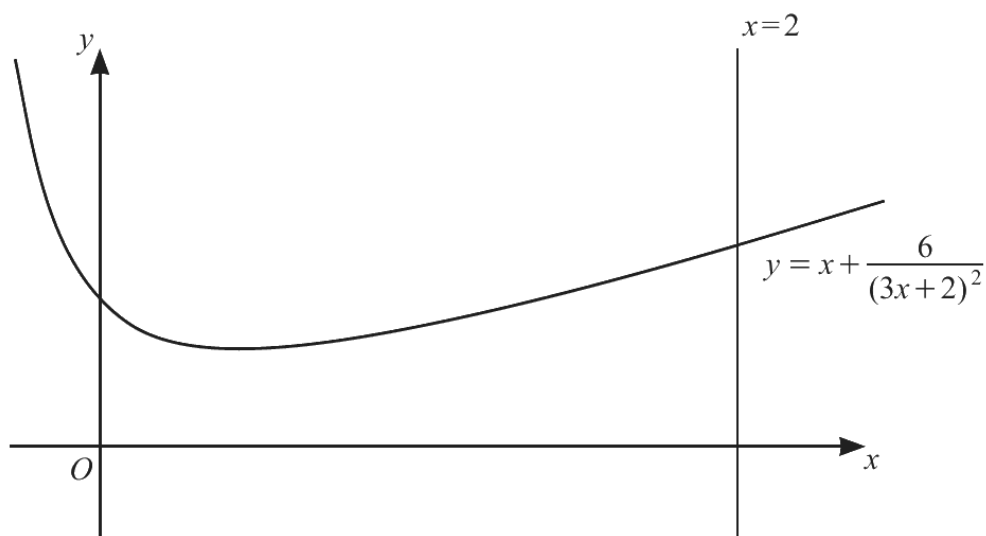
|        |                                                                          |               |                                                                                                                             |
|--------|--------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| 56 (i) | Quotient rule or product rule                                            | <b>M1</b>     |                                                                                                                             |
|        | $\frac{x - 2x \ln x}{x^4}$ or $\frac{x - \ln x \cdot 2x}{x^4}$ oe isw    | <b>A2/1/0</b> | Minus one each error. Allow unsimplified.                                                                                   |
| (ii)   | $x - 2x \ln x = 0$                                                       | <b>M1</b>     | Set $\frac{dy}{dx} = 0$ and attempt to solve. Must have two terms and obtain $\ln x = k$ only.                              |
|        | $x = 1.65$ awrt or $\sqrt{e}$                                            | <b>A1</b>     |                                                                                                                             |
|        | $y = 0.184$ awrt or $\frac{1}{2e}$                                       | <b>A1</b>     |                                                                                                                             |
| (iii)  | $\frac{\ln x}{x^2} = \int \frac{1}{x^3} - \frac{2 \ln x}{x^3} dx$        | <b>M1</b>     | Integrate <i>their</i> derivative from (i) which must have two terms. Condone omission of $dx$ .                            |
|        | $\frac{-1}{2x^2}$                                                        | <b>A1</b>     | Find $\int \frac{1}{x^3} dx$                                                                                                |
|        | $\int \frac{\ln x}{x^3} dx = -\frac{1}{4x^2} - \frac{\ln x}{2x^2} + (C)$ | <b>A1</b>     | oe Rearrange and complete                                                                                                   |
| (iv)   | Insert limits and subtract correctly                                     | <b>M1</b>     | <b>dep</b><br>Must be inserting into <b>two</b> terms in $x$ from (iii). Values explicitly seen if expression is incorrect. |
|        | $\frac{3}{16} - \frac{\ln 2}{8}$ or 0.101 awrt                           | <b>A1</b>     |                                                                                                                             |



272

 Calculator

Oct/Nov 2019 • Paper 23 • Q7



The diagram shows part of the curve  $y = x + \frac{6}{(3x+2)^2}$  and the line  $x = 2$ .

(i) Find, correct to 2 decimal places, the coordinates of the stationary point.

[6]

(ii) Find the area of the shaded region, showing all your working.

[4]

**272** Oct/Nov 2019 · Paper 23 · Q7

|        |                                                                     |           |                                                                                                            |
|--------|---------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------|
| 76 (i) | evidence of differentiation $(3x + 2)^{-3}$                         | <b>M1</b> |                                                                                                            |
|        | $-12(3x + 2)^{-3} \times 3$                                         | <b>A1</b> | may use PR or QR on fraction part                                                                          |
|        | +1                                                                  | <b>B1</b> |                                                                                                            |
|        | set <i>their</i> $\frac{dy}{dx} = 0$                                | <b>M1</b> | $1 - 36(3x + 2)^{-3} = 0$                                                                                  |
|        | $x = 0.43$ nfw                                                      | <b>A1</b> |                                                                                                            |
|        | $y = 0.98$ only                                                     | <b>A1</b> |                                                                                                            |
| (ii)   | $\frac{-2}{3x + 2}$ oe                                              | <b>B1</b> |                                                                                                            |
|        | $\frac{1}{2}x^2$                                                    | <b>B1</b> |                                                                                                            |
|        | $\left[ \frac{-2}{6 + 2} + 2 \right] - \left[ \frac{-2}{2} \right]$ | <b>M1</b> | insert correct limits into <i>their</i> two term integral and subtract two non-zero terms in correct order |
|        | 2.75 nfw                                                            | <b>A1</b> | 2.75 following B1 B1 implies M1                                                                            |

273



Calculator

Oct/Nov 2019 • Paper 23 • Q8

Given that a curve has equation  $y = x^3 - x^2 + bx + 24$  find, using your value of  $b$ , Note:  $b = -14$

(iii)  $\frac{dy}{dx}$ , [1]

(iv) the integer value of  $x$  for which the gradient of the curve is 2 and the corresponding value of  $y$ . [3]

The coordinates of the point  $P$  on the curve are given by the values of  $x$  and  $y$  found in **part (iv)**.

(v) Find the equation of the tangent to the curve at  $P$ . [1]

274



Calculator

May/June 2019 • Paper 21 • Q2

Two variables  $x$  and  $y$  are such that  $y = \frac{\ln x}{x^3}$  for  $x > 0$ .

(i) Show that  $\frac{dy}{dx} = \frac{1 - 3 \ln x}{x^4}$ . [3]

(ii) Hence find the approximate change in  $y$  as  $x$  increases from  $e$  to  $e + h$ , where  $h$  is small. [2]

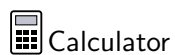
**273** Oct/Nov 2019 · Paper 23 · Q8

|          |                                             |           |                                                      |
|----------|---------------------------------------------|-----------|------------------------------------------------------|
| 77 (iii) | $\frac{dy}{dx} = 3x^2 - 2x - 14$            | <b>B1</b> | <b>FT</b> <i>their</i> numerical $b$ $3x^2 - 2x + b$ |
| (iv)     | set <i>their</i> $\frac{dy}{dx}$ equal to 2 | <b>M1</b> | <b>FT</b> <i>their</i> numerical $b$                 |
|          | $x = 2$                                     | <b>A1</b> |                                                      |
|          | $y = 40$ only                               | <b>A1</b> | no additional answers                                |
| (v)      | $y - 40 = 2(x + 2)$ ( $y = 2x + 44$ )       | <b>B1</b> |                                                      |

**274** May/June 2019 · Paper 21 · Q2

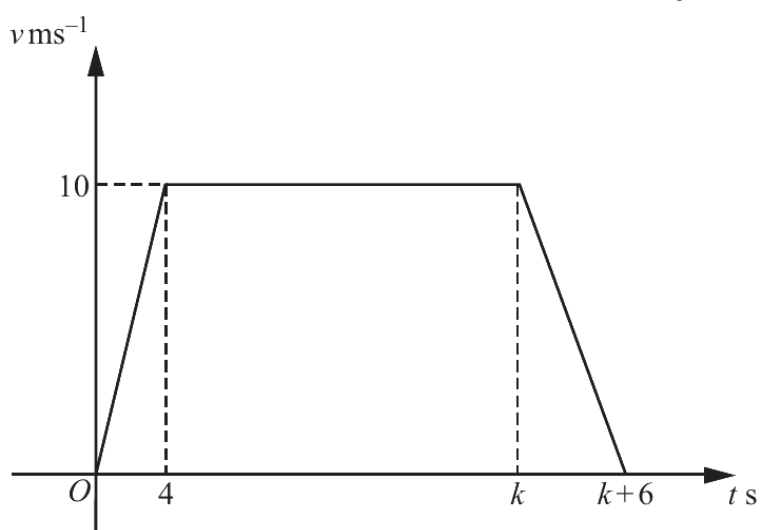
|        |                                                                                                                                         |           |  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|--|
| 10 (i) | $\frac{d}{dx}(\ln x) = \frac{1}{x}$ soi                                                                                                 | <b>B1</b> |  |
|        | $\frac{dy}{dx} = \frac{x^3 \left( \frac{1}{x} \right) - 3x^2 \ln x}{(x^3)^2}$ or $x^{-3} \left( \frac{1}{x} \right) + (-3x^{-4}) \ln x$ | <b>M1</b> |  |
|        | Completion to given answer:<br>$\frac{dy}{dx} = \frac{1 - 3 \ln x}{x^4}$                                                                | <b>A1</b> |  |
| (ii)   | $\left( \frac{1 - 3 \ln e}{e^4} \right) h$                                                                                              | <b>M1</b> |  |
|        | $-\frac{2h}{e^4}$ oe or $-0.0366h$ awrt                                                                                                 | <b>A1</b> |  |

275



Calculator

May/June 2019 · Paper 21 · Q5

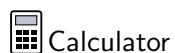


The velocity-time graph represents the motion of a particle travelling in a straight line.

- (i) Find the acceleration during the last 6 seconds of the motion. [1]
- (ii) The particle travels with constant velocity for 23 seconds. Find the value of  $k$ . [1]
- (iii) Using your answer to **part (ii)**, find the total distance travelled by the particle. [3]

**275** May/June 2019 · Paper 21 · Q5

|        |                                                                                                                                          |           |  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------|-----------|--|
| 11 (i) | $-\frac{10}{6}$ oe                                                                                                                       | <b>B1</b> |  |
| (ii)   | 27                                                                                                                                       | <b>B1</b> |  |
| (iii)  | Attempts to find total area                                                                                                              | <b>M1</b> |  |
|        | $\frac{1}{2}(23 + \text{their } k + 6) \times 10$<br>or $\frac{1}{2} \times 4 \times 10 + 23 \times 10 + \frac{1}{2} \times 6 \times 10$ | <b>M1</b> |  |
|        | 280                                                                                                                                      | <b>A1</b> |  |

**276****May/June 2019 • Paper 21 • Q7**

The variables  $x$ ,  $y$  and  $u$  are such that  $y = \tan u$  and  $x = u^3 + 1$ .

(i) State the rate of change of  $y$  with respect to  $u$ . [1]

(ii) Hence find the rate of change of  $y$  with respect to  $x$ , giving your answer in terms of  $x$ . [4]

**277****May/June 2019 • Paper 21 • Q12**

(a) Given that  $\int_0^a e^{2x} dx = 50$ , find the exact value of  $a$ . You must show all your working. [4]

(b) A curve is such that  $\frac{dy}{dx} = 3 - 2 \cos 5x$ . The curve passes through the point  $\left(\frac{\pi}{5}, \frac{8\pi}{5}\right)$ .

(i) Find the equation of the curve. [4]

(ii) Find  $\int y dx$  and hence evaluate  $\int_{\frac{\pi}{2}}^{\pi} y dx$ . [5]

**276** May/June 2019 · Paper 21 · Q7

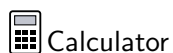
| Question | Answer                                                                                                                 | Marks     | Partial Marks                                                                                              |
|----------|------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------|
| 12 (i)   | $\sec^2 u$                                                                                                             | <b>B1</b> |                                                                                                            |
| (ii)     | Attempts $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$<br>or $\frac{dy}{dx} = \frac{dy}{du} \div \frac{dx}{du}$ | <b>M1</b> |                                                                                                            |
|          | $\frac{dy}{dx} = \frac{\text{their } \sec^2 u}{3u^2}$                                                                  | <b>A1</b> | <b>FT</b> <i>their</i> (i)                                                                                 |
|          | $u = \sqrt[3]{x-1}$ soi                                                                                                | <b>B1</b> |                                                                                                            |
|          | $\frac{\sec^2(\sqrt[3]{x-1})}{3(\sqrt[3]{x-1})^2}$ cao                                                                 | <b>A1</b> | final answer<br>If <b>B1</b> only then <b>SC1</b> for<br>$k(x-1)^{-\frac{2}{3}} \sec^2(x-1)^{\frac{1}{3}}$ |

**277** May/June 2019 · Paper 21 · Q12

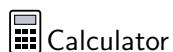
|        |                                                                                             |           |                                                        |
|--------|---------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------|
| 13 (a) | $\frac{e^{2x}}{2}$ seen                                                                     | <b>B1</b> |                                                        |
|        | $\frac{e^{2a}}{2} - \frac{1}{2} = 50$                                                       | <b>M1</b> | Uses limits correctly for their integral and sets = 50 |
|        | Rearranges and takes logs to base e:<br>$2a = \ln 101$ oe                                   | <b>M1</b> | Using <i>their</i> integral                            |
|        | $a = \frac{1}{2} \ln 101$ or $\ln \sqrt{101}$ final answer                                  | <b>A1</b> | Allow any exact equivalent                             |
| (b)(i) | $[y =] 3x - \frac{2}{5} \sin 5x [+c]$                                                       | <b>B2</b> | <b>B1</b> for $-k \sin 5x$ where $k > 0$               |
|        | $\frac{8\pi}{5} = \frac{3\pi}{5} - \frac{2}{5} \sin\left(5 \times \frac{\pi}{5}\right) + c$ | <b>M1</b> |                                                        |
|        | $y = 3x - \frac{2}{5} \sin 5x + \pi$                                                        | <b>A1</b> |                                                        |



|         |                                                                                                                                              |           |                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------|
| (b)(ii) | $\left[ \int y \, dx = \int \left( 3x - \frac{2}{5} \sin 5x + \pi \right) dx \right]$ $= \frac{3x^2}{2} + \frac{2}{25} \cos 5x + \pi x [+c]$ | <b>B3</b> | <b>B2</b> for $\frac{2}{25} \cos 5x$ oe nfw<br>and <b>B1FT</b> for $\frac{3x^2}{2} + \dots + \pi x [+c]$ |
|         | <i>their</i> $F(\pi) - \text{their } F\left(\frac{\pi}{2}\right)$                                                                            | <b>M1</b> |                                                                                                          |
|         | 16[.0] or 15.95 to 15.96 or<br>$\frac{13\pi^2}{8} - \frac{2}{25}$                                                                            | <b>A1</b> |                                                                                                          |

**278****May/June 2019 • Paper 22 • Q1**

Given that  $y = \frac{\sin x}{\ln x^2}$ , find an expression for  $\frac{dy}{dx}$ . [4]

**279****May/June 2019 • Paper 22 • Q4**

A circle has diameter  $x$  which is increasing at a constant rate of  $0.01 \text{ cm s}^{-1}$ . Find the exact rate of change of the area of the circle when  $x = 6 \text{ cm}$ . [5]

**280****May/June 2019 • Paper 22 • Q7**

(a) Solve  $\lg(x^2 - 3) = 0$ . [2]

(b) (i) Show that, for  $a > 0$ ,  $\frac{\ln a^{\sin(2x+5)} + \ln\left(\frac{1}{a}\right)}{\ln a}$  may be written as  $\sin(2x+5) + k$ , where  $k$  is an integer. [3]

(ii) Hence find  $\int \frac{\ln a^{\sin(2x+5)} + \ln\left(\frac{1}{a}\right)}{\ln a} dx$ . [3]

278 May/June 2019 · Paper 22 · Q1

| Question | Answer                                                                                                                     | Marks     | Partial Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------|-----------|---------------|
| 40       | <b>Either:</b><br>Quotient rule:<br>$\frac{d}{dx}(\sin x) = \cos x$ soi                                                    | <b>B1</b> |               |
|          | $\frac{d}{dx}(\ln x^2) = \frac{2}{x}$ oe soi                                                                               | <b>B1</b> |               |
|          | $\frac{(\ln x^2)(\text{their } \cos x) - (\sin x)\left(\text{their } \frac{2}{x}\right)}{(\ln x^2)^2}$<br>oe               | <b>M1</b> |               |
|          | $\frac{(\ln x^2)\cos x - (\sin x)\left(\frac{2}{x}\right)}{(\ln x^2)^2}$ oe isw                                            | <b>A1</b> |               |
|          | <b>Or:</b><br>Product rule on $y = (\sin x)(\ln x^2)^{-1}$<br>$\frac{d}{dx}(\sin x) = \cos x$ soi                          | <b>B1</b> |               |
|          | $\frac{d}{dx}((\ln x^2)^{-1}) = -(\ln x^2)^{-2} \times \frac{2}{x}$                                                        | <b>B1</b> |               |
|          | $(\sin x) \times \text{their} \left( -(\ln x^2)^{-2} \times \frac{2}{x} \right) + (\text{their } \cos x)(\ln x^2)^{-1}$ oe | <b>M1</b> |               |
|          | $(\sin x) \times \left( -(\ln x^2)^{-2} \times \frac{2}{x} \right) + (\cos x)(\ln x^2)^{-1}$ oe isw                        | <b>A1</b> |               |

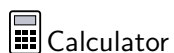
**279** May/June 2019 · Paper 22 · Q4

|    |                                                       |           |                                              |
|----|-------------------------------------------------------|-----------|----------------------------------------------|
| 41 | <b>Either:</b><br>$A = \frac{1}{4}\pi x^2$ oe, soi    | <b>B1</b> |                                              |
|    | $\frac{dA}{dx} = \frac{2}{4}\pi x$ oe, soi            | <b>B1</b> |                                              |
|    | $\frac{dx}{dt} = 0.01$ soi                            | <b>B1</b> |                                              |
|    | $\frac{2}{4}\pi(6) \times 0.01$                       | <b>M1</b> | <b>FT</b> their $\frac{dA}{dx}$ when $x = 6$ |
|    | 0.03 $\pi$ or exact equivalent                        | <b>A1</b> | mark final answer                            |
|    | <b>Or:</b><br>$A = \pi r^2$ and $r = \frac{x}{2}$ soi | <b>B1</b> |                                              |
|    | $\frac{dA}{dr} = 2\pi r$ oe soi                       | <b>B1</b> |                                              |
|    | $\frac{dr}{dt} = 0.005$ oe soi                        | <b>B1</b> |                                              |
|    | $2\pi(3) \times 0.005$                                | <b>M1</b> |                                              |
|    | 0.03 $\pi$ or exact equivalent                        | <b>A1</b> |                                              |

**280** May/June 2019 · Paper 22 · Q7

|        |                                                                                                                           |           |                                                                                                                              |
|--------|---------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------|
| 42 (a) | $\lg(x^2 - 3) = \lg 1$ soi                                                                                                | <b>M1</b> |                                                                                                                              |
|        | -2 and 2                                                                                                                  | <b>A1</b> | Implies <b>M1</b>                                                                                                            |
| (b)(i) | <b>Two separate terms in numerator:</b><br>$(\sin(2x + 5)) \ln a$ or<br>$\log_a a^{\sin(2x+5)} = (\sin(2x + 5)) \log_a a$ | <b>B1</b> | <b>Combines terms in numerator:</b><br>or $\ln\left(\frac{a^{\sin(2x+5)}}{a}\right)$                                         |
|        | $-\ln a$ or<br>$\log_a a^{-1} = -\log_a a$                                                                                | <b>B1</b> | or $(\sin(2x + 5) - 1) \ln a$<br><br>or $\frac{\ln\left(\frac{a^{\sin(2x+5)}}{a}\right)}{\ln a} = \log_a (a^{\sin(2x+5)-1})$ |
|        | $\sin(2x + 5) - 1$                                                                                                        | <b>B1</b> | <b>dep</b> all previous marks awarded;                                                                                       |

|         |                                             |                                                                                                                                                                                                                                                                                                          |
|---------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (b)(ii) | $-\frac{1}{2}\cos(2x+5) + (their -1)x (+c)$ | <p><b>B3</b> <b>FT</b> <i>their</i> numerical <math>k</math></p> <p><b>B2</b> for <math>-\frac{1}{2}\cos(2x+5)</math> seen</p> <p>or <b>B1</b> for <math>a\cos(2x+5)</math>, <math>a &lt; 0</math> or for <math>\frac{1}{2}\cos(2x+5)</math></p> <p>or for <math>-\frac{1}{2}\cos 2x + 5</math> seen</p> |
|---------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**281****May/June 2019 · Paper 22 · Q11**

The velocity,  $v \text{ m s}^{-1}$ , of a particle travelling in a straight line,  $t$  seconds after passing through a fixed point  $O$ , is given by  $v = \frac{4}{(t+1)^3}$ .

- (i) Explain why the direction of motion of the particle never changes. [1]
- (ii) Showing all your working, find the acceleration of the particle when  $t = 5$ . [3]
- (iii) Find an expression for the displacement of the particle from  $O$  after  $t$  seconds. [3]
- (iv) Find the distance travelled by the particle in the fourth second. [2]

**282****May/June 2019 · Paper 23 · Q2**

Differentiate  $\tan 3x \cos \frac{x}{2}$  with respect to  $x$ .

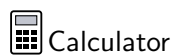
[4]

**281** May/June 2019 · Paper 22 · Q11

|        |                                                                                                                                                                                                                                       |           |                                                                                           |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|
| 43 (i) | $v \neq 0$ or $v > 0$ oe                                                                                                                                                                                                              | <b>B1</b> |                                                                                           |
| (ii)   | Differentiates :<br>$4 \times -3(t+1)^{-4}$ oe, isw                                                                                                                                                                                   | <b>B2</b> | <b>B1</b> for $k(t+1)^{-4}$ with $k \neq -12$                                             |
|        | $-\frac{1}{108}$ oe or $-0.00926$                                                                                                                                                                                                     | <b>B1</b> |                                                                                           |
| (iii)  | Integrates:<br>$[s =] -2(t+1)^{-2} + 2$ oe, isw                                                                                                                                                                                       | <b>B3</b> | <b>B2</b> for $-2(t+1)^{-2} [+c]$<br>or <b>B1</b> for $k(t+1)^{-2} [+c]$ with $k \neq -2$ |
| (iv)   | Finds <i>their</i> $s$ from (iii) when $t = 4$<br>or when $t = 3$<br><br>or finds<br>$\left[ \text{their} \left( -2(t+1)^{-2} [+2] \right) \right]_3^4$<br>$= \text{their} \left( -2(4+1)^{-2} - \left( -2(3+1)^{-2} \right) \right)$ | <b>M1</b> |                                                                                           |
|        | $\frac{9}{200}$ or $0.045$                                                                                                                                                                                                            | <b>A1</b> |                                                                                           |

**282** May/June 2019 · Paper 23 · Q2

|    |                                                                                                                                      |           |  |
|----|--------------------------------------------------------------------------------------------------------------------------------------|-----------|--|
| 61 | $\frac{d}{dx}(\tan 3x) = 3 \sec^2 3x$ soi                                                                                            | <b>B1</b> |  |
|    | $\frac{d}{dx} \left( \cos \frac{x}{2} \right) = -\frac{1}{2} \sin \frac{x}{2}$ soi                                                   | <b>B1</b> |  |
|    | Applies correct form of product rule                                                                                                 | <b>M1</b> |  |
|    | $\left[ \frac{dy}{dx} = \right]$<br>$-\frac{1}{2} \left( \sin \frac{x}{2} \right) \tan 3x + 3(\sec^2 3x) \cos \frac{x}{2}$<br>oe isw | <b>A1</b> |  |

**283****May/June 2019 • Paper 23 • Q6**

A curve has equation  $y = (3x - 5)^3 - 2x$ .

(i) Find  $\frac{dy}{dx}$  and  $\frac{d^2y}{dx^2}$ . [4]

(ii) Find the exact value of the  $x$ -coordinate of each of the stationary points of the curve. [2]

(iii) Use the second derivative test to determine the nature of each of the stationary points. [2]



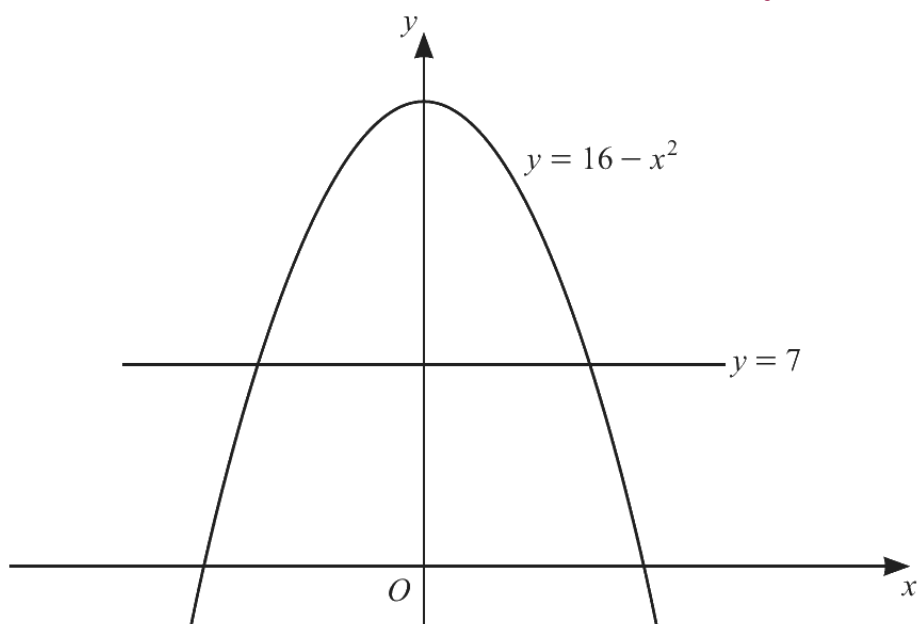
**283** May/June 2019 · Paper 23 · Q6

|        |                                                                                                                                                                                      |           |                                                |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------|
| 62 (i) | $9(3x-5)^2 - 2$ isw                                                                                                                                                                  | <b>B2</b> | <b>B1</b> for $k(3x-5)^2$ $k \neq 9$ seen      |
|        | $54(3x-5)^{[1]}$ isw                                                                                                                                                                 | <b>B2</b> | <b>B1</b> for $k(3x-5)^{[1]}$ $k \neq 54$ seen |
| (ii)   | Solves <i>their</i> $9(3x-5)^2 - 2 = 0$                                                                                                                                              | <b>M1</b> |                                                |
|        | $[x =] \frac{5}{3} \pm \frac{\sqrt{2}}{9}$ or exact equivalent                                                                                                                       | <b>A1</b> |                                                |
| (iii)  | Substitutes <i>their</i> $\frac{5}{3} + \frac{\sqrt{2}}{9}$<br>or <i>their</i> $\frac{5}{3} - \frac{\sqrt{2}}{9}$ into<br><i>their</i> $54(3x-5)^{[1]}$ and considers sign of result | <b>M1</b> |                                                |
|        | When $x = \frac{5}{3} + \frac{\sqrt{2}}{9}$ $\frac{d^2y}{dx^2} > 0$<br>so minimum<br><br>and when $x = \frac{5}{3} - \frac{\sqrt{2}}{9}$ $\frac{d^2y}{dx^2} < 0$<br>so maximum       | <b>A1</b> |                                                |

284

 Calculator

May/June 2019 · Paper 23 · Q9



The diagram shows the curve  $y = 16 - x^2$  and the straight line  $y = 7$ . Find the area of the shaded region. You must show all your working. [6]

**284** May/June 2019 · Paper 23 · Q9

| Question | Answer                                                                                    | Marks     | Partial Marks |
|----------|-------------------------------------------------------------------------------------------|-----------|---------------|
| 63       | Roots of curve: (4, 0) or (−4, 0) oe                                                      | <b>B1</b> |               |
|          | Intersections: (−3, 7) or (3, 7) oe                                                       | <b>B1</b> |               |
|          | Correct strategy for finding area                                                         | <b>B1</b> |               |
|          | $\int (16 - x^2) dx = 16x - \frac{x^3}{3}$<br>or $\int (9 - x^2) dx = 9x - \frac{x^3}{3}$ | <b>M1</b> |               |
|          | $F(b) - F(a)$                                                                             | <b>M1</b> |               |
|          | $\frac{148}{3}$ or $49\frac{1}{3}$ or 49.3[33...]                                         | <b>A1</b> |               |

**285**

Calculator

**May/June 2019 · Paper 23 · Q11**

A particle travelling in a straight line passes through a fixed point  $O$ . The displacement,  $x$  metres, of the particle,  $t$  seconds after it passes through  $O$ , is given by  $x = 5t + \sin t$ .

(i) Show that the particle is never at rest. [2]

(ii) Find the distance travelled by the particle between  $t = \frac{\pi}{3}$  and  $t = \frac{\pi}{2}$ . [2]

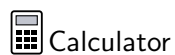
(iii) Find the acceleration of the particle when  $t = 4$ . [2]

(iv) Find the value of  $t$  when the velocity of the particle is first at its minimum. [2]

**285** May/June 2019 · Paper 23 · Q11

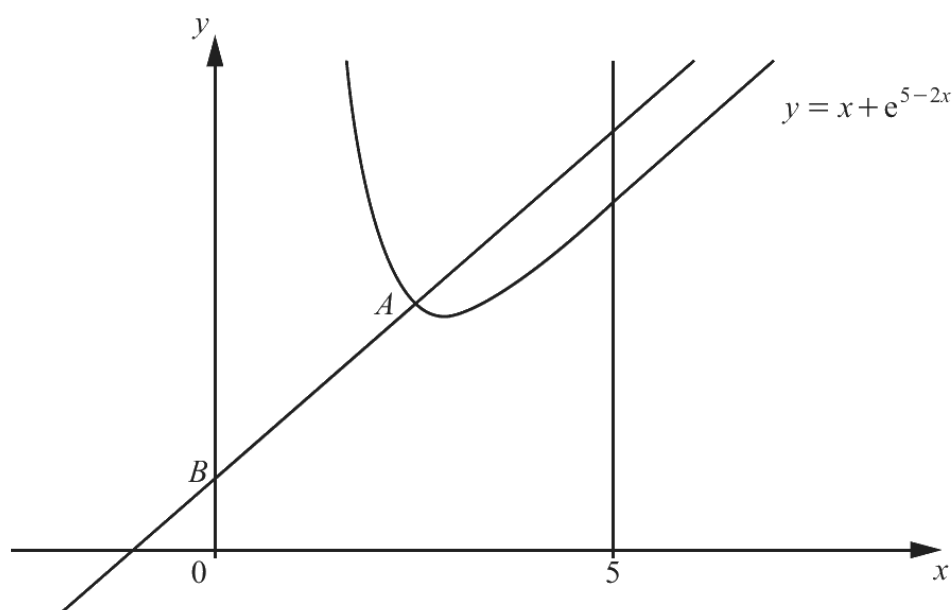
|        |                                                                                                                                                  |           |                                                                |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------|
| 64 (i) | $\left[ v = \frac{dx}{dt} = \right] 5 + \cos t$                                                                                                  | <b>B1</b> |                                                                |
|        | $5 + \cos t \neq 0$ (and so never at rest) oe                                                                                                    | <b>B1</b> |                                                                |
| (ii)   | $x = 5\left(\frac{\pi}{3}\right) + \sin\left(\frac{\pi}{3}\right)$ or<br>$x = 5\left(\frac{\pi}{2}\right) + \sin\left(\frac{\pi}{2}\right)$ seen | <b>M1</b> |                                                                |
|        | 2.75 to 2.752                                                                                                                                    | <b>A1</b> |                                                                |
| (iii)  | $\left[ a = \frac{dv}{dt} = \right] -\sin t$                                                                                                     | <b>M1</b> | <b>FT</b> <i>their</i> $v$ provided of the form $k \pm \cos t$ |
|        | $[t = 4, a = -\sin 4 =]$ 0.757 or<br>0.7568[024...]                                                                                              | <b>A1</b> |                                                                |
| (iv)   | Valid method soi e.g.<br><i>their</i> $(-\sin t) = 0$ or $\cos t = -1$<br>sketch of $v = 5 + \cos t$                                             | <b>M1</b> |                                                                |
|        | $t = \pi$                                                                                                                                        | <b>A1</b> |                                                                |

286



Calculator

Oct/Nov 2018 · Paper 21 · Q8



The diagram shows part of the curve  $y = x + e^{5-2x}$ , the normal to the curve at the point  $A$  and the line  $x = 5$ . The normal to the curve at  $A$  meets the  $y$ -axis at the point  $B$ . The  $x$ -coordinate of  $A$  is 2.5.

- (i) Find the equation of the normal  $AB$ .

[4]

- (ii) Showing all your working, find the area of the shaded region.

[6]

**286** Oct/Nov 2018 · Paper 21 · Q8

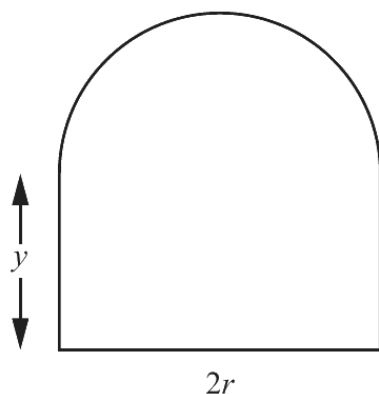
|        |                                                                   |           |                                     |
|--------|-------------------------------------------------------------------|-----------|-------------------------------------|
| 24 (i) | $\frac{dy}{dx} = 1 - 2e^{2-5x}$                                   | <b>B1</b> |                                     |
|        | $x = 2.5 \rightarrow \frac{dy}{dx} = -1$ and $y = 3.5$            | <b>B1</b> |                                     |
|        | Grad of normal = $\frac{-1}{\frac{dy}{dx}}$                       | <b>M1</b> |                                     |
|        | $y = x + 1$                                                       | <b>A1</b> | Equation of normal                  |
| (ii)   | Area of trapezium = $\frac{1}{2} \times 2.5 \times 4.5$           | <b>M1</b> |                                     |
|        | 5.625 sq units                                                    | <b>A1</b> |                                     |
|        | $\int_{2.5}^5 x + e^{(5-2x)} dx$                                  | <b>M1</b> | Area under curve                    |
|        | $= \left[ \frac{x^2}{2} - \frac{1}{2} e^{(5-2x)} \right]_{2.5}^5$ | <b>A1</b> |                                     |
|        |                                                                   | <b>M1</b> | insert limits and subtract (= 9.87) |
|        | Shaded area = 15.5                                                | <b>A1</b> | 5.625 + 9.87                        |

287

 Calculator

Oct/Nov 2018 • Paper 21 • Q9

In this question, all lengths are in metres.



The diagram shows a window formed by a semi-circle of radius  $r$  on top of a rectangle with dimensions  $2r$  by  $y$ . The total perimeter of the window is 5.

- (i) Find  $y$  in terms of  $r$ . [2]

- (ii) Show that the total area of the window is  $A = 5r - \frac{\pi r^2}{2} - 2r^2$ . [2]

- (iii) Given that  $r$  can vary, find the value of  $r$  which gives a maximum area of the window and find this area. (You are not required to show that this area is a maximum.) [5]



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|        |                                                                             |           |                                  |
|--------|-----------------------------------------------------------------------------|-----------|----------------------------------|
| 25 (i) | $2y + 2r + \pi r = 5$                                                       | <b>B1</b> |                                  |
|        | $y = \frac{5 - 2r - \pi r}{2}$                                              | <b>B1</b> | <b>Dep</b>                       |
| (ii)   | $A = 2yr + \frac{\pi r^2}{2}$                                               | <b>M1</b> |                                  |
|        | $= r(5 - 2r - \pi r) + \frac{\pi r^2}{2}$ $= 5r - 2r^2 - \frac{\pi r^2}{2}$ | <b>A1</b> |                                  |
| (iii)  |                                                                             | <b>M1</b> | differentiate                    |
|        | $\frac{dA}{dr} = 5 - \pi r - 4r$                                            | <b>A1</b> |                                  |
|        | $\frac{dA}{dr} = 0$                                                         | <b>M1</b> | set to zero and attempt to solve |
|        | $r = \frac{5}{\pi + 4} = 0.7$                                               | <b>A1</b> |                                  |
|        | $A = 1.75$                                                                  | <b>A1</b> |                                  |

**288**

Calculator

**Oct/Nov 2018 · Paper 22 · Q3**

A curve has equation  $y = \frac{x^3}{\sin 2x}$ . Find

(i)  $\frac{dy}{dx}$ , [3]

(ii) the equation of the tangent to the curve at the point where  $x = \frac{\pi}{4}$ . [3]

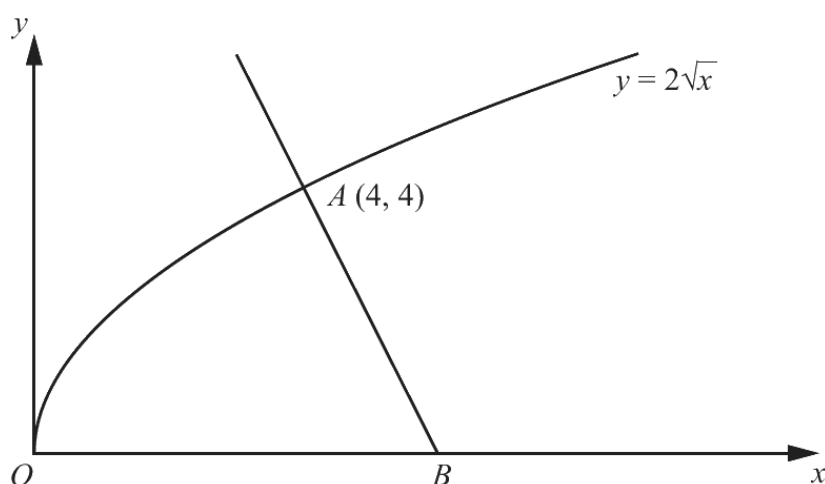
**288** Oct/Nov 2018 · Paper 22 · Q3

|        |                                                                     |           |                                                                      |
|--------|---------------------------------------------------------------------|-----------|----------------------------------------------------------------------|
| 52 (i) | $\frac{3x^2 \sin 2x - x^3 \times 2 \cos 2x}{(\sin 2x)^2}$           | <b>3</b>  | <b>M1</b> Quotient rule<br><b>A2/1/0</b> minus one each error<br>isw |
| (ii)   | $y = \frac{\pi^3}{64} \quad [=0.48\dots]$                           | <b>B1</b> |                                                                      |
|        | $\frac{dy}{dx} = \frac{3\pi^2}{16} \quad [=1.85] \text{ oe}$        | <b>B1</b> |                                                                      |
|        | $y = \frac{3\pi^2}{16}x - \frac{\pi^3}{32}$<br>$[y = 1.85x - 0.97]$ | <b>B1</b> | cao                                                                  |

289

 Calculator

Oct/Nov 2018 · Paper 22 · Q9



The diagram shows part of the curve  $y = 2\sqrt{x}$ . The normal to the curve at the point  $A(4, 4)$  meets the  $x$ -axis at the point  $B$ .

(i) Find the equation of the line  $AB$ . [4]

(ii) Find the coordinates of  $B$ . [1]

(iii) Showing all your working, find the area of the shaded region. [4]

**289** Oct/Nov 2018 · Paper 22 · Q9

|        |                                                                     |           |           |
|--------|---------------------------------------------------------------------|-----------|-----------|
| 53 (i) | $\frac{dy}{dx} = x^{-\frac{1}{2}}$                                  | <b>B1</b> |           |
|        | $x = 4 \rightarrow \frac{dy}{dx} = \frac{1}{2}$                     | <b>B1</b> |           |
|        | grad of normal = -2                                                 | <b>M1</b> |           |
|        | $\frac{y-4}{x-4} = -2 \rightarrow [y = -2x + 12]$                   | <b>A1</b> |           |
| (ii)   | (6, 0)                                                              | <b>B1</b> | <b>FT</b> |
| (iii)  | Area of triangle = $\frac{1}{2} \times 2 \times 4 = 4$              | <b>B1</b> | <b>FT</b> |
|        | Area under curve = $\int 2x^{\frac{1}{2}} dx$                       | <b>M1</b> |           |
|        | $= \frac{4}{3} x^{\frac{3}{2}}$                                     | <b>A1</b> |           |
|        | Total area = $14\frac{2}{3}$ [14.7]                                 | <b>A1</b> | <b>FT</b> |
|        | <b>OR</b>                                                           |           |           |
|        | Area of trapezium <i>OBAP</i><br>$= \frac{1}{2}(6+4) \times 4 = 20$ | <b>B1</b> | <b>FT</b> |
|        | Area between curve and y- axis<br>$= \int \frac{y^2}{4} dy$         | <b>M1</b> |           |
|        | $= \frac{y^3}{12}$                                                  | <b>A1</b> |           |
|        | Total area = $14\frac{2}{3}$ [14.7]                                 | <b>A1</b> | <b>FT</b> |

**290** Calculator**Oct/Nov 2018 • Paper 23 • Q4**

$$\frac{d^2y}{dx^2} = 2x + \frac{3}{(x+1)^4}$$

- (i) Find  $\frac{dy}{dx}$ , given that  $\frac{dy}{dx} = 1$  when  $x = 1$ . [3]

- (ii) Find  $y$  in terms of  $x$ , given that  $y = 3$  when  $x = 1$ . [3]

**291** Calculator**Oct/Nov 2018 • Paper 23 • Q10**

The equation of a curve is  $y = x^2\sqrt{3+x}$  for  $x \geq -3$ .

- (i) Find  $\frac{dy}{dx}$ . [3]

- (ii) Find the equation of the tangent to the curve  $y = x^2\sqrt{3+x}$  at the point where  $x = 1$ . [3]

- (iii) Find the coordinates of the turning points of the curve  $y = x^2\sqrt{3+x}$ . [4]

**290** Oct/Nov 2018 · Paper 23 · Q4

| Question | Answer                                                              | Marks | Partial Marks            |
|----------|---------------------------------------------------------------------|-------|--------------------------|
| 73 (i)   | integrate: increase in powers of at least one term                  | M1    | *                        |
|          | $\frac{dy}{dx} = x^2 - \frac{1}{(x+1)^3} + (C)$                     | A1    |                          |
|          | $C = \frac{1}{8}$                                                   | A1    |                          |
| (ii)     | integrate <i>their</i> (i): increase in powers of at least one term | M1    | Dep*                     |
|          | $y = \frac{1}{3}x^3 + \frac{1}{2(x+1)^2} + \frac{1}{8}x + (D)$      | A1    | two correct terms in $x$ |
|          | $D = \frac{29}{12}$                                                 | A1    |                          |

**291** Oct/Nov 2018 · Paper 23 · Q10

|        |                                                                                                                |    |                                                                                           |
|--------|----------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------------------|
| 74 (i) | $\frac{d}{dx}\sqrt{3+x} = \frac{1}{2}(3+x)^{-\frac{1}{2}}$                                                     | B1 |                                                                                           |
|        | correctly substitute <i>their</i> $\frac{1}{2}(3+x)^{-\frac{1}{2}}$<br>and <i>their</i> $2x$ into product rule | M1 |                                                                                           |
|        | $\frac{dy}{dx} = x^2 \times \frac{1}{2}(3+x)^{-\frac{1}{2}} + 2x(3+x)^{\frac{1}{2}}$                           | A1 |                                                                                           |
| (ii)   | $y = 2$                                                                                                        | B1 |                                                                                           |
|        | $\frac{dy}{dx} = \frac{17}{4}$                                                                                 | B1 |                                                                                           |
|        | $\frac{y-2}{x-1} = \frac{17}{4}$ ( $y = \frac{17}{4}x - \frac{9}{4}$ ) oe<br>or use $y = mx + c$ and find $c$  | B1 | FT on <i>their</i> 2 and <i>their</i> $\frac{17}{4}$ from<br><i>their</i> $\frac{dy}{dx}$ |
| (iii)  | set <i>their</i> $\frac{dy}{dx} = 0$                                                                           | M1 |                                                                                           |
|        | obtain correct quadratic equation<br>$5x^2 + 12x [= 0]$ soi                                                    | A1 |                                                                                           |
|        | (0, 0) and (−2.4, 4.46)                                                                                        | A2 | A1 for one point or two correct values of $x$                                             |

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Calculator

Oct/Nov 2018 · Paper 23 · Q11

A line with equation  $y = -5x + k + 5$  is a tangent to a curve with equation  $y = 7 - kx - x^2$ .

(i) Find the two possible values of  $k$ .

[5]

(ii) Find, for **each** of your values of  $k$ ,

- the equation of the tangent
- the equation of the curve
- the coordinates of the point of contact of the tangent and the curve.

[5]

(iii) Find the distance between the two points of contact.

[2]



**292** Oct/Nov 2018 · Paper 23 · Q11

|        |                                                                                          |           |                                                           |
|--------|------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------|
| 75 (i) | $-5x + k + 5 = 7 - kx - x^2$                                                             | <b>M1</b> | *                                                         |
|        | $b^2 - 4ac (=0) \rightarrow (k-5)^2 - 4(k-2) (=0)$                                       | <b>M1</b> | <b>Dep*</b>                                               |
|        | $k^2 - 14k + 33 (=0)$                                                                    | <b>A1</b> |                                                           |
|        | $(k-11)(k-3) (=0)$                                                                       | <b>M1</b> | <b>Dep dep *</b><br>solve quadratic in $k$                |
|        | $k = 11$ and $k = 3$                                                                     | <b>A1</b> |                                                           |
| (ii)   | $y = -5x + 16$ and $y = 7 - 11x - x^2$<br>$y = -5x + 8$ and $y = 7 - 3x - x^2$           | <b>B2</b> | <b>FT</b> <i>their k</i><br><b>B1</b> for any two correct |
|        | solve one tangent/curve pair for one variable from quadratic equation with repeated root | <b>M1</b> |                                                           |
|        | $(-3, 31)$ and $(1, 3)$                                                                  | <b>A2</b> | <b>A1</b> for one correct point or two correct $x$ values |
| (iii)  | find distance between any two points found in (ii)                                       | <b>M1</b> |                                                           |
|        | $\sqrt{800}$ oe                                                                          | <b>A1</b> |                                                           |

**293** Calculator**May/June 2018 · Paper 21 · Q2**

The variables  $x$  and  $y$  are such that  $y = \ln(3x - 1)$  for  $x > \frac{1}{3}$ .

(i) Find  $\frac{dy}{dx}$ . [2]

(ii) Hence find the approximate change in  $x$  when  $y$  increases from  $\ln(1.2)$  to  $\ln(1.2) + 0.125$ . [3]

**294** Calculator**May/June 2018 · Paper 21 · Q7**

Differentiate with respect to  $x$

(i)  $4x \tan x$ , [2]

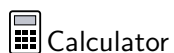
(ii)  $\frac{e^{3x+1}}{x^2 - 1}$ . [3]

**293** May/June 2018 · Paper 21 · Q2

|       |                                                                                                      |           |  |
|-------|------------------------------------------------------------------------------------------------------|-----------|--|
| 6 (i) | $k \times \frac{1}{3x-1}$                                                                            | <b>M1</b> |  |
|       | $3 \times \frac{1}{3x-1}$                                                                            | <b>A1</b> |  |
| (ii)  | $x = \frac{11}{15}$ soi                                                                              | <b>B1</b> |  |
|       | $0.125 \approx \text{their } \frac{dy}{dx} \Big _{x=\text{their } \frac{11}{15}} \times \delta x$ oe | <b>M1</b> |  |
|       | 0.05 nfww                                                                                            | <b>A1</b> |  |

**294** May/June 2018 · Paper 21 · Q7

|       |                                                                                   |           |                                                                                  |
|-------|-----------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------|
| 7 (i) | $4 \tan x + 4x \sec^2 x$ isw                                                      | <b>B2</b> | Fully correct<br><b>B1</b> for one correct term as part of e.g. a sum of 2 terms |
| (ii)  | $\frac{d}{dx}(e^{3x+1}) = 3e^{3x+1}$                                              | <b>B1</b> |                                                                                  |
|       | $\frac{(x^2-1)(\text{their } 3e^{3x+1}) - \text{their } (2x)e^{3x+1}}{(x^2-1)^2}$ | <b>M1</b> |                                                                                  |
|       | $\frac{(x^2-1)(3e^{3x+1}) - 2xe^{3x+1}}{(x^2-1)^2}$ oe isw                        | <b>A1</b> |                                                                                  |

**295****May/June 2018 · Paper 21 · Q10**

A particle moves in a straight line such that its displacement,  $s$  metres, from a fixed point  $O$  at time  $t$  seconds, is given by  $s = 4 + \cos 3t$ , where  $t \geq 0$ . The particle is initially at rest.

(i) Find the exact value of  $t$  when the particle is next at rest. [2]

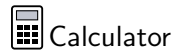
(ii) Find the distance travelled by the particle between  $t = \frac{\pi}{4}$  and  $t = \frac{\pi}{2}$  seconds. [3]

(iii) Find the greatest acceleration of the particle. [2]

**295** May/June 2018 · Paper 21 · Q10

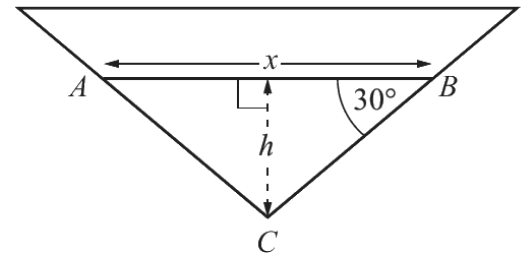
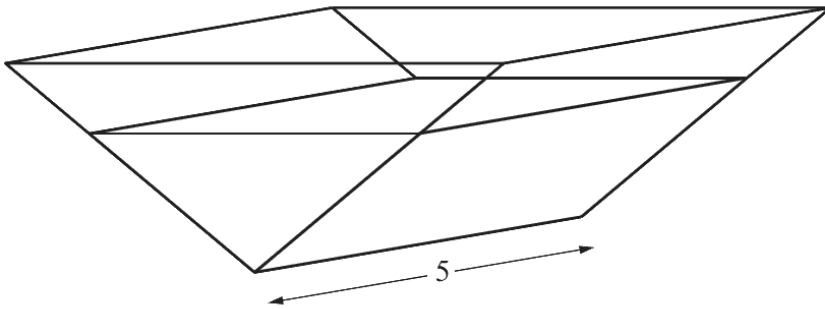
|       |                                                                        |           |                                    |
|-------|------------------------------------------------------------------------|-----------|------------------------------------|
| 8 (i) | $v = \frac{ds}{dt} = -3\sin 3t$                                        | <b>B1</b> |                                    |
|       | When $v = 0$ , $t = \frac{\pi}{3}$                                     | <b>B1</b> |                                    |
| (ii)  | Finding $s$ when<br>$t = \frac{\pi}{4}$ and $t = \frac{\pi}{2}$        | <b>M1</b> |                                    |
|       | Finding $s$ when<br>$t = \text{their } \frac{\pi}{3}$ and correct plan | <b>M1</b> | Using <i>their</i> (i) correctly   |
|       | 1.29 nfw                                                               | <b>A1</b> |                                    |
| (iii) | $a = \frac{dv}{dt} = -9\cos 3t$                                        | <b>B1</b> |                                    |
|       | 9                                                                      | <b>B1</b> | <b>FT</b> <i>their</i> $k \cos 3t$ |

296



May/June 2018 · Paper 21 · Q12

In this question all lengths are in metres.



A water container is in the shape of a triangular prism. The diagrams show the container and its cross-section. The cross-section of the water in the container is an isosceles triangle  $ABC$ , with angle  $ABC = \text{angle } BAC = 30^\circ$ . The length of  $AB$  is  $x$  and the depth of water is  $h$ . The length of the container is 5.

(i) Show that  $x = 2\sqrt{3}h$  and hence find the volume of water in the container in terms of  $h$ . [3]

(ii) The container is filled at a rate of  $0.5 \text{ m}^3$  per minute. At the instant when  $h$  is 0.25 m, find

(a) the rate at which  $h$  is increasing, [4]

(b) the rate at which  $x$  is increasing. [2]

**296** May/June 2018 · Paper 21 · Q12

|         |                                                                                                                         |           |                                    |
|---------|-------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------|
| 9 (i)   | $\tan 30 = \frac{h}{\frac{x}{2}}$ oe                                                                                    | <b>M1</b> |                                    |
|         | Correct completion to given answer                                                                                      | <b>A1</b> |                                    |
|         | $V = 5\sqrt{3} h^2$ isw                                                                                                 | <b>B1</b> |                                    |
| (ii)(a) | $\frac{dV}{dh} = \text{their } 10\sqrt{3}h$ or $\frac{5\sqrt{3}}{2}$                                                    | <b>B1</b> | <b>FT</b> <i>their</i> $V = k h^2$ |
|         | $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ soi                                                                | <b>M1</b> |                                    |
|         | $\frac{dh}{dt} = \frac{1}{\text{their} \left( \frac{dV}{dh} \right)} \times 0.5$                                        | <b>M1</b> |                                    |
|         | 0.115 or 0.11547 to 0.1155 oe                                                                                           | <b>A1</b> |                                    |
| (ii)(b) | $\left( \frac{dx}{dt} = \frac{dx}{dh} \times \frac{dh}{dt} = \right) 2\sqrt{3} \times \text{their} \frac{1}{5\sqrt{3}}$ | <b>M1</b> |                                    |
|         | $\frac{2}{5}$                                                                                                           | <b>A1</b> |                                    |

**297** Calculator**May/June 2018 · Paper 22 · Q9**

(i) Differentiate  $x^4(\sqrt{\sin x})$  with respect to  $x$ .

[4]

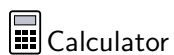
(ii) Hence find  $\int \left( x + \frac{x^4 \cos x}{\sqrt{\sin x}} + 8x^3(\sqrt{\sin x}) \right) dx$ . [3]



297 May/June 2018 · Paper 22 · Q9

| Question | Answer                                                                                                                                       | Marks     | Partial Marks                                                                                                                                                                                                                                                         |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 37 (i)   | $\frac{d}{dx}(\sqrt{\sin x}) = \frac{1}{2}(\sin x)^{-\frac{1}{2}}(\cos x)$ oe                                                                | <b>B2</b> | <b>B1</b> for $\frac{1}{2}(\sin x)^{-\frac{1}{2}} \times \dots$<br>or for $\frac{1}{2}(\sin x)^{-\frac{1}{2}}$<br>or for $\frac{1}{2}(\dots)^{-\frac{1}{2}} \times \cos x$<br>or for <i>their</i> $\frac{1}{2}(\sin x)^{\left(\frac{1}{2}\right)^{-1}} \times \cos x$ |
|          | <i>their</i> $(4x^3)\sqrt{\sin x}$<br>$+ x^4 \left( \text{their} \frac{1}{2}(\sin x)^{-\frac{1}{2}}(\cos x) \right)$ oe                      | <b>M1</b> | Applies correct form of product rule                                                                                                                                                                                                                                  |
|          | $4x^3\sqrt{\sin x} + x^4 \left( \frac{1}{2}(\sin x)^{-\frac{1}{2}}(\cos x) \right)$<br>oe isw                                                | <b>A1</b> | Not from wrong working                                                                                                                                                                                                                                                |
| (ii)     | $\int (4x^3\sqrt{\sin x}) dx$<br>$+ \int \left( x^4 \times \frac{1}{2}(\sin x)^{-\frac{1}{2}}(\cos x) \right) dx$<br>$= x^4\sqrt{\sin x}$ oe | <b>M1</b> | or $\int x dx + 2 \int \left( \frac{x^4 \cos x}{2\sqrt{\sin x}} + 4x^3\sqrt{\sin x} \right) dx$ oe<br><b>FT</b> <i>their</i> (i)                                                                                                                                      |
|          | $\frac{x^2}{2} + 2x^4\sqrt{\sin x} [+c]$                                                                                                     | <b>A2</b> | <b>A1</b> for $\int x dx + 2x^4\sqrt{\sin x}$                                                                                                                                                                                                                         |

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May/June 2018 · Paper 22 · Q11

(a) Find  $\int \sqrt[3]{2x-1} \, dx$ .

[2]

(b) (i) Find  $\int \sin 4x \, dx$ .

[2]

(ii) Hence evaluate  $\int_{\frac{\pi}{8}}^{\frac{\pi}{4}} \sin 4x \, dx$ .

[2]

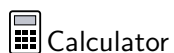
(c) Show that  $\int_0^{\ln 8} e^{\frac{x}{3}} \, dx = 3$ .

[5]

## 298 May/June 2018 · Paper 22 · Q11

|         |                                                                                                                                                             |           |                                                                                                                  |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------|
| 38 (a)  | $\frac{(2x-1)^{\frac{4}{3}}}{\frac{4}{3} \times 2} [+c]$ oe isw                                                                                             | <b>B2</b> | <b>B1</b> for $k \times \frac{(2x-1)^{\left(\frac{1}{3}+1\right)}}{\left(\frac{1}{3}+1\right)}$ where $k \neq 0$ |
| (b)(i)  | $k \cos 4x [+c]$ where $k < 0$ or $k = \frac{1}{4}$                                                                                                         | <b>M1</b> |                                                                                                                  |
|         | $-\frac{1}{4} \cos 4x [+c]$                                                                                                                                 | <b>A1</b> |                                                                                                                  |
| (b)(ii) | Sight of correct substitution of limits:<br>$-\frac{1}{4} \cos \frac{4\pi}{4} - \left(-\frac{1}{4} \cos \frac{4\pi}{8}\right)$ oe                           | <b>M1</b> | <b>FT</b> <i>their</i> $k \cos 4x$ from (b)(i)<br><b>dep</b> on <b>M1</b> awarded in (b)(i)                      |
|         | $\frac{1}{4}$                                                                                                                                               | <b>A1</b> | does <b>not</b> imply <b>M1</b>                                                                                  |
| (c)     | $\int e^{\frac{x}{3}} dx = ke^{\frac{x}{3}} [+c]$                                                                                                           | <b>M1</b> | $k$ any non-zero constant                                                                                        |
|         | $k = 3$                                                                                                                                                     | <b>A1</b> |                                                                                                                  |
|         | Sight of correct substitution of limits:<br>$\frac{\ln 8}{3}$<br><i>their</i> $ke^{\frac{\ln 8}{3}}$ – <i>their</i> $ke^0$ oe                               | <b>M1</b> | <b>dep</b> on first <b>M1</b>                                                                                    |
|         | Shows how to deal with the power of the first term e.g.<br>$\frac{\ln 8}{3} = \ln 8^{\frac{1}{3}}$ or $\frac{\ln 8}{3} = \ln 2$ or $3(\sqrt[3]{8})$<br>seen | <b>B1</b> |                                                                                                                  |
|         | $6 - 3 = 3$                                                                                                                                                 | <b>A1</b> | Not from wrong working                                                                                           |

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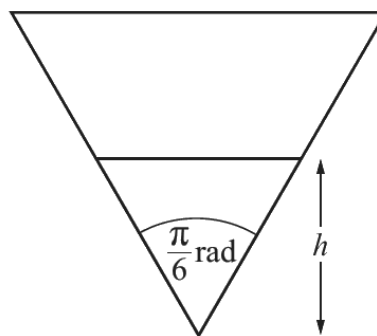
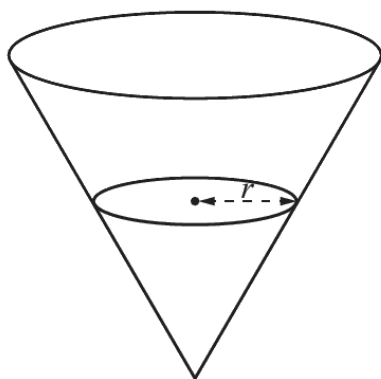
Calculator

May/June 2018 · Paper 22 · Q12

In this question all lengths are in centimetres.

The volume of a cone of height  $h$  and base radius  $r$  is given by  $V = \frac{1}{3}\pi r^2 h$ .

It is known that  $\sin \frac{\pi}{12} = \frac{\sqrt{6} - \sqrt{2}}{4}$ ,  $\cos \frac{\pi}{12} = \frac{\sqrt{6} + \sqrt{2}}{4}$ ,  $\tan \frac{\pi}{12} = 2 - \sqrt{3}$ .



A water cup is in the shape of a cone with its axis vertical. The diagrams show the cup and its cross-section. The vertical angle of the cone is  $\frac{\pi}{6}$  radians. The depth of water in the cup is  $h$ . The surface of the water is a circle of radius  $r$ .

- (i) Find an expression for  $r$  in terms of  $h$  and show that the volume of water in the cup is given by

$$V = \frac{\pi(7 - 4\sqrt{3})h^3}{3}. \quad [4]$$

- (ii) Water is poured into the cup at a rate of  $30 \text{ cm}^3 \text{ s}^{-1}$ . Find, correct to 2 decimal places, the rate at which the depth of water is increasing when  $h = 5$ . [4]

**299** May/June 2018 · Paper 22 · Q12

|        |                                                                                                                             |           |                                                                                                                                                          |
|--------|-----------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39 (i) | $\tan \frac{\pi}{12} = \frac{r}{h}$ oe                                                                                      | <b>M1</b> |                                                                                                                                                          |
|        | $r = h(2 - \sqrt{3})$ or $r = h \tan \frac{\pi}{12}$ oe                                                                     | <b>A1</b> |                                                                                                                                                          |
|        | $[V =] \frac{1}{3} \pi (2 - \sqrt{3})^2 h^2 \times h$ oe                                                                    | <b>M1</b> | Correctly uses <i>their</i> expression for $r$ in terms of $h$ in formula for volume of a cone dependent on finding an expression connecting $r$ and $h$ |
|        | $[V =] \frac{\pi(4 - 4\sqrt{3} + 3)h^3}{3}$ oe<br>correctly leading to<br>$[V =] \frac{\pi(7 - 4\sqrt{3})h^3}{3}$ <b>AG</b> | <b>A1</b> |                                                                                                                                                          |
| (ii)   | Correct derivative of $V$ e.g.<br>$\frac{3\pi(7 - 4\sqrt{3})h^2}{3}$ oe isw                                                 | <b>B1</b> |                                                                                                                                                          |
|        | $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$ soi                                                                    | <b>B1</b> |                                                                                                                                                          |
|        | $\frac{1}{\text{their} \left( \frac{dV}{dh} \right) \Big _{h=5}} \times 30$                                                 | <b>M1</b> | if correct implies <b>B1 B1</b> ;<br>if incorrect, a correct <b>FT</b> statement implies the second <b>B1</b>                                            |
|        | 5.32                                                                                                                        | <b>A1</b> |                                                                                                                                                          |

**300** Calculator**Oct/Nov 2017 • Paper 21 • Q5**

(i) Find  $\frac{d}{dx}\left(\frac{5}{3x+2}\right)$ .

[2]

(ii) Use your answer to part (i) to find  $\int \frac{30}{(3x+2)^2} dx$ .

[2]

(iii) Hence evaluate  $\int_1^2 \frac{30}{(3x+2)^2} dx$ .

[2]

**301** Calculator**Oct/Nov 2017 • Paper 21 • Q7**

Find  $y$  in terms of  $x$ , given that  $\frac{d^2y}{dx^2} = 6x + \frac{2}{x^3}$  and that when  $x = 1, y = 3$  and  $\frac{dy}{dx} = 1$ .

[6]

**300** Oct/Nov 2017 · Paper 21 · Q5

|        |                                                                        |           |                            |
|--------|------------------------------------------------------------------------|-----------|----------------------------|
| 20 (i) | $\frac{d}{dx}\left(\frac{5}{3x+2}\right) = -5(3x+2)^{-2} \times 3$     | <b>M1</b> | $-5(3x+2)^{-2}$            |
|        |                                                                        | <b>A1</b> | $\times 3$                 |
| (ii)   | $\int \frac{30}{(3x+2)^2} dx = \left[ \frac{-10}{(3x+2)} \right]$      | <b>M1</b> | $\frac{1}{(3x+2)}$         |
|        |                                                                        | <b>A1</b> | $\times -10$               |
| (iii)  | $\left[ \frac{-10}{(3x+2)} \right]_1^2 = -\frac{10}{8} + \frac{10}{5}$ | <b>M1</b> | Insert limits and subtract |
|        | $= \frac{3}{4}$                                                        | <b>A1</b> |                            |

**301** Oct/Nov 2017 · Paper 21 · Q7

|    |                                                               |           |                                                          |
|----|---------------------------------------------------------------|-----------|----------------------------------------------------------|
| 21 | $\frac{dy}{dx} = 3x^2 - \frac{1}{x^2} (+C)$                   | <b>B2</b> | <b>B1</b> for $3x^2$<br><b>B1</b> for $-\frac{1}{x^2}$ . |
|    | $x=1, \frac{dy}{dx}=1 \rightarrow C=-1$                       | <b>B1</b> |                                                          |
|    | $y = x^3 + \frac{1}{x} - x + D$<br>$x=1, y=3 \rightarrow D=2$ | <b>B2</b> | <b>B1</b> for two correct terms in $x$                   |
|    | $y = x^3 + \frac{1}{x} - x + 2$                               | <b>B1</b> |                                                          |

302



Calculator

Oct/Nov 2017 · Paper 21 · Q11

The line  $y = kx + 3$ , where  $k$  is a positive constant, is a tangent to the curve  $x^2 - 2x + y^2 = 8$  at the point  $P$ .

(i) Find the value of  $k$ . [4]

(ii) Find the coordinates of  $P$ . [3]

(iii) Find the equation of the normal to the curve at  $P$ . [2]



**302** Oct/Nov 2017 · Paper 21 · Q11

|        |                                                                  |           |               |
|--------|------------------------------------------------------------------|-----------|---------------|
| 22 (i) | $x^2 - 2x + (kx + 3)^2 = 8$                                      | <b>M1</b> | Eliminate $y$ |
|        | $(1 + k^2)x^2 + (6k - 2)x + 1 = 0$                               | <b>A1</b> |               |
|        | $b^2 - 4ac = 0 \rightarrow (6k - 2)^2 - 4(1 + k^2) = 0$          | <b>M1</b> |               |
|        | $k = \frac{3}{4}$                                                | <b>A1</b> | Answer given  |
| (ii)   | $x = \frac{-b}{2a} \rightarrow x = \frac{-2.5}{2 \times 1.5625}$ | <b>M1</b> |               |
|        | $= -0.8$                                                         | <b>A1</b> |               |
|        | $y = 0.75 \times -0.8 + 3 = 2.4$                                 | <b>A1</b> | <b>FT</b>     |
| (iii)  | Eqn of $PQ$ $\frac{y - 2.4}{x + 0.8} = \frac{-4}{3}$             | <b>M1</b> |               |
|        | $\rightarrow 3y = 4 - 4x$                                        | <b>A1</b> |               |

303



Calculator

Oct/Nov 2017 · Paper 21 · Q12

(i) Differentiate  $(\cos x)^{-1}$  with respect to  $x$ .

[2]

(ii) Hence find  $\frac{dy}{dx}$  given that  $y = \tan x + 4(\cos x)^{-1}$ .

[2]

(iii) Using your answer to part (ii) find the values of  $x$  in the range  $0 \leq x \leq 2\pi$  such that  $\frac{dy}{dx} = 4$ . [6]

**303** Oct/Nov 2017 · Paper 21 · Q12

|        |                                                                          |           |                                                         |
|--------|--------------------------------------------------------------------------|-----------|---------------------------------------------------------|
| 23 (i) | $\frac{d(\cos x)^{-1}}{dx} = \frac{1}{\cos^2 x} \times \sin x$           | <b>M1</b> | $\frac{1}{\cos^2 x}$                                    |
|        |                                                                          | <b>A1</b> | $\times \sin x$                                         |
| (ii)   | $\frac{dy}{dx} = \sec^2 x + \frac{4\sin x}{\cos^2 x}$                    | <b>B1</b> | $\sec^2 x$                                              |
|        |                                                                          | <b>B1</b> | $\frac{4\sin x}{\cos^2 x}$                              |
| (iii)  | $\frac{1}{\cos^2 x} + \frac{4}{\cos x} \times \frac{\sin x}{\cos x} = 4$ | <b>M1</b> | Equate <i>their</i> (i) to 4 and multiply by $\cos^2 x$ |
|        | $\rightarrow 1 + 4\sin x = 4\cos^2 x$                                    | <b>M1</b> | Use of identity and simplify                            |
|        | $4\sin^2 x + 4\sin x - 3 = 0$                                            | <b>A1</b> |                                                         |
|        | $(2\sin x - 1)(2\sin x + 3) = 0$                                         | <b>M1</b> | Solve                                                   |
|        | $x = \frac{\pi}{6}, \frac{5\pi}{6}$                                      | <b>A2</b> | <b>A1</b> for each                                      |

**304**

Calculator

**Oct/Nov 2017 · Paper 22 · Q6**

The volume of a closed cylinder of base radius  $x$  cm and height  $h$  cm is  $500 \text{ cm}^3$ .

(i) Express  $h$  in terms of  $x$ .

[1]

(ii) Show that the total surface area of the cylinder is given by  $A = 2\pi x^2 + \frac{1000}{x} \text{ cm}^2$ .

[2]

(iii) Given that  $x$  can vary, find the stationary value of  $A$  and show that this value is a minimum.

[5]

**305**

Calculator

**Oct/Nov 2017 · Paper 22 · Q7**

The gradient of the normal to a curve at the point with coordinates  $(x, y)$  is given by  $\frac{\sqrt{x}}{1-3x}$ .

(i) Find the equation of the curve, given that the curve passes through the point  $(1, -10)$ .

[5]

(ii) Find, in the form  $y = mx + c$ , the equation of the tangent to the curve at the point where  $x = 4$ .

[4]

**304** Oct/Nov 2017 · Paper 22 · Q6

|        |                                                                                                  |           |                                                                                                                                               |
|--------|--------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 47 (i) | $\pi x^2 h = 500 \rightarrow h = \frac{500}{\pi x^2}$                                            | <b>B1</b> | Ignore units Condone $r$ for $x$                                                                                                              |
| (ii)   | $A = 2\pi x^2 + 2\pi xh$                                                                         | <b>M1</b> | Correct expression for $A$ and insert for <i>their</i> $h$ .                                                                                  |
|        | $= 2\pi x^2 + 2\pi x \times \frac{500}{\pi x^2} = 2\pi x^2 + \frac{1000}{x}$                     | <b>A1</b> | Answer given<br>Condone $r$ for $x$ .                                                                                                         |
| (iii)  | Differentiate: at least one power reduced by 1                                                   | <b>M1</b> |                                                                                                                                               |
|        | $\frac{dA}{dx} = 4\pi x - \frac{1000}{x^2}$                                                      | <b>A1</b> |                                                                                                                                               |
|        | $\frac{dA}{dx} = 0 \rightarrow x = \sqrt[3]{\frac{1000}{4\pi}}$ isw or $(x = 4.3(0))$            | <b>A1</b> |                                                                                                                                               |
|        | $A = 2\pi(4.3)^2 + \frac{1000}{4.3} = 349 \text{ cm}^2$                                          | <b>A1</b> | awrt 349                                                                                                                                      |
|        | $\frac{d^2 A}{dx^2} = 4\pi + \frac{2000}{x^3} (> 0)$ or a positive value<br>( $\rightarrow$ min) | <b>B1</b> | Correct second differential (need not be evaluated) and conclusion.<br>or<br>Examine correct gradient either side of $x = 4.3$ and conclusion |

**305** Oct/Nov 2017 · Paper 22 · Q7

|        |                                                       |           |                                                    |
|--------|-------------------------------------------------------|-----------|----------------------------------------------------|
| 48 (i) | (Gradient or $\frac{dy}{dx}) = \frac{3x-1}{\sqrt{x}}$ | <b>B1</b> | Gradient = Negative reciprocal.<br>Can be implied. |
|        | $= 3x^{\frac{1}{2}} - x^{-\frac{1}{2}}$               | <b>B1</b> | $\pm$ One correct term                             |
|        | $y = 2x^{\frac{3}{2}} - 2x^{\frac{1}{2}} (+C)$        | <b>M1</b> | at least 1 fractional power increased by 1.        |
|        | $-10 = 2 - 2 + C \rightarrow C = -10$                 | <b>A1</b> | one term correct with simplified coefficients      |
|        | $y = 2x^{\frac{3}{2}} - 2x^{\frac{1}{2}} - 10$        | <b>A1</b> | For $C$ from correct working.                      |
| (ii)   | $x = 4 \rightarrow y = 16 - 4 - 10 = 2$               | <b>B1</b> |                                                    |

|  |                                                                      |           |                                                             |
|--|----------------------------------------------------------------------|-----------|-------------------------------------------------------------|
|  | $\rightarrow \frac{dy}{dx} = 6 - \frac{1}{2} = 5.5$                  | <b>B1</b> |                                                             |
|  | Eqn with <i>their</i> grad and point (4, ...)                        | <b>M1</b> |                                                             |
|  | Eqn of tangent: $\frac{y-2}{x-4} = 5.5 \rightarrow y = 5.5x - 20$ oe | <b>A1</b> | Must be in the form $y = mx + c$ but accept $2y = 11x - 40$ |

306

 Calculator

Oct/Nov 2017 · Paper 22 · Q9

(i) Find  $\frac{d}{dx}(x \ln x)$ .

[2]

(ii) Hence find  $\int \ln x \, dx$ .

[2]

(iii) Hence, given that  $k > 0$ , show that  $\int_k^{2k} \ln x \, dx = k(\ln 4k - 1)$ .

[4]

**306** Oct/Nov 2017 · Paper 22 · Q9

|        |                                                                                          |             |                                                                                                            |
|--------|------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------|
| 49 (i) | $\frac{d}{dx}(x \ln x) = x \times \frac{1}{x} + \ln x$ isw                               | <b>M1A1</b> | Product rule. One correct term + another term. Allow unsimplified.                                         |
| (ii)   | $\int 1 + \ln x dx = x \ln x$                                                            | <b>M1</b>   | Correct use of (i) and must be dealing with 2 terms. soi                                                   |
|        | $\int \ln x dx = x \ln x - x + (C)$                                                      | <b>A1</b>   | Correct answer with no working is fine.                                                                    |
| (iii)  | $\int_k^{2k} \ln x dx = [2k \ln 2k - 2k] - [k \ln k - k]$<br>$= k(2 \ln 2k - \ln k - 1)$ | <b>M1</b>   | Insert limits and subtract correctly using <i>their</i> result from (ii) which must contain an ln function |
|        | $= k(\ln(2k)^2 - \ln k - 1)$                                                             | <b>M1</b>   | Uses $n \ln a = \ln a^n$ somewhere oe                                                                      |
|        | $= k\left(\ln\left(\frac{4k^2}{k}\right) - 1\right)$                                     | <b>M1</b>   | Uses $\ln a - \ln b = \ln\left(\frac{a}{b}\right)$ or<br>$\ln a + \ln b = \ln ab$ somewhere                |
|        | $= k(\ln 4k - 1)$                                                                        | <b>A1</b>   | Answer given<br>Correct completion.                                                                        |



307



Calculator

Oct/Nov 2017 · Paper 22 · Q10

(i) Without using a calculator, solve the equation  $6c^3 - 7c^2 + 1 = 0$ .

[5]

It is given that  $y = \tan x + 6 \sin x$ .

(ii) Find  $\frac{dy}{dx}$ .

[2]

(iii) If  $\frac{dy}{dx} = 7$  show that  $6 \cos^3 x - 7 \cos^2 x + 1 = 0$ .

[2]

(iv) Hence solve the equation  $\frac{dy}{dx} = 7$  for  $0 \leq x \leq \pi$  radians.

[2]

**307** Oct/Nov 2017 · Paper 22 · Q10

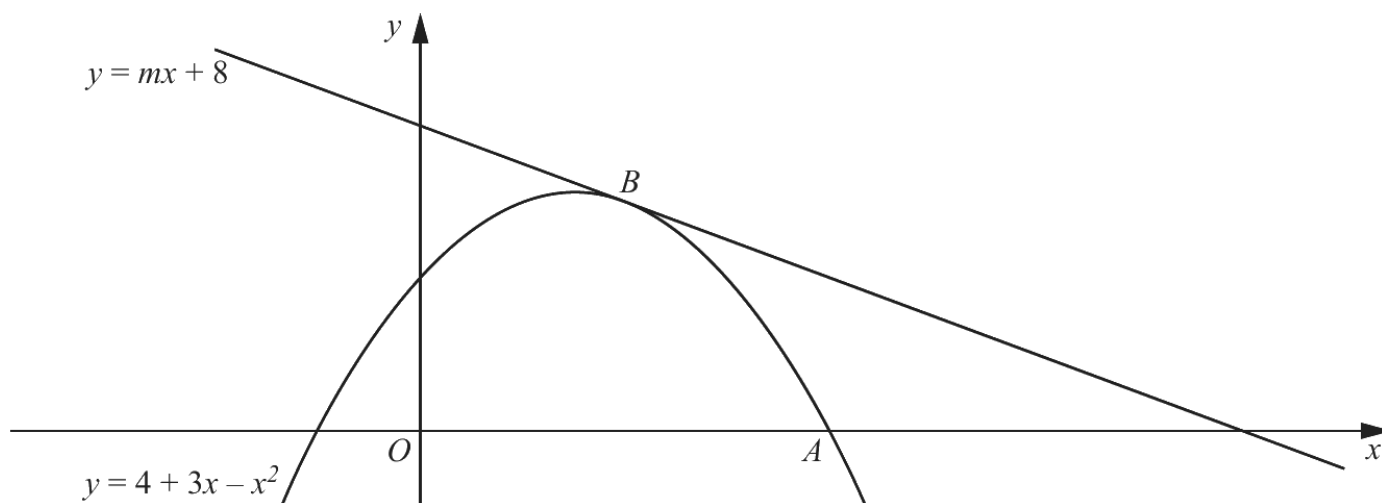
| Question | Answer                                                                                                             | Marks     | Partial Marks                                                                                                     |
|----------|--------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------|
| 50 (i)   | $c = 1 \rightarrow 6(1)^3 - 7(1)^2 + 1 = 0 \rightarrow (c - 1)$ is a factor.                                       | <b>B1</b> | Or correct division. Finding or using one correct factor.                                                         |
|          | Attempt to factorise or use long division to obtain $6c^2 \dots \pm 1$ or $6c^2 \pm c \dots$ respectively          | <b>M1</b> |                                                                                                                   |
|          | $(c - 1)(6c^2 - c - 1) = 0$                                                                                        | <b>A1</b> |                                                                                                                   |
|          | $(c - 1)(2c - 1)(3c + 1) = 0$                                                                                      | <b>A1</b> |                                                                                                                   |
|          | $c = 1, \frac{1}{2}, -\frac{1}{3}$                                                                                 | <b>A1</b> | <b>FT</b><br>From three different linear factors                                                                  |
| (ii)     | $\frac{dy}{dx} = \sec^2 x + 6\cos x$                                                                               | <b>B2</b> | <b>B1</b> for each term                                                                                           |
| (iii)    | $\frac{1}{\cos^2 x} + 6\cos x = 7$                                                                                 | <b>B1</b> | <b>B1dep</b><br>Replaces $\sec^2 x$ by $\frac{1}{\cos^2 x}$                                                       |
|          | $\rightarrow 6\cos^3 x - 7\cos^2 x + 1 = 0$                                                                        | <b>B1</b> | <b>B1dep</b><br>Answer given so all steps must be correct.                                                        |
| (iv)     | $\cos x = 1, \frac{1}{2}, -\frac{1}{3}$<br>$\rightarrow x = 0, 1.05 \left( \text{or } \frac{\pi}{3} \right), 1.91$ | <b>A2</b> | <b>A1</b> for 2 values awrt<br><b>A1</b> for third value and no others in range. No credit for answers in degrees |

308



Calculator

Oct/Nov 2017 · Paper 22 · Q11



The diagram shows the curve  $y = 4 + 3x - x^2$  intersecting the positive  $x$ -axis at the point  $A$ . The line  $y = mx + 8$  is a tangent to the curve at the point  $B$ . Find

(i) the coordinates of  $A$ , [2]

(ii) the value of  $m$ , [3]

(iii) the coordinates of  $B$ , [2]

(iv) the area of the shaded region, showing all your working. [5]

## 308 Oct/Nov 2017 · Paper 22 · Q11

|        |                                                                                                    |           |                                                                                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 51 (i) | $y = 0 \rightarrow (x - 4)(x + 1) = 0$                                                             | <b>M1</b> | Solve                                                                                                                                                                    |
|        | $\rightarrow A \text{ is } (4, 0) \text{ nfw}$                                                     | <b>A1</b> | Indication somewhere that $x = 4$ when $y = 0$                                                                                                                           |
| (ii)   | $4 + 3x - x^2 = mx + 8$<br>$x^2 + (m - 3)x + 4 = 0$                                                | <b>M1</b> | Eliminate $y$ .                                                                                                                                                          |
|        | $b^2 - 4ac (= 0) \rightarrow (m - 3)^2 = 16$                                                       | <b>M1</b> | <b>M1dep</b><br>Use of discriminant                                                                                                                                      |
|        | $m = -1$                                                                                           | <b>A1</b> | Do not award if $m = 7$ is not discarded                                                                                                                                 |
| (iii)  | Obtain quadratic $x^2 + (m - 3)x + 4 = 0$ using <i>their</i> $m$ and attempt to solve.             | <b>M1</b> | Working must be seen for any marks to be awarded. Must not be awarded if $m$ is not obtained correctly                                                                   |
|        | Point $B (2, 6)$                                                                                   | <b>A1</b> |                                                                                                                                                                          |
| (iv)   | Area under curve $= \int_2^4 (4 + 3x - x^2) dx$<br>Integrate powers increased in at least 2 terms  | <b>M1</b> |                                                                                                                                                                          |
|        | $= \left[ 4x + \frac{3}{2}x^2 - \frac{1}{3}x^3 \right]_2^4$                                        | <b>A1</b> |                                                                                                                                                                          |
|        | $= \left[ 16 + 24 - \frac{64}{3} \right] - \left[ 8 + 6 - \frac{8}{3} \right]$<br>$= 7\frac{1}{3}$ | <b>M1</b> | <b>M1dep</b><br>Insert limits of <i>their</i> 2 and 4 and subtract in correct order. May be implied by $18\frac{2}{3} - \dots$                                           |
|        | Intercept is $(8, 0)$ so area of triangle $= \frac{6 \times 6}{2} = 18$                            | <b>M1</b> | Area of triangle using<br>$\text{their } B = \frac{(\text{their } 8 - x_B)}{2} \times y_B$<br>or<br>Attempt to find other suitable areas to result in a complete method. |
|        | Shaded area $= 18 - 7\frac{1}{3} = 10\frac{2}{3}$                                                  | <b>A1</b> | Accept 10.7. Must not be awarded if point $B$ is not obtained correctly.                                                                                                 |

309



Calculator

Oct/Nov 2017 • Paper 23 • Q7

A particle moving in a straight line passes through a fixed point  $O$ . Its velocity,  $v \text{ ms}^{-1}$ ,  $t \text{ s}$  after passing through  $O$ , is given by  $v = 3 \cos 2t - 1$  for  $t \geq 0$ .

(i) Find the value of  $t$  when the particle is first at rest. [2]

(ii) Find the displacement from  $O$  of the particle when  $t = \frac{\pi}{4}$ . [3]

(iii) Find the acceleration of the particle when it is first at rest. [3]

**309** Oct/Nov 2017 · Paper 23 · Q7

|        |                                                                            |             |                                     |
|--------|----------------------------------------------------------------------------|-------------|-------------------------------------|
| 71 (i) | $v = 0 \rightarrow \cos 2t = \frac{1}{3}$                                  | <b>M1</b>   | set $v = 0$ and solve for $\cos 2t$ |
|        | $\rightarrow t = 0.615 \text{ or } 0.616$                                  | <b>A1</b>   |                                     |
| (ii)   | $s = \frac{3}{2} \sin 2t - t \quad (+c)$                                   | <b>M1A1</b> | <b>M1</b> for $\sin 2t$ and $\pm t$ |
|        | $t = \frac{\pi}{4} \rightarrow s = 1.5 - \frac{\pi}{4} \quad (= 0.715)$    | <b>A1</b>   |                                     |
| (iii)  | $a = -6 \sin 2t$                                                           | <b>M1A1</b> | <b>M1</b> for $-\sin 2t$            |
|        | $t = 0.615 \rightarrow a = -5.66 \text{ or } -5.65 \text{ or } -2\sqrt{8}$ | <b>A1</b>   | condone substitution of degrees     |

**310** Calculator**Oct/Nov 2017 • Paper 23 • Q9**

- (i) Show that  $\frac{d}{dx}\left(\frac{\ln x}{x^3}\right) = \frac{1 - 3 \ln x}{x^4}$ . [3]

- (ii) Find the exact coordinates of the stationary point of the curve  $y = \frac{\ln x}{x^3}$ . [3]

- (iii) Use the result from part (i) to find  $\int \left(\frac{\ln x}{x^4}\right) dx$ . [4]

**311** Calculator**May/June 2017 • Paper 21 • Q1**

Find the equation of the curve which passes through the point (2, 17) and for which  $\frac{dy}{dx} = 4x^3 + 1$ . [4]

**310** Oct/Nov 2017 · Paper 23 · Q9

|        |                                                                                                        |           |                                                                   |
|--------|--------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------|
| 72 (i) | $\frac{d}{dx}(\ln x) = \frac{1}{x}$ and<br>$\frac{d}{dx}x^3 = 3x^2$ or $\frac{d}{dx}x^{-3} = -3x^{-4}$ | <b>B1</b> | seen                                                              |
|        | Substitution of <i>their</i> derivatives into quotient rule                                            | <b>M1</b> |                                                                   |
|        | $\frac{d}{dx}\left(\frac{\ln x}{x^3}\right) = \frac{x^3 \times \frac{1}{x} - 3x^2 \ln x}{x^6}$ oe      | <b>A1</b> | correct completion                                                |
| (ii)   | $\frac{dy}{dx} = 0 \rightarrow 1 - 3\ln x = 0$ $\ln x = \frac{1}{3}$                                   | <b>M1</b> | equate given $\frac{dy}{dx}$ to zero and solve for $\ln x$ or $x$ |
|        | $x = e^{\frac{1}{3}}$                                                                                  | <b>A1</b> | seen                                                              |
|        | $y = \frac{1}{3e}$                                                                                     | <b>A1</b> | seen                                                              |
| (iii)  | $\frac{\ln x}{x^3} = \int \frac{1 - 3\ln x}{x^4} dx$ oe                                                | <b>M1</b> | use given statement in (i)                                        |
|        | $\int \frac{1}{x^4} dx = \frac{-1}{3x^3}$                                                              | <b>B1</b> | seen anywhere                                                     |
|        | $\int \frac{\ln x}{x^4} dx = -\frac{1}{9x^3} - \frac{\ln x}{3x^3}$ (+C) oe                             | <b>A2</b> | <b>A1</b> for each term                                           |

**311** May/June 2017 · Paper 21 · Q1

| Question | Answer                | Marks      | Guidance                                             |
|----------|-----------------------|------------|------------------------------------------------------|
| 3        | Integrates            | <b>M1</b>  | must be clear attempt to integrate at least one term |
|          | $[y =] x^4 + x (+c)$  | <b>A1</b>  | Both terms correct                                   |
|          | $17 = 2^4 + 2 + c$    | <b>DM1</b> | Substitution of $x = 2, y = 17$ to find $c$          |
|          | $y = x^4 + x - 1$ cao | <b>A1</b>  | must have $y =$                                      |



**312** Calculator**May/June 2017 • Paper 21 • Q3**

The variables  $x$  and  $y$  are such that  $y = \ln(x^2 + 1)$ .

- (i) Find an expression for  $\frac{dy}{dx}$ . [2]

- (ii) Hence, find the approximate change in  $y$  when  $x$  increases from 3 to  $3 + h$ , where  $h$  is small. [2]

**313** Calculator**May/June 2017 • Paper 21 • Q12**

A particle moves in a straight line so that,  $t$  seconds after passing a fixed point  $O$ , its displacement,  $s$  m, from  $O$  is given by

$$s = 1 + 3t - \cos 5t.$$

- (i) Find the distance between the particle's first two positions of instantaneous rest. [7]

- (ii) Find the acceleration when  $t = \pi$ . [2]

**314** Calculator**May/June 2017 • Paper 22 • Q4**

The point  $P$  lies on the curve  $y = 3x^2 - 7x + 11$ . The normal to the curve at  $P$  has equation  $5y + x = k$ . Find the coordinates of  $P$  and the value of  $k$ . [6]

**312** May/June 2017 · Paper 21 · Q3

|       |                                                                                  |           |                                                                                       |
|-------|----------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------|
| 4 (i) | $\frac{2x}{x^2+1}$ final answer                                                  | <b>B2</b> | <b>B1</b> for $\frac{1}{x^2+1} \times (ax+b)$ , $a$ or $b$ must be non-zero           |
| (ii)  | $\delta y = \text{their} \left( \frac{2(3)}{(3)^2+1} \right) \times h$ or better | <b>M1</b> | Substitutes $x = 3$ into <i>their</i> $\frac{dy}{dx}$<br><b>and</b> multiplies by $h$ |
|       | $\frac{6}{10}h$ oe                                                               | <b>A1</b> |                                                                                       |

**313** May/June 2017 · Paper 21 · Q12

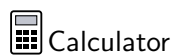
|       |                                                                   |            |                                                                                              |
|-------|-------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------|
| 5 (i) | $\left[ v = \frac{d(3t - \cos 5t + 1)}{dt} \right] 3 + 5 \sin 5t$ | <b>B2</b>  | <b>B1</b> for either with no other terms or for both with 1 extra                            |
|       | $\text{their} (3 + 5 \sin 5t) = 0$                                | <b>M1</b>  | Must be from an attempt to differentiate                                                     |
|       | awrt 0.76                                                         | <b>A1</b>  | 0.7570187525                                                                                 |
|       | awrt 1.13                                                         | <b>A1</b>  | 1.12793684                                                                                   |
|       | substitutes <i>their</i> $t$ values into $s$ (4.07..., 3.58...)   | <b>DM1</b> | must be two values                                                                           |
|       | 0.48 to 0.49 [m]                                                  | <b>A1</b>  | Final <b>A1</b> may imply earlier <b>A1</b> s                                                |
| (ii)  | $25 \cos 5t$                                                      | <b>M1</b>  | Differentiating <i>their</i> $v$ correctly providing at least 2 terms with one trig function |
|       | -25                                                               | <b>A1</b>  | Ignore +25 following -25                                                                     |

**314** May/June 2017 · Paper 22 · Q4

| Question | Answer                                                                    | Marks     | Partial Marks                                |
|----------|---------------------------------------------------------------------------|-----------|----------------------------------------------|
| 34       | $\frac{dy}{dx} = 6x - 7$ soi                                              | <b>B1</b> |                                              |
|          | $m_{\text{normal}} = -\frac{1}{5}$ soi                                    | <b>B1</b> | finds or uses correct gradient of normal     |
|          | $m_{\text{tangent}} = 5$ soi or $(6x-7)\left(-\frac{1}{5}\right) = -1$ oe | <b>M1</b> | uses $m_1 m_2 = -1$ with numerical gradients |
|          | $6x - 7 = 5$ oe $\Rightarrow x = 2$                                       | <b>A1</b> |                                              |
|          | $y = 9$                                                                   | <b>A1</b> |                                              |

|  |                                                  |           |  |
|--|--------------------------------------------------|-----------|--|
|  | $k = 47$                                         | <b>A1</b> |  |
|  | <b>Alternative method</b>                        |           |  |
|  | $m_{\text{normal}} = -\frac{1}{5}$               | <b>B1</b> |  |
|  | $m_{\text{tangent}} = 5$                         | <b>M1</b> |  |
|  | $3x^2 - 12x + 11 - c = 0$ oe                     | <b>A1</b> |  |
|  | solving $3x^2 - 12x + 12 = 0$ oe to find $x = 2$ | <b>A1</b> |  |
|  | $y = 9$                                          | <b>A1</b> |  |
|  | $k = 47$                                         | <b>A1</b> |  |

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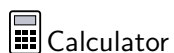
May/June 2017 · Paper 22 · Q5

- (i) Show that  $\frac{d}{dx}[0.4x^5(0.2 - \ln 5x)] = kx^4 \ln 5x$ , where  $k$  is an integer to be found. [2]
- (ii) Express  $\ln 125x^3$  in terms of  $\ln 5x$ . [1]
- (iii) Hence find  $\int (x^4 \ln 125x^3) dx$ . [2]

**315** May/June 2017 · Paper 22 · Q5

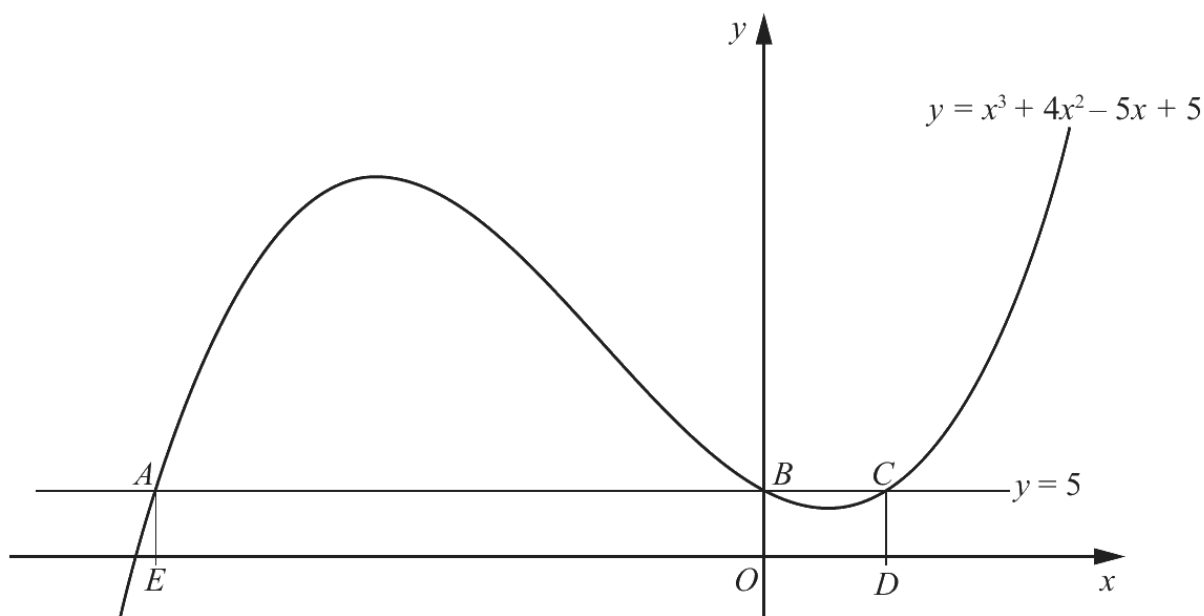
|        |                                                                                                                                                                                                                             |           |                                                                                                                                                                                                                                                                                                                                                                                        |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 35 (i) | $(\text{their } 2x^4)(0.2 - \ln 5x) + 0.4x^5 \left( \text{their } \frac{-5}{5x} \right)$ oe or<br>$\text{their } 0.4x^4 - \left( (\text{their } 2x^4) \ln 5x + 0.4x^5 \left( \text{their } \frac{5}{5x} \right) \right)$ oe | <b>M1</b> | clearly applies correct form of product rule                                                                                                                                                                                                                                                                                                                                           |
|        | $-2x^4 \ln 5x$ isw                                                                                                                                                                                                          | <b>A1</b> | nfww                                                                                                                                                                                                                                                                                                                                                                                   |
| (ii)   | $3 \ln 5x$ or $\ln 5x + \ln 5x + \ln 5x$                                                                                                                                                                                    | <b>B1</b> |                                                                                                                                                                                                                                                                                                                                                                                        |
| (iii)  | $\frac{3}{-2} \int (-2x^4 \ln 5x) dx$ oe                                                                                                                                                                                    | <b>M1</b> | <b>FT</b> $k = 2$ from (i)<br>allow for $\frac{3}{2} \int (2x^4 \ln 5x) dx$<br>or, when $k = -2$ , for<br>$\int (x^4 \ln 5x) dx = -0.2x^5(0.2 - \ln 5x)$<br>or $-\frac{2}{3} \int (3x^4 \ln 5x) dx = 0.4x^5(0.2 - \ln 5x)$ oe<br>or, when <b>FT</b> $k = 2$ , for<br>$\int (x^4 \ln 5x) dx = 0.2x^5(0.2 - \ln 5x)$<br>or $\frac{2}{3} \int (3x^4 \ln 5x) dx = 0.4x^5(0.2 - \ln 5x)$ oe |
|        | $-\frac{3}{2} (0.4x^5(0.2 - \ln 5x)) [+c]$ oe isw cao                                                                                                                                                                       | <b>A1</b> | nfww; implies M1<br>An answer of $0.6x^5(0.2 - \ln 5x)$ following $k = 2$ from (i) implies M1 A0                                                                                                                                                                                                                                                                                       |

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Calculator

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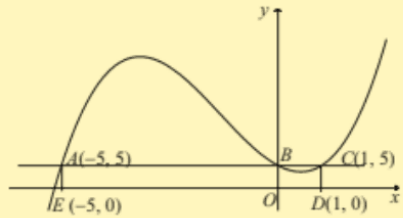
The diagram shows part of the curve  $y = x^3 + 4x^2 - 5x + 5$  and the line  $y = 5$ . The curve and the line intersect at the points  $A$ ,  $B$  and  $C$ . The points  $D$  and  $E$  are on the  $x$ -axis and the lines  $AE$  and  $CD$  are parallel to the  $y$ -axis.

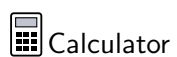
- (i) Find  $\int (x^3 + 4x^2 - 5x + 5) dx$ . [2]

- (ii) Find the area of each of the rectangles  $OEAB$  and  $OBCD$ . [4]

- (iii) Hence calculate the total area of the shaded regions enclosed between the line and the curve. You must show all your working. [4]

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|        |                                                                                                                                                                      |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 36 (i) | $\frac{x^4}{4} + \frac{4x^3}{3} - \frac{5x^2}{2} + 5x [+c]$ isw                                                                                                      | <b>B2</b> | <b>B1</b> for any 3 correct terms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| (ii)   | $x^3 + 4x^2 - 5x + 5 = 5$ and rearrange to $x(x^2 + 4x - 5) = 0$ oe soi                                                                                              | <b>B1</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|        | Solves <i>their</i> $x^2 + 4x - 5 [= 0]$ soi                                                                                                                         | <b>M1</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|        | $x = -5, x = 1$ soi                                                                                                                                                  | <b>A1</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|        | $OEAB = 25, OBCD = 5$                                                                                                                                                | <b>A1</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| (iii)  | Correct or correct <b>FT</b> substitution of 0, <i>their</i> $-5$ seen in $\left[ \frac{x^4}{4} + \frac{4x^3}{3} - \frac{5x^2}{2} + 5x \right]_{\text{their } -5}^0$ | <b>M1</b> | dependent on at least <b>B1</b> in (i)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|        | Correct or correct <b>FT</b> substitution of <i>their</i> 1, 0 seen in $\left[ \frac{x^4}{4} + \frac{4x^3}{3} - \frac{5x^2}{2} + 5x \right]_0^{\text{their } 1}$     | <b>M1</b> | dependent on at least <b>B1</b> in (i)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|        | <i>their</i> $\frac{1175}{12} - \text{their } OEAB + \text{their } OBCD - \text{their } \frac{49}{12}$ oe                                                            | <b>M1</b> | for the strategy needed to combine the areas; may be in steps;<br>$97.91\dot{6} - 25 + 5 - 4.08\dot{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|        | $\frac{886}{12}$ oe or $73\frac{5}{6}$ oe or $73.8\dot{3}$ rot to 3 or more sig figs                                                                                 | <b>A1</b> | all method steps must be seen; not from wrong working<br><br>If M0 then allow <b>SC3</b> for<br>$\int_{-5}^0 (x^3 + 4x^2 - 5x) dx - \int_0^1 (x^3 + 4x^2 - 5x) dx \text{ oe}$ $= \left[ \frac{x^4}{4} + \frac{4x^3}{3} - \frac{5x^2}{2} \right]_{-5}^0 - \left[ \frac{x^4}{4} + \frac{4x^3}{3} - \frac{5x^2}{2} \right]_0^1$ $= \left[ 0 - \left( \frac{625}{4} - \frac{500}{3} - \frac{125}{2} \right) \right] - \left[ \left( \frac{1}{4} + \frac{4}{3} - \frac{5}{2} \right) - 0 \right]$ $= \frac{443}{6} \text{ oe}$<br>or <b>SC2</b> for<br>$\int_{\text{their } (-5)}^0 (x^3 + 4x^2 - 5x) dx - \int_0^{\text{their } 1} (x^3 + 4x^2 - 5x) dx \text{ oe}$ $= \left[ \frac{x^4}{4} + \frac{4x^3}{3} - \frac{5x^2}{2} \right]_{\text{their } (-5)}^0 - \left[ \frac{x^4}{4} + \frac{4x^3}{3} - \frac{5x^2}{2} \right]_0^{\text{their } 1}$ $= [F(0) - F(\text{their } (-5))] - [F(\text{their } 1) - F(0)]$ |

**317****May/June 2017 · Paper 23 · Q7**Differentiate with respect to  $x$ ,

(i)  $(1 + 4x)^{10} \cos x$ , [4]

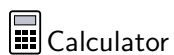
(ii)  $\frac{e^{4x-5}}{\tan x}$ . [4]



**317** May/June 2017 · Paper 23 · Q7

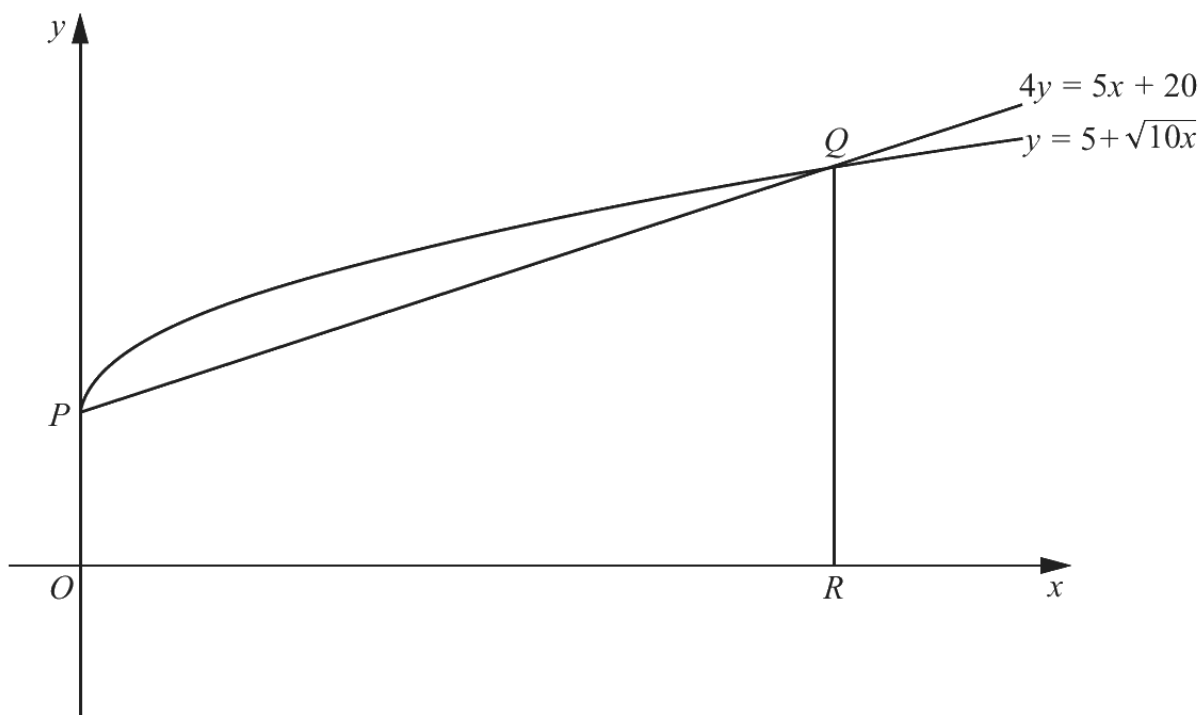
| Question | Answer                                                                                                                                   | Marks     | Guidance                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| 59 (i)   | $k(1+4x)^9$                                                                                                                              | <b>M1</b> |                                                                                                                                   |
|          | $4 \times 10(1+4x)^9$ or better                                                                                                          | <b>A1</b> |                                                                                                                                   |
|          | $(1+4x)^{10}(\text{their} - \sin x) + \cos x(\text{their}(4 \times 10 \times (1+4x)^9))$                                                 | <b>M1</b> | clearly applies product rule                                                                                                      |
|          | $(1+4x)^{10}(-\sin x) + \cos x(4 \times 10 \times (1+4x)^9)$                                                                             | <b>A1</b> | all correct                                                                                                                       |
| (ii)     | $\frac{d}{dx}(e^{4x-5}) = 4e^{4x-5}$ soi                                                                                                 | <b>B1</b> |                                                                                                                                   |
|          | $\frac{d}{dx}(\tan x) = \sec^2 x$ soi                                                                                                    | <b>B1</b> |                                                                                                                                   |
|          | clearly applies correct form of quotient rule<br>$\frac{\tan x(\text{their } 4e^{4x-5}) - e^{4x-5}(\text{their } \sec^2 x)}{(\tan x)^2}$ | <b>M1</b> | or correct form of product rule to<br>$e^{4x-5}(\tan x)^{-1}$<br>$4e^{4x-5}(\tan x)^{-1} + e^{4x-5}(\tan x)^{-2} \times \sec^2 x$ |
|          | $\frac{\tan x(4e^{4x-5}) - e^{4x-5}(\sec^2 x)}{(\tan x)^2}$ isw                                                                          | <b>A1</b> | all correct                                                                                                                       |

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Calculator

May/June 2017 · Paper 23 · Q11



The diagram shows part of the curve  $y = 5 + \sqrt{10x}$  and the line  $4y = 5x + 20$ . The line and curve intersect at the points  $P(0, 5)$  and  $Q$ . The line  $QR$  is parallel to the y-axis.

- (i) Find the coordinates of  $Q$ .

[4]

- (ii) Find the area of the shaded region. You must show all your working.

[6]

**318** May/June 2017 · Paper 23 · Q11

|        |                                                                                                                   |           |                                                                                                                                |
|--------|-------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------|
| 60 (i) | $5 + \sqrt{10x} = \frac{5x+20}{4} \rightarrow \cancel{20} + 4\sqrt{10x} = 5x + \cancel{20}$                       | <b>M1</b> | or better; equates and solves as far as clearing the fraction                                                                  |
|        | $\left[ \frac{x}{\sqrt{x}} \right] \sqrt{x} = \frac{4\sqrt{10}}{5}$ oe                                            | <b>M1</b> | Simplifies as far as $\sqrt{x} = \dots$                                                                                        |
|        | $x = 6.4$ cao                                                                                                     | <b>A1</b> | squares and simplifies to 6.4                                                                                                  |
|        | $[y = ]13$                                                                                                        | <b>B1</b> |                                                                                                                                |
| (ii)   | (area of trapezium = ) <i>their</i> 57.6                                                                          | <b>B1</b> | <b>FT</b> $x = \text{their } 6.4, y = \text{their } 13$ using any valid method                                                 |
|        | $\int_0^{6.4} (5 + \sqrt{10x}) dx$                                                                                | <b>M1</b> |                                                                                                                                |
|        | $\int (10x)^{\frac{1}{2}} dx = k (10x)^{\frac{3}{2}}$ or                                                          | <b>M1</b> | or $\int \sqrt{10x^{\frac{1}{2}}} dx = k \sqrt{10} (x)^{\frac{3}{2}}$                                                          |
|        | $\left[ 5x + \frac{2(10x)^{\frac{3}{2}}}{3 \times 10} \right]$                                                    | <b>A1</b> | or $\left[ 5x + \frac{2(10)^{\frac{1}{2}} (x)^{\frac{3}{2}}}{3} \right]$                                                       |
|        | <i>their</i> $\left[ 5(6.4) + \frac{2(10 \times 6.4)^{\frac{3}{2}}}{3 \times 10} \right] - \text{their } 57.6$ oe | <b>M1</b> | limits used correctly or correct <b>FT</b> and subtraction of trapezium;<br><i>their</i> $\frac{992}{15} - \text{their } 57.6$ |
|        | $\frac{128}{15}$ or 8.53 oe                                                                                       | <b>A1</b> | allow 8.5333333.... rot to 4 or more sf                                                                                        |

**319**

Calculator

**Oct/Nov 2016 · Paper 21 · Q4**

The number of bacteria,  $N$ , present in a culture can be modelled by the equation  $N = 7000 + 2000e^{-0.05t}$ , where  $t$  is measured in days. Find

(i) the number of bacteria when  $t = 10$ , [1]

(ii) the value of  $t$  when the number of bacteria reaches 7500, [3]

(iii) the rate at which the number of bacteria is decreasing after 8 days. [3]

**320**

Calculator

**Oct/Nov 2016 · Paper 21 · Q5**

The curve with equation  $y = x^3 + 2x^2 - 7x + 2$  passes through the point  $A(-2, 16)$ . Find

(i) the equation of the tangent to the curve at the point  $A$ , [3]

(ii) the coordinates of the point where this tangent meets the curve again. [5]

**319** Oct/Nov 2016 · Paper 21 · Q4

| Question | Answer                                                                                                                                                                  | Marks                                   | Part Marks                                                                                                    |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 16 (i)   | $t = 10 \rightarrow N = 7000 + 2000e^{-0.5}$<br>$= 8213$ or $8210$                                                                                                      | <b>B1</b>                               | Do not accept non integer responses.                                                                          |
| (ii)     | $N = 7500 \rightarrow 7500 = 7000 + 2000e^{-0.05t}$<br>$e^{-0.05t} = \frac{500}{2000}$<br>$-0.05t = \ln 0.25 \rightarrow t = \frac{\ln 0.25}{-0.05}$<br>$= 27.7$ (days) | <b>M1</b><br><br><b>M1</b><br><b>A1</b> | insert and make $e^{-0.05t}$ subject<br><br>take logs and make $t$ the subject<br>awrt 27.7                   |
| (iii)    | $\frac{dN}{dt} = -100e^{-0.05t}$<br>$t = 8 \rightarrow \frac{dN}{dt} = \pm 67$ (.0)                                                                                     | <b>M1</b><br><b>A1</b><br><b>A1</b>     | $ke^{-0.05t}$ where $k$ is a constant<br>$k = -100$ or $-0.05 \times 2000$<br>awrt $\pm 67$ mark final answer |

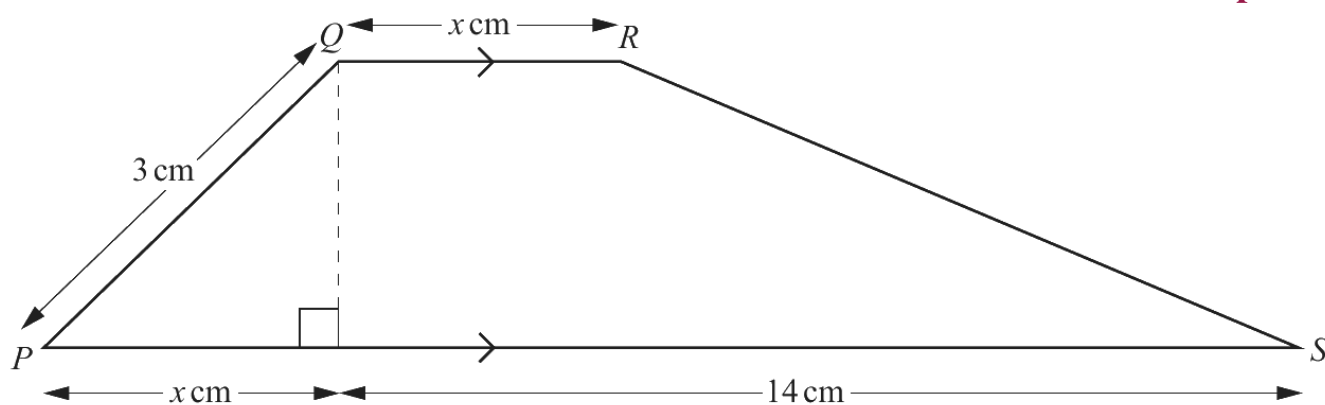
**320** Oct/Nov 2016 · Paper 21 · Q5

|        |                                                                                                                                                                      |                                                    |                                                                                                                                                                                                                                            |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17 (i) | $\frac{dy}{dx} = 3x^2 + 4x - 7$<br>$x = -2 \rightarrow \frac{dy}{dx} = 12 - 8 - 7 = -3$<br>Equation of tangent :<br>$\frac{y-16}{x+2} = -3 \rightarrow y = -3x + 10$ | <b>B1</b><br><br><b>M1</b><br><br><b>A1</b>        | insert $x = -2$ into <i>their</i> gradient and use $(-2, 16)$ and <i>their</i> gradient of tangent in equation of line.                                                                                                                    |
| (ii)   | Tangent cuts curve again<br>$x^3 + 2x^2 - 7x + 2 = -3x + 10$<br>$x^3 + 2x^2 - 4x - 8 = 0$<br>$(x+2)(x+2)(x-2) = 0$<br>$x = 2, \quad y = 4$                           | <b>M1</b><br><b>A1</b><br><b>M1</b><br><b>A1A1</b> | equate curve and <i>their</i> linear answer from (i).<br><br>factorise: $(x \pm 2)$ and a two or three term quadratic is sufficient. Allow long division withhold final <b>A1</b> if $(2, 4)$ not clearly identified as their sole answer. |

321

 Calculator

Oct/Nov 2016 · Paper 21 · Q7



- (i) Show that the area,  $A \text{ cm}^2$ , of the trapezium  $PQRS$  is given by  $A = (7 + x)\sqrt{9 - x^2}$ . [2]

- (ii) Given that  $x$  can vary, find the stationary value of  $A$ . [7]

**321** Oct/Nov 2016 · Paper 21 · Q7

| Question | Answer                                                                                                                                                                                                                                           | Marks                                                                                  | Part Marks                                                                                                                                                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18 (i)   | $h = \sqrt{9 - x^2}$ $A = \frac{\sqrt{9 - x^2}}{2}(14 + x + x) = \sqrt{9 - x^2}(7 + x)$                                                                                                                                                          | <b>B2/1/0</b>                                                                          | Must be clear that $\sqrt{9 - x^2}$ is the height of the trapezium. $14 + 2x$ oe must be seen AG                                                                                                                                                                                |
| (ii)     | $\frac{dA}{dx} = \sqrt{9 - x^2} + (7 + x) \frac{1}{2} (9 - x^2)^{-0.5} \times -2x$ $\frac{dA}{dx} = 0 \rightarrow 9 - x^2 = 7x + x^2$ $2x^2 + 7x - 9 = 0$ $x = 1$ $A = 16\sqrt{2} \text{ or } 8\sqrt{8} \text{ or } \sqrt{512} \text{ or } 22.6$ | <b>M1</b><br><b>A2/1/0</b><br><br><b>M1</b><br><b>A1</b><br><br><b>A1</b><br><b>A1</b> | product rule on correct function<br>minus 1 each error, allow unsimplified.<br><br>equate to 0 and simplify to a linear or quadratic equation.<br>correct three term quadratic obtained<br><br>Extra positive answer loses penultimate <b>A1</b> .<br>ignore negative solution. |

322

 Calculator

Oct/Nov 2016 • Paper 21 • Q8

The function  $f(x)$  is given by  $f(x) = \frac{3x^3 - 1}{x^3 + 1}$  for  $0 \leq x \leq 3$ .

(i) Show that  $f'(x) = \frac{kx^2}{(x^3 + 1)^2}$ , where  $k$  is a constant to be determined. [3]

(ii) Find  $\int \frac{x^2}{(x^3 + 1)^2} dx$  and hence evaluate  $\int_1^2 \frac{x^2}{(x^3 + 1)^2} dx$ . [4]

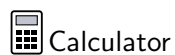
(iii) Find  $f^{-1}(x)$ , stating its domain. [4]



**322** Oct/Nov 2016 · Paper 21 · Q8

|        |                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19 (i) | $f'(x) = \frac{(x^3+1)9x^2 - (3x^3-1)3x^2}{(x^3+1)^2}$ $= \frac{12x^2}{(x^3+1)^2}$                                                                                     | <b>M1</b><br><b>A1</b> quotient rule or product rule<br>all correct<br><br><b>A1</b> www beware $9x^6 - 9x^6$ gets <b>A0</b>                                                                                                                                                                                   |
| (ii)   | $\int_1^2 \frac{x^2}{(x^3+1)^2} dx = \frac{1}{12} \left[ \frac{3x^3-1}{x^3+1} \right]_1^2$ $= \frac{1}{12} \left[ \frac{23}{9} - \frac{2}{2} \right]$ $= \frac{7}{54}$ | <b>M1</b><br>$c \times \frac{3x^3-1}{x^3+1}$<br><b>A1</b> <b>FT</b> $c = \frac{1}{\text{their}12}$<br><br><b>DM1</b> top limit – bottom limit in <i>their</i> integral.<br><br><b>A1</b> or 0.130 or 0.1296 or 0.12                                                                                            |
| (iii)  | $x = \frac{3y^3-1}{y^3+1}$ $y^3 = \frac{x+1}{3-x}$ $f^{-1}(x) = \sqrt[3]{\frac{x+1}{3-x}}$<br>Domain : $-1 \leq x \leq 2\frac{6}{7}$                                   | <br><br><br><b>B1</b> make $y^3$ or $x^3$ the subject<br><br><br><br><b>B1</b> <b>FT</b> take cube root (as long as $y^3$ or $x^3$ equals a fraction with terms in $x$ or $y$ only) oe<br><b>B1</b> <b>FT</b> change $x$ and $y$ – can be done at any time<br><b>B1</b> Allow upper limit of 2.86 . Do not isw |

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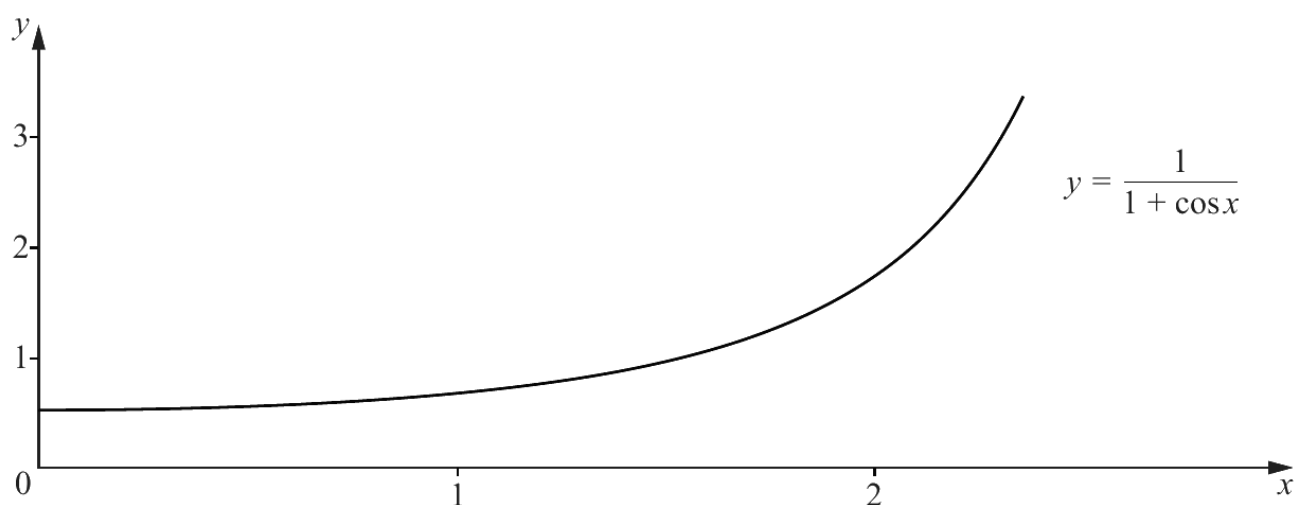
Calculator

Oct/Nov 2016 · Paper 23 · Q3

(i) Show that  $\frac{d}{dx}\left(\frac{\sin x}{1 + \cos x}\right) = \frac{1}{1 + \cos x}$ .

[4]

(ii)



The diagram shows part of the graph of  $y = \frac{1}{1 + \cos x}$ . Use the result from part (i) to find the area enclosed by the graph and the lines  $x = 0$ ,  $x = 2$  and  $y = 0$ .

[2]

**323** Oct/Nov 2016 · Paper 23 · Q3

|        |                                                                                                                                                                                                                    |                                                  |                                                                                                                                                   |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 68 (i) | $\frac{d}{dx} \left( \frac{\sin x}{1 + \cos x} \right) = \frac{(1 + \cos x) \cos x + \sin x \sin x}{(1 + \cos x)^2}$ $= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2}$ $= \frac{1 + \cos x}{(1 + \cos x)^2}$ | <b>M1</b><br><b>A1</b><br><b>B1</b><br><b>A1</b> | Quotient Rule (or Product Rule from<br>$(\sin x)(1 + \cos x)^{-1}$ )<br>correct unsimplified<br>use of $\sin^2 x + \cos^2 x = 1$ oe<br>completion |
| (ii)   | $\int_0^2 \left( \frac{1}{1 + \cos x} \right) dx = \left[ \frac{\sin x}{1 + \cos x} \right]_0^2$ awrt 1.56                                                                                                         | <b>M1</b><br><b>A1</b>                           | correct integrand                                                                                                                                 |

**324**

Calculator

**Oct/Nov 2016 · Paper 23 · Q6**

A curve has equation  $y = 7 + \tan x$ . Find

(i) the equation of the tangent to the curve at the point where  $x = \frac{\pi}{4}$ , [4]

(ii) the values of  $x$  between 0 and  $\pi$  radians for which  $\frac{dy}{dx} = y$ . [4]

**324** Oct/Nov 2016 · Paper 23 · Q6

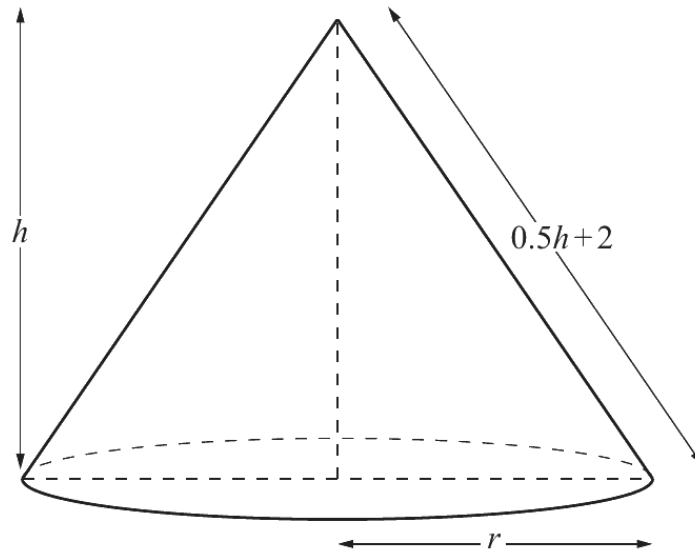
|        |                                                                                                                                                                                                                                                          |                                                      |                                                                                                                                                               |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 69 (i) | $\frac{dy}{dx} = \sec^2 x$<br>$x = \frac{\pi}{4} \rightarrow \frac{dy}{dx} = \sec^2 \frac{\pi}{4} = 2$<br>$y = 8$<br>Equation of tangent $\frac{y-8}{x-\frac{\pi}{4}} = 2$<br>$(4-2y = \pi - 16, y = 2x + 6.429\dots,$<br>$\frac{\pi}{4} = 0.7853\dots)$ | <b>B1</b><br><br><b>B1</b><br><b>B1</b><br><b>B1</b> | evaluated                                                                                                                                                     |
| (ii)   | $\sec^2 x = \tan x + 7$<br>$\tan^2 x - \tan x - 6 = 0$ oe<br>$(\tan x - 3)(\tan x + 2) = 0$<br>$\tan x = 3$ or $\tan x = -2$<br>$x = 1.25, \quad 2.03$                                                                                                   | <b>M1</b><br><br><b>M1</b><br><b>A1A1</b>            | use $\sec^2 x = 1 + \tan^2 x$ to obtain a 3 term quadratic in $\tan x$<br><br>solve three term quadratic for $\tan x$<br>extras in range lose final <b>A1</b> |

325

 Calculator

Oct/Nov 2016 • Paper 23 • Q7

In this question all lengths are in metres.



A conical tent is to be made with height  $h$ , base radius  $r$  and slant height  $0.5h + 2$ , as shown in the diagram.

- (i) Show that  $r^2 = 2h + 4 - 0.75h^2$ . [2]

The volume of the tent,  $V$ , is given by  $\frac{1}{3}\pi r^2 h$ .

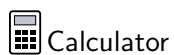
- (ii) Given that  $h$  can vary find, correct to 2 decimal places, the value of  $h$  which gives a stationary value of  $V$ . [5]

- (iii) Determine the nature of this stationary value. [2]

**325** Oct/Nov 2016 · Paper 23 · Q7

|        |                                                                                                                                                                                                 |           |                                                                  |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------|
| 70 (i) | $r^2 + h^2 = (0.5h + 2)^2$ oe<br>$r^2 = 0.25h^2 + 2h + 4 - h^2$<br>$r^2 = 2h + 4 - 0.75h^2$                                                                                                     | <b>M1</b> |                                                                  |
|        |                                                                                                                                                                                                 | <b>A1</b> | correct expansion and $r^2$ subject and completion www AG        |
| (ii)   | $V = \frac{1}{3}\pi r^2 h = \frac{\pi}{3}(2h^2 + 4h - 0.75h^3)$<br>$\frac{dV}{dh} = \frac{\pi}{3}(4h + 4 - 2.25h^2)$<br>$\frac{dv}{dh} = 0 \rightarrow 2.25h^2 - 4h - 4 = 0$<br>$h = 2.49$ only | <b>B1</b> | any correct form in terms of $h$ only                            |
|        |                                                                                                                                                                                                 | <b>M1</b> | differentiate $V$                                                |
|        |                                                                                                                                                                                                 | <b>A1</b> | correct differentiation                                          |
|        |                                                                                                                                                                                                 | <b>M1</b> | equate to 0 and solve 3 term quadratic                           |
|        |                                                                                                                                                                                                 | <b>A1</b> | cao                                                              |
| (iii)  | $\frac{d^2V}{dh^2} = \frac{\pi}{3}(4 - 4.5h)$ when $h = 2.49$<br>$(-7.545\dots) < 0$ so maximum                                                                                                 | <b>M1</b> | differentiate <i>their</i> 3 term $\frac{dV}{dh}$ and substitute |
|        |                                                                                                                                                                                                 | <b>A1</b> | <i>their</i> $h$<br>draw correct conclusion www                  |

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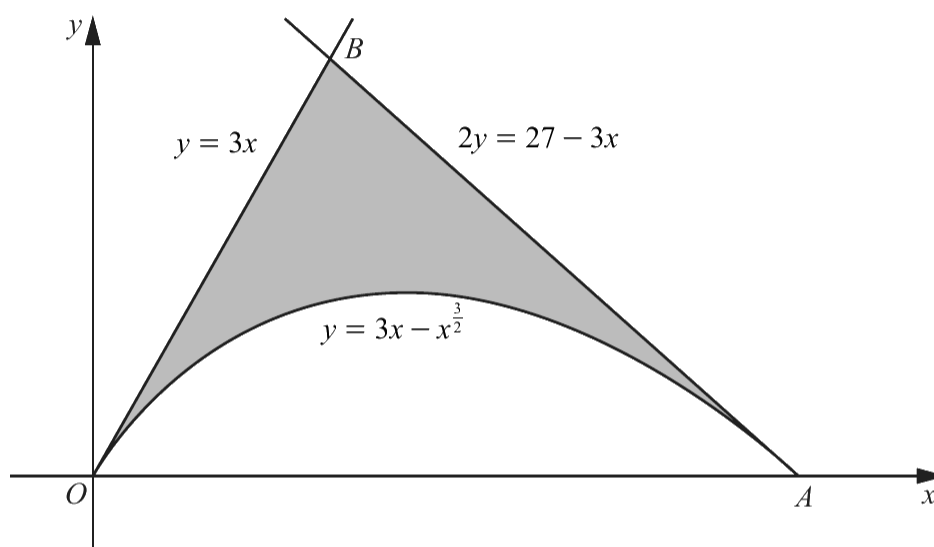
Calculator

May/June 2016 · Paper 21 · Q11

(i) Find  $\int (3x - x^{\frac{3}{2}}) dx$ .

[2]

The diagram shows part of the curve  $y = 3x - x^{\frac{3}{2}}$  and the lines  $y = 3x$  and  $2y = 27 - 3x$ . The curve and the line  $y = 3x$  meet the  $x$ -axis at  $O$  and the curve and the line  $2y = 27 - 3x$  meet the  $x$ -axis at  $A$ .



(ii) Find the coordinates of  $A$ .

[1]

(iii) Verify that the coordinates of  $B$  are  $(3, 9)$ .

[1]

(iv) Find the area of the shaded region.

[4]



**326** May/June 2016 · Paper 21 · Q11

|       |                                                                                                                                                                                                                                                                                              |                                                              |                                                                                                                                                                                                                                                         |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (i) | $\frac{3x^2}{2} - \frac{2x^{3/2}}{5} (+c)$ isw                                                                                                                                                                                                                                               | <b>B1+B1</b>                                                 |                                                                                                                                                                                                                                                         |
| (ii)  | (9, 0) oe                                                                                                                                                                                                                                                                                    | <b>B1</b>                                                    | Not just $x = 9$                                                                                                                                                                                                                                        |
| (iii) | Substitute (3, 9) into <b>both</b> lines<br><br>Or solves simultaneously ( $6x = 27 - 3x$ oe) to get $x = 3, y = 9$                                                                                                                                                                          | <b>B1</b>                                                    | $3 \times 3 = 9$ and $\frac{27 - 3 \times 3}{2} = 9$                                                                                                                                                                                                    |
| (iv)  | $\left[ \text{Area } AOB = \right] \frac{1}{2} \times 9 \times 9$ oe $\left( \frac{81}{2} \text{ or } 40.5 \right)$<br><br>$\text{their} \left[ \frac{3(9)^2}{2} - \frac{2(9)^{3/2}}{5} \right] - [0]$ (= 24.3)<br><br>$\text{their} \frac{81}{2} - \text{their} \frac{243}{10}$<br><br>16.2 | <b>M1</b><br><br><b>M1</b><br><br><b>M1</b><br><br><b>A1</b> | Uses <i>their</i> (ii). May split into 2 triangles (13.5 and 27). May integrate. Must be a complete method.<br><br>lower limit may be omitted but must be correct if seen<br><br>must be from genuine attempts at area of triangle and area under curve |

**327** Calculator**May/June 2016 · Paper 21 · Q12**

A curve has equation  $y = \frac{2x-5}{x-1} - 12x$ .

(i) Find  $\frac{dy}{dx}$ . [3]

(ii) Find  $\frac{d^2y}{dx^2}$ . [2]

(iii) Find the coordinates of the stationary points of the curve and determine their nature. [5]

**328** Calculator**May/June 2016 · Paper 22 · Q2**

Variables  $x$  and  $y$  are related by the equation  $y = \frac{5x-1}{3-x}$ .

(i) Find  $\frac{dy}{dx}$ , simplifying your answer. [2]

(ii) Hence find the approximate change in  $x$  when  $y$  increases from 9 by the small amount 0.07. [3]

## 327 May/June 2016 · Paper 21 · Q12

|                                                                                                                                                                                                                                                                                                                                                                                                                 |  |                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>2 (i)</p> $\left[ \frac{dy}{dx} \right] = \frac{2(x-1) - (2x-5)}{(x-1)^2}$ <p>– 12 isw</p> <p><b>ALT using</b> <math>y = \frac{-12x^2 + 14x - 5}{x-1}</math></p> <p>–24x + 14</p> $\left[ \frac{dy}{dx} \right] = \frac{(x-1)(-24x+14) - (-12x^2 + 14x - 5)}{(x-1)^2}$ <p>(ii)</p> $\left[ \frac{d^2y}{dx^2} \right] = k(x-1)^{-3}$ <p><math>k = -6</math> isw</p>                                           |  | <p><b>M1A1</b></p> <p><b>B1</b></p> <p><b>B1</b></p> <p><b>M1</b></p> <p><b>A1FT</b></p> <p><b>M1</b></p> <p><b>A1</b></p> | <p>Allow slips in <math>\frac{du}{dx}</math> and <math>\frac{dv}{dx}</math> but must be explicit.<br/>Allow <math>(x-1)^2 = x^2 - 2x + 1</math></p> <p><b>FT</b> on their derivative of 3 term quadratic</p> <p>No additional terms</p>                                                                                                                                                                                            |
| <p>(iii)</p> <p>their <math>\left[ \frac{3}{(x-1)^2} - 12 \right] = 0</math> and find a value for <math>x</math></p> <p><math>x = 0.5</math> and <math>x = 1.5</math></p> <p><math>y = 2</math> and <math>y = -22</math></p> <p><math>\frac{-6}{(-0.5)^3} &gt; 0</math> therefore min when <math>x = 0.5</math> oe</p> <p><math>\frac{-6}{(0.5)^3} &lt; 0</math> therefore max when <math>x = 1.5</math> oe</p> |  | <p><b>M1</b></p> <p><b>A1</b></p> <p><b>A1</b></p> <p><b>B1</b></p> <p><b>B1</b></p>                                       | <p><math>12x^2 - 24x + 9 = 0</math> oe<br/><math>(2x-3)(2x-1) = 0</math> oe</p> <p>if <b>A0 A0</b> then <b>A1</b> for a correct <math>(x, y)</math> pair</p> <p>or <math>\left[ \frac{-6}{(-0.5)^3} \right] = 48</math> therefore min when <math>x = 0.5</math> oe</p> <p>or <math>\left[ \frac{-6}{(0.5)^3} \right] = -48</math> therefore max when <math>x = 1.5</math> oe</p> <p><b>M1A1</b> is possible from other methods</p> |

**328** May/June 2016 · Paper 22 · Q2

|        |                                                                                                                                                                                                                                                |                                                                                                                                                                                                           |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 32 (i) | <p>Clear attempt at quotient rule or equivalent product rule</p> $\left[ \frac{dy}{dx} = \right] \frac{14}{(3-x)^2}$ <p>or <math>\left[ \frac{dy}{dx} = \right] \frac{14}{x^2 - 6x + 9}</math> cao</p> <p>or correct simplified equivalent</p> | <p><b>M1</b> condone omission of brackets</p> <p><b>A1</b> allow recovery from bracketing errors or omissions if implied in correct work to the correct answer</p>                                        |
| (ii)   | <p><math>[y = 9] x = 2</math></p> $\frac{0.07}{\delta x} \approx \left( \text{their } \frac{dy}{dx} \Big _{x=2} \right) \text{oe}$ <p>0.005 oe</p>                                                                                             | <p><b>B1</b></p> <p><b>M1</b> condone <math>\frac{0.07}{\delta x} = \left( \text{their } \frac{dy}{dx} \Big _{x=2} \right)</math></p> <p><b>A1</b> not from wrong working; answer only does not score</p> |

329

 Calculator

May/June 2016 · Paper 22 · Q9

(a) Find  $\int \frac{x^3 + x^2 + 1}{x^2} dx$ .

[3]

(b) (i) Find  $\int \sin(5x + \pi) dx$ .

[2]

(ii) Hence evaluate  $\int_{-\frac{\pi}{5}}^0 \sin(5x + \pi) dx$ .

[2]

**329** May/June 2016 · Paper 22 · Q9

|        |                                                                                                                                                   |                            |                                                                                                                                                                        |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 33 (a) | $\frac{x^2}{2} + x - \frac{1}{x} (+c)$ isw                                                                                                        | <b>B3</b>                  | <b>B1</b> for each term<br>allow $\frac{x^2}{2} + x + \frac{x^{-1}}{-1} (+c)$ isw for <b>B3</b>                                                                        |
| (b)(i) | $k \cos(5x + \pi)$ where $k < 0$<br>or $\frac{\cos(5x + \pi)}{5}$<br>$\frac{-\cos(5x + \pi)}{5} (+c)$                                             | <b>M1</b><br><br><b>A1</b> |                                                                                                                                                                        |
| (ii)   | $\frac{-\cos(5(0) + \pi)}{5} - \frac{-\cos(5(-\frac{7}{5}) + \pi)}{5}$<br>or $\frac{-\cos(\pi)}{5} - \left( \frac{-\cos(0)}{5} \right)$<br>0.4 oe | <b>M1</b><br><br><b>A1</b> | correct substitution of the given limits into <i>their</i> expression of the form $k \cos(5x + \pi)$ ,<br>dep on <b>M1</b> in (b)(i)<br><br>answer only does not score |